

Graded Multiplication Operators for Random-Matrix Moments

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Abstract

Polynomial random-matrix moments admit an exact representation by deterministic multiplication operators acting on the original probability space. We develop the estimates that make this representation useful: shared-factor Grams turn dimension–occupation products into additive bounds, connected-creation commutators preserve cancellations, and positive selector and collective row contractions retain coefficient geometry. Occupied coefficient spans and ordered child Grams specify what these estimates preserve through a further interaction. The organizing principle is that an exact representation becomes an effective moment bound by choosing a preserved domain and retaining the coefficient information that remains valid under the interactions still to come. A descent test decides when a retained description supports a further interaction and prices it by a Loewner inequality; live registers that keep a label through repeated visits and signed conditional cores with a certified returning remainder are exact and approximate instances. An ordered renewal identity retains the vacuum covariance spectrum before scalar relaxation and yields effective-rank refinements of compatible block-majorant bounds.

The same moment identity permits two admissible domains: complete reachable graded spaces and cones of actual walk polynomials preserved by multiplication. For independent entries with regular moment growth, sign parity gives orthogonal sectors while magnitude powers retain repeated visits to the same entry. Cauchy–Schwarz and moment doubling give a positive-polynomial comparison with factor $\sqrt{3}\alpha$ per degree; reconstruction into the random polynomial precedes Hölder’s inequality, and the strong/weak comparison that the thin-axis estimate needs is proved by an independent copy rather than quoted. Thin-row and thin-column contractions then yield the local moment estimate at logarithmic order. A deterministic deletion argument on the original rectangular matrix, with a cube-threshold screening and a centered Gram remainder, converts it into the sharp two-sided expected-norm scale expressed by variances and a shared-index deletion functional; the lower comparison is the known result.

The motivating applications are SparseStack, dense sign graph matrices, and two-round unitary sampling. Their estimates recover additive dimension–occupation dependence, the graph logarithmic exponent, and $O(\epsilon^{-2}[d + \log^2(d/\delta)])$ sampled rows. We distinguish coefficient, domain, and vacuum spectral-weight losses, retain the cost of distributional transfer, and state which continuations each contraction allows. Exact fourth-order and aggregate-moment identities supply original-law benchmarks, and explicit obstructions fix what a retained descriptor cannot discard. Complete proofs of the four application theorems are given here, the two sketching theorems by the direct occupation argument and the quadratic coefficient argument; earlier versions of the three original proof cores and of the upper half of the independent-entry theorem are formalized in Lean 4, and their records are identified. The analytic extensions are not asserted to have a complete formal verification.

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1 Introduction

1.1 The probabilistic question and the deterministic question

Let Z be a self-adjoint random $D \times D$ matrix with the moments used below finite. A standard route to a norm bound is

$$\mathbb{P}\{\|Z\| > t\} \leq t^{-2q} \mathbb{E} \operatorname{tr} Z^{2q}, \quad q \geq 1. \quad (1)$$

For a polynomial random matrix, expanding the trace produces a large sum of products of random variables and matrix coefficients. Independence eliminates many terms. The difficulty is to sum the surviving terms while preserving the relations that make their total small.

Put the scalar randomness into a probability space $\mathbf{X}$, regard Z as a matrix-valued function on that space, and let $\mathcal{M}_Z$ act by multiplication on $\mathbb{C}^D \otimes L^2(\mathbf{X})$. The constant function $\Omega = 1$ is a distinguished unit vector, called the *vacuum*. The exact identity

$$\mathbb{E} \operatorname{tr} Z^{2q} = \sum_{a=1}^D \|\mathcal{M}_Z^q(e_a \otimes \Omega)\|^2 \quad (2)$$

expresses a moment through the states reached by repeated multiplication. On a finite probability space, the full multiplication norm records the worst outcome. For unbounded entries that operator need not be bounded at all. Neither fact prevents us from estimating the finite sequence of vectors in (2).

A grading is an orthogonal decomposition into spaces indexed by a nonnegative integer. If multiplication increases grade by at most Δ , a walk of length q from the vacuum reaches only grades at most Δq . There are two useful domains on which to estimate the resulting maps. One is the entire permitted graded space, with any exact symmetry restrictions. The other is a family of polynomials preserved by the actual multiplication steps. An estimate on such a family is sufficient when it contains every vacuum walk.

Both choices preserve the original probability law. Coordinate multiplication operators commute. Creation, annihilation, and occupation operators express those same commuting multipliers in graded coordinates. Repeated visits use the same original random variable; there is no replacement by independent copies.

1.2 Preserved relations and two admissible domains

The representation and the contraction have different roles. Orthogonal polynomials and product spaces organize multiplication exactly. The useful bound comes from a relation among its coefficients: a shared external factor, connectivity of a graph fragment, positivity of effects, or orthogonality of a transform. A finite alphabet alone does not supply such a relation.

Three applications motivate the coefficient estimates. In fully independent SparseStack, a shared external factor turns a dimension–occupation product into a sum. In dense graph matrices, disjoint terms cancel in a creation commutator, leaving a cost governed by connected overlaps. In two-round row sampling, a collective coefficient map retains the relation between two sign registers before their occupation costs are counted. The commuting-transport identity in Section 3.2 connects the shared-factor and collective-row calculations through their actual cross Grams.

The same principle also governs a subsequent interaction. A projection insertion is measured by the weighted variance of its transported projections. Several exceptional coefficient directions require the span of the occupied coefficients, rather than one scalar marker. The cyclic examples show why original edge-coefficient products must stay coupled

to the Gram of the remaining registers. These objects give positive bounds that survive congruence on the stated input boundary. An inserted row map can change the output Gram, so its compatibility must be checked at that boundary. Section 7 turns this check into a test: a compressed description supports a proposed update precisely when the update kills the kernel of its encoding, and the quantitative price is a Loewner inequality between the description Grams before and after the interaction. Live registers and signed conditional cores are the exact and the approximate ways of passing that test.

Independent entries with regular moment growth reveal the second admissible domain. Write a symmetric entry as $Y_e = \varepsilon_e Z_e$ in law, with an independent sign and magnitude, and assume $\|Y_e\|_{L_{2p}} \leq \alpha \|Y_e\|_{L_p}$ for $p \geq 1$. Parity is an orthogonal coordinate, while every magnitude power stays inside its coefficient function. Creation raises parity grade and annihilation lowers it; both increase ordinary polynomial degree. Exact occupation counts restrict each interaction to few original rows or columns. The vacuum walks have nonnegative coefficients in the ordinary monomials of the original entries, and the parity projections and multiplication preserve this cone.

On that cone the scalar Cauchy–Schwarz inequality and moment doubling lead to the comparison

$$\|F\|_{L_4(\ell_2)} \leq (\sqrt{3}\alpha)^{\deg F} \|F\|_{L_2(\ell_2)}.$$

Here positivity concerns monomial coefficients, not the realized values of F . The conditional Fourier transform in the signs is an L_2 isometry only. Hölder is applied to the reconstructed random polynomial on the original joint probability space; no equality of higher L_p norms with occupation-amplitude norms is used. The strong/weak comparison needed by the thin-axis estimate is proved here using an independent copy of the coordinates. The Boolean hypercontractive input is classical: its L_2 -to- L_4 case is re-proved in Section 3.6, while the refined constants use the general theorem of Bonami and Beckner [55, 56] at the exponents actually compared. This cone estimate and the full-operator coefficient estimates are two forms of the same strategy: bound the domain visited by the moment while retaining the relation responsible for the contraction.

1.3 Four applications

Section 5 gives the precise hypotheses and constants for the following conclusions.

For fully independent real SparseStack, the sufficient scales are

$$m = O((d + \log(1/\delta))/\epsilon^2), \quad s = O(\log(d/\delta)/\epsilon).$$

The explicit reference constants use $q = \max\{1, \lceil \log_2(d/\delta) \rceil\}$; the direct estimates use $\log_4(d/\delta)$. Both prescriptions are proved here by the direct occupation argument of Appendix B; the coupled argument that first gave the reference constants is the formalized one. Constant comparisons retain these stated moment conventions and the applicable support model.

For the fixed dense sign graph model,

$$\mathbb{E}\|M_{H,n}\| = \Theta_H(n^{f(H)}(\log(2n))^{g(H)}).$$

A minimum separator determines $f(H)$, and its attached floating components determine $g(H)$. Xu, Fu, Qiao, and Tao give another proof of this expected-norm classification [30, Theorem 1.3]; our operator-based proof is in Appendix F. The graph and its boundary are fixed; the coefficient contraction does not remove their support hypotheses. Section 10.4 follows one shape, the overlapping triangle, into two other entry laws: for symmetric entries of regular moment growth the local moment theorem gives the sharp scale $\sqrt{n}\|X\|_{L_2[\log n]}$ with

explicit constants, and for sparse signed Bernoulli entries of density $\rho = n^{-\beta}$ an exact support-conditioned representation gives a phase diagram with a critical correction $\sqrt{\log n / \log \log n}$ at $\beta = \frac{1}{2}$, where the conditional neighborhood graphs reach constant mean degree.

For two-round row sampling, the Walsh theorem with cubic confidence allowance gives $O(\epsilon^{-2}[d + \log^3(d/\delta)])$ rows. The collective coefficient estimate gives the quadratic allowance $O(\epsilon^{-2}[d + \log^2(d/\delta)])$, capped at the ambient dimension. Since $\log^2 d = O(d)$, this is also $O(\epsilon^{-2}[d + \log^2(1/\delta)])$. Section 10 states the coherent-transform constants and the sampling conventions. The cubic statement identifies the formalized application; the quadratic and coherent estimates are analytic.

For an $n \times n$ matrix X with independent real centered entries and finite absolute moments of every order, assume $\|X_{ij}\|_{L_{2p}} \leq \alpha \|X_{ij}\|_{L_p}$ for $p \geq 1$. Set

$$M(X) = \max_i \left(\sum_j \mathbb{E} X_{ij}^2 \right)^{1/2} + \max_j \left(\sum_i \mathbb{E} X_{ij}^2 \right)^{1/2}$$

and

$$D(X) = \max_{1 \leq k \leq n} \min_{\substack{I \subset [n] \\ |I| \leq k}} \sup_{\|s\|_2, \|t\|_2 \leq 1} \left\| \sum_{i,j \notin I} X_{ij} s_i t_j \right\|_{L_{\log(k+1)}}.$$

Then Theorem 5.6 proves $\mathbb{E}\|X\| \asymp_\alpha M(X) + D(X)$, with the explicit upper constant $C_\alpha < 1.33 \cdot 10^6 \alpha^{10.63}$, and $418 \alpha^{10.63}$ for entries symmetric in law (Table 3 lists the trade-off between the prefactor and the exponent). At $\alpha = 1$ this covers weighted Rademacher matrices $(a_{ij} \varepsilon_{ij})$, for which Latała [42, Theorem 1.9] obtained the comparison up to a triple logarithm (Corollary 5.7), and Meller [33] has the general symmetric case up to a polyloglog factor. The same deterministic original set deletes both axes. The exponent $\log(k+1)$ includes $\log 2 < 1$, where L_p denotes a moment functional. The local positive-cone contraction is the method's contribution; a deterministic deletion argument on the original rectangular matrix, with a cube-threshold screening and a centered Gram remainder, turns it into the mean-norm estimate, and its upper half is formalized. The lower bound is the known result of Latała–Świątkowski, using disjoint witnesses on the same original matrix. This is an expected-norm comparison; no matching-quantile statement or inexpensive evaluation of the deletion functional is asserted.

What each application takes from here. The claims of this paper are theorem-specific. The paper organizes exact multiplication representations around domain-restricted and coefficient-sensitive estimates, and shows how those estimates produce particular sharp bounds, quantitative refinements and valid transfers; the representation itself, the trace method and the coefficient-flattening framework of Bandeira, Lucca, Nizić-Nikolac and van Handel [29] are established. For SparseStack, what is supplied here is the shared-factor Gram and the direct occupation proof with its constants (Appendix B); the coupled proof, which is the formalized one, is an earlier route to the same prescription. For dense sign graph matrices, Ahn, Medarametla, and Potechin already obtained the separator power of n , including isolated middle vertices, up to polylogarithmic factors [28, Theorems 2.24 and 6.1]. The systematic flattening approach of [29] supplies further comparisons. The sharp expected-norm classification also has the proof of Xu, Fu, Qiao, and Tao [30, Theorem 1.3]. Our contribution here is the specific operator argument: the connected creation commutator, critical factorization, and excursion lower argument (Appendix F), together with the one-shape analyses of Section 10.4. For two-round sampling, the collective coefficient estimate and the quadratic allowance are supplied here, and the cubic statement is the formalized one. For regular entries, the conjecture, the lower half and the deletion strategy are due to Latała and Świątkowski [48]; what is supplied here is the preserved-cone local theorem with explicit

constants and the elementary assembly of Section 5.4 and Appendix G. The descent test of Section 7 has established factorization precedents; its role here is to diagnose a proposed retained description, which is a matter of organization rather than a new inequality.

1.4 A map of the structural tools

Table 1 records the relations used in Section 3. The first four rows concern coefficient operators on their stated graded spaces; the last concerns the preserved family of walk polynomials. The comparisons in the table refer to those domains, not to interchangeable bounds for an arbitrary polynomial matrix. Figure 1 places both choices inside the common moment argument.

Principle	Coarse to structured	Relation preserved	Application	Sharpness or calibration
Shared-factor Gram	$d\ell$ to $d+\ell-1$; general $\tau + (\ell-1)\kappa$	Exact cross Grams of insertion maps	SparseStack	Schmidt-vector equality on labelled tensors
Connected creation	Separate product norms to a commutator recurrence	Disjoint terms cancel; overlaps are connected	Graph matrices	Matching graph exponent; elementary ladder model
Positive effects	Singleton bound qR to subset norm L_q	Positivity and actual sums over occupied selectors	Row sampling	Matrix Chernoff order; orthogonal effects
Collective coefficients	$d + 27q^3$ to $d + 12q^2$ in the two-round allowance	Parseval coefficients, collective maps, and a common reflection	Two-round transforms	Quadratic confidence exponent; explicit coefficient 12
Counts and positive coefficients	Ambient matrix to few rows or columns at degree q	Original parity, magnitude powers, and a preserved monomial cone	Independent entries	Mean norm $M(X) + D(X)$; explicit constants in α ; upper half formalized

Table 1: Structural estimates and their admissible domains. Coefficient constants, graph-shape constants, and equality claims have distinct scopes. The frame and positive-effects lemmas include the special cases used by the original formalized proofs. The polynomial-cone comparison is an analytic statement on its preserved coefficient family.

1.5 The losses are part of the method

We record the successive estimates as well as their final product. Table 2 gives three losses with concrete diagnostics. A fourth accounting issue, distributional transfer, is kept separate: an explicit coupling may change the moment root by a constant even when the reference operator estimate is sharp.

These distinctions determine what to retain in the next estimate. A crude coefficient bound, an unnecessary representation, and a small vacuum spectral mass require different repairs. The regular-entry proof illustrates another domain choice: it bounds the preserved walk cone without assuming a norm estimate on its entire linear span. Likewise, a positive

Loss	Checkable symptom	Worked example	What repairs it
Coefficient estimate	Same operator and same cutoff admit a smaller bound	Two-round allowance $27q^3 \rightarrow 12q^2$	Contract coefficient maps collectively
Symmetry sector	An excluded representation drives a channel norm	Four-bucket, three-site heavy-pair norm $1.20675 \rightarrow 1.08866$	Restrict to an exact vacuum symmetry
Vacuum spectral weight	A large eigenvalue remains in the cyclic space with small vacuum mass	Two-site spectrum $0, \pm 1$; even moment $1/b$	Keep moments or weighted paths; a uniform norm alone cannot recover the weight

Table 2: Three mechanisms and their corresponding repairs. The examples separate coefficient bounds, invariant sectors, and spectral mass.

Gram budget can be transported through an actual input congruence without claiming that the old output sectors survive an arbitrary coefficient insertion.

1.6 Precedents and a route through the paper

The representation has established precedents. Accardi and Bożejko [5] represented scalar moment functionals using interacting Fock spaces and Jacobi parameters. Hora and Obata [6], and work on quantum decomposition of graphs [7], developed related graded operator viewpoints in spectral graph theory. The classical trace method already turns random-matrix moments into constrained walks. We use the name *graded multiplication-operator method* for the combination of this representation with the compressions and contractions used here.

The sketching applications build on sparse Johnson–Lindenstrauss transforms [23], OSNAP and its parameter conjecture [21, 22], Cohen’s trace-inequality analysis [24], and the universality and decoupling results of Chenakkod, Dereziński, Dong, and Rudelson [25, 26, 27]. For graph matrices we build on the separator estimates and vertex partitioning of Ahn, Medarametla, and Potechin [28, Sections 5–6], and compare with the coefficient-flattening framework of Bandeira, Lucca, Nizić-Nikolac, and van Handel [29]. The sharp expected-norm question is recorded in [20]; another proof of its classification is given by Xu, Fu, Qiao, and Tao [30, Theorem 1.3]. Section 8 identifies the specific parameters supplied by our structural estimates.

For regular independent entries, the two-sided expected-norm conjecture, the lower bound, and the deterministic deletion strategy are due to Latała and Świątkowski [48], whose Remark 4.5 reduces the upper bound to an undeleted local estimate. The graded method supplies exactly that local estimate, through the preserved monomial cone. An earlier version of the argument quoted the strong/weak comparison of Latała and Strzelecka [45] and the variance-envelope theorem of Latała, van Handel, and Youssef [49]; the version presented here proves the specialized comparison it needs by an independent copy and replaces the envelope by an elementary Gram estimate. This is a self-contained proof of the required case within the established strong/weak-moment theory, not a claim of a new general comparison principle. The refined constants additionally use classical Boolean hypercontractivity [55, 56]; its L_2 -to- L_4 case is re-proved in Lemma 3.18. The complete argument, with explicit constants, is in Section 5.4 and Appendix G; the formal development of the earlier version is identified in Section 11, and Remark 5.8 records how the two versions differ.

The intended reader knows finite-dimensional linear algebra and basic probability. Tensor products are explained in coordinates when they first enter a proof. General entry laws require an L_2 probability space, but a fixed-horizon walk uses only finitely many original monomials. The scalar and vector moment comparisons used on the walk cone are proved in the form required here.

Sections 2–4 develop the representation, its structural contractions, and the conversion to moments. Section 2 starts with a three-state example, then introduces the parity–amplitude representation and its original count sectors. Figure 2 pictures the cutoff and returning paths, and Figure 3 the parity–degree lattice on which the walk cone lives. Section 3 places shared-factor and continued-Gram bounds beside connected creation, collective row estimates, and the positive-coefficient contraction. Section 4 explains scalar return-path bounds, the ordered renewal identity and covariance-spectrum comparison of Corollary 4.4, and direct iteration on the vacuum walk cone.

Section 5 applies these tools to the four models; for independent entries it isolates the local estimate that the cone contraction supplies and summarizes the deletion argument that converts it into the mean-norm theorem. Section 6 diagnoses the information discarded by the intermediate bounds. Section 7 states when a retained description supports a further interaction: an exact descent test, live registers, signed conditional cores with a certified remainder, an exact reduction for supplied coefficient algebras, and the obstructions that fix the limits of these descriptors. Sections 8 and 9 give calibrations, comparisons, and obstructions; the calibrations include an exact fourth-order energy identity for sums of independent components and an exact moment transfer for polynomials in few aggregate statistics. Section 10 and Appendices A–C supply the detailed transform and bucket calculations and their certificates, Appendix E collects the proofs and construction details behind the continuation rules and the aggregate transfer, and Appendices F and G give the complete proofs of the graph classification and of the deletion assembly, so that the four application theorems are proved within this paper. The twenty-seven-dimensional bucket calculation can be skipped on a first reading; it makes the symmetry-gap example explicit. For the longer model-specific arguments we identify each imported theorem and the structural step it supplies. Section 11 distinguishes the original proof certificates, the analytic estimates, and the limited scope of reported scalar formalization. The notation guide below is for consultation when moving between applications.

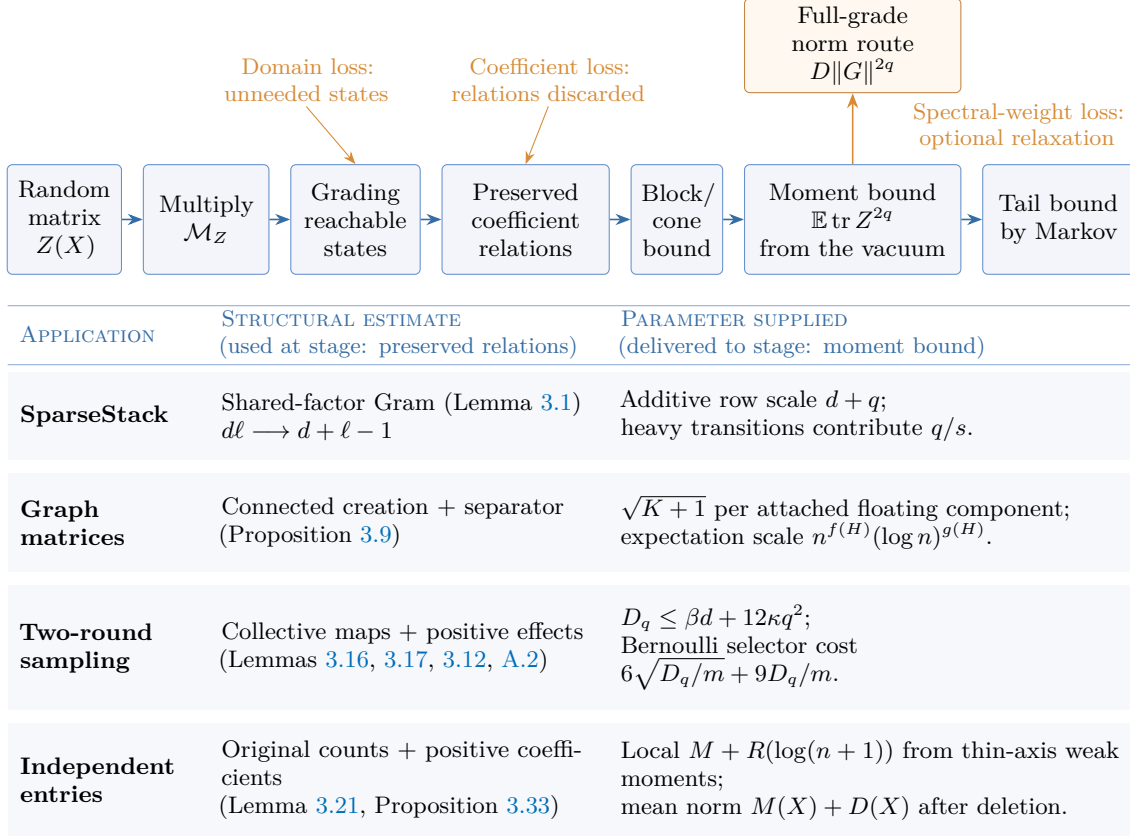


Figure 1: Two admissible-domain estimates inside one exact moment representation. The rows read across the pipeline: for each application, the middle column names the structural estimate applied at the stage *preserved coefficient relations*, together with the relation it retains, and the right column gives the parameter that this estimate delivers to the stage *moment bound from the vacuum*. For full graded spaces, $Z = Z^*$ is $D \times D$, sector 0 contains the vacuum, and a symmetric scalar matrix $G_{\alpha\beta} \geq \|P_\alpha M_q P_\beta\|$ gives $\mathbb{E} \text{tr } Z^{2q} \leq D(G^{2q})_{00}$. The orange relaxation replaces this vacuum entry by a full norm and may lose spectral-weight information. For a preserved walk cone, successive polynomial contractions directly bound each vector in (2); that route does not require a bound on all graded vectors. The application rows identify the relations carrying the parameters. The graph row gives the expectation and fixed- p moment scale. The two-round row uses $m = \rho n$ for Bernoulli sampling, before the factor-two transfer to a fixed row count. For independent entries, the moment argument gives the local estimate of Corollary 4.14; the deterministic deletion argument of Section 5.4 then gives the stated mean norm. Table 4 records the model-specific verification scope.

Notation and terminology

Application notation is local; the table records changes of meaning. Norms are Hilbert-space or operator norms unless specified otherwise. The adjoint is X^* ; $X \preceq Y$ means $Y - X$ is positive semidefinite.

Symbols	Meaning in the indicated section
Representation and moments: Sections 2 and 4	
$Z, \mathcal{M}_Z, \Omega$	Random self-adjoint matrix, its multiplication operator, and the constant function 1 (the vacuum). The operator acts on an external matrix factor tensored with the probability space.
$D; d$	D is the external matrix dimension in the trace identity, not the cutoff-space dimension. For the sketch errors, $D = d$, the embedded subspace dimension. For a graph matrix, dilation has dimension $(n)_{ U } + (n)_{ V }$ with injective indices, or $n^{ U } + n^{ V }$ after zero-padding to all boundary tuples; $(n)_a = n(n-1) \cdots (n-a+1)$. For an $n \times n$ independent-entry matrix the dilation dimension is $2n$.
q, Δ, K, Q_K, M_q	$2q$ is the even moment order. One multiplication increases grade by at most Δ ; K is a cutoff grade and Q_K its orthogonal projection. The exact moment cutoff is $M_q = Q_{\Delta q} \mathcal{M}_Z Q_{\Delta q}$.
$\mathcal{F}_k, P_\alpha, G$	Grade- k space, projection onto a finer sector α , and the nonnegative scalar matrix bounding the norms of the sector blocks. The vacuum sector is indexed by 0.
B, C, V, F_j, K_r (§4.3)	Blocks of the cutoff relative to the vacuum sector and the rest space, the covariance $V = B^*B = \mathbb{E}Z^2$, the ordered first-return operators $F_j = B^*C^{j-2}B$, and the vacuum blocks $K_r = PM^rP$ of the powers. In this subsection B, C are not the sign or creation operators of other sections.
$\mathcal{E}_j, W_j, B_0$ (Lemma 4.9)	Stage energies on the preserved cones, their weight operators, and the initial energy budget.
$Y_e = \varepsilon_e Z_e, \mathcal{U}, \mathcal{H}$ (§2.7)	Original symmetric entry, independent sign and magnitude, and the sign Fourier reconstruction from occupation amplitudes. $\mathcal{U}$ is an L_2 isometry; other L_p norms are norms of the reconstructed polynomial.
$\mathcal{P}_d^+, \mathcal{T}, \mathcal{C}_\pm$	Cone of nonnegative original monomial coefficients with ordinary degree at most d , exact multiplication, and parity creation/annihilation. Both parity directions multiply by the original magnitude and raise ordinary degree.
Structural estimates: Section 3	
F, τ, κ, ℓ (§3.1)	$F = \sum_i u_i u_i^*$ is the frame operator, $\tau = \text{tr } F$, $\kappa = \ F\ $, and ℓ is the number of labelled slots sharing one external factor. A Parseval frame has $F = I_d$.
Δ, r, k, K (§3.3)	Δ is the abstract creation step. In the graph construction it equals r , the fragment's edge count; k is the input edge occupation and K its cutoff. This r is distinct from the SRHT grade below.
A_j, Γ, L_q, K, D (§3.4)	Positive coefficient effects; $\Gamma = \ \sum_j A_j\ $ and $L_q = \max_{ T \leq q} \ \sum_{j \in T} A_j\ $. In this subsection only, K and D also denote the selector creation and occupancy operators, respectively.
$r, s, R_j^{\mu\nu}, R_T^{\mu\nu}$ (§3.5)	The two sign occupations have grades r, s . $R_j^{\mu\nu}$ is the row- j component changing them by $(\mu, \nu) \in \{\pm 1\}^2$; $R_T^{\mu\nu} z = (R_j^{\mu\nu} z)_{j \in T}$ retains the row labels.

Symbols	Meaning in the indicated section (continued)
$\Pi_q, G_q, \tau_q; L_{i,T}$ (§3.2.2)	Occupied coefficient-span projection, full exceptional output Gram, ordinary coefficient mass seen by that span, and the consumed-row child map. Ordered child products remain part of a continued Gram.
$H_\alpha, h_q, \chi_\alpha, \rho_q, \beta_q, \hat{\Theta}_q, L_\alpha$ (§3.6)	The positive-polynomial comparison factor $H_\alpha = \sqrt{3}\alpha$ and its order-adapted form $h_q(\alpha) = \sqrt{(4q-1)/(2q-1)}\alpha$, the strong/weak constant $\chi_\alpha = \frac{7}{3}2^{1/4}\alpha^2$, the spherical thin-axis factor $\rho_q(m)$ and the sign factor $\beta_q = \sqrt{(2q+1)/(2q-1)}$ of the parity-sector contraction, the parity constant $\hat{\Theta}_q < 2.31$ of the local moment theorem, and the local mean constant $L_\alpha = 2.31e^{1/4}\sqrt{\alpha}$. Here α is moment regularity, not a unitary coherence.
Retained state under continuation: Section 7	
$\Phi, \Gamma, T, \bar{T}$ (§7.1)	The encoding of descriptions into states, its Gram $\Gamma = \Phi^*\Phi$, a proposed update of descriptions, and the induced map on states when the descent test passes.
$V_a, D_a, g_a, \theta_a, F_I, \mu_I$ (§7.2.1)	Insertion and block transport of a live label, their squared and plain norm bounds, the partial frame operator of the internal labels, and their maximal leverage.
$\Pi_H, A, R, (a, e, n)$ (§7.3)	Projection onto a retained character group H , core $A = \Pi_H F$, remainder $R = F - A$, and the certified radii of a core, its remainder and the original polynomial at a stated exponent.
$\mathcal{A}, W, d_\lambda, \mu_\lambda, t_\lambda$ (§7.4)	A supplied coefficient algebra in sector form, the active space of its seeds, sector dimensions, multiplicities, and activated multiplicity dimensions.
Applications: Section 5 and their extensions	
n, d, U, S, E, m	For sketches, n is the input dimension, $U^*U = I_d$, S is the sketch, and $E = U^*S^*SU - I_d$. The target distortion and failure probability are ϵ, δ . The meaning of the row count m is specified below.
$s, b, m; L, H, \nu$ (§5.1)	SparseStack has s blocks, b buckets per block, and $m = sb$ rows. L, H count light and heavy sites; $\nu = L + 2H$ is their weighted grade. L_g, H_g are the counts in block g .
$H, h, e, U, V; s, b_*$ (§5.2)	H is the fixed graph shape, h its number of roles and e its edge count; U, V are its boundary lists. Here s is the minimum separator size. The count $b_{\text{att}}(S)$ records components of $H - S$ that miss both boundaries but have an edge to S ; b_* maximizes this count over minimum separators. Neither is a bucket count.
H, A, B, ρ, m, M, R_T (§5.3, §10.1)	Here H is the normalized Walsh matrix; A, B are its unitary replacements. The selector bias is ρ , $m = \rho n$ is the expected Bernoulli row count, and M is the fixed sample size. R_T without superscripts restricts coordinates to the sampled rows.
$\sigma_r(X), \sigma_c(X), M(X)$ $R_{X,I}(p), D(X)$ (§2.7)	Maximum row and column standard-deviation norms, their sum, the weak bilinear moment after deleting the same original set I on both axes, and the maximum over deletion budgets at orders $\log(k+1)$. For $p < 1$, L_p denotes a moment functional. Here $D(X)$ is a functional, not the trace dimension.
$D_q; \alpha, \beta, \kappa$	D_q is a scalar allowance, not a dimension: it is $d + 2q + 1$ in the original SparseStack estimate, $d + 27q^3$ in the original SRHT estimate, and $\beta d + (8\beta + 4\alpha)q^2 \leq \beta d + 12\kappa q^2$ in its extension. For the extension, α, β are the two squared synthesis bounds (unitary coherences), and $\kappa = \max(\alpha, \beta)$; this is distinct from the frame norm in Section 3.1.

Glossary of the operator bookkeeping. For readers coming from randomized numerical linear algebra (RandNLA), these terms describe functions, vectors, and matrix blocks under a classical probability law. They are grouped by their role in the proof.

Term	Meaning
The probability space as a vector space	
State; basis state	A state is a vector in the function space, possibly with an external vector coordinate; a basis state is a chosen basis vector. It is not a sampled outcome.
Vacuum	The unit vector $\Omega = 1$: the constant function, with every site at level zero. It starts the moment calculation and is not the zero vector.
Site; local factor	A site indexes one independent block X_j ; its local factor is $L^2(\mu_j)$. Coordinates within it may be dependent: a direct SparseStack site (g, i) contains the joint sign and hash (σ_{gi}, h_{gi}) , not independent bucket indicators.
Product space; tensor product	The full function space is $\bigotimes_j L^2(\mu_j)$. A tensor $\bigotimes_j f_j$ means $\prod_j f_j(X_j)$; inner products of such functions factor by independence.
Level; light; heavy	A level is a local subspace of a specified degree, possibly of dimension greater than one. Direct SparseStack's vacuum, light, and heavy levels have grades 0, 1, 2 and dimensions 1, b , $b - 1$.
Occupation; occupied site	An occupation specifies the local basis states. A site outside its vacuum level is occupied. Different patterns may have the same counts or total grade. With variable magnitudes, the occupation records sign parity while coefficient functions retain the repeated powers.
Count; grade	A count totals sites of a given type. A grade indexes an orthogonal decomposition, usually by a weighted sum of local levels: $\nu = L + 2H$ counts a heavy site twice. Distinct (L, H) can share a grade.
Register	A tensor factor, or group of factors, retaining orthogonal labels (rows, coordinates, or occupations). A passive register is carried unchanged through a map. It need not represent extra randomness.
External factor	The matrix-coordinate space $\mathbb{C}^D$ in $\mathbb{C}^D \otimes L^2(\mu)$, distinct from occupation labels. For sketch errors $D = d$. Tracing this factor supplies D in the moment bound.
Multiplication and its local transitions	
Multiplication operator	The deterministic map $(\mathcal{M}_Z f)(x) = Z(x)f(x)$. Repeated application computes random-matrix moments through the probability inner product.
Jacobi matrix	The tridiagonal matrix representing multiplication by one real random variable in its orthonormal polynomial basis. Its three diagonals raise, preserve, or lower polynomial degree.
Creation; annihilation; preservation	Parts raising, lowering, or preserving grade. For self-adjoint multiplication the first two are adjoints. Preservation keeps the grade, not necessarily the individual vector. Parity annihilation for regular entries still multiplies the magnitude and raises ordinary polynomial degree.
Hard-core occupation; particle; species	In a two-state sign register, inserting an already occupied site gives zero (the hard-core rule). A particle is one occupied site; species distinguish register types, such as SRHT's x and y signs.
Leg	One local operation in a tensor-product summand, such as creation, annihilation, or conversion between light and heavy states.

Term	Meaning (continued)
Channel; family	A channel is an operator component selected by specified local transitions or grade shifts, sometimes with paired orientations. A family is an indexed collection of vectors or maps. For the eight SparseStack operator families, summing the indexed terms gives the named channel operator.
Band	The sum of components with the same net grade change, or grade-vector change. Several channels can belong to one band.
Restricting the operator and retaining moments	
Cutoff; compression; reachability	A cutoff retains grades at most K ; compression gives $Q_K M Q_K$. After j steps from the vacuum, grade is at most Δj ; not every state below this bound need be reached. This reachability bound makes the moment cutoff exact, without making it invariant.
Sector; fixed-point sector	A subspace selected by counts, grades, or symmetries. A fixed-point sector consists of vectors unchanged by every symmetry; it is invariant when the symmetries commute with M , but can exceed the cyclic space. A count sector need not be invariant.
Preserved cone	A family closed under nonnegative scaling and addition in the chosen coefficient coordinates. The actual multiplication and masks must map the specified degree cone to its successor. A contraction on this family does not assert a bound on its full linear span.
Vacuum orbit; cyclic space	For a fixed finite operator M , the orbit consists of $\Omega, M\Omega, M^2\Omega, \dots$; their span is the cyclic space, the smallest M -invariant subspace containing Ω . This is a Krylov space. With an external factor, start from all $e_a \otimes \Omega$.
Vacuum matrix element; vacuum moment	The scalar $\langle \Omega, A\Omega \rangle$ tests A at the constant function; $\langle \Omega, \mathcal{M}_Z^{2q}\Omega \rangle = \mathbb{E} Z^{2q}$ for scalar Z . For matrix Z , sum over the starting vectors $e_a \otimes \Omega$ to obtain $\mathbb{E} \text{tr } Z^{2q}$.
Vacuum spectral measure; spectral weight	For finite self-adjoint M , put mass $\ P_\lambda \Omega\ ^2$ at each eigenvalue λ , where P_λ is its eigenspace projection. Then $\langle \Omega, M^{2q}\Omega \rangle$ is the $2q$ -th moment of these masses. Weight means mass as seen from the vacuum, not eigenvalue multiplicity.
Scalar majorant; returning path	A nonnegative matrix G bounding sector-block norms. The entry $(G^{2q})_{00}$ sums products of weights along length- $2q$ paths returning to sector 0. These paths index a matrix-power expansion, not extra random samples.
Keeping coefficient information in estimates	
Bra-ket notation	$ u\rangle\langle v $ is the outer-product operator uv^* , mapping x to $u\langle v, x \rangle$; $ u\rangle$ denotes the vector u .
Index contraction	Pairing and summing shared coordinates, as in an inner product. Here “contraction” can name this algebraic operation without asserting that its operator norm is at most one.
Positive effect	A positive semidefinite coefficient operator A_j , often a row Gram $R_j^* R_j$. Here “effect” imposes no normalization or upper bound $A_j \preceq I$ unless separately stated.
Coefficient kernel; occupation lift	A deterministic map on coordinate labels, before occupation operations. Its lift acts across the allowed occupied slots; its norm bound must account for insertion, removal, and repeated-site restrictions.
Marked	An index is retained to record which site or register participates in a contraction. A marked Gram estimate keeps that information when forming the adjoint product.

Term	Meaning (continued)
Continuation; retained budget	An actual input or output map appended to the operator. A positive Gram bound survives input congruence; a cone bound additionally needs the inserted map to preserve the specified coefficient domain. Whether a compressed description supports a proposed continuation at all is decided by the descent test of Section 7.
Live register; core; remainder	A live register keeps a label or an independent name as an orthogonal coordinate through every visit, so that label-dependent insertions act exactly. A core is the projection of a sign polynomial onto a retained group of characters; the remainder is what the projection discards, and its certified radius is carried through later products.
Collected	Indexed terms or labelled outputs are kept together in one linear map before a norm is taken. In the orthogonal row-register case, $R_T z = (R_j z)_{j \in T}$ gives $\ R_T\ ^2 = \ \sum_{j \in T} R_j^* R_j\ $; separate row estimates can discard this structure.

2 The representation and the exact cutoff

2.1 The complete argument in a three-state example

Example 2.1 (Two columns and one sketch block). Choose independent uniform hashes $h_1, h_2 \in [b]$ and signs $\sigma_1, \sigma_2 \in \{-1, 1\}$, with $b \geq 2$. Let the two columns of $S \in \mathbb{R}^{b \times 2}$ be $\sigma_i e_{h_i}$ and take $U = (1, 1)^*/\sqrt{2}$. The scalar Gram error is

$$Z = \|SU\|^2 - 1 = \sigma_1 \sigma_2 \mathbf{1}_{h_1=h_2}.$$

It is 1 or -1 with probability $1/(2b)$ each and zero otherwise, so $\mathbb{E}Z^{2q} = 1/b$ for every $q \geq 1$.

Here is the same calculation through multiplication. In the probability space of the four random choices, the three functions

$$\Omega = 1, \quad v_1 = \sqrt{b} Z, \quad v_2 = \frac{b \mathbf{1}_{h_1=h_2} - 1}{\sqrt{b-1}}$$

are orthonormal: the collision indicator has mean $1/b$, and averaging the signs kills every inner product involving one factor Z . Their span is preserved by multiplication by Z . In this basis its matrix is

$$M = \begin{pmatrix} 0 & b^{-1/2} & 0 \\ b^{-1/2} & 0 & \sqrt{(b-1)/b} \\ 0 & \sqrt{(b-1)/b} & 0 \end{pmatrix}. \quad (3)$$

For example, $Zv_1 = \sqrt{b} \mathbf{1}_{h_1=h_2} = b^{-1/2} \Omega + \sqrt{(b-1)/b} v_2$; the other two columns follow in the same way. Thus $\mathbb{E}Z^{2q} = \|M^q \Omega\|^2$.

Assign these basis vectors grades 0, 2, 4. Multiplication changes grade by two. For $q = 1$, compression to grades at most 2 gives the 2×2 upper-left block, whose norm is $b^{-1/2}$ and whose squared vacuum norm is exactly $1/b$. For $q \geq 2$ the cutoff contains all three states. Its eigenvalues are 0, 1, -1 , so the uniform norm bound gives $\mathbb{E}Z^{2q} \leq 1$, while the exact answer remains $1/b$. The vacuum has total spectral mass $1/b$ at the two nonzero eigenvalues. We have now seen representation, grade compression, and the choice between a norm and a vacuum moment in one calculation. Section 6 returns to this example to prove that further invariant-subspace restrictions cannot recover this spectral weight.

2.2 One variable: multiplication in an orthogonal basis

Let ξ have a finite law μ on the real line, and discard any zero-probability atoms. Its Hilbert space is $\mathcal{H} = L^2(\mu)$, with inner product $\langle f, g \rangle = \mathbb{E}[\overline{f(\xi)} g(\xi)]$. The constant function $p_0 = 1$ is a unit vector. Orthogonalizing $1, x, x^2, \dots$ gives an orthonormal polynomial basis $p_0, \dots, p_{a-1}$ when the support has a points.

Multiplication by ξ is represented by a real symmetric Jacobi matrix J . The three-term recurrence has the form

$$xp_k(x) = a_{k+1}p_{k+1}(x) + b_k p_k(x) + a_k p_{k-1}(x), \quad a_0 = a_a = 0. \quad (4)$$

To see why no lower terms occur, take an inner product with p_j for $j < k-1$ and move multiplication by x to p_j . The result vanishes by orthogonality. The upward, diagonal, and downward parts of J are called creation, preservation, and annihilation. These names describe matrix entries between polynomial degrees. We call an individual local transition, such as the upward map between two levels, a *leg*.

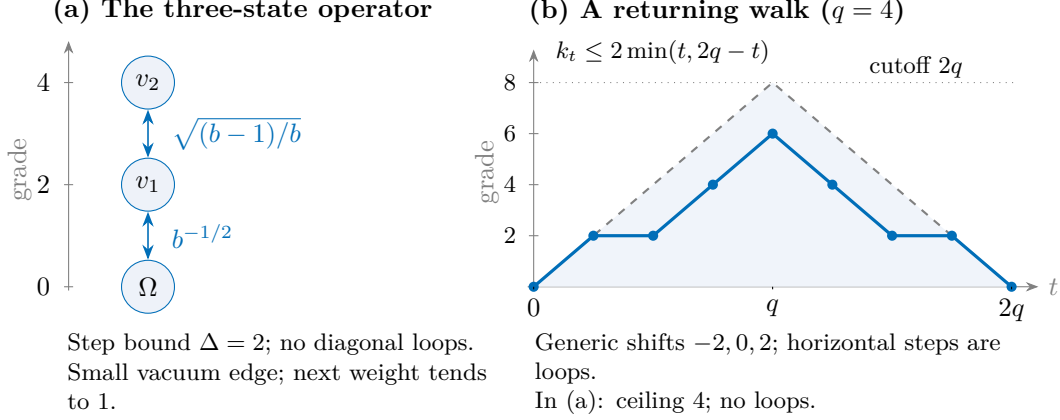


Figure 2: Transition weights and returning-path geometry. Panel (a) is the matrix in (3); each coefficient applies in both directions. The small vacuum edge accounts for the spectral weight in Figure 11. Panel (b) shows a generic even-grade walk. The exact q -step cutoff is $2q$, while return to the vacuum gives the tighter time-dependent envelope. In general, $k_t \leq \Delta \min(t, 2q - t)$. Dividing the plotted grades by two gives the indices in Lemma 4.1. That lemma assumes $T_{00} = 0$ (no vacuum loop); the general compression theorem does not.

Example 2.2 (Two local factors). A uniform sign has basis $(1, \xi)$ and

$$J = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

For $b > 1$, a variable with probabilities $\mathbb{P}(\xi = \pm 1) = 1/(2b)$ and $\mathbb{P}(\xi = 0) = 1 - 1/b$, use

$$p_0 = 1, \quad p_1 = \sqrt{b} \xi, \quad p_2 = \frac{b\xi^2 - 1}{\sqrt{b-1}}.$$

Multiplication by $\sqrt{b}\xi$ has matrix

$$J = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & \sqrt{b-1} \\ 0 & \sqrt{b-1} & 0 \end{pmatrix}.$$

We call the p_1 level light and the p_2 level heavy. The large heavy leg $\sqrt{b-1}$ is already visible in this exact local matrix. Replacing it by a generic bounded-variable norm would conceal which transitions carry that factor.

2.3 Independent blocks, including dependence within a block

Suppose $X = (X_1, \dots, X_N)$ has product law $\mu = \mu_1 \otimes \dots \otimes \mu_N$. Each X_j may itself be a vector of dependent coordinates. Then

$$L^2(\mu) = \bigotimes_{j=1}^N L^2(\mu_j), \quad \Omega = 1 \otimes \dots \otimes 1.$$

Here a tensor $f_1 \otimes \dots \otimes f_N$ is just the function $\prod_j f_j(X_j)$; the tensor inner product is the product of the local inner products because the blocks are independent. A tensor factor, or a specified group of factors carried together, is called a *register*. An *occupation* specifies

the local basis levels; a *count* records how many factors have a given type, and a grade is a chosen weighted sum of these local levels. Different occupation patterns can therefore have the same grade.

For direct SparseStack with $b \geq 2$, the independent site is (σ, h) , not the individual bucket indicators. It has $2b$ outcomes. The decomposition is

$$L^2(\sigma, h) = \mathbb{R}\Omega \oplus \mathbb{L} \oplus \mathbb{H}, \quad \dim \mathbb{L} = b, \quad \dim \mathbb{H} = b - 1,$$

with

$$e_a = \sqrt{b} \sigma \mathbf{1}_{h=a}, \quad f_a = \frac{b \mathbf{1}_{h=a} - 1}{\sqrt{b-1}}.$$

The e_a are orthonormal. The f_a form a regular simplex: $\langle f_a, f_c \rangle = (b\delta_{ac} - 1)/(b-1)$ and $\sum_a f_a = 0$. Multiplication by e_a is exactly

$$J_a = P_a^* + P_a + \sqrt{b-1}(R_a + R_a^*), \quad P_a^* = |e_a\rangle\langle\Omega|, \quad R_a = |f_a\rangle\langle e_a|. \quad (5)$$

Assign grades 0, 1, 2 to these three summands of the local space. At $b = 1$ the factor is the ordinary two-state sign space, with no heavy summand. The mutually exclusive bucket indicators are handled exactly inside one local factor. There is no independence assumption between them.

This observation also marks a limit. An arbitrary dependent law can always be treated as one enormous block, but that formal representation supplies neither a useful small grade nor a useful bound for its local maps.

2.4 An abstract moment-compression identity

Let $\mathcal{F} = \bigoplus_{k \geq 0} \mathcal{F}_k$ be an orthogonal grading of $L^2(\mu)$ with $\Omega \in \mathcal{F}_0$, $\|\Omega\| = 1$. Write Q_K for the orthogonal projection onto grades at most K , including the identity on any external matrix factor.

Theorem 2.3 (Exact finite moment compression). *Let $q, \Delta \geq 1$ be integers and let $Z(X)$ be a self-adjoint $D \times D$ matrix-valued function, with all products below square integrable. Suppose multiplication obeys*

$$\mathcal{M}_Z(\mathbb{C}^D \otimes \mathcal{F}_{\leq k}) \subseteq \mathbb{C}^D \otimes \mathcal{F}_{\leq k+\Delta}$$

for all integers $0 \leq k \leq \Delta q - \Delta$. Put $M_q = Q_{\Delta q} \mathcal{M}_Z Q_{\Delta q}$. Then

$$\mathbb{E} \operatorname{tr} Z^{2q} = \sum_{a=1}^D \|M_q^q(e_a \otimes \Omega)\|^2 \leq D \|M_q\|^{2q}. \quad (6)$$

The first equality is exact. The last inequality is an optional relaxation.

Proof. Multiplication gives $\mathcal{M}_Z^j(e_a \otimes \Omega)(x) = Z(x)^j e_a$. For $j \leq q$ this vector has grade at most $\Delta j \leq \Delta q$. An induction therefore gives $M_q^j(e_a \otimes \Omega) = \mathcal{M}_Z^j(e_a \otimes \Omega)$. Taking squared norms, summing over a , and using $Z = Z^*$ proves the identity. The norm inequality follows because each initial vector is a unit vector. No estimate of the dimension of the occupation space enters. $\square$

The factor D in (6) is the external trace dimension, not the often much larger dimension of $Q_{\Delta q} \mathcal{F}$. Taking the trace over the whole cutoff would generally introduce a spurious factor. In both sketching applications Z is a $d \times d$ Gram error, so $D = d$. For graph matrices the dilation instead has the combined row and column dimension. For a rectangular random

matrix A , one may use the self-adjoint dilation $\begin{pmatrix} 0 & A \\ A^* & 0 \end{pmatrix}$, or work directly with the Gram moment and its factor-boundary cutoff.

The identity and the full-space norm estimate in (6) are distinct statements. The identity only needs the actual prefixes $Z^j e_a$. One may therefore control these prefixes on a preserved cone, even when a bound on the entire graded space is unavailable. The parity–amplitude representation below gives such a domain; Lemma 4.8 states its moment conversion.

2.5 Exact symmetries before norm estimates

A *sector* is a subspace selected by specified counts or symmetry conditions. A count sector need not be invariant under multiplication; an invariant symmetry sector is preserved by the operator.

Proposition 2.4 (Invariant-sector reduction). *Let a finite group act measure preservingly on X , suppose $Z(gx) = Z(x)$, and suppose its unitary action preserves the grading. Its invariant-space projection Π commutes with $\mathcal{M}_Z$ and Q_K , fixes the vacuum, and may be inserted throughout (6) without changing the moment.*

Proof. For $T_g f(x) = f(g^{-1}x)$, invariance of Z gives $T_g \mathcal{M}_Z = \mathcal{M}_Z T_g$. Average over g to obtain Π . The grade assumption gives commutation with Q_K , and constants are fixed by every T_g . $\square$

The pointwise invariance assumption means $Z \circ g = Z$; it does not mean that the action fixes every probability-space point. This distinction allows nontrivial bucket permutations and sign transformations.

For direct SparseStack, flipping all signs in a block preserves its Gram matrix. Thus only even light count in each block is accessible. More strongly, simultaneously permuting buckets and flipping signs of individual bucket rows preserves each sum of terms with specified local legs. We call an operator component selected by its local transitions or grade shifts a *channel*. A *family* is an indexed collection of terms or maps; in the SparseStack formulas we also use this word for their summed channel, sometimes grouping adjoint orientations. A map is *collected* when the indexed terms or labelled outputs are kept together before taking its norm, as the formula specifies. The fixed sector of the signed bucket action requires even light count in each individual bucket. The full equality partition of occupied sites, together with heavy mean-zero constraints, describes this reduced invariant space; “same” and “different” suffice only in the simplest two-site pattern. Section 6 shows both gains and limits of this reduction.

2.6 Gaussians: finite moments do not require a finite alphabet

For a standard Gaussian, define $h_k(x) = (-1)^k e^{x^2/2} (d^k/dx^k) e^{-x^2/2} / \sqrt{k!}$. Repeated integration by parts shows $\mathbb{E} h_j(G) h_k(G) = \delta_{jk}$; differentiating this definition gives

$$x h_k(x) = \sqrt{k+1} h_{k+1}(x) + \sqrt{k} h_{k-1}(x).$$

If entries of $Z(G_1, \dots, G_N)$ have total polynomial degree at most Δ , multiplication increases total Hermite degree by at most Δ . Consequently Theorem 2.3 applies on the finite space of total degree at most Δq . This assertion concerns polynomials and their finite moments; no global boundedness of Gaussian multiplication is needed.

There is also an exactly equivalent finite law for these moments. Let J_K be the $K \times K$ Hermite Jacobi matrix on degrees $0, \dots, K-1$. Its spectral measure at e_0 is a probability law supported on at most K real nodes. For powers of order at most $2K-1$, a vacuum-returning nearest-neighbor walk cannot visit the omitted grade K : doing so and returning takes at least $2K$ steps. Hence this finite law matches Gaussian moments through degree $2K-1$.

Taking $K = \Delta q + 1$ and replacing coordinates independently preserves $\mathbb{E} \operatorname{tr} Z^{2q}$ exactly. This is the classical Gaussian quadrature construction from a Jacobi matrix, called Gauss–Hermite quadrature for this Gaussian law. The eigenvalue and eigenvector formula is the standard Jacobi-matrix construction; see Golub and Welsch [57] for its numerical realization. Its nodes are the eigenvalues λ_j of J_K and its weights are $|\langle e_0, v_j \rangle|^2$ for normalized eigenvectors v_j . The spectral theorem makes the weights nonnegative with sum one, and the preceding path argument proves every required matching identity. This gives a finite, exact interface for Gaussian polynomial moments.

2.7 Parity and magnitude amplitudes for regular entries

The Gaussian example has a convenient orthogonal polynomial recurrence. For a more general entry law, we can instead orthogonalize only its signs and retain every magnitude power in a coefficient function. This still represents the original multiplication exactly, while leaving the choice of an admissible domain to the contraction argument.

Let $X = (X_{ij})_{i,j \leq n}$ be a real matrix with independent mean-zero entries. We assume that every absolute moment is finite and that one constant $\alpha \geq 1$ satisfies

$$\|X_{ij}\|_{L_{2p}} \leq \alpha \|X_{ij}\|_{L_p}, \quad p \geq 1, \quad i, j \in [n]. \quad (7)$$

Here $[n] = \{1, \dots, n\}$ and $\|Z\|_{L_p} = (\mathbb{E}|Z|^p)^{1/p}$ for every $p > 0$. At $p < 1$ this notation denotes a moment functional, not a norm. Logarithms in these regular-entry quantities are natural; the stated base-two and base-four conventions for sketching remain unchanged. Matrix norms without a subscript are operator norms for the Euclidean norms on domain and range. For a scalar random variable we also use the abbreviation $\|Z\|_p = \|Z\|_{L_p}$; for a deterministic vector, the subscript 2 denotes its Euclidean norm.

Define the deterministic variance scales

$$\begin{aligned} \sigma_r(X) &= \max_i \left(\sum_j \mathbb{E} X_{ij}^2 \right)^{1/2}, \quad \sigma_c(X) = \max_j \left(\sum_i \mathbb{E} X_{ij}^2 \right)^{1/2}, \\ M(X) &= \sigma_r(X) + \sigma_c(X), \quad \sigma(X) = \max\{\sigma_r(X), \sigma_c(X)\}. \end{aligned} \quad (8)$$

For a deterministic set $I \subset [n]$ and $p > 0$, put

$$R_{X,I}(p) = \sup_{\|s\|_2, \|t\|_2 \leq 1} \left\| \sum_{i,j \notin I} X_{ij} s_i t_j \right\|_{L_p}, \quad R_X(p) = R_{X,\emptyset}(p). \quad (9)$$

The vectors in this supremum are deterministic. The deletion parameter is

$$D(X) = \max_{1 \leq k \leq n} \min_{I \subset [n], |I| \leq k} R_{X,I}(\log(k+1)). \quad (10)$$

One and the same original set I deletes both rows and columns. It is chosen for the law, before any realization is observed. The minimum is outside the vector supremum and is attained, since the collection of sets is finite. Deleting all indices gives zero, so the $k = n$ term is zero. The $k = 1$ term retains its exact exponent $\log 2 < 1$.

These deterministic quantities will be read by the contractions in Section 3 and assembled into the mean-norm theorem in Section 5.

2.7.1 The exact parity–amplitude representation

For this representation let $Y = (Y_{ij})_{i,j \leq n}$ have independent symmetric entries satisfying (7). There is one original label $e = (i, j)$ for each entry not identically zero; denote this finite

label set by $\mathcal{E}$. Keeping zero entries as zero multipliers would give the same formulas. Row vertices r_i and column vertices c_j belong to disjoint copies of $[n]$; write $\mathcal{V}$ for their union.

Choose independent magnitudes Z_e with the laws of $|Y_e|$ and independent Rademacher signs ε_e , all mutually independent, and represent the joint law by $Y_e = \varepsilon_e Z_e$. This remains valid at zero atoms: the product law on (ε, Z) has exactly the required image law for Y . The sign at (i, j) is distinct from that at (j, i) when both labels exist. Both directions of the single edge between r_i and c_j use the same variable Y_{ij} .

For an exponent vector $\nu \in \mathbb{N}_0^\mathcal{E}$, write $Y^\nu = \prod_{e \in \mathcal{E}} Y_e^{\nu_e}$ and $|\nu| = \sum_e \nu_e$. Ordinary total degree counts repeated powers. For a set of sign labels, $\varepsilon_S = \prod_{e \in S} \varepsilon_e$.

The space and its polynomial domain Let μ_Z be the joint magnitude law and define

$$\mathcal{H} = L_2(\mu_Z; \ell_2(\mathcal{V} \times 2^\mathcal{E})), \quad \|f\|_{\mathcal{H}}^2 = \mathbb{E}_Z \sum_{v \in \mathcal{V}} \sum_{S \subseteq \mathcal{E}} |f_{v,S}(Z)|^2. \quad (11)$$

Conditional Fourier transform in the signs is the isometry

$$(\mathcal{U}f)_v(\varepsilon, Z) = \sum_{S \subseteq \mathcal{E}} \varepsilon_S f_{v,S}(Z) \quad (12)$$

onto $L_2(\varepsilon, Z; \ell_2(\mathcal{V}))$. This transform preserves the L_2 norm, not general L_p norms. For example, with one sign and unit magnitude, amplitudes $f_\emptyset = f_{\{e\}} = 1$ have coefficient-vector norm $\sqrt{2}$, whereas $\mathcal{U}f = 1 + \varepsilon_e$ has L_4 norm $2^{3/4}$. Every non- L_2 norm used in the contraction below is therefore a norm of the reconstructed random polynomial $\mathcal{U}f$, rather than of its occupation-amplitude vector. All formulas below may be restricted to polynomial coefficients, for which every required moment is finite. No assertion about boundedness of the full multiplication operator on $\mathcal{H}$ is needed.

Define $\mathcal{P}_d^+$ to be the image under $\mathcal{U}^{-1}$ of vector polynomials in the original Y_e of degree at most d whose ordinary coefficients are nonnegative in each current-vertex coordinate. Thus a monomial $a_{v,\nu} Y^\nu$ contributes $a_{v,\nu} Z^\nu$ to the component (v, S) with $S = \{e : \nu_e \text{ is odd}\}$. This parity compatibility is part of the definition. It ensures that $\mathcal{U}f$ is a polynomial in the original variables. This is the coefficient condition used in Lemma 3.21.

The parity grade of (v, S) is $|S|$. Let P_k keep exactly grade k . For $f \in \mathcal{P}_d^+$, necessarily $P_k f = 0$ for $k > d$. Projection onto any collection of pairs (v, S) simply discards ordinary monomials, hence preserves $\mathcal{P}_d^+$. Distinct such collections are orthogonal in (11). Magnitude coefficients on different parity states may have common even powers, including powers of edges outside the occupied set S . They are not required to be orthogonal or independent.

Multiplication, creation, and annihilation The actual symmetric dilation is

$$H_Y = \begin{pmatrix} 0 & Y \\ Y^\top & 0 \end{pmatrix}. \quad (13)$$

For a scalar magnitude polynomial $h(Z)$ at state (v, S) , conjugating multiplication by H_Y through $\mathcal{U}$ gives

$$\mathcal{T}(h(Z)|v, S\rangle) = \sum_{e=vw} Z_e h(Z) |w, S \triangle \{e\}\rangle. \quad (14)$$

Here $e = vw$ means the original bipartite entry with endpoints v, w . Define $\mathcal{C}_+$ by keeping terms with $e \notin S$ and $\mathcal{C}_-$ by keeping terms with $e \in S$. Their sum is exactly $\mathcal{T}$, and they change parity grade by $+1$ and -1 .

Both maps multiply the magnitude coefficient by Z_e . On an ordinary monomial Y^ν , either action increases ν_e by one. Therefore

$$\mathcal{C}_+ \mathcal{P}_d^+ \subset \mathcal{P}_{d+1}^+, \quad \mathcal{C}_- \mathcal{P}_d^+ \subset \mathcal{P}_{d+1}^+, \quad \mathcal{T} \mathcal{P}_d^+ \subset \mathcal{P}_{d+1}^+. \quad (15)$$

In particular annihilation does not divide by a magnitude or reset its earlier powers. On Y_e^3 , its action gives Y_e^4 while removing the odd parity flag. This retains the full fourth moment of that same entry. Figure 3 shows the resulting lattice of reachable states.

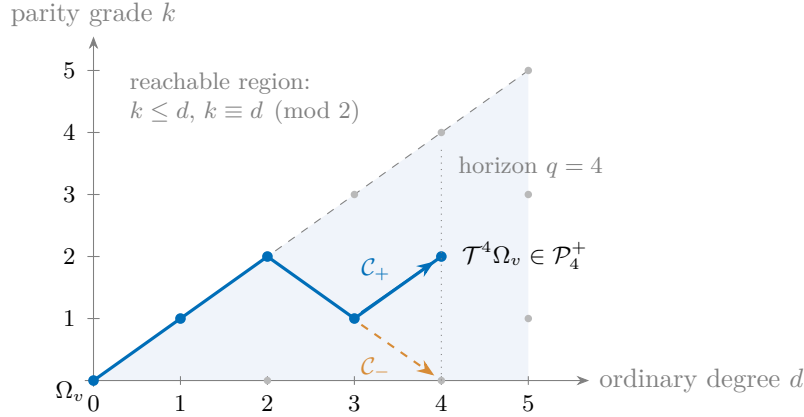


Figure 3: The parity–degree lattice for regular entries. Every step of $\mathcal{T} = \mathcal{C}_+ + \mathcal{C}_-$ raises the ordinary degree by one; creation raises the parity grade and annihilation lowers it, but neither divides by a magnitude. A vacuum walk of length q therefore stays in the triangle $k \leq d \leq q$, so the walk cone $\mathcal{P}_q^+$ is finite in both coordinates without any cutoff being imposed. Contrast Figure 2, where a Jacobi grading lets annihilation lower the degree. The parity grade indexes the orthogonal sectors of Lemma 2.5; the degree indexes the cone on which Proposition 3.33 is asserted.

2.7.2 Exact localization by the two original count vectors

For $S \subset \mathcal{E}$, define

$$r_i(S) = |S \cap (\{i\} \times [n])|, \quad c_j(S) = |S \cap ([n] \times \{j\})|.$$

The vectors $r(S), c(S)$ count occupied original parity labels, not total magnitude degrees. Write δ_i for a coordinate vector in the relevant row or column count space.

Lemma 2.5 (Two exact count decompositions). *Consider maps from a current row to a current column, at input parity grade k . Each of the following gives orthogonal partitions of both the input states and the output states of the specified map.*

1. *For creation, label input (r_i, S) by $d = r(S) + \delta_i$ and output (c_j, U) by $d = r(U)$. Put $I = \{i : d_i > 0\}$. Then $|I| \leq k + 1$ and the entire sector is an input/output parity-count compression of multiplication by $Y[I, :]^\top$.*
2. *For annihilation, label input (r_i, S) by $d = c(S)$ and output (c_j, U) by $d = c(U) + \delta_j$. Put $J = \{j : d_j > 0\}$. Then $|J| \leq k$ and the entire sector is an input/output parity-count compression of multiplication by $Y[:, J]^\top$. At $k = 0$ the map is zero.*

The reverse current-side direction transposes the statements. Every magnitude variable remains available inside the coefficients.

Proof. Creation along $e = (i, j) \notin S$ produces $U = S \cup \{(i, j)\}$, so $r(U) = r(S) + \delta_i = d$. For fixed d , every input current row lies in I , and all parity flags on input and output lie in $I \times [n]$. The sum of the coordinates of d is $k + 1$, giving $|I| \leq k + 1$. Multiplication by $Y[I, :]^\top$ therefore includes every actual transition, with exactly its original coefficient.

More explicitly, let P_d select the input label $r(S) + \delta_i = d$ and Q_d select output grade $k + 1$ with $r(U) = d$. The sector map is $Q_d \mathcal{U}^{-1} M_{Y[I,:]}^\top \mathcal{U} P_d$, with the natural current-coordinate inclusions understood. Output grade removes every annihilation term. Conversely every selected creation term satisfies the same original input and output conditions. This proves equality of maps, rather than a comparison of their coefficients.

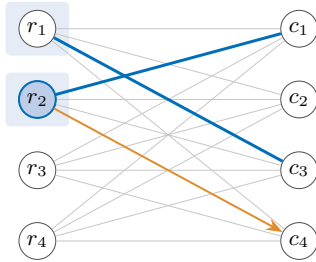
For annihilation, $e = (i, j) \in S$ produces $U = S \setminus \{(i, j)\}$, so $c(U) + \delta_j = c(S) = d$. Every parity flag lies in $[n] \times J$ and every output current column lies in J , although the input current row may be arbitrary. Now $\sum_j d_j = k$. Select input label $c(S) = d$ and output grade $k - 1$ with $c(U) + \delta_j = d$. The analogous compression of $M_{Y[:,J]}^\top$ retains exactly the annihilation terms. Creation terms have grade $k + 1$ and cannot survive the output projection.

Each input label is a function of (v, S) alone, and so is each output label. Distinct labels thus give disjoint components of (11), hence orthogonal decompositions. The two current-side directions have orthogonal domains and ranges. Different input grades also have different output grades within each of $\mathcal{C}_+$ and $\mathcal{C}_-$. No amplitude restriction was used: even powers outside the selected row or column set remain in $f_{v,S}(Z)$ throughout. $\square$

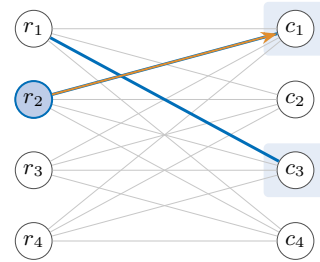
The distinction between the two labels is necessary. The creator is controlled through a small set of original rows in its output counts; the annihilator is controlled through a small set of original columns in its input counts. We will estimate the latter directly, not infer it by applying an adjoint argument to a cone-restricted norm inequality. Figure 4 shows the two labels on a small example.

(a) Creation: sector fixed by row counts

(b) Annihilation: sector fixed by column counts



State: current vertex r_2 , flags $S = \{(1, 3), (2, 1)\}$, grade $k = 2$. The step (orange) creates the flag $(2, 4)$. Label $d = r(S) + \delta_2 = (1, 2, 0, 0)$; rows $I = \{1, 2\}$ (shaded), $|I| \leq k + 1$. The sector is a compression of multiplication by $Y[I, :]^\top$; the sphere average is over $\mathbb{R}^I$ only.



Same state and current vertex. The step (orange) travels along the flag $(2, 1)$ and removes it. Label $d = c(S) = (1, 0, 1, 0)$; columns $J = \{1, 3\}$ (shaded), $|J| \leq k$. The sector is a compression of multiplication by $Y[:, J]^\top$; the sphere average is over $\mathbb{R}^J$ only.

Figure 4: The two count decompositions of Lemma 2.5 on a 4×4 example. A state is a current vertex together with the set S of edges whose sign is currently odd (thick); magnitude powers stay in the coefficient and are not drawn. Creation (a) adds a flag at the current row, so the sector labelled by the row-count vector involves at most $k + 1$ original rows and is a compression of multiplication by the thin matrix $Y[I, :]^\top$. Annihilation (b) removes a flag, which must already lie in the column set J of the input, so its sector is a compression of $Y[:, J]^\top$ with $|J| \leq k$. In both cases the directional average of Lemma 3.31 is taken only on the small axis, and the sectors for different labels are orthogonal, so their squared bounds add without a factor for their number.

Parity grade records odd original-edge exponents; it does not truncate the magnitude coefficients. The ordinary-degree domain and its preserved cone supply the estimates in

Section 3. Thus the exact moment identity remains available, while its optional full-cutoff norm relaxation in Theorem 2.3 need not be used.

2.8 Transfer is a separate, priced operation

Lemma 2.6 (Conditional barycenter transfer). *Suppose self-adjoint random matrices Z and Y are coupled so that $\mathbb{E}[Y \mid \mathcal{A}] = aZ$ for an $\mathcal{A}$ -measurable Z and $a > 0$. Then*

$$\mathbb{E} \operatorname{tr} Z^{2q} \leq a^{-2q} \mathbb{E} \operatorname{tr} Y^{2q}.$$

Proof. We first check the scalar convexity used by Jensen. For self-adjoint H, K , differentiating the trace twice and diagonalizing H gives

$$\left. \frac{d^2}{dt^2} \operatorname{tr}(H + tK)^{2q} \right|_{t=0} = 2q \sum_{i,j} |K_{ij}|^2 \sum_{\ell=0}^{2q-2} \lambda_i^\ell \lambda_j^{2q-2-\ell} \geq 0.$$

For unequal eigenvalues the inner sum is $(\lambda_i^{2q-1} - \lambda_j^{2q-1})/(\lambda_i - \lambda_j)$, which is nonnegative because an odd power is increasing. For equal eigenvalues it is $(2q-1)\lambda_i^{2q-2}$. Applying this at every point of each affine line proves convexity of $H \mapsto \operatorname{tr} H^{2q}$. Conditional Jensen gives $a^{2q} \operatorname{tr} Z^{2q} \leq \mathbb{E}[\operatorname{tr} Y^{2q} \mid \mathcal{A}]$, and expectation proves the lemma. $\square$

A *hollow* Gram sum omits terms with equal input-column indices. Suppose the column couplings $(X_i, \hat{X}_i)$ are independent and $\mathbb{E}[\hat{X}_i \mid X_i] = a_0 X_i$. For $i \neq j$, conditional independence gives

$$\mathbb{E}[\langle \hat{X}_i, \hat{X}_j \rangle \mid (X_k)_k] = a_0^2 \langle X_i, X_j \rangle.$$

Thus the hollow matrix barycenter is $a_0^2 Z$, and the moment-root cost is a_0^{-2} . It is not a_0^{-1} . The original SparseStack proof uses $a_0 \geq 1 - e^{-1}$ to transfer an independent-entry estimate. The direct model (5) avoids this transfer, while the stratified extension uses it to cover new support laws.

Classical sampling comparisons of Hoeffding and Gross–Nesme provide related transfer tools [8, 9]. Their hypotheses and the actual reference law must be checked: a Bernoulli selector array, a fixed number of independent draws, and a uniform fixed-size subset are different probability spaces. The operator core and the coupling layer should therefore remain distinct in both the theorem statement and its constant ledger.

3 Structural estimates: preserving the right algebra

3.1 Several particles sharing one external factor

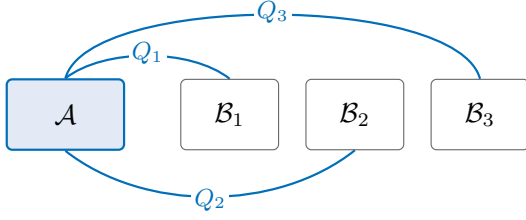
Takeaway. In a Parseval frame, ℓ insertions sharing one external factor have a Gram bound $d + \ell - 1$, instead of $d\ell$.

The dimension term in the SparseStack argument comes from a contraction between an occupied site and the external space carrying the matrix coefficients. A single contraction can have squared norm d . Bounding ℓ such contractions separately would therefore cost $d\ell$. The useful observation is that two different contractions overlap through a frame Gram matrix of norm one. Keeping those overlaps gives $d + \ell - 1$ instead. We prove the mechanism for an arbitrary frame, separating its algebraic content from the sampling model; Figure 5 pictures it.

Let $n \geq 1$ and let $u_1, \dots, u_n$ belong to a nonzero finite-dimensional Hilbert space $\mathcal{A}$. Set

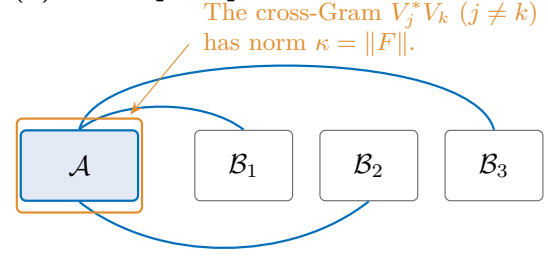
$$F = \sum_i u_i u_i^*, \quad \tau = \operatorname{tr} F, \quad \kappa = \|F\|, \quad \omega = \sum_i u_i \otimes e_i \in \mathcal{A} \otimes \mathbb{C}^n. \quad (16)$$

(a) Bounded separately



$\|Q_j\| = \tau$ each, so $\|\sum_j Q_j\| \leq \ell \tau$.
For a Parseval frame, $\tau = d$: cost $d\ell$.

(b) Overlaps kept



$\|\sum_j Q_j\| \leq \tau + (\ell - 1)\kappa$ (Lemma 3.1).
For a Parseval frame, $\kappa = 1$: cost $d + \ell - 1$.

Figure 5: The shared-factor Gram estimate. Each $Q_j = |\omega\rangle\langle\omega|$ acts on the external factor $\mathcal{A}$ and on one slot $\mathcal{B}_j$, and all ℓ of them use the same external factor. Bounding them separately (a) costs τ per slot. Factor $Q_j = V_j V_j^*$ as in Lemma 3.1. The off-diagonal insertion Grams $V_j^* V_k$ in (b), not the products $Q_j Q_k$, have norm $\kappa = \|\sum_i u_i u_i^*\|$. The resulting block-Gram estimate costs one τ plus κ for each further slot. In SparseStack the slots are light sites, the external factor carries the sketch coefficients, and $\sum_i u_i u_i^* = I_d$ makes $\tau = d$ and $\kappa = 1$; this is the origin of the additive row scale $d + q$ in Figure 1.

Here (e_i) is the standard orthonormal basis, and inner products are conjugate-linear in the first argument. All statements also hold over $\mathbb{R}$. On the labelled tensor product $\mathcal{A} \otimes (\mathbb{C}^n)^{\otimes \ell}$, let Q_j act as $|\omega\rangle\langle\omega|$ on the external factor and slot j , and as the identity on the other slots. Thus all Q_j use the same external factor; they are not operators on mutually disjoint tensor factors.

When $F = I_d$, the nonzero Schmidt coefficients of ω are all one, so a change of bases on its active supports identifies it with the unnormalized vector $\sum_{r=1}^d e_r \otimes e_r$. These shared-factor operators, up to normalization, already occur in quantum cloning and singlet monogamy [40]. Nechita, Pellegrini, and Rochette [41, Section 3.3, Proposition 3.8 and Theorem 3.11] analyze their positively weighted sums; their maximal-eigenvalue formula includes $d + \ell - 1$ and the Parseval case of Corollary 3.2. We use these established special cases in random-matrix moment bounds and prove the arbitrary-frame formulas below.

Lemma 3.1 (Shared-factor Gram estimate). *For every integer $\ell \geq 1$,*

$$0 \preceq \sum_{j=1}^{\ell} Q_j \preceq [\tau + (\ell - 1)\kappa]I. \quad (17)$$

The upper bound is attained on the full labelled tensor product. Every isometric compression satisfies the same upper bound. A compression onto a nonzero space attains the bound precisely when its range contains a maximizing vector of the full operator; this criterion applies in particular to symmetry and distinct-site restrictions.

Proof. Write $\mathcal{B}_k = \mathbb{C}^n$ for slot k and $\mathcal{D}_j = \bigotimes_{k \neq j} \mathcal{B}_k$. Define the linear map $V_j : \mathcal{D}_j \rightarrow \mathcal{A} \otimes \bigotimes_k \mathcal{B}_k$ on elementary tensors by

$$V_j \left(\bigotimes_{k \neq j} x_k \right) = \sum_i u_i \otimes x_1 \otimes \cdots \otimes e_i^{(j)} \otimes \cdots \otimes x_\ell.$$

These insertion maps have norm $\sqrt{\tau}$ and share the external factor. Direct contraction gives

$$V_j V_j^* = Q_j, \quad V_j^* V_j = \tau I.$$

For $j \neq k$, the only moving register in $V_j^* V_k$ is slot j in the domain, which becomes slot k in the range. In a Schmidt decomposition

$$\omega = \sum_a \sqrt{\lambda_a} f_a \otimes g_a,$$

the transferred vector x becomes $\sum_a \lambda_a g_a \langle g_a, x \rangle$; the other registers remain unchanged. The numbers λ_a are the nonzero eigenvalues of F , so $\|V_j^* V_k\| = \kappa$ when $F \neq 0$. In the original coordinate basis this map is the transpose of the frame Gram matrix $(\langle u_i, u_j \rangle)_{ij}$; the Schmidt formula keeps the complex conjugations explicit.

Form the block row $V = [V_1 \cdots V_\ell]$. For a vector $z = (z_1, \dots, z_\ell)$ in its domain,

$$\begin{aligned} \langle z, V^* V z \rangle &\leq \tau \sum_j \|z_j\|^2 + \kappa \sum_{j \neq k} \|z_j\| \|z_k\| \\ &\leq [\tau + (\ell - 1)\kappa] \sum_j \|z_j\|^2. \end{aligned}$$

Since $VV^* = \sum_j Q_j$, this proves the upper bound. Positivity and isometric compression preserve it.

For attainment, suppose $F \neq 0$, order the Schmidt values so that $\lambda_1 = \kappa$, and put

$$w_j = \omega_{0j} \otimes \bigotimes_{k \neq j} g_1^{(k)}.$$

Here 0 denotes the external factor. Direct contraction gives $Q_j w_j = \tau w_j$ and $Q_j w_k = \kappa w_j$ for $j \neq k$. Thus $\sum_j w_j$ is an eigenvector with eigenvalue $\tau + (\ell - 1)\kappa$. It is nonzero: its coefficient at $f_1 \otimes g_1^{\otimes \ell}$ is $\ell\sqrt{\kappa} > 0$. If $F = 0$, all the operators vanish. For a compression by an isometry J , equality in the upper bound is equivalent to $\langle Jx, \{[\tau + (\ell - 1)\kappa]I - \sum_j Q_j\} Jx \rangle = 0$ for some unit x . The bracket is positive, so equality holds exactly when Jx is a maximizing eigenvector. The zero operator satisfies the same statement on every nonzero compressed space. $\square$

For a Parseval frame, $F = I_d$, so $(\tau, \kappa) = (d, 1)$ and (17) becomes $d + \ell - 1$. For a general frame the same estimate separates the trace scale $\text{tr } F$ from the interaction scale $\|F\|$. The improvement over the direct sum of norms is $(\ell - 1)(\tau - \kappa)$; it disappears for a rank-one frame. This is the precise structure preserved by the argument. Merely counting occupied particles would discard it.

Corollary 3.2 (Weighted shared factors). *For $a_1, \dots, a_\ell \geq 0$, define*

$$v = (\sqrt{a_1}, \dots, \sqrt{a_\ell})^\top, \quad H_a = (\tau - \kappa) \text{diag}(a_1, \dots, a_\ell) + \kappa v v^*.$$

On the full labelled tensor product,

$$\left\| \sum_j a_j Q_j \right\| = \lambda_{\max}(H_a) \leq (\tau - \kappa) \max_j a_j + \kappa \sum_j a_j. \quad (18)$$

After isometric compression the equality is replaced by an upper bound.

Proof. Use the insertion row $W = [\sqrt{a_1} V_1 \cdots \sqrt{a_\ell} V_\ell]$. Its diagonal Gram blocks are $\tau a_j I$; its off-diagonal blocks have norms at most $\kappa \sqrt{a_j a_k}$. The preceding quadratic-form argument bounds $W^* W$ by the scalar matrix H_a , applied to the vector of block norms. Hence $\|W W^*\| \leq \lambda_{\max}(H_a)$.

For the reverse inequality, take the Schmidt vector g_1 used above. In the domain of each V_j , let h_j be the unit tensor with g_1 in every remaining register. The subspace spanned by

these vectors in their separate domain summands is invariant under W^*W : on it the diagonal entries are τa_j , and the off-diagonal entries are exactly $\kappa \sqrt{a_j a_k}$. Its matrix is H_a . This proves equality, including zero weights; the zero-frame case is immediate. Finally $\tau \geq \kappa$, so the triangle inequality applied to the diagonal and rank-one parts of H_a gives the last bound. $\square$

If $\tau = \kappa$, the final scalar upper bound in (18) is always an equality. If $\tau > \kappa > 0$ and at least one weight is positive, equality holds precisely when all positive weights are equal. Indeed, equality in the sum of the diagonal and rank-one norm bounds requires their maximizing eigenvectors to coincide. The rank-one vector is v , and it belongs to the top eigenspace of $\text{diag}(a_j)$ exactly under the stated condition. If all weights vanish or $F = 0$, both sides are zero.

Auxiliary labels cause no loss when every one-slot contraction carries them unchanged. In the cross Gram they contribute only a register permutation, whose norm is one. Such labels should be retained until the coefficient contraction has been estimated.

3.2 A bridge through commuting coefficient transports

Takeaway. The shared-factor estimate survives coordinate-dependent transports in a common commutative algebra. Its cross Gram is also the positive coefficient kernel in two-transform row sampling. Keeping this kernel through an occupation reflection sharpens the SRHT contraction.

Continue with the frame in (16), and write $P_{il} = \langle u_i, u_l \rangle$. Let $\mathcal{H}$ be a nonzero finite-dimensional Hilbert space and let M_i act on $\mathcal{H}$. On $\mathcal{A} \otimes \mathcal{H} \otimes \bigotimes_{k=1}^{\ell} \mathcal{B}_k$, where $\mathcal{B}_k = \mathbb{C}^n$, define marked insertion maps by

$$V_j \left(h \otimes \bigotimes_{k \neq j} x_k \right) = \sum_i u_i \otimes M_i h \otimes x_1 \otimes \cdots \otimes e_i^{(j)} \otimes \cdots \otimes x_\ell, \quad Q_j = V_j V_j^*. \quad (19)$$

Every insertion uses the same external factor and the same auxiliary factor $\mathcal{H}$. Direct contraction gives

$$V_j^* V_j = \left(\sum_i \|u_i\|^2 M_i^* M_i \right) \otimes I. \quad (20)$$

For $j \neq k$, identify the input slot j with the output slot k ; the other particle labels are passive. On the moving slot and $\mathcal{H}$, the cross Gram $V_j^* V_k$ has blocks

$$K_{li} = P_{il} M_i^* M_l, \quad \text{output label } l, \quad \text{input label } i. \quad (21)$$

The order of the two transports is part of this identity.

Lemma 3.3 (Commuting-transport bridge). *Suppose that the M_i are normal and commute pairwise. Choose a common orthonormal eigenbasis (h_a) , with $M_i h_a = m_i(a) h_a$, and put*

$$F_a = \sum_i |m_i(a)|^2 u_i u_i^*, \quad \tau_a = \text{tr } F_a, \quad \kappa_a = \|F_a\|, \quad b = \max_i \|M_i\|.$$

For every integer $\ell \geq 1$, on the full labelled tensor product,

$$\left\| \sum_{j=1}^{\ell} Q_j \right\| = \max_a \{ \tau_a + (\ell - 1) \kappa_a \} \leq b^2 [\tau + (\ell - 1) \kappa]. \quad (22)$$

The sum is positive. Every isometric compression satisfies the same upper bounds; equality with the full norm holds precisely when its nonzero range contains a maximizing vector of the full operator. After ordering the moving label before $\mathcal{H}$, the cross kernel is

$$K = D(P^\top \otimes I_{\mathcal{H}}) D^* \succeq 0, \quad D = \text{diag}_i(M_i), \quad \|K\| = \max_a \kappa_a \leq b^2 \kappa. \quad (23)$$

Proof. Commuting normal matrices in finite dimension admit simultaneous unitary diagonalization. Each auxiliary mode h_a reduces every insertion and its adjoint. On this mode, (19) is the ordinary shared-factor insertion with coefficient vectors $m_i(a)u_i$. Lemma 3.1, including its attainment argument, gives the exact norm $\tau_a + (\ell - 1)\kappa_a$. Taking the maximum over modes proves the equality in (22). Since $F_a \preceq b^2 F$, the trace and norm bounds give the stated inequality. Positivity and the maximizing-vector criterion follow as in Lemma 3.1.

The (l, i) block of the factored kernel in (23) is $P_{il}M_lM_i^*$. Simultaneous diagonalization gives $M_lM_i^* = M_i^*M_l$, so this is exactly (21). The matrix $P^\top$ is positive Hermitian, including for complex frames. On mode a the kernel is the transpose Gram of $(m_i(a)u_i)_i$, whose norm is κ_a . This proves the remaining assertions. $\square$

What survives a continuation. The retained data are the joint scalar marks $(m_i(a))_i$ in each mode, the actual frame coefficients, and all uncontracted labels and occupation constraints. The three scalars τ, κ, b alone do not retain the sharper mode-dependent information. The following updates are certified:

1. Products, adjoints, and linear combinations of the transports inside the same supplied commutative $*$ -algebra update their scalar marks explicitly. For $M_i \mapsto N_iM_i$, use $m_i(a) \mapsto n_i(a)m_i(a)$ and recompute F_a . A coarse bound charges the product of the maximum transport norms, squared in (22). This update applies to each insertion leg's emitted coordinate. An operation controlled by one fixed ambient slot across the entire sum is a different operation: other insertion legs carry that slot as a spectator.
2. Appending one shared labelled insertion with the same marked coefficient family changes the allowance in mode a by exactly κ_a . Passive factors and one common unitary change of auxiliary basis cost one.
3. A common external map L replaces F_a by LF_aL^* ; the coarse squared-norm cost is $\|L\|^2$. A terminal isometric compression of the whole positive sum is contractive.

Compression does not justify replacing a product by the product of its compressed factors. In particular, compressed transports need not be normal or commute. To continue the same algebraic rule after a restriction, its range must reduce the transports, or the full ambient operators and actual intervening projections must be retained and reanalyzed. Variable identification or coefficient aggregation requires the new exact cross Gram and frame; the old parameters cannot simply be reused. No closure under arbitrary identification is asserted. Lemma 7.6 gives the exact form of this warning for a projection onto retained sign characters: the product of two projected factors misses precisely the term in which two discarded frequencies combine into a retained one.

The next elementary order statement retains more than a terminal norm. It will preserve the budget of the SRHT cross kernel through a common row compression and subsequent ordered interactions.

Lemma 3.4 (Positive budget through a commuting reflection). *Let $X \succeq 0$, and let $S = S^*$ satisfy $\|S\| \leq 1$ and $XS = SX$. For any linear map J into their common space, put*

$$B_J = J^*XJ, \quad H_J = J^*XSJ.$$

Then

$$-B_J \preceq H_J \preceq B_J. \tag{24}$$

There is a self-adjoint contraction C_J on $\text{supp } B_J$ such that $H_J = B_J^{1/2}C_JB_J^{1/2}$. Consequently, for every conformable ordered sandwich,

$$\|LH_JR\| \leq \|LB_J^{1/2}\| \|B_J^{1/2}R\|. \tag{25}$$

The order budget also survives any further common congruence.

Proof. Commutation gives $XS = X^{1/2}SX^{1/2}$, so $-X \preceq XS \preceq X$. Congruence by J proves (24). Positivity of $B_J \pm H_J$ implies that H_J annihilates $\ker B_J$. Invert $B_J^{1/2}$ on its support and set $C_J = B_J^{-1/2}H_JB_J^{-1/2}$ there. The same order bound gives $-I \preceq C_J \preceq I$. Factorization and submultiplicativity prove (25). A further common congruence preserves (24). $\square$

The budget in (25) keeps the actual boundary Grams. It does not say that J^*SJ is still a reflection or that its product with B_J equals H_J .

Recovery in two models and scope. For SparseStack, $M_i = I$ recovers $d + \ell - 1$ for a Parseval frame. The marked version permits further coefficient transports in a common commutative algebra, provided the particular occupation channel is realized by the stated labelled maps and its actual compression. For two-transform SRHT, Section 10.1 supplies

$$M_i = \text{diag}_a(\sqrt{n} \overline{B_{ai}}), \quad g_{ji} = M_i A^* e_j, \quad \max_i \|M_i\|^2 \leq \beta.$$

Taking $u_i = U^* e_i$ gives $P = UU^*$ and a Parseval frame. The identity

$$P_{il} \langle g_{ji}, g_{kl} \rangle = P_{il} \langle e_j, AM_i^* M_l A^* e_k \rangle$$

identifies the one-particle row kernel with a row-unitary conjugate of (21). Thus its positivity and norm bound follow from the displayed coefficient factorization. Appendix A retains this kernel through the reflection $I - 2N_{y,a}$ and improves the flat allowance from $d + 14q^2$ to $d + 12q^2$. The quadratic exponent already comes from the collective contractions; this is a constant refinement.

The ingredients are simultaneous diagonalization, Gram factorization, and the shared-factor estimate, whose Parseval precedents were discussed above [40, 41]. The bridge supplies a checkable continuation rule and a useful contraction in two models; no novelty claim is needed for these ingredients. Its hypotheses do not imply general connected-creation cancellation or separator factorization. Section 3.3.3 gives a specified two-column family shared by the frame and graph maps; the connected-family estimates retain their own hypotheses. Nor does this operator statement assert economical enumeration of moment states, matching lower bounds for the random matrix, or a quantile comparison.

Example 3.5 (Unitary transports without commutation). Let $G = S_3$, $\mathcal{H} = \ell^2(G)$, and $M_i e_j = e_{ji}$, with juxtaposition denoting permutation composition. Put $P = I_6 - \frac{1}{6} \mathbf{1}\mathbf{1}^\top$, $\mathcal{A} = \text{range } P$, and $u_i = P e_i$. The frame is Parseval of dimension five, and every M_i is unitary. Nevertheless their cross kernel, with output (j, l) and input (k, i) , is

$$K_{(j,l),(k,i)} = P_{il} \mathbf{1}_{\{ji=kl\}} = (I - L)_{(j,l),(k,i)}, \quad L_{(j,l),(k,i)} = \frac{1}{6} \mathbf{1}_{\{ji=kl\}}.$$

Order G by the one-line image lists (012), (021), (102), (120), (201), (210); these are not cycle notation. Flatten the following array by rows in the (j, l) indexing:

$$z = \begin{pmatrix} 0 & 0 & 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & -1 \\ -1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & -1 & 0 \\ 0 & -1 & 1 & 0 & 0 & 0 \end{pmatrix}.$$

Summing the six entries satisfying $ji = kl$ gives $6\mathbf{L}z = -3z$, so $\mathbf{K}z = \frac{3}{2}z$ and $z \neq 0$. For two insertion slots, (20) and (21) give the joint Gram

$$[V_1 \ V_2]^*[V_1 \ V_2] = \begin{pmatrix} 5I & \mathbf{K} \\ \mathbf{K} & 5I \end{pmatrix}.$$

Its eigenvector (z, z) has eigenvalue $13/2 > 5 + 2 - 1$. Thus even unitary transports can destroy the additive Parseval allowance if commutation is omitted. The same family $g_{ji} = M_i e_j = e_{ji}$ has orthonormal rows and columns, yet its continued cross kernel has norm at least $3/2$. Marginal Bessel bounds alone therefore do not imply the SRHT kernel bound. This is not an instance of the two-unitary sampling theorem: its common diagonal coefficient factorization is absent.

A projection between two creations need not preserve the transport algebra. Its error can be retained as a positive variance on the complete occupation grade. The next two estimates do not require the commuting marks used in Lemma 3.3.

Here the label set is $[n]$, $c_i = a_i^*$ creates an original hard-core flag, and $n_i = c_i c_i^*$ is its occupation projection. Distinct original labels commute, and repeated use of a label reuses the same flag. Each stage prepends its emitted coefficient factor; all auxiliary factors and their order remain part of the map. A supplied mark is additional deterministic operator data, rather than a property inferred from an unmodified sampling law.

3.2.1 Projection variance controls full input superpositions

Let $u_i \in E$ form a Parseval frame, $\nu_i = \|u_i\|^2$, and let N_i be unitary on H . Define

$$C = \sum_i u_i \otimes N_i^* \otimes c_i, \quad Q_i = N_i Q N_i^*$$

for an orthogonal projection Q . Each stage prepends its coefficient factor. On input occupation S of grade r , let $F = S^c$ be the free labels and put

$$\tau_F = \sum_{i \in F} \nu_i, \quad D_F = \sum_{i \in F} \nu_i Q_i, \quad B_F = D_F / \tau_F, \quad V_F = D_F - D_F^2 / \tau_F.$$

When $\tau_F = 0$, set $B_F = V_F = 0$. Write B_{occ} and V_{occ} for the corresponding occupation-diagonal operators. They retain the actual auxiliary matrices: $0 \preceq B_{\text{occ}} \preceq I$ and $V_{\text{occ}} \succeq 0$.

Proposition 3.6 (Full-grade projection repair). *Let $T_Q = C_{\text{next}} Q C$, $T_I = C_{\text{next}} C$, and let g_{r+1} be a proved bound for $\|C_{\text{next}}|_{r+1}\|^2$. For $\epsilon > 0$ and $0 \leq r \leq n - 2$,*

$$\begin{aligned} T_Q^* T_Q &\preceq (1 + \epsilon) B_{\text{occ}} T_I^* T_I B_{\text{occ}} \\ &\quad + (1 + \epsilon^{-1}) g_{r+1} (r + 1) V_{\text{occ}}. \end{aligned} \tag{26}$$

In particular $\|T_Q\| \leq \|T_I\| + \sqrt{g_{r+1}(r+1)} \|V_{\text{occ}}\|$. No pairwise commutation of the N_i is required for this proposition.

Proof. For every self-adjoint B , direct expansion in the original order gives

$$\sum_{i \in F} \nu_i (Q_i - B)^2 = V_F + \tau_F (B - B_F)^2. \tag{27}$$

Thus V_F is positive. The first-stage residual $E_r = QC - CB_{\text{occ}}$ creates $i \notin S$ with coefficient $u_i \otimes N_i^* (Q_i - B_F)$. Keep both S and the last-created label i before merging them. The retained-label map has exact input Gram V_{occ} by (27). The merge map $|i, S\rangle \mapsto |S \cup \{i\}\rangle$

has squared norm $r + 1$ because each output has that many predecessors. Consequently $E_r^* E_r \preceq (r + 1) V_{\text{occ}}$ on the complete grade, including interference between different initial occupations. The exact identity $T_Q = T_I B_{\text{occ}} + C_{\text{next}} E_r$ and the operator Young inequality prove (26); the triangle inequality gives its norm consequence. $\square$

The first term is the actual congruence $B_{\text{occ}} T_I^* T_I B_{\text{occ}}$; it cannot be replaced by $T_I^* T_I$ merely because B_{occ} is a contraction. An elementary available choice is $g_{r+1} = (r + 2) \sum_i \nu_i$, by retaining and then merging the created label. A sharper coefficient estimate already proved for the actual family may be substituted.

Removing free labels improves the variance in positive order. If $z \in F$ and $\tau_F > \nu_z$, expansion yields

$$V_F - V_{F \setminus \{z\}} = \frac{\tau_F \nu_z}{\tau_F - \nu_z} (Q_z - B_F)^2 \succeq 0. \quad (28)$$

A zero weight changes nothing; if it is the sole positive weight, both variances are zero. Similarly, for disjoint positive-weight families F, G ,

$$V_{F \cup G} = V_F + V_G + \frac{\tau_F \tau_G}{\tau_F + \tau_G} (B_F - B_G)^2.$$

These formulas prove deletion and addition rules without scalarizing projection disagreement. In particular $V_{\text{occ}} \preceq I_{\text{occ}} \otimes V_{[n]}$, after the fixed tensor permutation. If all positive-leverage Q_i coincide with one projection Q_* , the residual vanishes and $T_Q = T_I Q_*$. For a sparse exceptional orbit, retaining its actual labels can avoid the full predecessor-merging cost, as follows.

When supplied exceptional labels depart from a common projection, their occupied coefficient span sharpens the residual. The consumed-row version below retains that same span and charges its child Gram.

3.2.2 Several exceptional directions: retain the occupied span

Let $u_i \in E$ be a Parseval frame, $\sum_i u_i u_i^* = I_E$, and put $P_{ij} = u_i^* u_j$. Supply unitaries N_i on an auxiliary space H , projections Q, Q_* , and a set J such that

$$\Delta_i = N_i Q N_i^* - Q_*, \quad \Delta_i = 0 \quad (i \in O := J^c).$$

These are hypotheses on the supplied marks; no commutation is required. Using the original hard-core flags, define

$$F = \sum_{j \in J} u_j \otimes M_j \otimes c_j, \quad M_j = N_j^* \Delta_j, \quad C_O = \sum_{i \in O} u_i \otimes N_i^* \otimes c_i.$$

Fix total input grade r , exceptional count $q \geq 1$, and ordinary count $t = r - q$. The cross $K_q = F_{q-1}^* C_{O,q}$ has shared output counts $(q, t + 1)$; its other input has counts $(q - 1, t + 1)$. Unavailable sectors give zero maps.

Let $\mathfrak{F}_{q-1}$ denote F on exceptional flags alone and set

$$G_q = \mathfrak{F}_{q-1} \mathfrak{F}_{q-1}^* = \sum_{j,l \in J} u_j u_l^* \otimes M_j M_l^* \otimes c_j c_l^*, \quad \eta_q = \|\mathfrak{F}_{q-1}\|^2. \quad (29)$$

For an occupied exceptional support A , write $U_A = [u_j]_{j \in A}$, $P_A = U_A^* U_A$, and

$$p_A = U_A P_A^\dagger U_A^*, \quad \Pi_q = \bigoplus_{|A|=q} p_A \otimes I_H \otimes |A\rangle\langle A|.$$

Here $P_A^\dagger$ is the Moore–Penrose inverse, so p_A is the orthogonal projection onto the occupied coefficient span, including when its columns are dependent. Ordinary flags carry an identity. Set

$$S_J = P_{JO}P_{OJ} = P_J - P_J^2, \quad \tau_A = \sum_{i \in O} \|p_A u_i\|^2 = \text{tr}(P_A^\dagger(S_J)_{AA}), \quad \tau_q = \max_{|A|=q} \tau_A. \quad (30)$$

Proposition 3.7 (Occupied-span cross). *On the stated sectors,*

$$K_q^* K_q = \eta_q C_{O,q}^* \Pi_q C_{O,q} - C_{O,q}^* (\eta_q \Pi_q - G_q) C_{O,q}, \quad (31)$$

$$\|K_q\|^2 \leq \eta_q(t+1)\tau_q. \quad (32)$$

The subtracted matrix in (31) is positive. Moreover, for $z > 0$, the full positive boundary obeys

$$\begin{pmatrix} 0 & K_q \\ K_q^* & 0 \end{pmatrix} \preceq \text{diag}(z F_{q-1}^* F_{q-1}, z^{-1} C_{O,q}^* \Pi_q C_{O,q}). \quad (33)$$

Proof. At output support A , every summand of F lies in $\text{ran } p_A$. Thus $\Pi_q F = F$ and $0 \preceq G_q \preceq \eta_q \Pi_q$, proving (31). Ordinary creation preserves A . At an ordinary output support V of size $t+1$, apply

$$(t+1) \sum_{i \in V} \|x_i\|^2 - \left\| \sum_{i \in V} x_i \right\|^2 = \sum_{\{i,l\} \subset V} \|x_i - x_l\|^2$$

to $x_i = (p_A u_i \otimes N_i^*) f(V \setminus \{i\})$. Summation over V gives $C_{O,q}^* \Pi_q C_{O,q} \preceq (t+1)\tau_q I$. Parseval gives (30); equivalently $\tau_A = \text{tr}[p_A(I - U_J U_J^*)]$. In particular $\tau_A \leq \min(\text{rank } P_A, \text{rank } P_J - \text{tr } P_J)$. Finally the difference in (33) is the Gram of $\sqrt{z} F_{q-1} x - z^{-1/2} \Pi_q C_{O,q} y$. $\square$

The full Gram matters: its (A, B) block sums over $j \in A, l \in B$ with $A \setminus \{j\} = B \setminus \{l\}$. It therefore contains coherent occupation swaps and is generally not a scalar projection. If desired, a coefficient certificate for η_q is

$$\eta_q \leq \rho_q d_q, \quad \nu_j = \|u_j\|^2, \\ \rho_q = \max_{|A|=q} \|[P_{jl}/\sqrt{\nu_j \nu_l}]_{j,l \in A}\|, \quad d_q = \max_{|S|=q-1} \left\| \sum_{j \in J \setminus S} \nu_j \Delta_j^2 \right\|.$$

Zero-leverage labels are omitted. This follows by synthesizing each output through its normalized coefficient columns and then summing the squared input amplitudes; $M_j^* M_j = \Delta_j^2$.

3.2.3 The original consumed rows give the same profile with a charged child factor

Supply unitary transforms $A_0, B_0 \in U(n)$ with B_0 flat and an original row set T . For child input grade s , retain

$$L_{i,T} = \sum_a \overline{(B_0)_{ai}} c_{y,a} \langle a | A_0^* I_T, \quad \mathcal{F}_T = \sum_{j \in J} u_j \otimes M_j \otimes c_{x,j} \otimes L_{j,T},$$

and define $\mathcal{C}_{O,T}$ similarly with N_i^* and $i \in O$. The full creator is $\mathcal{C}_T = \sum_i u_i \otimes N_i^* \otimes c_{x,i} \otimes L_{i,T}$; its next-stage version uses the incremented x and child grades and places the newly emitted coefficient factor before the preceding one. These maps consume $\ell^2(T)$; no row-output reshuffle is assumed. Their actual shared output Gram is

$$\mathcal{G}_{q,T} = \sum_{j,l \in J} u_j u_l^* \otimes M_j M_l^* \otimes c_{x,j} c_{x,l}^* \otimes L_{j,T} L_{l,T}^*.$$

With all rows, $A_0^* A_0 = I$ gives $L_j L_l^* = \sum_a \overline{(B_0)_{aj}} (B_0)_{al} n_{y,a}$. At a child output support W , this is a sum of $s+1$ exceptional Grams, each with coefficient phases from one row of B_0 . An occupation-diagonal unitary removes those phases, and flatness gives squared norm η_q/n for each summand. Row restriction is an input compression. Hence, with $\tilde{\Pi}_q = \Pi_q \otimes I_{Y_{s+1}}$,

$$0 \preceq \mathcal{G}_{q,T} \preceq \frac{s+1}{n} \eta_q \tilde{\Pi}_q.$$

For the ordinary leg, factor through $A_0^* I_T$, the separate row modes, and child insertion. The preceding insertion calculation bounds each mode by $(t+1)\tau_q/n$ in squared norm; child insertion costs $s+1$. Thus the actual cross satisfies

$$\|\mathcal{F}_{q-1,T}^* \mathcal{C}_{O,q,T}\|^2 \leq \frac{(s+1)^2}{n^2} \eta_q (t+1) \tau_q. \quad (34)$$

The positive gaps and the full boundary (33) remain available before this scalar readout. The loss has no auxiliary-dimension or row-cardinality factor, but flatness is an explicit hypothesis.

A two-stage cross retains the ordinary–exceptional mass For a more specific continued residual, suppose $J = \{0, 1\}$ and the unitary marks commute pairwise. The differences Δ_i need not commute with them. Supply two original row inputs, in order T_2, T_1, H, X_r, Y_s , and form $\mathcal{R} = \mathcal{C}_{T_2, \text{next}} \mathcal{F}_{T_1}$. At initial exceptional count zero or one its nonzero blocks are

$$\mathcal{R} = \begin{pmatrix} \mathcal{U} & 0 \\ \mathcal{B} & \mathcal{D} \end{pmatrix}, \quad \mathcal{R}^* \mathcal{R} = \begin{pmatrix} \mathcal{U}^* \mathcal{U} + \mathcal{B}^* \mathcal{B} & \mathcal{B}^* \mathcal{D} \\ \mathcal{D}^* \mathcal{B} & \mathcal{D}^* \mathcal{D} \end{pmatrix}.$$

The two output rows have exceptional counts one and two. The initial one-occupied sectors are ordered $\{0\}, \{1\}$; the saturated residual is zero. Thus the cross $\mathcal{L} = \mathcal{B}^* \mathcal{D}$ cannot be removed by taking a maximum over initial exceptional counts.

Retain the actual ordered child maps

$$F_{ba} = L_{b,T_2}^{(s+1)} (I_{T_2} \otimes L_{a,T_1}^{(s)}), \quad \|F_{ba}\|^2 \leq p_s := \frac{(s+1)(s+2)}{n^2}, \quad 0 \leq s \leq n-2. \quad (35)$$

Put $a = \nu_0 \nu_1$, $b = |P_{01}|^2$, $V_0 = \Delta_0 \otimes F_{10}$, and $V_1 = \Delta_1 \otimes F_{01}$. Directly multiplying the two ways to create the pair gives

$$\mathcal{B}^* \mathcal{B} = (a+b)(V_0^* V_0 + V_1^* V_1) - b(V_0 - V_1)^* (V_0 - V_1). \quad (36)$$

Indeed their ordered coefficient outputs are $u_1 \otimes u_0$ and $u_0 \otimes u_1$; their inner product is b . Both auxiliary products have the same unitary left factor $(N_0 N_1)^*$, which cancels in the Gram. This proves the formula without moving a difference Δ_i through a mark.

There is also a sharper cross bound. Set $\kappa_J = \|P_J\|$, $\delta = \max_i \|\Delta_i\|$, $s_{\max} = \max(S_J)_{ii}$, and define the ordinary rows

$$\mathfrak{R}_l = \sum_{j \in O} P_{lj} N_j^* \otimes c_{x,j} \otimes L_{j,T_2}^{(s+1)}, \quad \|\mathfrak{R}_l\|^2 \leq \frac{r(s+2)}{n} (S_J)_{ll}.$$

The estimate factors through the row transform, a diagonal coefficient family with square sum $(S_J)_{ll}/n$, ordinary insertion with r predecessors, and child insertion with $s+2$ predecessors. If J_2 swaps the two exceptional coordinates, the actual cross factors, with those row inputs in their stated order, as

$$\begin{aligned} \mathcal{L} = & [\Delta_1 \otimes F_{01}^*, \Delta_0 \otimes F_{10}^*] \text{diag}(\mathfrak{R}_0, \mathfrak{R}_1) (J_2 P_J J_2 \otimes I) \\ & \cdot \text{diag}(N_0 \Delta_1 \otimes (I \otimes L_{1,T_1}^{(s)}), N_1 \Delta_0 \otimes (I \otimes L_{0,T_1}^{(s)})). \end{aligned}$$

Expansion selects the common ordinary creation in the two lower blocks of $\mathcal{R}$; their coefficient contractions are P_{lj} . The first factor recreates the two-row boundary, the middle row consumes T_2 , and the last factor consumes T_1 . The squared allowances for the three factors are respectively $p_s \|\Delta_0^2 + \Delta_1^2\|$, $r(s+2)s_{\max}/n$, and $\delta^2(s+1)/n$. It follows that

$$\|\mathcal{L}\| \leq p_s \kappa_J \delta \sqrt{r s_{\max} \|\Delta_0^2 + \Delta_1^2\|}. \quad (37)$$

The full frame dimension is absent. With the child maps removed the same multiplication gives the primitive bound without p_s . These are bounds for the stated cross; the complete continued Gram and its negative square remain the boundary for a further operation.

3.3 Connected creation: cancel before counting

Takeaway. After noninteracting add–remove pairs cancel, one connected interior costs a single square root of the occupation cutoff at leading order.

We estimate the represented operator by comparing two orders of the same operation. Estimating creation and annihilation separately counts many pairs of embeddings. Subtracting their two Gram operators first cancels the pairs that do not interact. Connectedness then forces every surviving nontrivial pair to use an already occupied label. That restriction saves a power of the ambient size.

3.3.1 The elementary recurrence behind the argument

Let $\mathcal{H} = \bigoplus_{k \geq 0} \mathcal{H}_k$ be a finite graded Hilbert space, and suppose $C\mathcal{H}_k \subseteq \mathcal{H}_{k+\Delta}$ for an integer $\Delta \geq 1$. Write C_k for this block, and set negative-grade blocks equal to zero, and let $P_{\leq K}$ project onto $\bigoplus_{k=0}^K \mathcal{H}_k$. The following statement separates the universal linear algebra from the model-specific cancellation.

Lemma 3.8 (From a graded commutator to creation norms). *Suppose nonnegative numbers a_k satisfy*

$$C_k^* C_k - C_{k-\Delta} C_{k-\Delta}^* \preceq a_k I_{\mathcal{H}_k}. \quad (38)$$

Then

$$\|C_k\|^2 \leq \sum_{j=0}^{\lfloor k/\Delta \rfloor} a_{k-j\Delta}. \quad (39)$$

In particular, if $a_k \leq a(K)$ for $0 \leq k \leq K$, then

$$\|CP_{\leq K}\|, \|C^*P_{\leq K}\| \leq \sqrt{(\lfloor K/\Delta \rfloor + 1)a(K)}. \quad (40)$$

Proof. Taking the largest eigenvalue in (38) gives $\|C_k\|^2 \leq \|C_{k-\Delta}\|^2 + a_k$. Iterate until the grade is negative. Distinct source grades have distinct target grades, so the norm of $CP_{\leq K}$ is the largest block norm. The blocks of $C^*P_{\leq K}$ are adjoints of $C_{k-\Delta}$ with $k \leq K$, and obey the same bound. $\square$

The hypothesis controls every vector of the stated grade. Its useful scale comes from cancellation in the difference of the two Gram operators. We now prove that cancellation for a connected embedding family. In this graph application, the grade increment is $\Delta = r = |E(J)|$, the number of edges created by one embedding.

3.3.2 A connected embedding family supplies the hypothesis

Let J be a fixed finite simple graph whose roles are divided into roots R and a nonempty interior C . Assume that $J[C]$ is connected and every interior role has an incident edge. Remove edges between two roots. Put $c = |C|$ and $r = |E(J)| \geq 1$. Assign pairwise disjoint label pools, each of size at most n , to the roles, and fix the root labels. These pools ensure that two different roles cannot receive the same label.

Let $\mathcal{E}$ be the set of primitive unordered label pairs. The occupation basis is $\{|\omega\rangle : \omega \subseteq \mathcal{E}\}$, with grade $|\omega|$. For an embedding ϕ respecting the pools, write E_ϕ for its r primitive edges. Define

$$\mathcal{C}|\omega\rangle = \sum_{\phi: E_\phi \cap \omega = \emptyset} |\omega \cup E_\phi\rangle. \quad (41)$$

The source state may contain arbitrary primitive edges, including edges unrelated to J . The estimate therefore applies on the entire occupation sector appearing in a factorization.

Proposition 3.9 (Connected creation estimate). *With $A_c = 2^c + c$, the blocks of (41) satisfy*

$$\mathcal{C}_k^* \mathcal{C}_k - \mathcal{C}_{k-r} \mathcal{C}_{k-r}^* \preceq \left[n^c + A_c n^{c-1} (1 + 2k)^{c+1} \right] I. \quad (42)$$

Consequently, both $\|\mathcal{C}P_{\leq K}\|$ and $\|\mathcal{C}^*P_{\leq K}\|$ are at most

$$\sqrt{\lfloor K/r \rfloor + 1} \left[n^{c/2} + \sqrt{A_c} n^{(c-1)/2} (1 + 2K)^{(c+1)/2} \right]. \quad (43)$$

Figure 6 illustrates the three pair types that enter the proof, using a rooted path with two interior roles.

Proof. Fix $|\omega| = k$. A term in the first Gram operator adds E_ϕ and then removes E_ψ ; a term in the second removes E_ψ and then adds E_ϕ . If the interior images of ϕ and ψ are disjoint, their edge sets are disjoint: every edge has an interior endpoint. In this case the two orders have identical admissibility conditions and give the same target state. They cancel, with their multiplicities.

Role pools let us record a surviving pair by $T = \{x \in C : \phi(x) \neq \psi(x)\}$. The case $T = C$ has just canceled. If $T = \emptyset$, the embeddings coincide. Their diagonal contribution is the number of addable embeddings minus the number of removable ones, at most n^c .

Consider $\emptyset \neq T \subsetneq C$. In the add–remove order, every ψ -edge incident to a changed role belongs to ω : it differs from its corresponding ϕ -edge, and disjoint role pools prevent its coincidence with a different edge role. Thus all changed ψ -labels lie in the vertex support of ω , a set of size at most $2k$. Connectedness gives an interior edge from T to $C \setminus T$. Its unchanged endpoint also lies in that support. Choose this endpoint deterministically from T . One otherwise free ϕ -label is now restricted to at most $2k$ choices. For this T there are at most

$$n^{c-1} (2k)^{|T|+1} \leq n^{c-1} (1 + 2k)^{c+1}$$

pairs. There are at most 2^c choices of T .

In the remove–add order, every interior label of ψ belongs to the same vertex support, since every interior role has an incident edge. There are at most $(2k)^c$ choices for ψ . The embeddings share an interior label; choosing one shared role leaves at most cn^{c-1} choices for ϕ . This contributes at most $cn^{c-1} (1 + 2k)^c$. A pair determines the target, so these are absolute row-sum bounds, including multiplicities.

After subtracting the diagonal part, the remainder of the commutator is self-adjoint. Its absolute row sums, and hence its norm, are bounded by $A_c n^{c-1} (1 + 2k)^{c+1}$. This proves (42). Apply Lemma 3.8 and $\sqrt{x+y} \leq \sqrt{x} + \sqrt{y}$ to finish. $\square$

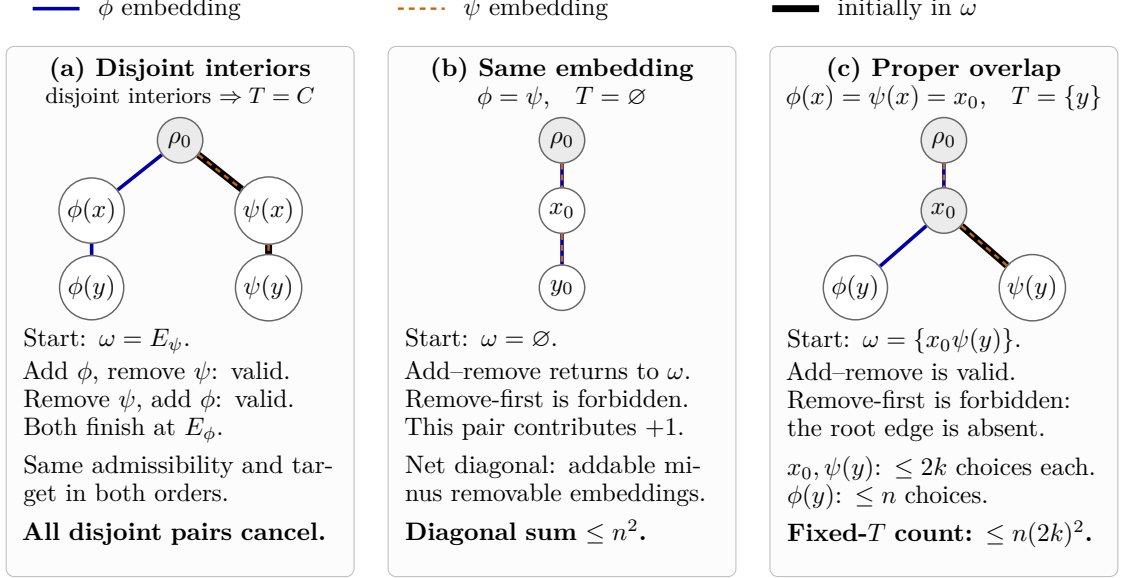


Figure 6: Embedding pairs in the commutator proof of Proposition 3.9, for the rooted path $\rho - x - y$ with fixed root label ρ_0 and $c = 2$. Thick black base edges belong to the initial occupied set ω . In (c), the occupied edge $x_0\psi(y)$ pins the unchanged label x_0 as well as the changed label $\psi(y)$ to the vertex support of ω . The fixed- T bound counts admissible add-remove pairs before further cancellation; it does not assert that every proper overlap survives.

For fixed J and $K = O(\log n)$, the error in the squared commutator bound is smaller than its leading term by $O_J(n^{-1}(\log n)^{c+1})$. The leading creation cost is therefore $n^{c/2}\sqrt{K+1}$. Its dependence on K counts one connected interior, regardless of how many roles that interior contains. The number of roles still affects the power of n and the error polynomial.

Remark 3.10 (Why the square root is a real scale). For a single root joined to one interior role, take a pool of size n . The operator is $C = \sum_{i=1}^n a_i^*$ on n hard-core edge sites: a_i^* inserts edge i if absent and is zero if that edge is already occupied. The normalized uniform vector v_k over all k -element occupation sets satisfies

$$Cv_k = \sqrt{(k+1)(n-k)} v_{k+1}.$$

Thus the $\sqrt{n(K+1)}$ leading scale is attained up to constants for $K \ll n$. This example separates the necessary square-root occupation factor from the extra factors introduced by estimating overlaps separately.

We use the proof as a cancellation test: identify pairs for which adding and removing commute, cancel them before counting, and show that every remaining nontrivial pair fixes at least one otherwise free label. In another model, these three properties and a uniform bound on the remaining pairs are the hypotheses to establish.

The commutator argument contracts a whole connected family. A small explicit family also exposes how its graph output Gram agrees with a marked frame cross, and which child products must survive when the interaction is extended. We keep the same original edges in both maps.

3.3.3 A normalized two-column graph gives one collective operator

Consider $K_{m,2}$ with original edges $j_i = (i, 0)$, $o_i = (i, 1)$, real weights a_i, b_i satisfying $a_i^2 + b_i^2 = 1$, and $n = 2m$. Use the hard-core flags and prepend convention from Section 3.2.1.

Here J will denote exceptional edge labels, rather than a graph template. The edge-indexed vectors $u_{j_i} = a_i e_i$, $u_{o_i} = b_i e_i$ form a Parseval frame. On $H = \mathbb{C}_{\text{current}}^2 \otimes \mathbb{C}_{\text{orbit}}^2$, let X be the coordinate flip and supply

$$Q = I \otimes \text{diag}(1, 0), \quad \Delta = I - 2Q, \quad N_{j_i} = I \otimes X, \quad N_{o_i} = X \otimes I.$$

Set $Q_* = Q$, $J = \{j_i\}$, and $C = \sum_e u_e \otimes N_e^* \otimes c_e$. Preparing the orbit input in $|+\rangle$ and projecting the output current factor onto $|0\rangle$ gives exactly the graph creator

$$|t, S\rangle \mapsto \sum_{i:(i,t) \notin S} b_{it} |i, S \cup \{(i,t)\}\rangle, \quad b_{i0} = a_i, \quad b_{i1} = b_i.$$

Indeed the current input t permits precisely the corresponding edge mark. No original edge labels have been identified.

Let F and C_O be the residual and ordinary creators of Section 3.2.2 for these supplied data. Writing $\Delta_o = I - 2\text{diag}(1, 0)$, orthogonality of different e_i gives the exact marked primitive cross

$$F^* C_O = V \otimes \sum_i \kappa_i c_{j_i}^* c_{o_i}, \quad V = X \otimes (\Delta_o X), \quad \kappa_i = a_i b_i, \quad V^* V = I. \quad (44)$$

For the reverse direction the graph creator is

$$|i, S\rangle \mapsto \sum_{t \in \{0,1\}: (i,t) \notin S} b_{it} |t, S \cup \{(i,t)\}\rangle.$$

These same rung products occur in its output Gram. We retain the diagonal as well as the cross. For an output (t, T) let $\eta = \partial T \triangle \{t\}$, where ∂T is the set of vertices of odd occupied degree. Fix η , the half-rung set A , and the full-rung set F_{full} . Write $R \subset A$ for half-rungs oriented toward column one. Column parity forces the current column; each nontrivial block pairs a layer $|R| = h$ at current column zero with $|S| = h + 1$ at column one. Its exact matrix is

$$\begin{pmatrix} D_0 & I_h^* \\ I_h & D_1 \end{pmatrix}, \quad (I_h)_{S,R} = \begin{cases} \kappa_i, & S = R \cup \{i\}, \\ 0, & \text{otherwise,} \end{cases} \quad (45)$$

where

$$D_0(R) = \sum_{i \in F_{\text{full}}} a_i^2 + \sum_{i \in A \setminus R} a_i^2, \quad D_1(S) = \sum_{i \in F_{\text{full}}} b_i^2 + \sum_{i \in S} b_i^2.$$

Every occupied edge incident to the current column gives one predecessor, which proves the diagonal. Two distinct outputs share a predecessor exactly when they create the two edges of the same empty input rung; their coefficient product is κ_i . A full output rung has only one such predecessor continuation, so contributes only to the diagonal. This proves (45); isolated endpoint states retain their diagonal. The return restriction is the actual output projection onto occupied current degree at least two.

Fix the full and empty rungs, and let I_0 be the M half-rungs, each carrying exactly one original flag. Let q count their exceptional flags. Full and empty rungs contribute zero to (44); on this sector its nontrivial part is

$$D_q(\kappa) = \sum_{i \in I_0} \kappa_i \sigma_i^- : \ell^2 \left(\begin{pmatrix} I_0 \\ q \end{pmatrix} \right) \longrightarrow \ell^2 \left(\begin{pmatrix} I_0 \\ q-1 \end{pmatrix} \right).$$

The total number M of half-rungs stays fixed. Complementing the half-rung labels, and taking an adjoint when needed, identifies this with the inclusion cross I_h in (45).

$$\begin{aligned} \|D_q(\kappa)\|^2 &\leq \min(q, M - q + 1) \sum_i \kappa_i^2, \\ \|D_q(\kappa)\|^2 &= \kappa^2 q (M - q + 1) \quad \text{if all } |\kappa_i| = \kappa. \end{aligned} \quad (46)$$

For the first bound, weighted Cauchy at each output and summation count each input q times. Unweighted Cauchy instead costs $M - q + 1$ per output and leaves $\sum_i \kappa_i^2 n_i \preceq \sum_i \kappa_i^2 I$. For equal magnitudes an occupation sign gauge removes their signs; the inclusion matrix has row sum $M - q + 1$ and column sum q , and its two uniform layer vectors attain the corresponding norm.

For the consumed maps of Section 3.2.3, put $Z_i = L_{j_i, T}^* L_{o_i, T}$. The same calculation gives, on this half-rung sector,

$$\mathcal{K}_T = V \otimes \sum_i \kappa_i \sigma_i^- \otimes Z_i, \quad (47)$$

$$\mathcal{K}_T^* \mathcal{K}_T = \sum_i \kappa_i^2 n_i \otimes Z_i^* Z_i + \sum_{i \neq l} \kappa_i \kappa_l \sigma_i^+ \sigma_l^- \otimes Z_i^* Z_l. \quad (48)$$

At an output S , $|S| = q - 1$, set $z_i = \kappa_i Z_i f(S \cup \{i\})$. The difference-square identity in the proof of Proposition 3.7, with $M - q + 1$ terms, proves the positive budget

$$\mathcal{K}_T^* \mathcal{K}_T \preceq (M - q + 1) \sum_i \kappa_i^2 n_i \otimes Z_i^* Z_i.$$

Since $\|L_{i, T}\|^2 \leq (s + 1)/n$, the two Cauchy arguments also give $\|\mathcal{K}_T\|^2 \leq ((s + 1)/n)^2 \min(q, M - q + 1) \sum_i \kappa_i^2$. At child vacuum with all rows,

$$Z_i = \frac{1}{n} A_0 \text{diag}_a(\zeta_i(a)) A_0^*, \quad |\zeta_i(a)| = 1.$$

In each row mode multiply the occupation $|S\rangle$ by $\prod_{i \in S} \overline{\zeta_i(a)}$. This removes the phases in (47), proving the stronger exact identity

$$\|\mathcal{K}_{[n]}|_{s=0}\|^2 = \|D_q(\kappa)\|^2 / n^2. \quad (49)$$

Row restriction gives its upper-bound version. At positive child grade, the ordered $Z_i^* Z_l$ in (48) remain the interface.

3.3.4 The same square survives another creation and reads its child pair

Take $m = 2$, with $(j_0, o_0, j_1, o_1) = (0, 1, 2, 3)$ and matching supports $\mathcal{B}_0 = \{j_0, o_1\}$, $\mathcal{B}_1 = \{o_0, j_1\}$. The two lowering paths meet at $\{o_0, o_1\}$, so their ordered cycle word is $W = |\mathcal{B}_1\rangle \langle \mathcal{B}_0|$ on these masks. Thus $W^2 = 0$ and the corresponding term of (48) is

$$\kappa_0 \kappa_1 (W \otimes Z_1^* Z_0 + W^* \otimes Z_0^* Z_1).$$

Its norm is $|\kappa_0 \kappa_1| \|Z_1^* Z_0\|$: squaring the two-by-two off-diagonal block proves this without a factor-two triangle loss. Different squares still require collective assembly.

Now form the actual two-creation residual $Q \mathcal{C}_{T_2, \text{next}} \mathcal{C}_{T_1} - \mathcal{C}_{T_2, \text{next}} \mathcal{C}_{T_1} Q$. Restrict inputs to $\mathcal{B}_0, \mathcal{B}_1$, output flags to all four edges, and the ordered coefficient output to $e_1 \otimes e_0$. Exactly two paths remain: o_0 then j_1 , and j_0 then o_1 , respectively. Their auxiliary residual is the same unitary. With

$$F_{ba} = L_{b, T_2}^{(s+1)} (I_{T_2} \otimes L_{a, T_1}^{(s)}), \quad d_0 = b_0 a_1, \quad d_1 = a_0 b_1,$$

the selected map is therefore $[d_0 F_{2,1}, d_1 F_{3,0}]$ up to that unitary. Notice $d_0 d_1 = \kappa_0 \kappa_1$.

For equal row sets apply the actual isometry $I_{\text{eq}}|j\rangle = |j, j\rangle$. Put $h_{j,i} = \sum_a (A_0)_{ja} (B_0)_{ai} c_{y,a}$; then $F_{ba} I_{\text{eq}} = \mathcal{H}_{ba} := \sum_j h_{j,b} h_{j,a} |j\rangle$. Commutation equates $\mathcal{H}_{ba}$ and $\mathcal{H}_{ab}$ for the same unordered pair, but not different pairs. Use all rows, child vacuum, and $(A_0)_{ab} = (B_0)_{ab} =$

$(-1)^{a \cdot b}/2$ for binary two-bit indices. Direct multiplication gives $\mathcal{H}_{2,1} = -\mathcal{H}_{3,0} = -H$, where the only nonzero output rows of H are $\{0, 3\}$ and $\{1, 2\}$:

$$H = \frac{1}{8} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \begin{pmatrix} 1 & -1 & -1 & 1 \end{pmatrix}, \quad \|H\|^2 = \frac{1}{8}.$$

Indeed a child pair survives precisely when its binary difference is 3; its two chronological terms then have magnitude $1/8$. The actual row-identified map and its retained negative square are

$$R_{\text{eq}} = [-d_0 H, d_1 H], \quad R_{\text{eq}}^* R_{\text{eq}} = 2 \text{diag}(d_0^2 H^* H, d_1^2 H^* H) - [d_0 I, d_1 I]^* H^* H [d_0 I, d_1 I].$$

Take $a_0 = a_1 = 3/5$, $b_0 = 4/5$, $b_1 = -4/5$. The primitive selected row $[d_0, d_1] = [12/25, -12/25]$ kills $(|\mathcal{B}_0\rangle + |\mathcal{B}_1\rangle)/\sqrt{2}$. For the same matching input and unit row vector $(1, -1, -1, 1)^T/2$, the actual consumed continuation has squared norm $2(12/25)^2/8 = 36/625$. Consequently no scalar multiple of that primitive pair Gram, tensored with the row identity, can dominate the continued Gram. The missing information is its distinct-pair child product, not an identification of original random entries.

3.4 Positive effects and a subset norm

Takeaway. Positivity bounds selector creation by the square root of the product of the total coefficient norm and the largest small-subset norm.

The selector step in the two-round sketch uses a different structural fact: its coefficient operators are positive. Positivity allows an entire selected set of coefficients to be measured together. The resulting quantity is the norm of the selected sum. We prove the selector estimate directly from this quantity.

Let $A_1, \dots, A_n \succeq 0$ act on a d -dimensional Hilbert space $\mathcal{A}$. For an integer $q \geq 1$, set

$$C = \sum_j A_j, \quad \Gamma = \|C\|, \quad L_q = \max_{|T| \leq \min(q, n)} \left\| \sum_{j \in T} A_j \right\|. \quad (50)$$

These are deterministic coefficient parameters. In an application the space carrying the effects may itself be a compressed occupation space.

Lemma 3.11 (Positive row inequality). *For positive operators $(A_j)_{j \in T}$ and arbitrary vectors (x_j) ,*

$$\left\| \sum_{j \in T} A_j x_j \right\|^2 \leq \left\| \sum_{j \in T} A_j \right\| \sum_{j \in T} \langle x_j, A_j x_j \rangle. \quad (51)$$

Proof. Define $B : \bigoplus_{j \in T} \mathcal{A} \rightarrow \mathcal{A}$ by $B(y_j)_j = \sum_j A_j^{1/2} y_j$. Its adjoint is $B^* z = (A_j^{1/2} z)_j$, so $BB^* = \sum_j A_j$. Applying $\|By\|^2 \leq \|BB^*\| \|y\|^2$ to $y = (A_j^{1/2} x_j)_j$ proves the claim. This includes empty sets and zero effects, for which both sides vanish. $\square$

The usual scalar Cauchy–Schwarz estimate would place $|T|$ before $\sum_j \|A_j x_j\|^2$. Equation (51) keeps a positive quadratic form and the actual norm of the subset sum. This distinction is what makes the next estimate reusable.

Lemma 3.12 (Selector creation and occupancy). *On*

$$\mathcal{H}_{\leq q} = \bigoplus_{J \subseteq [n], |J| \leq q} \mathcal{A},$$

define compressed creation K and occupancy D by

$$(Kx)_T = \sum_{j \in T} A_j x_{T \setminus \{j\}}, \quad (Dx)_J = \left(\sum_{j \in J} A_j \right) x_J.$$

The creation formula is used for $1 \leq |T| \leq q$ and is zero at the empty set. Then

$$\|K\| \leq \sqrt{\Gamma L_q}, \quad \|D\| \leq L_q. \quad (52)$$

Proof. Creation is zero on the top retained grade. For $t < \min(q, n)$, it maps grade t to grade $t + 1$, so different input grades have orthogonal outputs. On one grade, apply Lemma 3.11 and sum over its output sets:

$$\begin{aligned} \sum_{|T|=t+1} \|(Kx)_T\|^2 &\leq L_q \sum_{|T|=t+1} \sum_{j \in T} \langle x_{T \setminus \{j\}}, A_j x_{T \setminus \{j\}} \rangle \\ &= L_q \sum_{|J|=t} \left\langle x_J, \left(\sum_{j \notin J} A_j \right) x_J \right\rangle \leq \Gamma L_q \sum_{|J|=t} \|x_J\|^2. \end{aligned}$$

The last inequality uses positivity and $\sum_{j \notin J} A_j \preceq C$. The occupancy operator is block diagonal, and each block has norm at most L_q . $\square$

Theorem 3.13 (A small-subset moment bound). *Let $\eta_1, \dots, \eta_n$ be independent Bernoulli variables of mean $0 < \rho < 1$, and put*

$$E_\rho = \sum_j (\eta_j / \rho - 1) A_j, \quad \sigma = \sqrt{\frac{1 - \rho}{\rho}}, \quad \theta = \frac{1 - 2\rho}{\rho}.$$

Then

$$\mathbb{E} \operatorname{tr} E_\rho^{2q} \leq d \left[2\sigma \sqrt{\Gamma L_q} + |\theta| L_q \right]^{2q}. \quad (53)$$

The same conclusion holds for $2q$ -wise independent selectors with the same Bernoulli marginals.

Proof. Use the orthonormal basis $p_0 = 1$, $p_1 = (\eta - \rho) / \sqrt{\rho(1 - \rho)}$. For $\xi = \eta / \rho - 1$, direct expectation gives $\mathbb{E}\xi = 0$, $\mathbb{E}[\xi p_1] = \sigma$, and $\mathbb{E}[\xi p_1^2] = \theta$. Multiplication by ξ is therefore

$$\begin{pmatrix} 0 & \sigma \\ \sigma & \theta \end{pmatrix}.$$

Consequently the error multiplication operator, compressed to selector degree at most q , is

$$\mathcal{M}_q = \sigma(K + K^*) + \theta D.$$

Lemma 3.12 gives $\|\mathcal{M}_q\| \leq 2\sigma \sqrt{\Gamma L_q} + |\theta| L_q$. Starting at the vacuum, each multiplication increases selector degree by at most one, so the first q iterates are preserved by the compression. For an external orthonormal basis $(e_a)_{a=1}^d$,

$$\mathbb{E} \operatorname{tr} E_\rho^{2q} = \sum_{a=1}^d \|\mathcal{M}_q^q(e_a \otimes \Omega)\|^2 \leq d \|\mathcal{M}_q\|^{2q}.$$

The trace moment is a polynomial involving at most $2q$ distinct selector coordinates in each monomial, which proves the limited independence assertion by moment matching. $\square$

Remark 3.14 (Chernoff calibration and its limitation). Suppose $C = I$, $0 < \varepsilon \leq 1$, $0 < \delta < 1$, and $q \geq \max\{1, \lceil \log_4(d/\delta) \rceil\}$. Writing $m = \rho n$ for the expected number of selected effects, the moment root in (53) is at most

$$2\sqrt{nL_q/m} + nL_q/m.$$

Thus $m \geq 25nL_q/\varepsilon^2$, when feasible, makes this root at most $0.44\varepsilon < \varepsilon/2$ and gives failure probability at most $d4^{-q} \leq \delta$. If the condition requires $m > n$, taking all effects gives the exact answer.

The selector norm estimate has an equality case: if the nonzero A_j are mutually orthogonal projections, then $\Gamma = L_q = 1$. A unit vector x in one of their ranges, placed at the vacuum, satisfies $\|Kx\|^2 = \sum_j \|A_j x\|^2 = 1$, so $\|K\| = \sqrt{\Gamma L_q}$.

The singleton hypothesis $\|A_j\| \leq R$ implies $L_q \leq \min\{1, qR\}$. Substitution recovers the familiar order $m = O(nR \log(d/\delta)/\varepsilon^2)$ of matrix Chernoff sampling [10]. An improved order requires a better estimate of the actual subset norm. For example, mutually orthogonal projections have $L_q = 1$ even when $qR = q$, but omitting one causes unit error in its range. The sufficient condition then correctly gives no useful high-accuracy subsample. This example calibrates the hypothesis against the geometry of the coefficient family.

The selector law enters after the positive-operator estimate. We transfer a moment to another law by proving either the required joint moment identities or a conditional barycenter identity, as in Lemma 2.6. This keeps the coefficient estimate and the sampling comparison separate.

An input isometry preserves a positive budget even when it mixes the count sectors used to prove that budget.

3.4.1 A Parseval input insertion permits an unpinched positive budget

Count blocks need not survive an additional coefficient map exactly. Let C_{graph} denote the row-to-column graph creator of Section 3.3.3, and supply an analysis isometry $R : E_0 \rightarrow \ell^2(\text{rows})$, $R^*R = I$. The inserted creator $C_{\text{graph}}(R \otimes I)$ has actual output Gram

$$C_{\text{graph}}(RR^* \otimes I)C_{\text{graph}}^* \preceq C_{\text{graph}}C_{\text{graph}}^*. \quad (54)$$

This follows from $RR^* \preceq I$. A supplied stronger majorant $RR^* \preceq \text{diag}(g_i)$ may be used in the same place.

To see why output-count pinching is different, keep the same normalized $K_{m,2}$ family with $a_i = b_i = 1/\sqrt{2}$, start at vacuum, and set $R1 = (1, \dots, 1)^T/\sqrt{m}$. Its $2m$ output states are $|t, \{(i, t)\}\rangle$. The inserted vector has every coordinate $1/\sqrt{2m}$, so its output Gram is $J_{2m}/(2m)$ with norm one. Pinching by the original occupied-row count gives m separate blocks $J_2/(2m)$, whose combined norm is $1/m$. Thus pinching loses a factor m , although (54) remains valid. This insertion is explicitly supplied; it is not inferred from the unmodified SparseStack or SRHT distribution.

Sections 3.2.2–3.3.4 identify the coefficients, occupation masks and child products read by these maps. Their full positive budgets survive actual input congruences. Arbitrary coefficient changes require their new Gram; these interface results alone assert no new complete SRHT sampling theorem or general separator-reuse theorem.

3.5 Retaining coefficient maps before inserting occupations

Takeaway. Keeping the row label and the two first-layer transitions intact lowers the creation-containing contribution from a product of three counts to a sum of products of two.

The occupation representation separates two operations: a deterministic coefficient map chooses a coordinate, and an insertion or extraction map changes the occupied set. Estimating these operations in the wrong order can introduce a factor of the current degree that the coefficient map already controls. We now prove two contraction lemmas under abstract Bessel bounds. Their useful feature is the order of the factorizations: we retain the coefficient register until its Parseval identity has been used.

Let $\mathcal{X}_r$ and $\mathcal{Y}_s$ be the Hilbert spaces with orthonormal bases indexed by the r - and s -subsets of $[n]$, with the zero space at grades outside $0, \dots, n$. A removal $a_{x,i}$ deletes i when present and is zero otherwise; its adjoint inserts i when absent. Write $X_i = a_{x,i} + a_{x,i}^*$, so that $X_i^* X_i = I$ on the complete x occupation space, since it exchanges a subset with its symmetric difference with $\{i\}$. We use the same notation for the y family. The inner product is conjugate linear in its first argument, and

$$a_y(g) = \sum_a \bar{g}_a a_{y,a}, \quad Y(g) = a_y(g) + a_y(\bar{g})^* = \sum_a \bar{g}_a (a_{y,a} + a_{y,a}^*). \quad (55)$$

The conjugation in the creation term is necessary for complex coefficients.

Let $c_i : \mathbb{C}^d \rightarrow \mathbb{C}$ be row functionals with $\sum_i c_i^* c_i = I_d$, and let $g_{ji} \in \mathbb{C}^n$ for $i, j \in [n]$. Assume that, for constants $\alpha, \beta \geq 0$,

$$\left\| \sum_i z_i g_{ji} \right\|^2 \leq \alpha \sum_i |z_i|^2 \quad \text{for each } j \text{ and all } z \in \mathbb{C}^n, \quad (56)$$

$$\sum_j |g_{ji}\rangle \langle g_{ji}| \preceq \beta I_n \quad \text{for each } i. \quad (57)$$

Here $|g\rangle \langle g|$ is the rank-one positive operator $v \mapsto g \langle g, v \rangle$. Define

$$R_j = \frac{1}{\sqrt{n}} \sum_i c_i \otimes X_i \otimes Y(g_{ji}). \quad (58)$$

On the input space $\mathbb{C}^d \otimes \mathcal{X}_r \otimes \mathcal{Y}_s$, let $R_j^{\mu\nu}$ be the component changing the two grades by $(\mu, \nu) \in \{+1, -1\}^2$; we call each such component a *channel*. The first sign always refers to the x grade and the second to the y grade. For $T \subseteq [n]$, the *collected* map $R_T^{\mu\nu}$ retains the row label as an orthogonal output register, rather than summing its output vectors: $R_T^{\mu\nu} z = (R_j^{\mu\nu} z)_{j \in T}$.

Lemma 3.15 (Occupation insertion and extraction). *The map*

$$E_s |S\rangle = \sum_{a \in S} e_a \otimes |S \setminus \{a\}\rangle$$

from $\mathcal{Y}_s$ to $\mathbb{C}^n \otimes \mathcal{Y}_{s-1}$ satisfies $E_s^ E_s = sI$. The insertion $J_s(e_a \otimes |S\rangle) = a_{y,a}^* |S\rangle$ has norm at most $\sqrt{s+1}$. The corresponding assertions hold for the x occupation.*

Proof. Each input subset produces s distinct orthogonal basis vectors, and the outputs of different input subsets are disjoint. This proves the first identity. Insertion is E_{s+1}^* , proving the second assertion. Forbidden insertions and removals are zero, so the bounds also cover empty and boundary grade spaces. $\square$

Lemma 3.16 (Collecting rows inside the mixed coefficient map). *For every input grade and every $T \subseteq [n]$,*

$$n \|R_T^{+-}\|^2 \leq \beta(r+1)s. \quad (59)$$

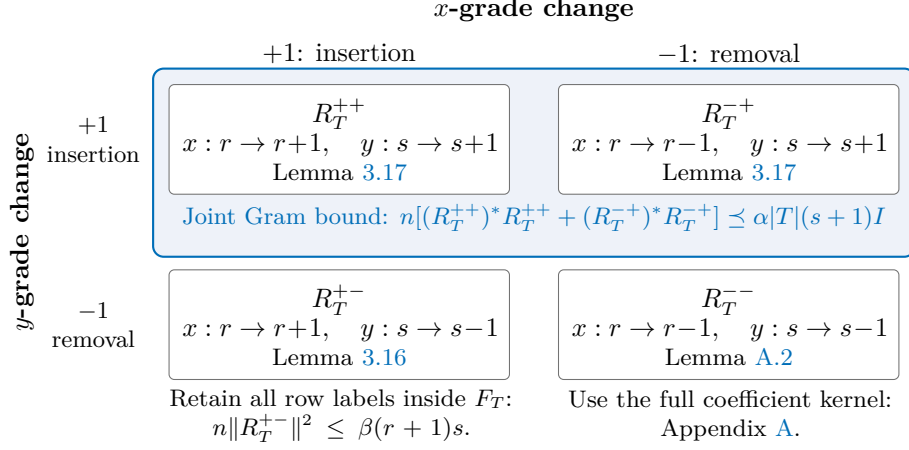


Figure 7: The four row channels on input grade (r, s) . The blue box marks the pair estimated together in Lemma 3.17: both create a y occupation, while their x transitions recombine into $X_i = a_{x,i} + a_{x,i}^*$. We preserve the coefficient sum inside this map until applying its Bessel bound and $X_i^* X_i = I$. The cross terms between the two boxed channels then vanish because their output x grades differ. Estimating their norms independently would lose this shared contraction. The lower-left channel instead uses the collected coefficient map of Lemma 3.16.

Proof. On the external factor and one extracted y coordinate, define

$$F_T : \mathbb{C}^d \otimes \mathbb{C}^n \longrightarrow \ell^2(T) \otimes \mathbb{C}^n, \quad (F_T z)_{j,i} = (c_i \otimes \langle g_{ji} |) z.$$

The output basis vectors labeled by distinct pairs (j, i) are orthogonal. Taking the adjoint and summing those coordinates therefore gives

$$F_T^* F_T = \sum_i c_i^* c_i \otimes \sum_{j \in T} |g_{ji}\rangle \langle g_{ji}| \preceq \beta \sum_i c_i^* c_i \otimes I = \beta I.$$

The map $\sqrt{n} R_T^{+-}$ is the following composition: extract one y coordinate, apply F_T , and insert its coordinate i into the x occupation. The other occupation registers are carried along unchanged. For clarity, on an elementary input $z \otimes |S\rangle \otimes |V\rangle$ the resulting row- j output is

$$\sum_i \sum_{a \in V} c_i(z) \overline{(g_{ji})_a} a_{x,i}^* |S\rangle \otimes |V \setminus \{a\}\rangle.$$

This is the defining coefficient expansion of $\sqrt{n} R_j^{+-}$; in particular, repeated occupation coordinates are handled by the zero insertion rule. Lemma 3.15 bounds the two occupation maps by $\sqrt{s}$ and $\sqrt{r+1}$, while $\|F_T\| \leq \sqrt{\beta}$. Their product proves (59). In particular, collecting the rows has not introduced an additional factor $\sqrt{|T|}$. $\square$

Lemma 3.17 (Keeping the two first-layer transitions together). *For every input grade and every $T \subseteq [n]$,*

$$n \left[(R_T^{++})^* R_T^{++} + (R_T^{-+})^* R_T^{-+} \right] \preceq \alpha|T|(s+1)I. \quad (60)$$

Proof. For a fixed row j , introduce a new one-particle y register and put

$$G_j z = \sum_i \overline{g_{ji}} \otimes (c_i \otimes X_i \otimes I_{y_s}) z.$$

The coefficient synthesis bound (56) also holds for conjugated vectors: conjugate the scalar coefficients and the output of its defining inequality. It remains valid with a passive Hilbert

space: expand vectors in an orthonormal basis of that space, apply the scalar inequality to each basis coefficient, and sum. Consequently

$$\|G_j z\|^2 \leq \alpha \sum_i \|(c_i \otimes X_i \otimes I)z\|^2 = \alpha \langle z, (\sum_i c_i^* c_i \otimes I)z \rangle = \alpha \|z\|^2.$$

Insert its new y coordinate using J_s . By (55), the resulting map is exactly $\sqrt{n}(R_j^{++} + R_j^{-+})$. Its norm is at most $\sqrt{\alpha(s+1)}$. For an input of fixed x grade r , the two summands have orthogonal output grades $r+1$ and $r-1$. Thus their positive Grams add without cross terms. Summing this operator inequality over $j \in T$ proves (60). $\square$

The stronger bounds follow from two different retained structures. In Lemma 3.16, the row register stays inside a contraction. In Lemma 3.17, each X_i stays attached to its Parseval coefficient c_i , and its two grade transitions remain together until orthogonality is used. When the fixed- j coefficient vectors are orthonormal, G_j is an isometry before the final insertion. Thus we identify the exact isometric step separately from the subsequent norm bound on insertion.

For comparison, with $\alpha = \beta = 1$, estimating each of the three channels and each row separately gave

$$n\|Q_T\| \leq |T|[(r+1)(s+1) + (r+1)s + r(s+1)], \quad Q_T = \sum_{(\mu,\nu) \neq (-1,-1)} (R_T^{\mu\nu})^* R_T^{\mu\nu}.$$

Lemmas 3.16 and 3.17 give, in general,

$$n\|Q_T\| \leq \alpha|T|(s+1) + \beta(r+1)s. \quad (61)$$

For flat coefficient families this replaces a product of three occupation parameters by products of two. The two Bessel bounds are the complete hypotheses of these creation-containing channel estimates. The remaining double-annihilation channel uses additional coefficient structure; Appendix A proves its bound for the two-unitary factorization used in Theorem 10.1.

3.6 Contractions on a preserved polynomial cone

The preceding estimates control complete graded spaces or positive operator budgets. A second possibility is to retain a smaller family of vectors that the actual multiplication preserves. In the parity–amplitude representation of Section 2.7, this family is $\mathcal{P}_d^+$: positive coefficients in the original entry monomials, with ordinary degree at most d . The coefficients encode all repeated magnitudes; parity supplies the orthogonal sectors.

Three scalar and vector estimates convert that algebraic condition into a useful L_2 contraction: a Boolean fourth-moment inequality for the sign register, a product-moment bound for the magnitudes, and a strong/weak comparison for a vector with independent coordinates. All three are proved here from the entry hypothesis (7) alone. An earlier version of this argument quoted the comparison from Latała–Strzelecka [45, Theorem 1.1]; Section 3.6.3 proves the independent-coordinate case directly, by an independent copy, which is all the thin-matrix estimate uses. The consequences of that change are collected in Remark 5.8.

3.6.1 A Boolean polynomial estimate

On independent signs $\varepsilon_1, \dots, \varepsilon_N$, write $\varepsilon_S = \prod_{e \in S} \varepsilon_e$ and let N_ρ be the noise operator defined on Fourier characters by $N_\rho \varepsilon_S = \rho^{|S|} \varepsilon_S$. The L_2 -to- L_4 case of Boolean hypercontractivity is proved here and suffices for the reference constants of the earlier version of the argument; the

refined constants below also use the classical general Boolean hypercontractivity theorem of Bonami and Beckner [55, 56], at the exponents actually compared. We use the formulation in O'Donnell's [47, Theorem 1.1] and [46, Chapter 10]; these are expository sources, not the original attribution.

Lemma 3.18 (Hilbert-valued Boolean polynomials). *If $F = \sum_{|S| \leq d} \varepsilon_S v_S$ takes values in a real Euclidean space, then*

$$\|F\|_{L_4(\ell_2)} \leq 3^{d/2} \|F\|_{L_2(\ell_2)}. \quad (62)$$

Proof. For fixed real Hilbert vectors a, b ,

$$\mathbb{E}_\varepsilon \|a + \varepsilon b\|_2^4 = (\|a\|_2^2 + \|b\|_2^2)^2 + 4\langle a, b \rangle^2 \leq \|a\|_2^4 + 6\|a\|_2^2 \|b\|_2^2 + \|b\|_2^4.$$

If A, B depend on other signs, integrate this inequality and use Cauchy–Schwarz in the mixed term. Taking a square root gives

$$\|A + \varepsilon B\|_{L_4(\ell_2)}^2 \leq \|A\|_{L_4(\ell_2)}^2 + 3\|B\|_{L_4(\ell_2)}^2. \quad (63)$$

Set $\rho = 1/\sqrt{3}$ and induct on the number of signs. Write $F = A + \varepsilon_N B$, where A, B depend on the first $N - 1$ signs. The noise operator on those signs is denoted by N'_ρ . The induction hypothesis and (63) yield

$$\|N_\rho F\|_{L_4(\ell_2)}^2 \leq \|N'_\rho A\|_{L_4(\ell_2)}^2 + 3\rho^2 \|N'_\rho B\|_{L_4(\ell_2)}^2 \leq \|A\|_{L_2(\ell_2)}^2 + \|B\|_{L_2(\ell_2)}^2 = \|F\|_{L_2(\ell_2)}^2.$$

The base case of zero signs is equality. Finally set $G = \sum_{|S| \leq d} \rho^{-|S|} \varepsilon_S v_S$, so $N_\rho G = F$. Hilbert Fourier orthogonality gives

$$\|F\|_{L_4(\ell_2)} \leq \|G\|_{L_2(\ell_2)} = \left(\sum_{|S| \leq d} \rho^{-2|S|} \|v_S\|_2^2 \right)^{1/2} \leq \rho^{-d} \|F\|_{L_2(\ell_2)}.$$

□

Lemma 3.19 (Hilbert-valued hypercontractivity at general exponents). *If $F = \sum_{|S| \leq d} \varepsilon_S v_S$ takes values in a real Euclidean space, then for every $u \geq 2$,*

$$\|F\|_{L_u(\ell_2)} \leq (u - 1)^{d/2} \|F\|_{L_2(\ell_2)}. \quad (64)$$

Proof. Let $\rho = (u - 1)^{-1/2}$ and $G = \sum_S \rho^{-|S|} \varepsilon_S v_S$, so that $N_\rho G = F$. The noise operator is an average over a probability measure, $N_\rho G(x) = \mathbb{E}_y G(y)$ with y a ρ -correlated copy of x , so $\|N_\rho G(x)\|_2 \leq N_\rho(\|G\|_2)(x)$ pointwise. The scalar hypercontractive inequality $\|N_\rho h\|_{L_u} \leq \|h\|_{L_2}$ for $\rho \leq (u - 1)^{-1/2}$ [47, Theorem 1.1], applied to $h = \|G\|_2$, gives $\|F\|_{L_u(\ell_2)} \leq \|G\|_{L_2(\ell_2)} \leq \rho^{-d} \|F\|_{L_2(\ell_2)}$ by Fourier orthogonality. At $u = 4$ this is Lemma 3.18. □

3.6.2 Positive original coefficients control polynomial moments

Fix independent symmetric variables $W = (W_e)_{e \leq N}$ satisfying (7), and put $Z_e = |W_e|$. For an exponent vector $\nu \in \mathbb{N}_0^N$, write $Z^\nu = \prod_e Z_e^{\nu_e}$ and $|\nu| = \sum_e \nu_e$. Polynomial degree below always means this ordinary total degree, including all repeated powers.

Lemma 3.20 (A product-moment bound). *For $m_e(u) = \mathbb{E} Z_e^u$, $u \in \mathbb{N}_0$, and $m_e(0) = 1$,*

$$m_e(u + v) \leq \alpha^{u+v} m_e(u) m_e(v), \quad u, v \in \mathbb{N}_0. \quad (65)$$

Consequently, every scalar polynomial $G(Z) = \sum_\nu a_\nu Z^\nu$ with $a_\nu \geq 0$ and degree at most D satisfies

$$\mathbb{E} G(Z)^2 \leq \alpha^{2D} (\mathbb{E} G(Z))^2. \quad (66)$$

Proof. For positive integers u, v , Cauchy–Schwarz and moment doubling give

$$\begin{aligned} m_e(u+v) &\leq \sqrt{m_e(2u)m_e(2v)}, \\ m_e(2u) &= \|W_e\|_{L_{2u}}^{2u} \leq \alpha^{2u} \|W_e\|_{L_u}^{2u} = \alpha^{2u} m_e(u)^2. \end{aligned}$$

Apply the second formula also with v to prove (65). If either exponent is zero, the assertion follows from $\alpha \geq 1$; this does not apply moment doubling at zero. Identically zero variables cause no exception.

For two monomials of G , independence gives

$$\mathbb{E}Z^{\nu+\mu} = \prod_e m_e(\nu_e + \mu_e) \leq \alpha^{|\nu|+|\mu|} \prod_e m_e(\nu_e) \prod_e m_e(\mu_e).$$

Multiply by $a_\nu a_\mu \geq 0$, sum over ν, μ , and use $|\nu| + |\mu| \leq 2D$ to obtain (66). $\square$

Lemma 3.21 (Positive-coefficient polynomial comparison). *Let $F(W)$ be a polynomial taking values in a fixed finite-dimensional real Euclidean space. Suppose every coordinate has nonnegative coefficients in the ordinary monomial basis of the original W_e , and $\deg F \leq d$. With*

$$H_\alpha = \sqrt{3}\alpha, \tag{67}$$

one has

$$\|F\|_{L_4(\ell_2)} \leq H_\alpha^d \|F\|_{L_2(\ell_2)}, \tag{68}$$

$$\|F\|_{L_{2p/(p-2)}(\ell_2)} \leq H_\alpha^{4d/p} \|F\|_{L_2(\ell_2)}, \quad p \geq 4. \tag{69}$$

Proof. Represent $W_e = \varepsilon_e Z_e$ in law with independent signs and independent magnitudes. Group ordinary monomials by their exponent parity:

$$F(\varepsilon Z) = \sum_S \varepsilon_S F_S(Z).$$

Every coordinate of each F_S still has nonnegative coefficients. Its monomials have exponent parity S , degree at most d , and hence $|S| \leq d$. Conditional on Z , Lemma 3.18 gives

$$\mathbb{E}_\varepsilon \|F(\varepsilon Z)\|_2^4 \leq 3^{2d} \left(\sum_S \|F_S(Z)\|_2^2 \right)^2. \tag{70}$$

The scalar polynomial $G(Z) = \sum_S \|F_S(Z)\|_2^2$ has nonnegative coefficients and degree at most $2d$. Conditional sign orthogonality gives $\mathbb{E}G = \mathbb{E}\|F\|_2^2$. Thus (66), with $D = 2d$, proves

$$\mathbb{E}\|F\|_2^4 \leq 3^{2d} \alpha^{4d} (\mathbb{E}\|F\|_2^2)^2.$$

Its fourth root is (68). For $u = 2p/(p-2) \in [2, 4]$, the interpolation identity $1/u = (1-4/p)/2 + (4/p)/4$ proves (69). Degree zero gives equality in both comparisons. $\square$

The hypothesis concerns coefficients, not pointwise positivity of $F(W)$. A coordinate such as $W_1 + W_2$ is allowed although its value may be negative. Nor does the lemma apply to every polynomial merely because the law is symmetric: arbitrary negative monomial coefficients may create cancellations in the L_2 norm. We will only project or multiply polynomials in ways that preserve the stated coefficient cone.

Corollary 3.22 (Even-order doubling for linear forms). *Let $W = (W_e)$ be independent symmetric variables satisfying (7), let a_e be arbitrary real coefficients, and put $S = \sum_e a_e W_e$. For every integer $q \geq 1$,*

$$\|S\|_{L_{4q}} \leq h_q(\alpha) \|S\|_{L_{2q}}, \quad h_q(\alpha) = \left(\frac{4q-1}{2q-1}\right)^{1/2} \alpha \leq H_\alpha. \quad (71)$$

At $q = 1$ the constant is $h_1(\alpha) = H_\alpha$; $h_q(\alpha)$ decreases to $\sqrt{2}\alpha$.

Proof. Since the W_e are symmetric, replacing a_e by $|a_e|$ changes neither side, so assume $a_e \geq 0$, and represent $W_e = \varepsilon_e Z_e$ in law as in Section 2.7.1. Conditional on Z , $S = \sum_e a_e Z_e \varepsilon_e$ is a homogeneous polynomial of degree one in the signs, on which the noise operator acts by $T_\rho S = \rho S$. The hypercontractivity theorem for uniform signs [47, Theorem 1.1], [46, Chapter 10], $\|T_\rho f\|_{L_u} \leq \|f\|_{L_p}$ for $1 \leq p \leq u$ and $\rho \leq \sqrt{(p-1)/(u-1)}$, applied with $p = 2q$ and $u = 4q$, gives

$$\mathbb{E}_\varepsilon |S|^{4q} \leq \left(\frac{4q-1}{2q-1}\right)^{2q} (\mathbb{E}_\varepsilon |S|^{2q})^2.$$

The conditional moment $G(Z) = \mathbb{E}_\varepsilon S^{2q}$ is a polynomial in the magnitudes with nonnegative coefficients and degree $2q$: expanding S^{2q} and averaging the signs discards every monomial with an odd exponent and leaves coefficients that are products of multinomial coefficients and powers of the $a_e \geq 0$. Lemma 3.20 with $D = 2q$ gives $\mathbb{E} G^2 \leq \alpha^{4q} (\mathbb{E} G)^2$, and $\mathbb{E} G = \mathbb{E} S^{2q}$. Hence $\mathbb{E} S^{4q} = \mathbb{E}_Z \mathbb{E}_\varepsilon S^{4q} \leq ((4q-1)/(2q-1))^{2q} \alpha^{4q} (\mathbb{E} S^{2q})^2$; take $4q$ -th roots. The weaker constant H_α for every q also follows from Lemma 3.21 applied to S^q in one dimension, since S^q has nonnegative coefficients and degree q . $\square$

This is the only consequence of the cone comparison needed for scalar sums, and it replaces the scalar moment-growth estimate that an external strong/weak theorem would otherwise supply. Only the homogeneity of S in the signs is used for the sign factor; the cone polynomials of Section 3.6.4 are not homogeneous, and there the L_2 -based form of Lemma 3.19 is used instead.

3.6.3 A strong/weak comparison from an independent copy

The thin-matrix estimate below compares the L_{4q} moment of the Euclidean norm of a random vector with its variance and its weak moments. For a vector with *independent* coordinates this can be proved directly, and that is all the matrix argument uses. No comparison for general Banach norms or general index sets is required.

Lemma 3.23 (Symmetrized squares of a regular linear form). *Let $V = \sum_e a_e W_e$ be a linear form in independent symmetric variables satisfying (7), and let V' be an independent copy of V . Then for every integer $r \geq 1$,*

$$\|V^2 - (V')^2\|_{L_{2r}} \leq \chi_\alpha \|VV'\|_{L_{2r}}, \quad \chi_\alpha = \chi_0 \alpha^2, \quad \chi_0 = \frac{7}{3} 2^{1/4} < 2.775. \quad (72)$$

Proof. Since $|x^2 - y^2| \leq \max(x^2, y^2)$, one has $|V^2 - (V')^2|^{2r} \leq V^{4r} + (V')^{4r}$, so Corollary 3.22 and the factorization $\|VV'\|_{L_{2r}} = \|V\|_{L_{2r}}^2$ of the independent copy give

$$\|V^2 - (V')^2\|_{L_{2r}} \leq 2^{1/(2r)} \|V\|_{L_{4r}}^2 \leq 2^{1/(2r)} \frac{4r-1}{2r-1} \alpha^2 \|V\|_{L_{2r}}^2 = 2^{1/(2r)} \frac{4r-1}{2r-1} \alpha^2 \|VV'\|_{L_{2r}}.$$

Both factors in front of α^2 decrease in r , so for $r \geq 2$ the constant is at most its value $2^{1/4} \cdot \frac{7}{3} = \chi_0$ at $r = 2$. At $r = 1$ expand the fourth moment instead: with $v_e = \mathbb{E} W_e^2$ and $a_e \geq 0$,

$$\mathbb{E} V^4 = \sum_e a_e^4 \mathbb{E} W_e^4 + 3 \sum_{e \neq f} a_e^2 a_f^2 v_e v_f \leq \max\{\alpha^4, 3\} \left(\sum_e a_e^2 v_e \right)^2 = \max\{\alpha^4, 3\} (\mathbb{E} V^2)^2,$$

by (7) at $p = 2$. Hence $\mathbb{E}(V^2 - (V')^2)^2 = 2\mathbb{E}V^4 - 2(\mathbb{E}V^2)^2 \leq 2(\max\{\alpha^4, 3\} - 1)(\mathbb{E}V^2)^2$, while $\mathbb{E}(VV')^2 = (\mathbb{E}V^2)^2$, so the ratio of the two L_2 norms is at most $\sqrt{2(\max\{\alpha^4, 3\} - 1)} \leq 2\alpha^2 < \chi_0\alpha^2$ for $\alpha \geq 1$. Zero variance makes both sides vanish. $\square$

Theorem 3.24 (Independent-coordinate strong/weak comparison). *Let $(V_j)_{j \in J}$ be a finite family of independent real random variables, each symmetric in law, with finite moments of order $4q$ for an integer $q \geq 1$, let (V'_j) be an independent copy of the family, and suppose $\gamma \geq 0$ satisfies $\|V_j^2 - (V'_j)^2\|_{L_{2r}} \leq \gamma \|V_j V'_j\|_{L_{2r}}$ for $j \in J$ and $1 \leq r \leq q$. Put*

$$R = \left(\sum_j V_j^2 \right)^{1/2}, \quad \sigma^2 = \sum_j \mathbb{E}V_j^2, \quad W_s = \sup_{\sum_j t_j^2 \leq 1} \left\| \sum_j t_j V_j \right\|_{L_s}.$$

Then

$$\|R\|_{L_{4q}} \leq \frac{\gamma W_{2q}}{2} + \left(\sigma^2 + \frac{\gamma^2 W_{2q}^2}{4} \right)^{1/2} \leq \sigma + \gamma W_{2q}. \quad (73)$$

The hypothesis holds with $\gamma = 2\beta^2$ whenever $\|V_j\|_{L_{4r}} \leq \beta \|V_j\|_{L_{2r}}$ for $1 \leq r \leq q$, by Minkowski's inequality and the factorization $\|V_j V'_j\|_{L_{2r}} = \|V_j\|_{L_{2r}}^2$; and it holds with $\gamma = \chi_\alpha = \chi_0\alpha^2$ whenever every V_j is a linear form in independent symmetric variables satisfying (7), by Lemma 3.23.

Proof. Put $A_j = V_j^2 - (V'_j)^2$ and $B_j = V_j V'_j$. Both families are independent in j and symmetric in law. Expand $\mathbb{E}(\sum_j A_j)^{2q}$ and $\mathbb{E}(\sum_j B_j)^{2q}$ by the multinomial formula. By symmetry and independence only the monomials in which every exponent is even survive, and the hypothesis compares those term by term; hence $\|\sum_j A_j\|_{L_{2q}} \leq \gamma \|\sum_j B_j\|_{L_{2q}}$. Write $Q = \sum_j V_j^2$ and Q' for its copy, so $\mathbb{E}Q' = \sigma^2$. Jensen in the copy coordinate gives $\|Q - \sigma^2\|_{L_{2q}} \leq \|Q - Q'\|_{L_{2q}} = \|\sum_j A_j\|_{L_{2q}}$. Conditioning on the copy, $\sum_j B_j$ is the linear form with the deterministic coefficient vector V' , so $\|\sum_j B_j\|_{L_{2q}} \leq W_{2q} \|R\|_{L_{2q}}$. Put $x = \|R\|_{L_{4q}}$ and $y = \|R\|_{L_{2q}} \leq x$. Since $\|Q\|_{L_{2q}} = x^2$, Minkowski gives $x^2 \leq \sigma^2 + \gamma W_{2q} y \leq \sigma^2 + \gamma W_{2q} x$, and the quadratic inequality $x^2 \leq \sigma^2 + ax$ with $x, \sigma, a \geq 0$ forces $x \leq a/2 + \sqrt{\sigma^2 + a^2/4} \leq \sigma + a$. $\square$

Two features of (73) matter later. The weak moment enters at order $2q$, half the order of the strong moment on the left, and there is no dimension factor. Both are used at the logarithmic moment order of the local theorem.

The deletion argument of Section 5.4 also compares weak moments across orders, down to the subunit order $\log 2$. Corollary 3.22 supplies this with an explicit constant and no restriction to orders above one.

Proposition 3.25 (Uniform real-order doubling). *Let $S = \sum_e a_e W_e$ be as in Corollary 3.22 and put*

$$J_\alpha = H_\alpha^7. \quad (74)$$

Then $\|S\|_{L_{2p}} \leq J_\alpha \|S\|_{L_p}$ for every real $p \geq \frac{1}{2}$, in particular for every $p \geq \log 2$.

Proof. Hölder with the interpolation $2 = \frac{4}{7} \cdot \frac{1}{2} + \frac{3}{7} \cdot 4$ and the case $q = 1$ of (71) give $\mathbb{E}|S|^2 \leq (\mathbb{E}|S|^{1/2})^{4/7} (\mathbb{E}|S|^4)^{3/7} \leq H_\alpha^{12/7} (\mathbb{E}|S|^{1/2})^{4/7} (\mathbb{E}|S|^2)^{6/7}$, hence $\|S\|_{L_2} \leq H_\alpha^6 \|S\|_{L_{1/2}}$. For $\frac{1}{2} \leq p \leq 2$, monotonicity of the moment functional gives $\|S\|_{L_{2p}} \leq \|S\|_{L_4} \leq H_\alpha \|S\|_{L_2} \leq H_\alpha^7 \|S\|_{L_{1/2}} \leq H_\alpha^7 \|S\|_{L_p}$. For $p > 2$ put $r = \lfloor p/2 \rfloor \geq 1$, so $2r \leq p$ and $2p \leq 8r$; two applications of (71) give $\|S\|_{L_{2p}} \leq \|S\|_{L_{8r}} \leq H_\alpha^2 \|S\|_{L_{2r}} \leq H_\alpha^2 \|S\|_{L_p}$. $\square$

The assembly of Section 5.4 compares weak moments across widely separated orders, from the order at which the deletion parameter controls a weak moment to the order at which a block is estimated. Applying Proposition 3.25 once per doubling charges H_α^7 each time; the following order-adapted form charges the magnitudes exactly once per doubling of the order and takes the sign cost at the terminal exponent.

Definition 3.26 (Interpolated doubling exponent). For $t \geq 1$ let $m = \lfloor \log_2 t \rfloor$ and

$$\eta(t) = m + 2 - \frac{2^{m+1}}{t}. \quad (75)$$

It is continuous and increasing in t and equals m at $t = 2^m$.

Lemma 3.27 (Magnitude growth at a specified ratio). *Let W satisfy (7). Then $\|W\|_{L_{tu}} \leq \alpha^{\eta(t)} \|W\|_{L_u}$ for all $u \geq 1$ and $t \geq 1$. Consequently, if $G(Z) = \sum_{\nu} c_{\nu} Z^{\nu}$ has nonnegative coefficients and every monomial has degree b , in independent magnitudes $Z_e = |W_e|$ of variables satisfying (7), then $\|G\|_{L_t} \leq \alpha^{b\eta(t)} \mathbb{E}G$ for $t \geq 1$.*

Proof. With $m = \lfloor \log_2 t \rfloor$, m and $m+1$ applications of (7) give $\|W\|_{2^m u} \leq \alpha^m \|W\|_u$ and $\|W\|_{2^{m+1}u} \leq \alpha^{m+1} \|W\|_u$; Hölder's inequality between the orders $2^m u$ and $2^{m+1}u$ with the weight $\theta = 2 - 2^{m+1}/t$, the solution of $1/t = (1-\theta)/2^m + \theta/2^{m+1}$, gives $\|W\|_{tu} \leq \alpha^{m+\theta} \|W\|_u$. For a monomial, independence gives $\|Z^{\nu}\|_{L_t} = \prod_{e: \nu_e > 0} \|Z_e\|_{L_{t\nu_e}}^{\nu_e} \leq \alpha^{\eta(t) \sum_e \nu_e} \prod_e \mathbb{E} Z_e^{\nu_e}$ by the first part with $u = \nu_e \geq 1$. Minkowski's inequality at $t \geq 1$ and the nonnegative coefficients give the second claim. $\square$

Proposition 3.28 (Doubling from an even anchor to the terminal order). *Let $S = \sum_e a_e W_e$ be as in Corollary 3.22, let $b \geq 2$ be an even integer and $P \geq b$ real. Then*

$$\|S\|_{L_P} \leq \left(\frac{P-1}{b-1} \right)^{1/2} \alpha^{\eta(P/b)} \|S\|_{L_b}. \quad (76)$$

If $0 < p_0 \leq b$, then with $\epsilon = 2b/p_0 - 2$,

$$\|S\|_{L_b} \leq h_{b/2}(\alpha)^{\epsilon} \|S\|_{L_{p_0}} = \left(\frac{2b-1}{b-1} \right)^{\epsilon/2} \alpha^{\epsilon} \|S\|_{L_{p_0}}, \quad (77)$$

and if $b \leq p_0$ then $\|S\|_{L_b} \leq \|S\|_{L_{p_0}}$.

Proof. Assume $a_e \geq 0$ and represent $W_e = \varepsilon_e Z_e$. Conditional on Z , hypercontractivity from L_b to L_P for the homogeneous degree-one sign polynomial S , exactly as in the proof of Corollary 3.22, gives $\mathbb{E}_{\varepsilon} |S|^P \leq ((P-1)/(b-1))^{P/2} G(Z)^{P/b}$ with $G(Z) = \mathbb{E}_{\varepsilon} S^b$, a polynomial with nonnegative coefficients all of whose monomials have degree b , because b is even. Lemma 3.27 at $t = P/b$ gives $\mathbb{E}_Z G^{P/b} \leq \alpha^{P\eta(P/b)} (\mathbb{E}G)^{P/b}$ with $\mathbb{E}G = \mathbb{E}S^b$; take P -th roots. For (77), Hölder between the orders p_0 and $2b$ with $1/b = (1-\theta)/p_0 + \theta/(2b)$, that is $\theta/(1-\theta) = \epsilon$, and Corollary 3.22 at $q = b/2$ give $\|S\|_b \leq \|S\|_{p_0}^{1-\theta} (h_{b/2} \|S\|_b)^{\theta}$, which rearranges to (77). Hölder's inequality is log-convexity of the moment functional in the reciprocal exponent and is valid for every positive exponent, including $p_0 < 1$. The last statement is monotonicity of the moment functional. $\square$

Both bounds use only (7), the hypercontractivity theorem for signs, and Hölder's inequality. With $b = 2$ and $P = 2^{m+1}$, (76) reads $\|S\|_{L_P} \leq \sqrt{P-1} \alpha^m \|S\|_{L_2}$: at $\alpha = 1$ this is the hypercontractive Khintchine bound, and each doubling of the order costs one factor α in the magnitudes.

Let Y be a matrix with independent entries that are symmetric in law and satisfy (7). Taking bilinear suprema over deterministic unit vectors on a fixed deleted submatrix, Proposition 3.25 gives

$$R_{Y,I}(2p) \leq J_{\alpha} R_{Y,I}(p), \quad p \geq \log 2, \quad (78)$$

for every deterministic I , since each bilinear form $\sum_{i,j \notin I} Y_{ij} s_i t_j$ is a deterministic linear combination of independent symmetric entries. Symmetry is essential here; it is the reason

the deletion argument is run on a symmetrized matrix. In the same way Proposition 3.28 gives, for even $b \geq 2$, $P \geq b$ and $0 < p_0 \leq b$,

$$R_{Y,I}(P) \leq \left(\frac{P-1}{b-1}\right)^{1/2} \alpha^{\eta(P/b)} R_{Y,I}(b), \quad R_{Y,I}(b) \leq \left(\frac{2b-1}{b-1}\right)^{\epsilon/2} \alpha^\epsilon R_{Y,I}(p_0), \quad \epsilon = \frac{2b}{p_0} - 2, \quad (79)$$

and $R_{Y,I}(b) \leq R_{Y,I}(p_0)$ when $b \leq p_0$; these are the forms used in Section 5.4.

3.6.4 Thin matrices and the cone contraction

Let Y be an $n \times n$ matrix with independent symmetric-in-law α -regular entries. Use $\sigma_r, \sigma_c, \sigma$ and R_Y from (8)–(9), and use $\mathcal{U}, \mathcal{T}, \mathcal{C}_\pm, \mathcal{P}_d^+$ from Section 2.7.1. Matrix symmetry is not assumed. Write

$$\Lambda_q(Y) = \sigma(Y) + \chi_\alpha R_Y(2q), \quad \chi_\alpha = \chi_0 \alpha^2, \quad \chi_0 = \frac{7}{3} 2^{1/4}, \quad (80)$$

and $\Lambda_{q,r}(Y), \Lambda_{q,c}(Y)$ for the same expression with $\sigma_r(Y), \sigma_c(Y)$ in place of $\sigma(Y)$.

Lemma 3.29 (Directional moments control a random matrix). *Let A be a random real $m \times N$ matrix, $m \geq 1$, let $p > 0$, and let U be uniform on the unit sphere of $\mathbb{R}^m$ and independent of A . Then*

$$(\mathbb{E}\|A\|^p)^{1/p} \leq (\mathbb{E}|U_1|^p)^{-1/p} \sup_{\|s\|_2=1} (\mathbb{E}\|A^\top s\|_2^p)^{1/p}, \quad (81)$$

the supremum being over deterministic unit vectors. For $p = 4q$,

$$\rho_q(m) := (\mathbb{E}|U_1|^{4q})^{-1/(4q)} = \left[\prod_{r=0}^{2q-1} \frac{m+2r}{1+2r} \right]^{1/(4q)}, \quad (82)$$

so that $\rho_q(1) = 1$ and $\rho_q(m)$ is increasing in m .

Proof. Fix a realization of A and a unit top eigenvector u of $AA^\top$. Then $\|A^\top s\|_2^2 = s^\top AA^\top s \geq \|A\|^2 \langle s, u \rangle^2$ for every s , so by rotational invariance $\mathbb{E}_U \|A^\top U\|_2^p \geq \|A\|^p \mathbb{E}_U |\langle U, u \rangle|^p = \|A\|^p \mathbb{E}|U_1|^p$. Take the expectation over A , exchange the order of integration, and bound the average over U by the supremum over deterministic directions. For (82), write a standard Gaussian vector $g \in \mathbb{R}^m$ as $g = \|g\|_2 U$ with $\|g\|_2$ and U independent; then $\mathbb{E}g_1^{4q} = \mathbb{E}\|g\|_2^{4q} \mathbb{E}U_1^{4q}$ with $\mathbb{E}g_1^{4q} = \prod_{r < 2q} (1+2r)$ and $\mathbb{E}\|g\|_2^{4q} = \prod_{r < 2q} (m+2r)$. The formula also holds at $m = 1$, where the sphere is $\{\pm 1\}$. $\square$

The lemma is pointwise in A : no independent copy of any entry and no covering of the sphere is used, and the auxiliary direction is integrated out before the estimate meets a walk polynomial.

Lemma 3.30 (An exponential envelope for the spherical factor). *For $q \geq 1$ and $0 \leq k \leq q$, with $\phi(t) = \frac{1}{4}[(2 + \frac{t}{2}) \log(2 + \frac{t}{2}) - \frac{t}{2} \log \frac{t}{2} - 2 \log 2]$ for $t > 0$ and $\phi(0) = 0$,*

$$\log \rho_q(k+1) \leq \phi(k/q) \leq \frac{1}{2} \log \frac{17}{16} + \frac{\log 17}{8} \frac{k}{q}. \quad (83)$$

In particular $\rho_q(k+1) \leq \frac{\sqrt{17}}{4} 17^{k/(8q)}$, and $\rho_q(q) \leq e^{\phi(1)} < 1.37$.

Proof. For $t = k/q > 0$, $\log \rho_q(k+1) = \frac{1}{4q} \sum_{r=0}^{2q-1} \log(1 + \frac{t}{2x_r})$ with $x_r = (2r+1)/(2q)$ the midpoints of the $2q$ intervals of length $1/q$ that partition $[0, 2]$. The function $x \mapsto \log(1+t/(2x))$ is convex on $(0, \infty)$ and integrable at 0, so each midpoint value is at most the average of the function over its interval, and $\log \rho_q(k+1) \leq \frac{1}{4} \int_0^2 \log(1 + \frac{t}{2x}) dx = \phi(t)$, the integral being evaluated by parts. Differentiation gives $\phi'(t) = \frac{1}{8} \log(1 + 4/t)$, so ϕ is concave, and its tangent line at $t = \frac{1}{4}$, with slope $\frac{\log 17}{8}$ and intercept $\phi(\frac{1}{4}) - \frac{\log 17}{32} = \frac{1}{2} \log \frac{17}{16}$, lies above it. At $t = 0$ the logarithm and $\phi(0)$ vanish while the tangent has the positive intercept $\frac{1}{2} \log \frac{17}{16}$, so the inequality holds there as well. Finally $\phi(1) = \frac{1}{4}[\frac{5}{2} \log \frac{5}{2} - \frac{1}{2} \log \frac{1}{2} - 2 \log 2] < 0.313$. $\square$

Lemma 3.31 (Thin matrices: only the small axis costs). *Let $q \geq 1$ be an integer. For a fixed original row set I and a fixed original column set J ,*

$$(\mathbb{E}\|Y[I, :]\|^{4q})^{1/(4q)} \leq \rho_q(|I|) \Lambda_{q,r}(Y), \quad (84)$$

$$(\mathbb{E}\|Y[:, J]\|^{4q})^{1/(4q)} \leq \rho_q(|J|) \Lambda_{q,c}(Y). \quad (85)$$

Empty submatrices give zero.

Proof. Fix a deterministic unit vector $s \in \mathbb{R}^I$ and consider the random vector $Y[I, :]^\top s$, whose j -th coordinate is $V_j = \sum_{i \in I} s_i Y_{ij}$. Distinct coordinates use disjoint sets of original entries, so the family $(V_j)_j$ is independent; each V_j is a linear form in independent symmetric variables satisfying (7), hence symmetric in law. Independence and zero means give

$$\sum_j \mathbb{E} V_j^2 = \sum_{i \in I} s_i^2 \sum_j \mathbb{E} Y_{ij}^2 \leq \sigma_r(Y)^2. \quad (86)$$

For a deterministic unit vector t , $\sum_j t_j V_j = \sum_{i \in I, j} s_i t_j Y_{ij}$ is one of the bilinear forms in the definition (9) of R_Y , evaluated at s extended by zero and at t ; hence $W_{2q} \leq R_Y(2q)$. Theorem 3.24 with $\gamma = \chi_\alpha$, supplied by Lemma 3.23, therefore gives

$$\| \|Y[I, :]^\top s\|_2 \|_{L_{4q}} \leq \sigma_r(Y) + \chi_\alpha R_Y(2q) = \Lambda_{q,r}(Y). \quad (87)$$

Lemma 3.29 with $A = Y[I, :]$, $m = |I|$ and $p = 4q$ proves (84). Transposing the matrix exchanges the roles of rows and columns and proves (85) with the maximum column variance. $\square$

The weak moment entering (84) is taken at order $2q$, half the order of the strong moment on the left. This is what makes the whole chain close at the logarithmic order: the local theorem needs its weak input at $2q$, not at $4q$. The factor $\rho_q(m)$ replaces the factor $2 \cdot 5^{m/(4q)}$ of a $\frac{1}{2}$ -net argument and equals one on a single row.

The contraction is stated on the parity sectors of the cone, with the parity grade k of Section 2.7.1 and the ordinary degree j kept apart, because the two costs of the estimate, the sign cost and the thin-axis cost, depend on k and not on j . Put

$$\beta_q = \left(\frac{2q+1}{2q-1} \right)^{1/2}, \quad (88)$$

which is at most $3^{1/(2q)}$ and decreases to one.

Lemma 3.32 (Cone comparison at the actual exponents). *Let $F(W)$ be as in Lemma 3.21: Euclidean-valued, with nonnegative coefficients in every coordinate and ordinary degree at most d ; suppose moreover that every monomial of F has at most k odd exponents. Then for every $p \geq 4$,*

$$\|F\|_{L_{2p/(p-2)}(\ell_2)} \leq \left(\frac{p+2}{p-2} \right)^{k/2} \alpha^{4d/p} \|F\|_{L_2(\ell_2)}, \quad (89)$$

and $\|F\|_{L_4(\ell_2)} \leq 3^{k/2} \alpha^d \|F\|_{L_2(\ell_2)}$.

Proof. The two losses come from different degrees. The sign comparison sees only the odd exponents, whereas the magnitude comparison sees every occurrence of an entry.

1. Condition on the magnitudes and group by parity.

Write $W = \varepsilon Z$ in law and collect monomials with the same odd set:

$$F(\varepsilon Z) = \sum_S \varepsilon_S F_S(Z), \quad F_S = 0 \quad \text{when } |S| > k.$$

Put $u = 2p/(p-2) \in (2, 4]$ and

$$Q(Z) = \sum_S \|F_S(Z)\|_2^2.$$

For each fixed Z , the function of ε is a Hilbert-valued Boolean polynomial of degree at most k . Lemma 3.19 and Walsh orthogonality therefore give

$$\mathbb{E}_\varepsilon \|F(\varepsilon Z)\|_2^u \leq (u-1)^{uk/2} Q(Z)^{u/2}, \quad Q(Z) = \mathbb{E}_\varepsilon \|F(\varepsilon Z)\|_2^2.$$

2. Estimate the positive magnitude polynomial.

Every coordinate of every F_S has nonnegative coefficients. Thus Q is a scalar polynomial with nonnegative coefficients and ordinary degree at most $2d$. Equation (66) gives

$$\|Q\|_{L_2} \leq \alpha^{2d} \|Q\|_{L_1}.$$

The reciprocal of $u/2 = p/(p-2)$ is $1 - 2/p$. Interpolation between L_1 and L_2 consequently uses weight $4/p$ on L_2 :

$$\|Q\|_{L_{u/2}} \leq \|Q\|_{L_1}^{1-4/p} \|Q\|_{L_2}^{4/p} \leq \alpha^{8d/p} \|Q\|_{L_1}.$$

All norms here are taken in the original magnitude law.

3. Combine the two comparisons.

Integrating the conditional sign estimate and using $\mathbb{E}Q = \mathbb{E}\|F\|_2^2$ yields

$$\|F\|_{L_u(\ell_2)}^2 \leq (u-1)^k \|Q\|_{L_{u/2}} \leq (u-1)^k \alpha^{8d/p} \|F\|_{L_2(\ell_2)}^2.$$

Take square roots and use $u-1 = (p+2)/(p-2)$. The asserted L_4 bound is the case $p=4$. The proof preserves the coefficient cone throughout; it makes no claim about its full linear span. $\square$

Proposition 3.33 (Contraction on the preserved cone, by parity sector). *Let $q \geq 1$, $0 \leq j \leq q-1$, and $f \in \mathcal{P}_j^+$. For every parity grade k ,*

$$\begin{aligned} \|\mathcal{C}_+ P_k f\|_{\mathcal{H}} &\leq \rho_q(k+1) \beta_q^k \alpha^{j/q} \Lambda_q(Y) \|P_k f\|_{\mathcal{H}}, \\ \|\mathcal{C}_- P_k f\|_{\mathcal{H}} &\leq \rho_q(k) \beta_q^k \alpha^{j/q} \Lambda_q(Y) \|P_k f\|_{\mathcal{H}} \quad (k \geq 1), \end{aligned} \tag{90}$$

and $\mathcal{C}_- P_0 f = 0$. In particular, since $k \leq j \leq q-1$ on $\mathcal{P}_j^+$,

$$\|\mathcal{C}_\pm f\|_{\mathcal{H}} \leq 1.37 H_\alpha \Lambda_q(Y) \|f\|_{\mathcal{H}}, \quad \|\mathcal{T}f\|_{\mathcal{H}} \leq 2.74 H_\alpha \Lambda_q(Y) \|f\|_{\mathcal{H}}. \tag{91}$$

Proof. Fix $p = 4q$. We first prove the row-to-column creation bound. The other three directional bounds require the count sectors specified below.

1. Localize to a deterministic count sector.

By Lemma 2.5, decompose

$$P_k f = \sum_d f_d$$

into row-count sectors, with orthogonal inputs and corresponding orthogonal outputs. The projection defining f_d only discards ordinary monomials, so $f_d \in \mathcal{P}_j^+$. Moreover, every monomial of the reconstructed polynomial $\mathcal{U}f_d$ has exactly k odd exponents: its parity state is its odd set.

For each fixed label d , the sector map is an input/output compression of multiplication by $Y[I, :]^\top$, where the deterministic set I satisfies $|I| \leq k + 1$. First reconstruct using $\mathcal{U}$; the output compression is then an L_2 contraction. This is the point at which the matrix multiplier is identified with the original random matrix.

2. Apply Hölder on the original joint probability space.

The thin-axis estimate and Lemma 3.32 give

$$\begin{aligned} \|\mathcal{C}_{+,d} f_d\|_{\mathcal{H}} &\leq \|Y[I, :]^\top \mathcal{U}f_d\|_{L_2(\ell_2)} \\ &\leq (\mathbb{E}\|Y[I, :]\|^p)^{1/p} \|\mathcal{U}f_d\|_{L_{2p/(p-2)}(\ell_2)} \\ &\leq \rho_q(k+1) \Lambda_{q,r}(Y) \beta_q^k \alpha^{j/q} \|f_d\|_{\mathcal{H}}. \end{aligned} \tag{92}$$

Here ρ_q is increasing, $((p+2)/(p-2))^{1/2} = \beta_q$, and $4j/p = j/q$. Hölder does not require independence between the multiplier and the polynomial: they use the same signs and the same accumulated magnitude powers. The higher norm is a norm of $\mathcal{U}f_d$, not of occupation amplitudes. The reconstruction is used as an isometry only at L_2 .

3. Sum squares before removing the count label.

The bound in (92) is uniform in d . Since the input and output sectors are orthogonal,

$$\|\mathcal{C}_+ P_k f\|_{\mathcal{H}}^2 = \sum_d \|\mathcal{C}_{+,d} f_d\|_{\mathcal{H}}^2 \leq [\rho_q(k+1) \Lambda_{q,r}(Y) \beta_q^k \alpha^{j/q}]^2 \sum_d \|f_d\|_{\mathcal{H}}^2.$$

The last sum is $\|P_k f\|_{\mathcal{H}}^2$. No factor for the number of count sectors appears.

4. Prove the other directions on their own sectors.

For row-to-column annihilation use the column-count decomposition and (85). Its thin set has size at most k , giving $\rho_q(k)$ instead of $\rho_q(k+1)$; at $k=0$ the map vanishes. Transposition gives the two reverse-current-side bounds. The two current-side directions have orthogonal domains and ranges. Taking the larger directional variance budget replaces $\Lambda_{q,r}$ or $\Lambda_{q,c}$ by Λ_q , proving (90). No annihilation estimate has been inferred by taking the adjoint of a cone-restricted inequality.

5. Obtain the optional coarse bound.

On the specified cone, $k \leq j \leq q-1$. Lemma 3.30 and the bound on β_q imply

$$\rho_q(k+1) \leq \rho_q(q) < 1.37, \quad \beta_q^k \alpha^{j/q} \leq 3^{(q-1)/(2q)} \alpha \leq H_\alpha.$$

Distinct input grades remain orthogonal under creation separately and under annihilation separately. Sum their squared estimates to obtain the two coarse bounds for $\mathcal{C}_\pm$. Finally use $\mathcal{T} = \mathcal{C}_+ + \mathcal{C}_-$ and the triangle inequality. $\square$

There is no random maximum over row or column sets in this argument. Each set is the support of a deterministic count label, and the same unconditional inequality bounds every corresponding squared sector norm before summation. Nor is (90) a bound on the entire polynomial linear span: it is asserted only on the specified cone at the specified total degree. The crude form (91) discards the parity information; the local theorem of Section 4.4.1 keeps it by following the parity grade along the walk.

3.6.5 Constant magnitudes permit a full-space variance Gram

For constant magnitudes, Boolean hypercontractivity controls every coefficient vector on a parity grade. The same occupied-edge bookkeeping therefore gives a full-space contraction for arbitrary weighted signs, without a graph-degree hypothesis. The resource is an exact partition of the output Gram by original occupation counts. It reduces each block to a matrix with few rows or few columns before any probability estimate is made.

Let $B = (b_{ij})$ be a real $a \times b$ matrix and $X_B = (b_{ij}\varepsilon_{ij})$, with one independent sign per original entry. The two vertex roles are disjoint even when $a = b$. Set

$$\sigma_r^2 = \max_i \sum_j b_{ij}^2, \quad \sigma_c^2 = \max_j \sum_i b_{ij}^2, \quad \sigma = \max(\sigma_r, \sigma_c),$$

$$R_B(p) = \sup_{\|s\|_2, \|t\|_2 \leq 1} \left\| \sum_{ij} b_{ij} \varepsilon_{ij} s_i t_j \right\|_{L_p}.$$

On every current-vertex, k -edge occupation state define the original creator

$$C_k|v, S\rangle = \sum_{e=vw \notin S} b_e|w, S \cup \{e\}\rangle.$$

The diagonal vertex-variance operator is $V_B(v, S) = \sum_{e \ni v} b_e^2$.

Theorem 3.34 (Arbitrary-support sign contraction). *There is an absolute constant A such that, for all integers $q \geq 1$ and $0 \leq k < q$,*

$$\|C_k\| \leq A\{\sigma + R_B(4q)\}, \tag{93}$$

$$C_k^* C_k \preceq 4A^2\{V_B + R_B(4q)^2 I\}. \tag{94}$$

One may take $A = 2\sqrt{e} 5^{1/4} C_{\text{SW}}$, where C_{SW} is the absolute Rademacher strong/weak moment constant.

Proof. Write $C_k^{r \rightarrow c}$ for creation from a current row to a current column. Two outputs $(j, T), (j', T')$ have a common predecessor only if

$$T = S \cup \{(i, j)\}, \quad T' = S \cup \{(i, j')\}, \quad (i, j), (i, j') \notin S$$

for the same original row i and k -edge set S . Thus their complete row-count vectors $d(T) = (|T \cap (\{i\} \times [b])|)_i$ agree. The output Gram is exactly block diagonal in $d(T)$. In one such block put $I = \{i : d_i(T) > 0\}$. Then $|I| \leq k + 1$, and all output flags, predecessor flags, and predecessor current rows belong to the original submatrix $B[I, :]$. The entire block factor is an output compression of its creator. Hence

$$\|C_k^{r \rightarrow c}(B)\| \leq \max_{|I| \leq k+1} \|C_k^{r \rightarrow c}(B[I, :])\|. \tag{95}$$

This is a direct-sum maximum of deterministic operators, not a union bound over random subsets. The other direction uses column counts.

For any fixed unit vector s supported on I , the established vector moment comparison [45, Theorem 1.1] gives, for $p \geq 1$,

$$\|X_B[I, :]^* s\|_2 \leq C_{\text{SW}}\{\sigma_r + R_B(p)\}.$$

Indeed its expected Euclidean norm is at most σ_r by the second moment, and every scalar dual functional has p -moment at most $R_B(p)$. A $1/2$ -net in the $|I|$ -dimensional unit sphere has at most $5^{|I|}$ points. The pointwise net bound and the p -moment of a finite maximum yield

$$\|X_B[I, :]\|_{L_p} \leq 2 \cdot 5^{|I|/p} C_{\text{SW}}\{\sigma_r + R_B(p)\}. \quad (96)$$

Only the thin axis enters the net. Theorem 3.24, applied with $\gamma = \chi_1 = \chi_0 < 2.775$ by Lemma 3.23, gives (96) at $p = 4q$ with the explicit value $C_{\text{SW}} = \chi_0$ and the weak moment already at order $2q$, and Lemma 3.29 replaces the net factor $2 \cdot 5^{|I|/p}$ by $(\mathbb{E}|U_1|^p)^{-1/p}$; we keep the general- p form here.

Boolean Fourier transform identifies the creator with $P_{k+1} M_{X_B[I, :]}^* P_k$. For a Hilbert-valued sign polynomial f of degree at most k , Boolean hypercontractivity gives

$$\|f\|_{L_u(\ell_2)} \leq (u-1)^{k/2} \|f\|_{L_2(\ell_2)}, \quad u \geq 2.$$

The vector form has no dimension cost: the positive noise operator dominates the norm pointwise, and Fourier orthogonality bounds the L_2 norm of the inverse noise scaling by its largest degree multiplier. The scalar input is [47, Theorem 1.1]. For $p > 2$, Hölder and the orthogonal output projection consequently give

$$\|C_k^{r \rightarrow c}(B[I, :])f\|_2 \leq \left(\frac{p+2}{p-2}\right)^{k/2} \|X_B[I, :]\|_{L_p} \|f\|_2.$$

Take $p = 4q$. Since $|I| \leq q$ and $k < q$, the net costs at most $5^{1/4}$ and the degree factor at most $\sqrt{e}$. Equations (95)–(96) give (93). The two directional creators have orthogonal domains and ranges, so their combination takes the larger directional norm.

To retain a positive Gram, put $s_0 = R_B(4q)$. If $B = 0$ there is nothing to prove. Otherwise $s_0 > 0$, and for row-current inputs define

$$D_i = \sum_j b_{ij}^2 + s_0^2, \quad b'_{ij} = b_{ij}/\sqrt{D_i}.$$

The maximum row variance and weak $4q$ -moment of B' are at most one: $R_{B'}(4q) \leq \|D^{-1/2}\| R_B(4q) \leq 1$. Here D acts on the current input row, identically on every occupation state. The directional argument therefore gives $\|C_k^{r \rightarrow c}(B)D^{-1/2}\| \leq 2A$. Congruence yields the row-current part of (94); scaling original columns yields its other part. No new variables or changed identifications enter.

□

The retained Gram has a direct continuation rule. For every bounded actual input map L ,

$$(C_k L)^*(C_k L) \preceq 4A^2\{L^* V_B L + R_B(4q)^2 L^* L\}.$$

An output contraction can be inserted as well. These statements preserve the transported variance budget; they do not declare that L preserves the old count blocks. Applied to a chosen heavy coefficient submatrix, the result supplies its input Gram without a matching, forest, girth, or bounded-degree assumption. Choosing q of logarithmic order makes this weak-moment scale available to the vacuum-moment argument in Section 4.

This is an analytic gain uniform over a potentially enormous occupation space. It does not promise inexpensive state enumeration. For variable magnitudes, Proposition 3.33 controls the preserved positive walk cone instead. The stronger whole-span conclusion is specific to the sign-polynomial estimate used here. Section 9.1 states the resulting continuation domains.

4 From operator estimates to moments

4.1 A nonnegative matrix that remembers the vacuum

After choosing grades, and perhaps finer counts, decompose the cutoff as $\bigoplus_{\alpha} \mathcal{K}_{\alpha}$. If

$$\|P_{\alpha} M_q P_{\beta}\| \leq G_{\alpha\beta}$$

for a nonnegative symmetric matrix G , the vector of norms of the components of $M_q v$ is componentwise at most G times the corresponding vector for v . Induction gives

$$\sum_a \|M_q^a(e_a \otimes \Omega)\|^2 \leq D \|G^q e_0\|^2 = D(G^{2q})_{00}, \quad (97)$$

when the initial vacuum lies in sector 0; e_0 is the corresponding coordinate unit vector. This retains more information than $D\|G\|^{2q}$, while involving only a small scalar matrix. A tridiagonal G sums walks with steps $-1, 0, 1$, often called Motzkin-type walks. A two-count G retains, for example, light and heavy occupations separately.

Symmetric block Schur supplies the coarser inequality $\|G\| \leq \max_{\alpha} \sum_{\beta} G_{\alpha\beta}$. It is often useful, but it replaces the whole vacuum path by the largest allowed row sum. The next lemma quantifies what can be kept instead.

4.2 Returning paths and a geometric mean

Lemma 4.1 (Triangular row-sum product). *Let T be a nonnegative symmetric tridiagonal matrix on $0, \dots, q$, with $T_{00} = 0$ and $T_{01} \leq a$. Suppose row sums are bounded by an increasing sequence $R_0, \dots, R_q$. For $q \geq 2$,*

$$(T^{2q})_{00} \leq a^2 R_1 R_q \prod_{k=2}^{q-1} R_k^2. \quad (98)$$

For $q = 1$ the left side is T_{01}^2 .

Proof. Every returning walk begins and ends with the edge $0 \leftrightarrow 1$. Remove those two edges. At full time t the index is at most $\min(t, 2q - t)$, as in Figure 2(b) after dividing its grades by two. Restrict each intermediate index to this reachable range; this removes no contributing path. Summing the nonnegative weights of the remaining $2q - 2$ transitions is bounded by the product of their maximal row sums, at times $t = 1, \dots, 2q - 2$. The available grades in this list are $1, 2, \dots, q, q - 1, \dots, 2$, giving exactly (98). Loops at positive indices and repeated returns to the vacuum are allowed; only nonnegativity and tridiagonality were used. $\square$

Lemma 4.2 (A geometric-mean envelope). *Let T satisfy Lemma 4.1, in particular nonnegativity, symmetry, tridiagonality, and $T_{00} = 0$, and let $\epsilon > 0$. Suppose the row sums of T/ϵ are at most $\Psi((2k + 1)/(2q + 1))$, where Ψ is positive, increasing, and concave on $[0, 1]$, and $T_{01}/\epsilon \leq a$. Put*

$$J = \exp \left(\int_0^1 \log \Psi(u) du \right).$$

If $a^2 J \leq \Psi(0)^3$, then for every $q \geq 2$, $[(T^{2q})_{00}/\epsilon^{2q}]^{1/(2q)} \leq J$.

Proof. If $T_{01} = 0$, the vacuum is disconnected and the assertion is immediate. Otherwise the repeated $0 \leftrightarrow 1$ walk has positive weight, so $(T^{2q})_{00} > 0$ and $a > 0$. Set $f = \log \Psi$ and $u_k = (2k + 1)/(2q + 1)$. The preceding lemma gives

$$\frac{1}{2} \log \frac{(T^{2q})_{00}}{\epsilon^{2q}} \leq \log a + \frac{1}{2} f(u_1) + \sum_{k=2}^{q-1} f(u_k) + \frac{1}{2} f(1).$$

The function f is increasing and concave. Its trapezoidal sum is at most $(q + 1/2) \int_{u_1}^1 f$. Since $(q + 1/2)u_1 = 3/2$, this is at most $q \log J + \frac{1}{2} \log J - \frac{3}{2} \log \Psi(0)$. The assumed boundary condition cancels the last two terms with $\log a$. $\square$

This is an all- q argument. A finite certificate for its integral follows from midpoint concavity:

$$J \leq \left[\prod_{j=0}^{N-1} \Psi \left(\frac{2j+1}{2N} \right) \right]^{1/N}. \quad (99)$$

Bounding radicals upward by rationals reduces the right side to an exact rational product inequality. The proof covers every moment order through the concave envelope. Section 10 uses this argument for the direct SparseStack constants. For a chosen finite q , one can instead evaluate (97) by the positive recurrence $v^{(0)} = e_0$, $v^{(j+1)} = Gv^{(j)}$; upward rounding at every addition and multiplication preserves a componentwise upper bound.

4.3 Ordered first returns and the covariance spectrum

Takeaway. Expanding a vacuum moment by its ordered first returns to the vacuum sector retains the spectrum of the covariance $V = \mathbb{E}Z^2$ before any scalar relaxation. When the rest-space majorant respects the block structure, this replaces the external dimension D in (97) by the effective rank $\text{tr } V / \|V\|$, and it does so without assuming that the return operators commute.

The scalar majorant (97) treats the vacuum sector as a single index. Every time a walk leaves that sector it pays the same scalar weight, and the trace over the external factor contributes the factor D . The exact block structure retains more. Let M be a finite self-adjoint matrix on a space containing the vacuum sector $\mathcal{K}_0 = \mathbb{C}^D \otimes \Omega$, which we identify with $\mathbb{C}^D$, and let P be the orthogonal projection onto it. Suppose $M^j(e_a \otimes \Omega) = \mathcal{M}_Z^j(e_a \otimes \Omega)$ for $0 \leq j \leq q$ and every a , as the cutoff M_q of Theorem 2.3 does. Writing $r = i + j$ with $i, j \leq q$ and using self-adjointness,

$$PM^rP = \mathbb{E}Z^r, \quad 0 \leq r \leq 2q. \quad (100)$$

We assume $\mathbb{E}Z = 0$, as holds for every centered error in this paper, and write, relative to the decomposition into $\mathcal{K}_0$ and its orthogonal complement, the *rest space*,

$$M = \begin{pmatrix} 0 & B^* \\ B & C \end{pmatrix}, \quad V = B^*B, \quad K_r = PM^rP, \quad F_j = B^*C^{j-2}B \quad (j \geq 2). \quad (101)$$

Under (100), $V = \mathbb{E}Z^2$ and $\text{tr } K_{2q} = \mathbb{E} \text{tr } Z^{2q}$. The operator F_j sums the walks that leave the vacuum sector, spend $j - 2$ steps in the rest space, and return; it is the ordered *first-return operator* of length j . Neither the F_j nor V are assumed to commute with one another. The letters B, C, V are local to this subsection. The recurrence and resolvent identity below are the standard block Schur-complement calculation, expanded by first returns; see [58, Chapter 1]. We include the proof to fix the order of the noncommuting factors. The subsequent covariance-spectrum comparison is a separate estimate.

Theorem 4.3 (Ordered renewal). *For the block matrix (101), $K_0 = I_D$, $K_1 = 0$, and*

$$K_r = \sum_{j=2}^r F_j K_{r-j}, \quad r \geq 2. \quad (102)$$

As formal power series in z ,

$$\sum_{r \geq 0} K_r z^r = [I - z^2 B^* (I - zC)^{-1} B]^{-1}. \quad (103)$$

Let $B = WV^{1/2}$ be the polar decomposition, with W a partial isometry, and let $h_0, h_1, \dots$ be nonnegative numbers with $h_j \geq \|W^* C^j W\|$ and generating function $h(z) = \sum_{j \geq 0} h_j z^j$. Then, for every $r \geq 0$,

$$|\operatorname{tr} K_r| \leq [z^r] \operatorname{tr}(I - z^2 h(z) V)^{-1}, \quad (104)$$

where $[z^r]$ extracts the coefficient of z^r .

Proof. Put $L_r = (I - P)M^r P$, so that $L_0 = 0$. Writing $M^r = M \cdot M^{r-1}$ and multiplying the blocks gives, for $r \geq 1$,

$$K_r = B^* L_{r-1}, \quad L_r = B K_{r-1} + C L_{r-1},$$

because the vacuum block of M is zero. Unrolling the second recurrence, $L_{r-1} = \sum_{i=1}^{r-1} C^{i-1} B K_{r-1-i}$; substituting into the first and putting $j = i + 1$ proves (102). Multiplying (102) by z^r and summing gives $K(z) = I + F(z)K(z)$ with $F(z) = \sum_{j \geq 2} F_j z^j = z^2 B^* (I - zC)^{-1} B$, which is (103): the right side is the vacuum compression $P(I - zM)^{-1} P$ of the resolvent, the inverse of the Schur complement of the rest block of $I - zM$.

Iterating (102) writes K_r as the sum of the ordered products $F_{j_1} \cdots F_{j_t}$ over all compositions $j_1 + \cdots + j_t = r$ with parts $j_i \geq 2$; the empty composition gives $K_0 = I$. Put $N_i = W^* C^{j_i-2} W$, so that $F_{j_i} = V^{1/2} N_i V^{1/2}$ and, by cyclicity of the trace, $\operatorname{tr}(F_{j_1} \cdots F_{j_t}) = \operatorname{tr}(V N_1 V N_2 \cdots V N_t)$. Hölder's inequality for Schatten norms, together with the ideal property $\|XY\|_{S_t} \leq \|X\|_{S_t} \|Y\|$ [50, 51], gives

$$|\operatorname{tr}(V N_1 \cdots V N_t)| \leq \prod_{i=1}^t \|V N_i\|_{S_t} \leq \prod_{i=1}^t \|V\|_{S_t} \|N_i\| = \operatorname{tr}(V^t) \prod_{i=1}^t h_{j_i-2},$$

where $\|X\|_{S_t} = (\operatorname{tr} |X|^t)^{1/t}$ and $\|V\|_{S_t}^t = \operatorname{tr} V^t$ because $V \succeq 0$. For $t = 1$ this is the trace-norm bound $|\operatorname{tr}(V N_1)| \leq \operatorname{tr}(V) \|N_1\|$. Summing over compositions,

$$|\operatorname{tr} K_r| \leq \sum_{t \geq 0} \sum_{\substack{j_1 + \cdots + j_t = r \\ j_i \geq 2}} \operatorname{tr}(V^t) \prod_{i=1}^t h_{j_i-2},$$

and the right side is the coefficient of z^r in $\sum_{t \geq 0} \operatorname{tr}(V^t) (z^2 h(z))^t = \operatorname{tr}(I - z^2 h(z) V)^{-1}$. $\square$

The recurrence retains the ordered first-return operators exactly; only the trace majorant replaces them by the numbers h_j . The following corollary is the form used for calibration. It keeps the covariance spectrum until the last step, where a single scalar relaxation remains optional.

Corollary 4.4 (Covariance-spectrum comparison). *Suppose the rest space is the orthogonal sum of sectors $\mathcal{H}_1, \dots, \mathcal{H}_A$ with projections $P_1, \dots, P_A$, that the range of B lies in $\mathcal{H}_1$, and that a nonnegative symmetric $A \times A$ matrix G satisfies $\|P_i C P_j\| \leq G_{ij}$. For $v \geq 0$ let*

$$T(v) = \begin{pmatrix} 0 & \sqrt{v} e_1^T \\ \sqrt{v} e_1 & G \end{pmatrix}$$

act on $\mathbb{R} \oplus \mathbb{R}^A$, with e_1 the first coordinate vector of $\mathbb{R}^A$ and index 0 for the first summand. If $\lambda_1, \dots, \lambda_D$ are the eigenvalues of V , then for every $q \geq 1$,

$$\operatorname{tr} K_{2q} \leq \sum_{a=1}^D (T(\lambda_a)^{2q})_{00} \leq \frac{\operatorname{tr} V}{v} (T(v)^{2q})_{00}, \quad v \geq \|V\|, \quad v > 0. \quad (105)$$

Proof. The range of the partial isometry W is the range of B , so $W^*C^jW = W^*P_1C^jP_1W$, and induction on j with the block bounds gives $\|P_1C^jP_1\| \leq (G^j)_{11}$. Take $h_j = (G^j)_{11}$. For a scalar $\lambda \geq 0$, the matrix $T(\lambda)$ has the block form (101) with $B = \sqrt{\lambda}e_1$, $C = G$, and first returns $\lambda(G^{j-2})_{11} = \lambda h_{j-2}$, so its renewal series is $(1 - z^2h(z)\lambda)^{-1}$ and $(T(\lambda)^{2q})_{00} = [z^{2q}](1 - z^2h(z)\lambda)^{-1}$. Diagonalizing V in (104) proves the first inequality. For $q \geq 1$ the coefficient $[z^{2q}](1 - z^2h(z)\lambda)^{-1} = \sum_{t \geq 1} c_t \lambda^t$ is a polynomial in λ with nonnegative coefficients and zero constant term. For $0 \leq \lambda \leq v$ one has $\lambda^t \leq \lambda v^{t-1}$, so $\sum_t c_t \lambda_a^t \leq (\lambda_a/v) \sum_t c_t v^t$, and summing over a proves the second inequality. $\square$

If G is the rest-space part of a scalar majorant whose vacuum row contains $\|B\| = \|V\|^{1/2}$, the right side of (97) is $D(T(\|V\|)^{2q})_{00}$. The corollary replaces D by the effective rank $\text{tr } V/\|V\| \leq D$, and before that relaxation it retains every eigenvalue of the covariance. This is best possible from V and G alone: on $\mathbb{R}^A \otimes \mathbb{C}^D$ take $Bx = e_1 \otimes V^{1/2}x$ and $C = G \otimes I_D$; in an eigenbasis of V the matrix is the direct sum of the scalar $T(\lambda_a)$, and the first inequality is an equality. Further improvements must retain more than the covariance and the sector norms.

Example 4.5 (The rare-collision weight through arbitrarily many returns). In Example 2.1, the cyclic matrix (3) has $B = b^{-1/2}e_1$ and $C = \sqrt{(b-1)/b}(e_1e_2^\top + e_2e_1^\top)$. Its first returns are $F_{2j} = b^{-1}(1 - b^{-1})^{j-1}$ and $F_{2j+1} = 0$, so (103) gives

$$\sum_r K_r z^r = \frac{1 - (1 - b^{-1})z^2}{1 - z^2}, \quad K_{2q} = \frac{1}{b} \quad (q \geq 1).$$

The renewal calculation retains the rare-event weight $1/b$ through every return, whereas the norm bound of Proposition 6.1 gives 1. Charging a new collision at every return would give b^{-q} and fail this test in the other direction: the walk returns to the vacuum on the same collision event, and the first-return operator records exactly that.

Remark 4.6 (What the refinement buys for SparseStack). For the SparseStack error, (117) gives $V = \mathbb{E}E^2 = m^{-1}[(d+1)I - 2\sum_i \ell_i u_i u_i^*]$, so $(d+1 - 2\mu)/m \leq \|V\| \leq (d+1)/m$ with $\mu = \max_i \ell_i \leq 1$ and

$$\frac{\text{tr } V}{\|V\|} \geq \frac{d(d+1) - 2\sum_i \ell_i^2}{d+1} \geq \frac{d(d-1)}{d+1}.$$

The covariance is nearly isotropic, so the effective rank is close to d . In the direct certificate (226) the vacuum edge is $\alpha_0^2 = (d+1)/m \geq \|V\|$, and the argument of the corollary gives

$$\mathbb{E} \text{tr } E^{2q} \leq \left(d - \frac{2\sum_i \ell_i^2}{d+1}\right)(T_q^{2q})_{00},$$

which improves the prefactor d only by the frame term $2\sum_i \ell_i^2/(d+1) < 2$.

The prefactor is not the whole story, since the corollary retains every eigenvalue before that relaxation; the following argument bounds the gain of the complete spectral refinement in the moment root, uniformly in q . Fix the rest-space majorant and write $f_q(t) = (T(t)^{2q})_{00} = \sum_{j=1}^q c_j t^j$ with $c_j \geq 0$, as in the proof of the corollary; the degree is at most q because every first return uses at least two of the $2q$ steps. Put $v_0 = (d+1)/m$ and $r = 1 - 2\mu/(d+1)$, which is positive for $d \geq 2$. The covariance identity gives $rv_0 \leq \lambda_a \leq v_0$ for every eigenvalue, so $f_q(\lambda_a) \geq \sum_j c_j r^j v_0^j \geq r^q f_q(v_0)$, and therefore

$$d r^q f_q(v_0) \leq \sum_{a=1}^d f_q(\lambda_a) \leq d f_q(v_0), \quad \left[\frac{d f_q(v_0)}{\sum_a f_q(\lambda_a)} \right]^{1/(2q)} \leq \sqrt{\frac{d+1}{d+1-2\mu}}.$$

With the rest-space majorant held fixed, retaining the complete covariance spectrum can improve the moment root of the direct certificate by at most the factor $\sqrt{(d+1)/(d+1-2\mu)} = 1 + O(\mu/d)$, for every q . This is a useful calibration result in the sense of Section 6: it shows that the covariance spectrum is not where that certificate loses, and it separates four quantities that the exact small-matrix diagnostics of that section can compare on one model, namely the exact moment, the covariance-spectrum bound, the scalar block majorant, and the final norm relaxation. It does not rule out gains from retaining more of the rest-space dynamics C , from sharper coefficient estimates, or from a better domain.

Example 4.7 (Positive return kernels do not give a Loewner recurrence). The trace comparison in Theorem 4.3 cannot be replaced by an operator inequality between the ordered products, even when every first return is positive and every odd return vanishes. The scalar renewal with $h_{2j} = c^{2j}$ and no odd returns gives $\lambda(\lambda + c^2)^{q-1}$, which suggests the operator recurrence $K_{2q} \preceq V(V + c^2 I)^{q-1}$ with $c = \|C\|$. Take

$$V = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \quad \Delta = \frac{1}{4} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad A_0 = V - \Delta \succeq 0,$$

and on the rest space $(\mathbb{R}^2 \oplus \mathbb{R}^2) \oplus \mathbb{R}^2$ put $Bx = (A_0^{1/2}x, \Delta^{1/2}x; 0)$ and $C(y, z; w) = (w, 0; y)$. Then $B^*B = V$, $\|C\| = 1$, $F_2 = V$, $F_{2j} = A_0$ for $j \geq 2$, and every odd return vanishes. The recurrence gives $K_6 = V^3 + VA_0 + A_0V + A_0$, and

$$V(V + I)^2 - K_6 = \Delta + V\Delta + \Delta V = \frac{1}{4} \begin{pmatrix} 3 & 4 \\ 4 & 5 \end{pmatrix},$$

whose determinant is $-1/16$. Here $F_{2j} = A_0 \preceq V = F_2$ for $j \geq 2$, yet the smaller return operator cannot be replaced by the larger one inside the noncommuting polynomial $V^3 + VA_0 + A_0V + A_0$: positivity of every ingredient does not permit monotone substitution in an ordered product. Trace Hölder, block row and column contractions, and the pointwise Gram multiplication used in Section 3 remain legitimate; they prove different statements. The example is an obstruction at the level of finite block operators and is not asserted to arise from a SparseStack input.

If $\mathbb{E}Z \neq 0$, the vacuum block $PMP = \mathbb{E}Z$ adds the term $(\mathbb{E}Z)K_{r-1}$ to (102); none of the applications needs it. The renewal identity is also a natural entry point for an asymptotic analysis: it describes the vacuum spectral measure through first-return kernels and the compressed resolvent, rather than through an assumed scalar process on occupation counts.

4.4 Iteration on a preserved family

The nonnegative path matrix controls all admissible grade transitions. When a sharper estimate is available only on a preserved cone, the exact vacuum identity can instead be bounded prefix by prefix. The following form retains a different allowance for each interaction.

Lemma 4.8 (Moment conversion on preserved cones). *Let Z be a self-adjoint $D \times D$ random matrix and let $\mathcal{U} : \mathcal{H} \rightarrow L_2(\Omega; \mathbb{C}^D)$ be a linear isometry. On a domain containing the vectors below, suppose $\mathcal{U}\mathcal{T}f = Z\mathcal{U}f$. Fix an integer $q \geq 1$. Let $\mathcal{K}_0, \dots, \mathcal{K}_q$ be cones in that domain such that, for $0 \leq j < q$,*

$$\mathcal{T}\mathcal{K}_j \subseteq \mathcal{K}_{j+1}, \quad \|\mathcal{T}f\|_{\mathcal{H}} \leq a_j \|f\|_{\mathcal{H}} \quad (f \in \mathcal{K}_j), \quad a_j \geq 0.$$

Suppose $\omega_1, \dots, \omega_D \in \mathcal{K}_0$ and $\mathcal{U}\omega_b = e_b$, and assume the displayed products are square integrable. Then

$$\mathbb{E} \operatorname{tr} Z^{2q} = \sum_{b=1}^D \|\mathcal{T}^q \omega_b\|_{\mathcal{H}}^2 \leq D \prod_{j=0}^{q-1} a_j^2. \quad (106)$$

Proof. Every prefix $\mathcal{T}^j \omega_b$ lies in $\mathcal{K}_j$. Iteration bounds its final norm by $\prod_j a_j$, since the isometry gives $\|\omega_b\|_{\mathcal{H}} = 1$. The intertwining identity gives $\mathcal{U} \mathcal{T}^q \omega_b = Z^q e_b$. Sum their squared L_2 norms and use self-adjointness of Z . $\square$

No bound on $\text{span } \mathcal{K}_j$ or bounded extension of $\mathcal{T}$ is needed. The norm used to reconstruct each prefix is always L_2 . Any other moment comparison enters the proof of the one-step allowances, on the reconstructed random polynomial.

The cone lemma bounds every prefix by its norm. A weighted version retains a different nonnegative energy at each stage, quadratic forms being the important special case; it is the abstract shape of the moment arguments in this paper, and Section 7 explains, for quadratic energies and linear encodings, which choices of forms are admissible after a further interaction.

Lemma 4.9 (Weighted energy transport). *In the setting of Lemma 4.8, with numbers $a_j \geq 0$ in place of the norm allowances, let $\mathcal{E}_0, \dots, \mathcal{E}_q$ be nonnegative functions on the cones $\mathcal{K}_0, \dots, \mathcal{K}_q$ such that, for $0 \leq j < q$ and $f \in \mathcal{K}_j$,*

$$\mathcal{T} \mathcal{K}_j \subseteq \mathcal{K}_{j+1}, \quad \mathcal{E}_{j+1}(\mathcal{T}f) \leq a_j^2 \mathcal{E}_j(f),$$

that $\sum_{b=1}^D \mathcal{E}_0(\omega_b) \leq B_0$, and that $\|f\|_{\mathcal{H}}^2 \leq \mathcal{E}_q(f)$ for $f \in \mathcal{K}_q$. Then

$$\mathbb{E} \text{tr } Z^{2q} \leq B_0 \prod_{j=0}^{q-1} a_j^2. \quad (107)$$

Proof. Every prefix $\mathcal{T}^j \omega_b$ lies in $\mathcal{K}_j$, so $\|\mathcal{T}^j \omega_b\|_{\mathcal{H}}^2 \leq \mathcal{E}_j(\mathcal{T}^j \omega_b) \leq \prod_{j=0}^j a_j^2 \mathcal{E}_0(\omega_b)$. Sum over b and use the identity in (106). $\square$

Lemma 4.8 is the case $\mathcal{E}_j = \|\cdot\|^2$ and $B_0 = D$. The scalar majorant (97) is the case $\mathcal{E}_j(f) = \|G^{q-j} n(f)\|_2^2$, where $n(f)$ is the vector of sector norms of f : the componentwise inequality $n(\mathcal{T}f) \leq G n(f)$ and the monotonicity of the nonnegative matrix G give every $a_j = 1$, the terminal condition is Parseval, and $B_0 = D(G^{2q})_{00}$. For quadratic forms $\mathcal{E}_j(f) = \langle f, W_j f \rangle$ with $W_j \succeq 0$ on the whole space, the tautological choice $W_q = I$, $W_j = \mathcal{T}^* W_{j+1} \mathcal{T}$ also has every $a_j = 1$ and hides the entire moment in $B_0 = \sum_b \langle \omega_b, W_0 \omega_b \rangle$. The content of an energy interface is therefore a choice of forms that are simpler to estimate than the exact iterates, informative, and closed under the actual insertions and occupation masks. When a quadratic form is computed from a linear compressed description of the state, it must first be well defined on the state; Proposition 7.1 gives the test, and shows that the one-step inequality above is then a Loewner inequality between the Grams of the descriptions. Energies that are not quadratic, such as the sector-norm majorant of Section 4.4.1, are covered by the lemma but not by that interpretation.

4.4.1 Vacuum walks and the local moment theorem

The sector bounds of Proposition 3.33 depend on the parity grade, and the vacuum walk visits the grades in a specific pattern: creation raises the grade by one and annihilation lowers it by one. Recording the norms of the parity components of each prefix in a vector and majorizing one step by a nonnegative matrix keeps this information.

Definition 4.10 (Spherical parity majorant). For $q \geq 1$ let $\widehat{G}_q$ be the nonnegative $(q+1) \times (q+1)$ matrix indexed by the grades $0, \dots, q$, with

$$(\widehat{G}_q)_{k+1,k} = \rho_q(k+1) \beta_q^k \quad (0 \leq k < q), \quad (\widehat{G}_q)_{k-1,k} = \rho_q(k) \beta_q^k \quad (1 \leq k \leq q),$$

and all other entries zero, and put $\widehat{\Theta}_q = \|\widehat{G}_q e_0\|_2^{1/q}$.

Theorem 4.11 (Symmetric regular-law local moment bound). *Let Y be a real $n \times n$ matrix with independent symmetric-in-law entries satisfying (7). For every integer $q \geq 1$,*

$$(\mathbb{E}\|Y\|^{2q})^{1/(2q)} \leq n^{1/(2q)} \widehat{\Theta}_q \alpha^{(q-1)/(2q)} \Lambda_q(Y), \quad \Lambda_q(Y) = \sigma(Y) + \chi_\alpha R_Y(2q), \quad (108)$$

and $\widehat{\Theta}_q < 2.31$ for every q (Theorem 4.13). The same bound holds with $\sigma(Y)$ replaced by any B that dominates both the row and the column variance scale, in particular by $M(Y)$.

Proof. The proof tracks parity along actual vacuum prefixes. It does not estimate multiplication on the entire polynomial span.

1. The vacuum iterates stay in the coefficient cone.

For a current vertex $v \in \mathcal{V}$, let Ω_v have coefficient one at $(v, \emptyset)$ and zero elsewhere. Then

$$\|\Omega_v\|_{\mathcal{H}} = 1, \quad \mathcal{U}\Omega_v = e_v, \quad \mathcal{U}\mathcal{T}^j\Omega_v = H_Y^j e_v.$$

Every coordinate of $H_Y^j e_v$ is a sum of monomials indexed by original ordered walks of length j . Their coefficients are nonnegative integers, irrespective of the signs of the realized entries. Revisiting an edge increases its power in that same monomial. Hence $f_j := \mathcal{T}^j\Omega_v \in \mathcal{P}_j^+$.

2. Record norms on parity grades.

For $0 \leq j \leq q$, put

$$a_j = (\|P_k f_j\|_{\mathcal{H}})_{k=0}^q \in \mathbb{R}_+^{q+1}.$$

Then $a_0 = e_0$. Creation raises the grade and annihilation lowers it, so, with nonexistent grades interpreted as zero,

$$P_k \mathcal{T} f_j = \mathcal{C}_+ P_{k-1} f_j + \mathcal{C}_- P_{k+1} f_j.$$

Apply Proposition 3.33 and the triangle inequality within each output grade. By the definition of $\widehat{G}_q$, this yields the componentwise inequality

$$a_{j+1} \leq \alpha^{j/q} \Lambda_q(Y) \widehat{G}_q a_j, \quad 0 \leq j < q.$$

3. Iterate the nonnegative majorant.

Since $\widehat{G}_q$ has nonnegative entries, componentwise inequalities can be composed. The degree factors give

$$\prod_{j=0}^{q-1} \alpha^{j/q} = \alpha^{(q-1)/2}, \quad a_q \leq \alpha^{(q-1)/2} \Lambda_q(Y)^q \widehat{G}_q^q e_0.$$

Orthogonality of the parity grades therefore gives

$$\|\mathcal{T}^q \Omega_v\|_{\mathcal{H}}^2 = \sum_k a_{q,k}^2 \leq \alpha^{q-1} \Lambda_q(Y)^{2q} \widehat{\Theta}_q^{2q}. \quad (109)$$

This is a positive-vector induction. In particular, $\|\widehat{G}_q^{q-j} a_j\|_2^2$ need not be a quadratic form in f_j ; no quadratic-energy interpretation is used here.

4. Recover the original trace and then the norm.

The dilation H_Y is real symmetric, and its two diagonal blocks in an even power have equal traces. Consequently

$$\begin{aligned} 2\mathbb{E} \operatorname{tr}(Y Y^\top)^q &= \mathbb{E} \operatorname{tr} H_Y^{2q} \\ &= \sum_{v \in \mathcal{V}} \mathbb{E} \|H_Y^q e_v\|_2^2 = \sum_{v \in \mathcal{V}} \|\mathcal{T}^q \Omega_v\|_{\mathcal{H}}^2 \\ &\leq 2n \alpha^{q-1} \Lambda_q(Y)^{2q} \widehat{\Theta}_q^{2q}. \end{aligned} \quad (110)$$

Divide by two, use $\|Y\|^{2q} \leq \text{tr}(YY^\top)^q$, and take the $2q$ -th root. This proves (108). Only the common upper bound on the row and column standard-deviation norms entered the contraction, so any such bound B may replace $\sigma(Y)$. $\square$

The trace normalization is the original dimension $2n$, rather than the size of the parity state space. Every equality uses the original joint law. All expectations are finite by the entry moment assumptions. The proof requires neither polynomial completeness in an infinite L_2 space nor a bounded extension of the full multiplication operator. At q comparable to $\log n$, the factor $n^{1/(2q)}$ is absolute. The moment identity remains that of the original input law.

The formal development described in Section 11 formalizes the earlier form of (108), whose analytic constant is $4 \cdot 5^{1/4} H_\alpha$ from the crude contraction and a $\frac{1}{2}$ -net in place of $\hat{\Theta}_q \alpha^{(q-1)/(2q)}$, with $2H_\alpha^2$ in place of χ_α , and with the factor $(2n)^{1/(2q)}$, counting both sides of the dilation rather than comparing its two diagonal blocks; the literal constant of the formal statement is fixed there, and the present form is analytic only.

Remark 4.12 (The first step is exact). $\|\mathcal{T}\Omega_v\|_{\mathcal{H}}^2 = \sum_{e \ni v} \mathbb{E} Z_e^2$ is the variance sum at the vertex v , and it lies entirely in grade one; since $\rho_q(1) = 1$, also $\hat{G}_q e_0 = e_1$. Starting the iteration at $j = 1$ and summing the two-sided variance $2V_{\text{tot}}$, $V_{\text{tot}} = \sum_{ij} \mathbb{E} Y_{ij}^2$, over the vertices gives

$$(\mathbb{E}\|Y\|^{2q})^{1/(2q)} \leq V_{\text{tot}}^{1/(2q)} \hat{\Theta}_q \alpha^{(q-1)/(2q)} \Lambda_q(Y)^{1-1/q},$$

which improves (108) because $V_{\text{tot}} \leq n\sigma(Y)^2 \leq n\Lambda_q(Y)^2$, and reduces at $q = 1$ to the Frobenius bound $\mathbb{E}\|Y\|^2 \leq V_{\text{tot}}$.

Theorem 4.13 (A uniform bound on the parity constant). *Put $s_\infty = \frac{1}{2} + \frac{\log 17}{8}$ and*

$$C_{\text{sph}} = \frac{\sqrt{17}}{2} \exp\left(\int_0^1 \log \cosh(s_\infty u) du\right). \quad (111)$$

Then $\hat{\Theta}_q \leq C_{\text{sph}}$ for every integer $q \geq 1$, and $2.309978 < C_{\text{sph}} < 2.309981 < 2.31$.

Proof. 1. Majorize each transition by an exponential weight.

Let

$$A = \frac{\sqrt{17}}{4}, \quad s_q = q \log \beta_q + \frac{\log 17}{8}.$$

Lemma 3.30 and $\rho_q(k) \leq \rho_q(k+1)$ imply that every nonzero entry in column k of $\hat{G}_q$ is at most $A \exp(s_q k/q)$.

2. Bound the sum over nonnegative paths by a product.

Use $\|z\|_2 \leq \|z\|_1$ for $z = \hat{G}_q^q e_0 \geq 0$. Its ℓ_1 norm is the sum of the weights of all nonnegative ± 1 paths $k_0 = 0, k_1, \dots, k_q$. Each path has weight at most

$$A^q \exp\left(\frac{s_q}{q} \sum_{j=0}^{q-1} k_j\right).$$

Enlarge this sum to all 2^q unrestricted increment choices $\xi_0, \dots, \xi_{q-1}$. Since $\sum_{j < q} k_j = \sum_{i < q} (q-1-i)\xi_i$, the enlarged sum factors, giving

$$\hat{\Theta}_q \leq 2A \left[\prod_{m=0}^{q-1} \cosh(s_q m/q) \right]^{1/q}.$$

This is an inequality, not an exact formula for the original weighted path sum.

3. Compare every finite-order sum with one integral.

The elementary bounds for $\log(1+x)$ and $\operatorname{artanh} x$ give

$$\frac{1}{2} \leq q \log \beta_q \leq \frac{1}{2} + \frac{1}{4q-2}.$$

Writing $s_\infty = \frac{1}{2} + \frac{\log 17}{8}$, for $0 \leq m < q$ we have

$$(q \log \beta_q - \frac{1}{2})m \leq \frac{q-1}{4q-2} < \frac{1}{4} < \frac{s_\infty}{2}.$$

Therefore $s_q m/q \leq s_\infty(m + \frac{1}{2})/q$. The function $u \mapsto \log \cosh(s_\infty u)$ is increasing and convex. Its midpoint sum is bounded above by its integral, so

$$\frac{1}{q} \sum_{m=0}^{q-1} \log \cosh(s_q m/q) \leq \int_0^1 \log \cosh(s_\infty u) du.$$

Together with $2A = \sqrt{17}/2$, this proves $\hat{\Theta}_q \leq C_{\text{sph}}$ for every integer $q \geq 1$.

4. Certify the one remaining numerical inequality.

The midpoint and trapezoid rules bracket the integral of this convex function. Evaluating both finite sums on 256 subintervals with outward-rounded interval arithmetic gives the displayed bracket for C_{sph} ; see Appendix D. The uniformity in q was proved in Step 3. It is not inferred from a numerical sweep over moment orders. $\square$

The computed values of $\hat{\Theta}_q$, finite nonnegative matrix products, are 1, 1.412, 1.710, 1.936, 2.088, 2.234, 2.295 and 2.299 at $q = 1, 2, 4, 8, 16, 64, 1024, 4096$. With every entry of $\hat{G}_q$ replaced by one, $\|\hat{G}_q^q e_0\|_2^2$ counts pairs of nonnegative ± 1 paths of length q with a common endpoint and equals the Catalan number C_q , the q -th moment of the Marchenko–Pastur law at aspect ratio one. This is an exact calibration of the bare path count at each fixed order; it does not assert that the weighted majorant, or the passage from it to C_{sph} through the ℓ_1 comparison, the unrestricted paths and the envelope, is sharp on the matrix family, and fixed-order agreement with a limiting trace moment is not a statement about orders growing with the dimension.

Corollary 4.14 (The local mean at the exact logarithmic order). *Let Y be as in Theorem 4.11 with $n \geq 1$, put $q = \lceil 2 \log(n+1) \rceil$ and*

$$L_\alpha = 2.31 e^{1/4} \sqrt{\alpha} < 2.967 \sqrt{\alpha}. \quad (112)$$

Then

$$\mathbb{E}\|Y\| \leq L_\alpha \{M(Y) + \chi_\alpha \kappa_\alpha(n) R_Y(\log(n+1))\}, \quad (113)$$

where $\kappa_\alpha(n)$ is the explicit factor of Proposition 3.28 that carries R_Y from the order $\log(n+1)$ to the order $2q$, and

$$\kappa_\alpha(n) \leq 13.8 \alpha^{4.78} \quad (n \geq 1), \quad \kappa_\alpha(n) \leq 5.6 \alpha^{3.31} \quad (n \geq 2), \quad \kappa_\alpha(n) \leq 3.9 \alpha^3 \quad (n \geq 7).$$

Proof. Since $q \geq 2 \log(n+1)$, $n^{1/(2q)} \leq e^{1/4}$ and $\alpha^{(q-1)/(2q)} \leq \sqrt{\alpha}$; with $\hat{\Theta}_q < 2.31$, $\sigma(Y) \leq M(Y)$ and $\mathbb{E}\|Y\| \leq (\mathbb{E}\|Y\|^{2q})^{1/(2q)}$, (108) gives $\mathbb{E}\|Y\| \leq L_\alpha \{M(Y) + \chi_\alpha R_Y(2q)\}$. Put $p_0 = \log(n+1)$ and $P = 2q \in [4p_0, 4p_0 + 2)$. For $p_0 < 4$ apply (79) with the anchor $b = 2$, preceded by the bridge when $p_0 < 2$: $\kappa_\alpha(n) = 3^{\epsilon/2} \sqrt{P-1} \alpha^{\epsilon+\eta(P/2)}$ with $\epsilon = \max(4/p_0 - 2, 0)$. At $n = 1$ this is $3^{2/\log 2 - 1} \sqrt{3} \alpha^{4/\log 2 - 1} < 13.75 \alpha^{4.771}$; for $2 \leq n \leq 6$ the exponent is at most its value $4/\log 3 - \frac{1}{3} < 3.31$ at $n = 2$ and the prefactor at most $3^{2/\log 3 - 1} \sqrt{5} < 5.51$; for

$7 \leq n \leq 53$, $\epsilon = 0$ and $P \leq 16$, so $\kappa_\alpha(n) \leq \sqrt{15} \alpha^{\eta(8)} = \sqrt{15} \alpha^3$. For $p_0 \geq 4$ take the anchor $b = 2\lfloor p_0/2 \rfloor \leq p_0$: for $4 \leq p_0 < 6$, $b = 4$ and $P \leq 24$, so $\kappa_\alpha(n) \leq \sqrt{23/3} \alpha^{\eta(6)} < 2.8 \alpha^{2.67}$; for $6 \leq p_0 < 8$, $b = 6$ and $P \leq 32$, so $\kappa_\alpha(n) \leq \sqrt{31/5} \alpha^{\eta(16/3)} < 2.5 \alpha^{2.5}$; and for $p_0 \geq 8$, $b \geq p_0 - 2$ gives $P/b \leq 17/3$ and $(P-1)/(b-1) \leq 33/5$, so $\kappa_\alpha(n) \leq \sqrt{33/5} \alpha^{\eta(17/3)} < 2.6 \alpha^{2.6}$. The three displayed bounds collect these cases, taking the largest prefactor and the largest exponent separately, which is admissible for $\alpha \geq 1$. $\square$

Corollary 4.14 is the local estimate that the graded method supplies for independent entries: an expected-norm bound in terms of the variance scale and the *undelated* weak moment at the exact order $\log(n+1)$. Section 5.4 places it inside the mean-norm theorem and summarizes the deletion argument that replaces $R_Y(\log(n+1))$ by the smaller parameter $D(Y)$, a matrix problem of a different kind.

4.5 Tail conversion and limited independence

If (97) is at most $D(\epsilon/2)^{2q}$, Markov gives failure at most $D4^{-q}$. This explains the useful choice $q = \max\{1, \lceil \log_4(D/\delta) \rceil\}$. Different displayed choices of logarithm base must be accounted for when comparing constants.

Proposition 4.15 (Moment matching). *Suppose each scalar summand in the trace expansion of Z^{2q} depends on at most K independent blocks under a reference product law. Any other law with the same marginals on every set of at most K blocks has the same trace moment and inherits its moment tail bound.*

Proof. Each summand has the same expectation under the two laws. Sum the finite expansion and apply (1). $\square$

For a polynomial of degree Δ in independent scalar coordinates, $2\Delta q$ -wise matching is a general sufficient condition. A sharper count can separate sign and hash coordinates, or count whole columns; Corollary 10.12 carries out such a count for the degree-three triangle of Section 10.4, where squared root factors reduce $6q$ to $4q$, and to $2q$ for signs. The original SparseStack theorem uses full independence. The proposition extends its moment conclusion to matching block marginals; constructing such a law with a prescribed seed length is a separate sampling problem.

5 Applications of the common estimates

5.1 SparseStack: why dimension and occupation add

Let $U \in \mathbb{R}^{n \times d}$ have orthonormal columns, with rows u_i^* . Stack s independent CountSketch blocks, each with b buckets:

$$S_{(g,a),i} = s^{-1/2} \sigma_{gi} \mathbf{1}_{h_{gi}=a}, \quad m = sb, \quad E = S_U^* S_U - I_d, \quad S_U = S U.$$

All (σ_{gi}, h_{gi}) are mutually independent, with uniform signs and uniform hashes. Every column has squared norm one, so the error is exactly hollow in the input-column indices:

$$E = \frac{1}{s} \sum_g \sum_{i \neq j} \sigma_{gi} \sigma_{gj} \mathbf{1}_{h_{gi}=h_{gj}} u_i u_j^*. \quad (114)$$

This deterministic absence of a diagonal fluctuation is part of the model. It would be false for independent Bernoulli supports without exact column normalization.

Theorem 5.1 (SparseStack). *Let $d \geq 1$, $0 < \epsilon \leq 1$, $0 < \delta \leq 1$, and set*

$$q = \max\{1, \lceil \log_2(d/\delta) \rceil\}, \quad D_q = d + 2q + 1, \quad c_* = (1 - e^{-1})^{-1}, \quad \Lambda = 90c_*^2.$$

Choose

$$s = \left\lceil \frac{\Lambda(2q+1)}{\epsilon} \right\rceil, \quad b = \left\lceil \frac{\Lambda^2 D_q}{s\epsilon^2} \right\rceil, \quad m = sb.$$

Then

$$\mathbb{P}\{\|E\| > \epsilon\} \leq \delta, \quad s < 1356q/\epsilon, \quad m < 306456(d+q)/\epsilon^2.$$

Proof. The failure bound is the last assertion of Corollary B.1, proved in Appendix B by the direct occupation argument on the actual signed bucket variables. The size bounds follow from the ceilings: with $\Lambda = 90c_*^2 < 225.25$, $s < \Lambda(2q+1)/\epsilon + 1 \leq (3\Lambda+1)q/\epsilon < 1356q/\epsilon$, and $m = sb < \Lambda^2 D_q/\epsilon^2 + s \leq \{\Lambda^2(d+3q) + 677q\}/\epsilon^2 < 306456(d+q)/\epsilon^2$, since $\Lambda^2 < 50760$. $\square$

The original proof of this prescription, through a conditional barycenter comparison and a band envelope, is the one formalized in Lean 4 (Section 11); none of the four application theorems depends on it, and its independent-entry operator bound enters this paper once, in Section 10.2. Its architecture explains the scales more clearly than the rounded constants, and we record it. First, a conditional barycenter compares (114) to the hollow Gram matrix of independent three-point entries. It costs c_b^2 in the moment root, where

$$c_b = [1 - (1 - 1/b)^b]^{-1} \leq c_*.$$

The local three-level matrix in Example 2.2 then gives an exact operator with grade $\nu = L+2H$ and shifts $-2, 0, 2$. Here L and H count light and heavy sites. Grouping all terms by their net grade change gives three *bands*: lowering, preserving, and raising. Each band may contain several channels with different local legs.

The coefficient relation is Parseval: $\sum_i u_i u_i^* = I_d$. On ℓ light sites, the creation Gram contracts each site against the same external $\mathbb{R}^d$ factor. Lemma 3.1 bounds this sum by $d + \ell - 1$, rather than $d\ell$. This is the structural reason the eventual row requirement contains $d + q$. Heavy conversions carry the local factor $b - 1$; after division by $m = sb$, their scale is q/s . These two mechanisms produce the separate row and sparsity requirements.

More explicitly, the band-by-band norm bounds of that proof give a common *band envelope*, meaning a number that bounds the norm of each of these three bands on the moment cutoff:

$$\beta_q = (3 + \sqrt{2}) \left[\sqrt{\frac{D_q}{m}} + \frac{D_q}{m} + \frac{2q+1}{s} \right]. \quad (115)$$

Expanding a vacuum matrix element of the $2q$ -th power chooses one of the three bands at each step: a *band word* of length $2q$. Only words that return to grade zero contribute, but counting all 3^{2q} words is a valid upper bound. This gives

$$\mathbb{E} \operatorname{tr} E^{2q} \leq d(3c_b^2 \beta_q)^{2q}. \quad (116)$$

The displayed parameter choices make $3c_b^2 \beta_q < \epsilon/2$, and Markov and integer rounding give the prescription of Theorem 5.1 along this route as well. The factor 3, the coupling cost, the band coefficients, and the shared-factor estimate have different origins; only the last one creates additive dimension dependence.

There is an immediate low-moment calibration. If $\ell_i = \|u_i\|^2$, direct sign and collision counting gives

$$\mathbb{E} E^2 = \frac{1}{m} \left[(d+1)I - 2 \sum_i \ell_i u_i u_i^* \right], \quad \mathbb{E} \operatorname{tr} E^2 = \frac{d(d+1) - 2 \sum_i \ell_i^2}{m}. \quad (117)$$

Indeed the two off-diagonal pairs in the squared error must agree up to reversal, and their collision has probability $1/b$. This equals the independent-entry hollow second moment. Thus the coupling constant is already unnecessary at $q = 1$, although that observation alone does not prove a coupling-free comparison for higher moments.

The analytic direct-model extension uses the $2b$ -dimensional local factor (5) and the exact multiplication operator

$$\mathcal{M} = \frac{1}{m} \sum_{g,a} \sum_{i \neq j} u_i u_j^* \otimes J_{a,gi} J_{a,gj}.$$

Each summand combines two of the local legs in (5) at distinct sites of one block. Figure 8 shows their action on the light and heavy counts; Appendix B gives the full family definitions. This analytic proof reaches the same probability law directly. In Section 10 we retain block parity and returning-path geometry to obtain the uniform pair (26.5, 179), with a complete derivation and rational certificate. The gain combines the direct local factor with sharper sector and path estimates; the explicit transfer cost remains the separately identified factor c_b^2 . Remark 8.4 compares the resulting constants, on the lower edge that they share, with the injectivity theorem obtained from Tropp’s extreme-eigenvalue comparison.

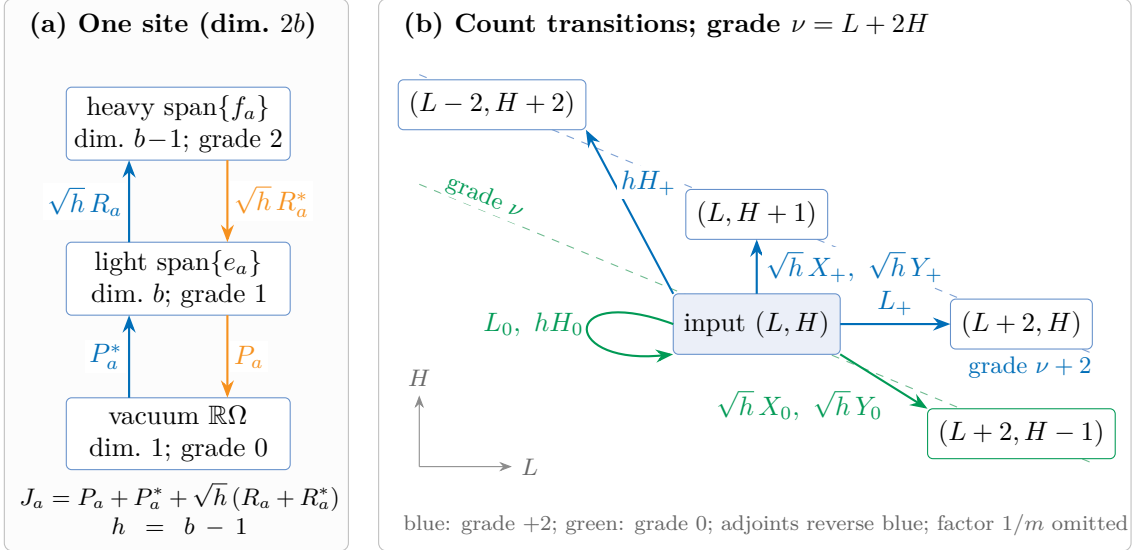


Figure 8: Direct-model SparseStack, the analytic extension proved in Theorem 10.6. Panel (a) is one site: the local multiplier J_a of (5) moves between the three levels along the four legs shown; light vectors are orthonormal and heavy vectors form a simplex. Panel (b) is the lattice of total counts (L, H) , with L horizontal and H vertical, so that the dashed lines are the grade isolines $\nu = L + 2H$. Each term of the error multiplier combines two legs at two distinct sites of one block, which moves the counts along one arrow; the eight families and the three bands are defined in Appendix B, (214)–(215). The green loop represents L_0 and hH_0 , which preserve the counts while changing which sites carry each level. Forbidden occupations and negative counts give zero, and every block’s light parity $L_g \bmod 2$ is preserved.

This application identifies three useful proof responsibilities. The exact-column SparseStack law makes the diagonal exactly deterministic; the reference comparison applies to the resulting hollow error. The shared-factor Gram creates the optimal order in d and q . The coupling, band summation, and rounding determine substantial parts of the explicit numerical constants. Only after these roles are separated does constant improvement become a well-posed local question.

5.2 The logarithmic exponent of dense sign graph matrices

The connected creation estimate predicts which graph matrices acquire a logarithm in their norm. We begin with one edge, where the distinction is visible before any operator calculation.

5.2.1 The model and the scale

Fix a finite simple graph $H = (\mathcal{V}, E)$, with $h = |\mathcal{V}|$ and $e = |E|$, and two ordered boundary lists U, V of distinct roles. The lists may overlap or be empty. Let W_{ij} , $1 \leq i < j \leq n$, be independent uniform signs, set $W_{ji} = W_{ij}$, and set $W_{ii} = 0$. For injective boundary assignments a, b , define

$$M_{H,n}[a, b] = \sum_{\substack{\phi: \mathcal{V} \hookrightarrow [n] \\ \phi|_U = a, \phi|_V = b}} \prod_{xy \in E} W_{\phi(x)\phi(y)}. \quad (118)$$

The injection is global, including across components. Incompatible assignments give zero, and an empty boundary indexes a one-dimensional space. All asymptotics below keep the entire shape fixed.

Example 5.2 (One edge, two boundary choices). Take H to be the edge rx . If $U = (r)$ and $V = (x)$, then $M_{H,n} = W$: its expected norm is of order $\sqrt{n}$. Instead take $U = V = (r)$. The common role forces the row and column labels to agree, so the matrix is diagonal:

$$M_{H,n}[i, i] = \sum_{j \neq i} W_{ij}, \quad M_{H,n}[i, k] = 0 \quad (i \neq k).$$

Its expected norm is of order $\sqrt{n \log n}$. The same edge now defines a fluctuating row sum, and the norm takes its maximum over the boundary label i . Deleting the common boundary role r leaves one connected component, the singleton x , attached to r . This component is the source of the extra square root of a logarithm. The classification below places both examples in one formula.

A separator S meets every path from U to V ; it may contain boundary roles and must contain $U \cap V$, because a common role gives a path of length zero. Write s for the minimum separator size and $\mathcal{S}_{\min}$ for the family of minimum separators. For $S \in \mathcal{S}_{\min}$, call a component of $H - S$ *attached floating* if it misses both boundaries and has an edge to S . Write $b_{\text{att}}(S)$ for their number, and define

$$b_* = \max_{S \in \mathcal{S}_{\min}} \#\{\text{attached floating components of } H - S\}, \quad T_{H,n} = n^{(h+i_0-s)/2} (\log(2n))^{b_*/2}, \quad (119)$$

where i_0 counts isolated roles outside $U \cup V$.

Theorem 5.3 (Graph classification). *For the model (118) and every fixed $p > 0$,*

$$(\mathbb{E} \|M_{H,n}\|^p)^{1/p} = \Theta_{H,p}(T_{H,n}).$$

If t is the number of original boundary-free components containing an edge, then, for every fixed $A > 0$,

$$(\mathbb{E} \|M_{H,n}\|^p)^{1/p} \leq C_{H,A} T_{H,n} p^{t/2}, \quad 1 \leq p \leq A \log(2n).$$

The constants depend on the fixed shape and the displayed moment parameters, and are uniform in n .

The full theorem is proved in Appendix F. The separator power of n , including the isolated-interior correction, appears already in Ahn, Medarametla, and Potechin [28, Theorems 2.24 and 6.1], up to polylogarithmic factors; see also the systematic matrix-chaos treatment in [29]. The sharp classification specifies the attached-component logarithmic factor and a matching lower bound.

Another proof and the scope of the overlap. Xu, Fu, Qiao, and Tao [30, Theorem 1.3] give another proof of the expected-norm classification for the same fixed simple, globally injective dense-sign model. Their statistics (v, h, s, a_*) correspond to (h, i_0, s, b_*) here. This is an overlap of theorem statements, not merely a related bound for a different model. Their Appendix F.6 also discusses the logarithmic moment window; each paper’s probability and moment assertions retain their stated quantifiers.

Role coloring and sign-Fourier projection are common reduction tools; vertex partitioning already appears in [28, Section 5]. Our proof uses occupation operators, connected creation, and an *LDR* factorization with a column gauge. The proof in [30] uses replica-partition counting and conditional flattening. We compare these arguments without assigning priority or claiming independent discovery. The companion [2] records the provenance and formalization of our graph argument.

The same preprint also treats fixed-law typed factor chaoses with moment growth $\|\xi_e\|_{L_q} \asymp q^{\theta_e/2}$, $0 \leq \theta_e \leq 2$ [30, Theorem 5.2]. Its finite-profile trace upper bound has an expected-norm consequence under a weighted-cut stability hypothesis [30, Proposition 5.4]; it is not an unrestricted sparse-law classification. Section 10.4 states the comparison relevant to the one-shape results below.

5.2.2 The critical factorization

Give different roles disjoint pools of ground labels. For a shape with $h > 0$, independent uniform role colors give the exact identity $M_{H,n} = h^h \mathbb{E}_{\text{colors}} M_{H,n}^{\text{col}}$. Thus a uniform colored upper estimate transfers at a shape-dependent constant cost. In Walsh coordinates, a monomial toggles its primitive edge set: $|\omega\rangle \mapsto |\omega \triangle E_\phi\rangle$. For the self-adjoint dilation $\mathcal{W}$ of the colored matrix, the moment identity is

$$\mathbb{E} \operatorname{tr}[(M^{\text{col}}(M^{\text{col}})^*)^q] = \sum_a \langle a, \emptyset | (P_{\leq eq} \mathcal{W} P_{\leq eq})^{2q} | a, \emptyset \rangle. \quad (120)$$

Every returning walk has grade at time j at most $e \min(j, 2q - j)$, which explains the exact cutoff.

We first treat a *core shape*: every original connected component meets U or V . In particular, there are no isolated interior roles. For the factorization, remove edges internal to either boundary; their signs are diagonal row or column factors of norm one, and this removal preserves the separator parameters.

In a forward transition from the U boundary to the V boundary, let $F \subseteq E$ record the abstract edges that are already occupied. Thus an embedding contributes precisely when $E_\phi \cap \omega = \phi(F)$; the step removes those edges and adds the others. Write $\mathcal{A}_F$ for the part of this forward operator with occupied-edge pattern F . A new role incident to an occupied edge must take its label from the vertex support of ω , which has at most $2K$ labels at grade K . A role with no incident occupied edge can instead have n label choices. The reverse transition interchanges occupied and unoccupied edges.

The *free-label signature* records these two kinds of unrestricted roles. Write $I = \mathcal{V} \setminus (U \cup V)$, $B = U \cap V$, and let $\delta_H(x)$ be the edges incident to x . The reverse-free and forward-free sets, respectively, are

$$\begin{aligned} R_F &= \{x \in (U \setminus B) \cup I : \delta_H(x) \subseteq F\}, \\ Z_F &= \{x \in (V \setminus B) \cup I : \delta_H(x) \cap F = \emptyset\}, \quad S_F = \mathcal{V} \setminus (R_F \cup Z_F). \end{aligned}$$

The two free sets are disjoint, and no edge joins them: such an edge would have to belong both to F and to its complement. Every U -role outside S_F is in R_F , and every V -role outside S_F is in Z_F . Hence S_F separates the boundaries. An elementary row-and-column count has

leading factor $n^{(|R_F|+|Z_F|)/2} = n^{(h-|S_F|)/2}$, with a fixed polynomial in K for the remaining labels. If $|S_F| > s$, it therefore saves at least $n^{1/2}$. We call a transition *critical* when $|S_F| = s$: its free labels have reached the largest possible power of n , so the occupation estimate must supply the finer information.

For such a transition, each component of $H - S_F$ lies wholly in one free set. A component meeting U belongs to the left side; one meeting V belongs to the right side. The remaining components miss both boundaries and are attached to S_F because the shape is a core. Figure 9 shows this decomposition.

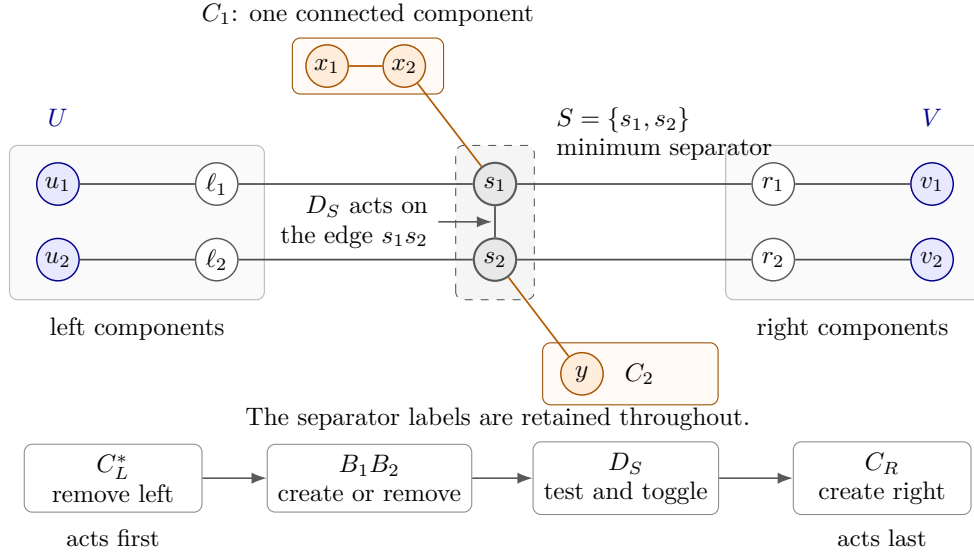


Figure 9: A minimum separator $S = \{s_1, s_2\}$ splits two boundary paths and leaves two attached floating components. The two vertex-disjoint U – V paths certify $|S| = 2$ is minimum. The two-role component C_1 supplies one factor B_1 , just as the singleton C_2 supplies one factor B_2 . The vertical edge s_1s_2 is internal to the separator and is tested and toggled by D_S . The drawn shape has $h = 13$, $i_0 = 0$, $s = 2$, and $b_* = 2$; Theorem 5.3 therefore gives $\mathbb{E}\|M_{H,n}\| \asymp n^{11/2} \log n$ as $n \rightarrow \infty$. The operator stages are shown in their order of action; their product is read right to left.

For this picture, every minimum separator removes exactly one vertex from each horizontal boundary path. Each surviving segment of either path still contains a boundary endpoint. Thus a boundary-free component can only be one of the two pendant branches, and it detaches precisely when its root s_i is removed. Consequently $b_{\text{att}}(S') = |S' \cap \{s_1, s_2\}| \leq 2$ for every minimum separator S' , with equality only for the displayed separator. This verifies the logarithmic factor directly from the drawing.

The factorization first removes the left-side edges and keeps the chosen separator labels in a register. Next it creates or removes each entire floating component, without changing that register. It then tests and toggles the edges internal to the separator, and finally creates the right-side edges. In symbols,

$$\mathcal{A}_F = C_R D_{S_F} \left(\prod_{j=1}^{b_{\text{att}}(S_F)} B_j \right) C_L^*. \quad (121)$$

Here C_L creates the left side from separator labels, so C_L^* removes it; C_R creates the right side. The separator map D_{S_F} has norm at most one. For the j th floating component, B_j is annihilation if it lies in R_F and creation if it lies in Z_F . Disjoint role pools make the factors'

primitive edge families disjoint, so their product preserves every occupancy test and every embedding multiplicity. Intermediate grades need only increase the cutoff from K to $K + e$.

The side-creation estimate, Lemma F.13, has leading norm $n^{\ell/2}$ when the side adds ℓ roles outside its input separator. Its error is bounded by $2^{v/2}(1 + 2K)^v n^{(\ell-1)/2}$ for a side with v roles. Proposition 3.9 gives $n^{c_j/2} \sqrt{K+1}$ for a floating component with c_j interior roles, again with an $n^{-1/2}$ relative error times a fixed polynomial. Multiplying the factors in (121), and then summing the finitely many signatures, gives

$$\|P_{\leq K} \mathcal{W} P_{\leq K}\| \leq C_H n^{(h-s)/2} (K+1)^{b_*/2} + C_H n^{(h-s-1)/2} (K+1)^{D_H}. \quad (122)$$

Here $D_H = 6h + 4$ suffices. The largest floating count appears because we sum over finitely many critical signatures. At $q \asymp \log n$, (120) absorbs the boundary dimension factor $n^{|U|}$ after taking a $2q$ -th root, and the second term of (122) is lower order. This proves the core upper scale. We handle original boundary-free components by scalar moment estimates, and isolated interiors by counting their remaining injective labels, as detailed in Section F.8.

5.2.3 Why the logarithm is present, and what sharpness means

For the lower bound, we choose a minimum separator attaining b_* . Conditional moment concentration supplies simultaneous excursions of its attached component fields across many separator labels. Normalized side columns and a Fourier projection transfer this colored lower estimate to the original matrix. These two ingredients provide the separate lower-bound argument of Section F.7.

In Example 5.2, opposite boundaries give $s = 1$ and $b_* = 0$, whereas the common boundary gives $s = 1$ and $b_* = 1$. This is exactly the distinction between $\sqrt{n}$ and $\sqrt{n \log n}$: one connected fluctuating field has many boundary labels over which to realize an extreme.

The matching uncolored lower bound is in expectation, in every fixed positive L^p , and at a positive probability bounded away from zero. The strong upper moments yield polynomial upper tails at $T_{H,n}$ for core shapes, and at $T_{H,n}(\log(2n))^{t/2}$ in general. Their upper-only transfer needs $O_H(\log n)$ -wise independent signs; the matching lower argument uses full independence.

Remark 5.4 (Uniformity in the moment order). The logarithmic exponent is determined by the estimate (122) uniformly at $q \asymp \log n$, together with the separate lower bound. Connected cancellation controls this growing range of moments, while the separator factorization identifies the components that contribute a square root of q . All shape-dependent constants stay fixed because the dense simple shape and its globally injective model stay fixed as n grows.

5.3 Two-round Walsh sampling

The one-round SRHT analysis of Tropp [15] is a predecessor for this sketching model, not the two-round theorem below. The companion [3] records the earlier cubic proof and its formalization; the quadratic refinement is proved here.

Let n be a power of two, let H be the normalized Walsh matrix, and let $U \in \mathbb{R}^{n \times d}$ have orthonormal columns. Let the $2n$ entries of the sign diagonals D_x, D_y be independent and unbiased, and set

$$W = H D_y H D_x U, \quad w_j^* = e_j^* W.$$

The following statement, with its cubic allowance, is the one formalized (Section 11); it is a consequence of the quadratic theorem proved in this paper.

Theorem 5.5 (Two-round Walsh). *For independent Bernoulli selectors of mean $0 < \rho < 1$, independent of the signs, put*

$$E_\rho = \rho^{-1} W^* \text{diag}(\eta) W - I_d, \quad m = \rho n, \quad D_q^{\text{old}} = d + 27q^3.$$

For every integer $q \geq 1$,

$$\mathbb{E} \operatorname{tr}(E_\rho^{2q}) \leq d[6\sqrt{D_q^{\text{old}}/m} + 9D_q^{\text{old}}/m]^{2q}.$$

For $0 < \epsilon < 1$, $0 < \delta < 1/2$, choose $q = \lceil \log_4(4d/\delta) \rceil$ and

$$M = \min\{n, \lceil 8192\epsilon^{-2}D_q^{\text{old}} \rceil\}.$$

If T is a uniform M -element subset independent of the signs and $S = \sqrt{n/M} R_T H D_y H D_x$, then

$$\mathbb{P}\{\|U^* S^* S U - I_d\| > \epsilon\} \leq \delta.$$

The Bernoulli assertion also holds for sign families with $\min(n, 4q)$ -wise independent unbiased marginals and selectors with $\min(n, 2q)$ -wise independent Bernoulli marginals, provided the three families are mutually independent. In the fixed-size assertion, the sign families may have the same limited independence and T retains its uniform law.

Proof. For $A = B = H$ the coherence parameters of Theorem 10.1 are $\alpha = \beta = 1$, so its allowance is $D_q = d + 12q^2 \leq d + 27q^3 = D_q^{\text{old}}$, and the right side of (180) is increasing in the allowance; this gives the moment bound. The fixed-size argument of Theorem 10.1 uses only $M \geq 1024D_q/\epsilon^2$, which $M = \lceil 8192\epsilon^{-2}D_q^{\text{old}} \rceil$ satisfies, and the case $M = n$ is exact. The limited-independence clauses are the same degree count. $\square$

The original proof of the cubic statement, formalized in Lean 4, runs through the four row channels of Section 3.5: a normal-ordering identity controls double annihilation, positivity assembles the resulting row effects, Lemma 3.12 controls selector creation, and the rectangular cutoff retains the first q vacuum iterates; its fixed-size sampling comparison costs four in the moment root, which produces the constant 8192.

Our collective coefficient contractions strengthen this argument. Theorem 10.1 proves the resulting extension for arbitrary complex unitary pairs, including the Walsh and Fourier cases, with a quadratic confidence term. We give the extension's coefficient and sampling proofs in full below; that refinement is analytic.

For perspective, Section A.4 gives a scalar fixed-distortion obstruction: along $n = L^2$ and $\delta = 2^{-4L}$, two Walsh rounds require $M \geq \frac{3}{4}(n - L + 1)$. The extension attains the quadratic confidence exponent of this family. The lower bound concerns that confidence regime, while the sufficient theorem also specifies dimension and distortion dependence. Remark 10.2 states the catalog form of the two-round question, reads off the constant that Theorems 5.5 and 10.1 give for it, and compares it with another proof.

5.4 Independent entries with regular moment growth

The preceding applications use complete-grade contractions. For general independent entries, the preserved-cone contraction supplies the local estimate of Corollary 4.14, and a deterministic deletion argument assembles it at the appropriate moment orders. Use the entry regularity and the quantities $M(X)$, $D(X)$ of Section 2.7. The conjectured comparison, its lower bound, and the local-to-global reduction are from Latała–Świątkowski [48, Theorem 4.1, Conjecture 4.3, and Remark 4.5]; the earlier formalized upper proof is recorded in the companion [4].

Reader path and structural scale. The regular-entry argument runs through Section 2.7 (the original law, the parity–amplitude representation, and the count sectors), Section 3.6 (the preserved cone, the spherical thin-axis estimate, and the scalar comparisons), Section 4.4.1

(the parity moment theorem), and the assembly below (variance where variance suffices, the deletion parameter where it is needed). At its structural scale the local statement is

$$(\mathbb{E}\|Y\|^{2q})^{1/(2q)} \lesssim n^{1/(2q)} \sqrt{\alpha} \{\sigma(Y) + \alpha^2 R_Y(2q)\},$$

with the operator part supplying the exact representation, the preserved domain, the thin sectors and the parity iteration, and the scalar part supplying the order-sensitive comparisons; the explicit constants 2.31, χ_0 and those of (124) are quantitative refinements of this scale, not its content.

Theorem 5.6 (Independent entries with regular moment growth). *For every $\alpha \geq 1$ there are constants $c_\alpha, C_\alpha > 0$ such that every $n \geq 1$ and every real $n \times n$ matrix X with independent centered entries, finite absolute moments of every order, and (7) obey*

$$c_\alpha \{M(X) + D(X)\} \leq \mathbb{E}\|X\| \leq C_\alpha \{M(X) + D(X)\}. \quad (123)$$

The constants depend only on α , not on the dimension, the individual entry distributions, or the deterministic deletion sets. For entries symmetric in law the assembly of Section 5.4 gives the two-coefficient bound $\mathbb{E}\|Y\| \leq A_\alpha M(Y) + B_\alpha D(Y)$ of (133), and, for each value of the moment-policy parameter $c \in \{2, 1, \frac{1}{2}, \frac{1}{4}\}$ of (126), one may take

$$C_\alpha^{\text{sym}} = C_c \alpha^{\zeta_c}, \quad (C_c, \zeta_c) = (455, 11.628), (418, 10.629), (496, 9.629), (993, 8.635) \\ \text{for } c = 2, 1, \frac{1}{2}, \frac{1}{4} \text{ respectively,} \quad (124)$$

so that $C_\alpha^{\text{sym}} \leq 418 \alpha^{10.629}$ and $C_\alpha = 2C_{2\alpha}^{\text{sym}} \leq 1.33 \cdot 10^6 \alpha^{10.629}$ at $c = 1$; the unrounded values are recorded in Table 3. The explicit constants are established by the analytic argument of Section 5.4 and Appendix G. An earlier upper theorem is separately formalized with the explicit Lean constant identified in Section 11. The analytic reference formula (134) is recorded for comparison; it is not identified with that Lean term.

The estimate is Conjecture 4.3 of Latała and Świątkowski [48]; its lower half is their Theorem 4.1. The earlier formalized upper theorem is distinguished from the current analytic constants in Section 11. The constants (124) are established by the analytic argument of this section, completed in Appendix G. This section explains what the graded method contributes to that proof and summarizes the rest, so that the two parts can be told apart.

The lightest-tailed case deserves a separate statement, because it is the one for which the comparison was previously known only up to an iterated logarithm.

Corollary 5.7 (Weighted Rademacher matrices). *Let $X = (a_{ij}\varepsilon_{ij})_{i,j \leq n}$ with deterministic real weights a_{ij} and independent Rademacher signs ε_{ij} . Then*

$$c \{M(X) + D(X)\} \leq \mathbb{E}\|X\| \leq C_1^{\text{sym}} \{M(X) + D(X)\}$$

with absolute constants: c from [48, Theorem 4.1] at $\alpha = 1$, and $C_1^{\text{sym}} = 418$ from (124) at $\alpha = 1$, $c = 1$.

Proof. Every entry satisfies $\|a_{ij}\varepsilon_{ij}\|_{L_{2p}} = |a_{ij}| = \|a_{ij}\varepsilon_{ij}\|_{L_p}$, so (7) holds with $\alpha = 1$. The entries are symmetric in law, so the symmetrization step of Theorem 5.6 is not needed and the constant is C_1^{sym} rather than $2C_2^{\text{sym}}$. $\square$

Latała [42] proved this comparison up to a factor $\log \log \log n$ for arbitrary weights, and without that factor for $\{0, 1\}$ weights, with a weak-moment order convention that differs from (10) only at small k ; by Proposition 3.25 the two conventions are comparable within absolute

constants. Meller [33] proved the comparison for independent symmetric κ -regular entries, $\kappa > 1$, normalized by $\mathbb{E}|X_{ij}| = 1$ and weighted by a deterministic matrix, up to a factor $(\text{Log Log } n)^{C(\kappa)}$, with the order $\text{Log } k = \log(k \vee e)$; since $\frac{1}{2} \text{Log } k \leq \log(k+1) \leq 2 \text{Log } k$ for every $k \geq 1$ and Proposition 3.25 applies at every real order $p \geq \frac{1}{2}$, the two deletion functionals are within the factor $J_\alpha = H_\alpha^7$ of each other on symmetric laws, and the normalization changes M by at most the factor κ . Neither comparison specifies its constants. The Rademacher case is also the one in which the parity–amplitude representation is a complete Walsh grading: the magnitudes are constant, so the cone restriction of Proposition 3.33 is unnecessary and Theorem 3.34 supplies the local contraction on the whole graded space. The deletion argument below is unchanged.

What the cone contraction supplies Remark 4.5 of [48] observes that, to obtain the upper half of (123), it suffices to prove for entries symmetric in law the *undeleted* local estimate

$$\mathbb{E}\|Y\| \leq C_\alpha \{M(Y) + R_Y(\log(n+1))\}. \quad (125)$$

Corollary 4.14 is exactly this statement, with explicit constants, and it is the new local content of the theorem. Its origin in the method is worth stating once more in the language of the earlier applications. The exact representation is the parity–amplitude space of Section 2.7, in which only the sign register is graded and every magnitude power stays inside a coefficient function (Figure 3). The sector decomposition is Lemma 2.5: creation from a given parity state involves at most $k+1$ original rows and annihilation at most k original columns, so each sector is a compression of multiplication by a thin submatrix. The preserved relation is positivity of the ordinary monomial coefficients, which the vacuum walks have and which every parity projection and multiplication step keeps; on that cone Lemma 3.32, Lemma 3.29 and Theorem 3.24 bound each sector, and Proposition 3.33 sums the squared sector bounds with no factor for the number of sectors. Finally the parity majorant of Theorem 4.11 converts the sector allowances, followed along the grades of the walk, into the moment bound. The distinctive feature, compared with the finite deterministic blocks of the earlier applications, is that the sector maps here are multiplication operators by random thin matrices, whose operator norms need not be finite; the contraction is asserted only on the cone and is proved by moment inequalities on the reconstructed random polynomial.

From the local estimate to the deletion parameter Corollary 4.14 controls $\mathbb{E}\|Y\|$ by the undeleted weak moment $R_Y(\log(n+1))$, which may far exceed $D(Y)$. The reduction that closes this gap is deterministic matrix decomposition; the graded method re-enters only through the local theorem applied to a family of principal submatrices. We give the reduction in full; the six elementary verifications it uses, the schedule count and screening property, the mask partition, the monotonicity of weak moments, the maximum lemma, the centered Gram estimate and the symmetrization, are proved in Appendix G. Throughout, Y has independent entries symmetric in law satisfying (7), $M = M(Y)$, $D = D(Y)$, and $v_{ij} = \mathbb{E}Y_{ij}^2$, so that $\sum_j v_{ij} \leq M^2$ and $\sum_i v_{ij} \leq M^2$.

Schedule and nested family

Put

$$b_k = 2^{5^k} - 1, \quad \pi_k = \log(b_k + 1) = 5^k \log 2, \quad t_k = b_k^3,$$

so $b_0 = 1$, $b_1 = 31$ and $b_{k+1} + 1 = (b_k + 1)^5$. If $M = 0$ the matrix vanishes almost surely and there is nothing to prove, so let $M > 0$. For each budget $k < n$ let I_k be a deterministic minimizer of $I \mapsto R_{Y,I}(\log(k+1))$ over $|I| \leq k$, so that $R_{Y,I_k}(\log(k+1)) \leq D$, and put $I_k = [n]$ for $k \geq n$, an admissible minimizer with zero residual weak moment; the norm of an

empty block is zero. Starting from $S_0 = I_1$, enlarge S_k by every row i with $(t_k + 1)v_{ij} > M^2$ for some column $j \in S_k$, by every column with the transposed property, and by I_{b_k} ; call the result S_{k+1} . A column admits at most t_k such rows, so $|S_{k+1}| \leq 2|S_k|(1 + t_k) + b_k$, and the schedule gives $|S_k| \leq b_k$ and $S_{n+1} = [n]$. Two properties result. An entry between S_r and the complement of S_{r+1} , in either orientation, has variance at most $M^2/(b_r^3 + 1)$; and $R_{Y, S_{a-1}}(\pi_{(a-2)+}) \leq D$ for every $a \geq 1$, because S_{a-1} contains the minimizer $I_{b_{a-2}}$ and enlarging a deleted set decreases the weak moment. The minimizer inserted at step k carries the budget b_k , one stage behind the set that contains it; this shift is what keeps the recorded orders comparable to the scheduled ones below.

Levels and five masks

Let ℓ_i be the first r with $i \in S_r$, so $\{i : \ell_i \leq r\} = S_r$ has at most b_r elements. Split the entries of Y by the five deterministic masks of Figure 10: nearLow ($\ell_i, \ell_j \leq 1$), nearHigh ($\lfloor \ell_i/2 \rfloor = \lfloor \ell_j/2 \rfloor \geq 1$), nearCross ($\lfloor (\ell_i + 1)/2 \rfloor = \lfloor (\ell_j + 1)/2 \rfloor$ and $\ell_i \neq \ell_j$), and the two far masks $\ell_j + 1 < \ell_i$ and $\ell_i + 1 < \ell_j$. Exactly one condition holds at each pair of levels, so the triangle inequality bounds $\|Y\|$ by the five masked norms.

Near blocks at scheduled orders

The nearLow mask is supported on $S_1 \times S_1$ with $|S_1| \leq 31$; its Frobenius norm gives $\sqrt{31} M$. The two other near masks are block diagonal. The block at index m is the principal submatrix on $V_a = \{i : \ell_i \in \{a, a+1\}\}$ with $a = 2m$ for nearHigh and $a = 2m - 1$ for nearCross, in the second case restricted off the two diagonal shell blocks, which a conjugation by $\text{diag}(\pm 1)$ shows costs nothing. Here the method re-enters. The order at which a block is estimated is a choice; fix a moment-policy parameter c and put

$$q_a = \lceil c \pi_{a+1} \rceil, \quad P_a = 2q_a, \quad c \in \{2, 1, \frac{1}{2}, \frac{1}{4}\}, \quad (126)$$

so that $2^{m_c+1} = 256c$ with $m_c = 7 + \log_2 c \in \{8, 7, 6, 5\}$. Since $|V_a| \leq b_{a+1} = e^{\pi_{a+1}} - 1$ and $P_a \geq 2c \pi_{a+1}$, the dimension factor is $|V_a|^{1/P_a} \leq e^{1/(2c)}$, and Theorem 4.11 applied to the block at $q = q_a$, with $\hat{\Theta}_{q_a} < 2.31$, $\alpha^{(q_a-1)/(2q_a)} \leq \sqrt{\alpha}$ and $\sigma(Y[V_a, V_a]) \leq M$, yields

$$(\mathbb{E}\|Y[V_a, V_a]\|^{P_a})^{1/P_a} \leq \ell_c \sqrt{\alpha} \{M + \chi_\alpha R_{Y[V_a, V_a]}(P_a)\}, \quad \ell_c = 2.31 e^{1/(2c)}. \quad (127)$$

Nothing in the local theorem prefers one horizon over another; a smaller c costs a larger dimension factor and saves doublings of the weak-moment order below. The weak moment of the block is carried to the order P_a from one of the two scales already present in the theorem. For $a \in \{1, 2\}$ it is carried from the variance: by independence and centering,

$$R_{Y, I}(2)^2 = \sup_{s, t} \sum_{i, j \notin I} \mathbb{E} Y_{ij}^2 s_i^2 t_j^2 = \max_{i, j \notin I} \mathbb{E} Y_{ij}^2 \leq (M/2)^2$$

for every deterministic I , with an empty maximum interpreted as zero. Thus (79) with the anchor $b = 2$ gives, since $P_1 < P_2 = 2\lceil 125c \log 2 \rceil$ and the factor is increasing in P ,

$$R_{Y[V_a, V_a]}(P_a) \leq \frac{\sqrt{P_2 - 1}}{2} \alpha^{\nu_c} M, \quad \nu_c = \eta(P_2/2), \quad a \in \{1, 2\}. \quad (128)$$

For $a \geq 3$ it is carried from the deletion parameter: since $V_a \cap S_{a-1} = \emptyset$, the weak moment of the block at any order is at most $R_{Y, S_{a-1}}$ at that order, and $R_{Y, S_{a-1}}(\pi_{a-2}) \leq D$. With $p = \pi_{a-2} = 5^{a-2} \log 2$, the even anchor $b = 2\lceil p/2 \rceil$, the bridge in (79) with $\epsilon = 2b/p - 2$, and the anchor bound from b to $P_a = 2\lceil 125cp \rceil$,

$$R_{Y[V_a, V_a]}(P_a) \leq \left(\frac{2b-1}{b-1}\right)^{\epsilon/2} \left(\frac{P_a-1}{b-1}\right)^{1/2} \alpha^{\epsilon+\eta(P_a/b)} D \leq K_c \alpha^{\tau_c} D, \quad a \geq 3, \quad (129)$$

where

$$K_c = \left(\frac{7}{3}\right)^{\delta/2} \left(\frac{P_3 - 1}{3}\right)^{1/2}, \quad \tau_c = \delta + \eta(P_3/4), \quad \delta = \frac{8}{5 \log 2} - 2, \quad P_3 = 2 \lceil 625c \log 2 \rceil, \quad (130)$$

are the values at $a = 3$, where $p = 5 \log 2$ and $b = 4$.

Why the first deletion-controlled block dominates

For the remaining blocks, for $a \geq 4$, $p \geq 25 \log 2 > 17$, $p \leq b < p + 2$ and $250cp \leq P_a < 250cp + 2$, so $2^{m_c} < 250cp/(p+2) < P_a/b \leq 250c + 2/p < 2^{m_c+1}$, $\eta(P_a/b) = m_c + 2 - 256cb/P_a$, and the exponent

$$\begin{aligned} \epsilon + \eta(P_a/b) &= m_c + \frac{b}{p} \left(2 - \frac{256cp}{P_a}\right) \leq m_c + \left(1 + \frac{2}{p}\right) \left(2 - \frac{256}{250 + 2/(cp)}\right) \\ &\leq m_c + \frac{19}{17} \left(2 - \frac{256}{250 + 8/17}\right) < m_c + 1.094, \end{aligned}$$

the middle expression increasing in $2/p$ and in $1/(cp)$; while $\tau_c - m_c = \delta + 2 - 1024c/P_3 > 1.11$ for each admitted c (Table 3). The prefactor is at most $(\frac{2p-1}{p-1})^{2/p} (\frac{250cp+1}{p-1})^{1/2}$, which decreases in p and is below K_c at $p = 17$ for each admitted c (the two values are compared in the table). Thus every block satisfies

$$(\mathbb{E} \|Y[V_a, V_a]\|^{P_a})^{1/P_a} \leq L := \ell_c \sqrt{\alpha} \left\{ \left(1 + \frac{\chi_0 \sqrt{P_2 - 1}}{2} \alpha^{2+\nu_c}\right) M + \chi_0 K_c \alpha^{2+\tau_c} D \right\}. \quad (131)$$

The expectation of the maximum over the blocks of one mask is bounded by integrating the union of the Markov tails above the level L : for nonnegative Z_i with $\|Z_i\|_{L^{p_i}} \leq L$ and $p_i > 1$,

$$\mathbb{E} \max_i Z_i \leq L + \sum_i \int_L^\infty (L/t)^{p_i} dt = L \left(1 + \sum_i \frac{1}{p_i - 1}\right).$$

No independence assumption is used. Each index $a \geq 1$ occurs in exactly one of the two masks and $P_a - 1 \geq (50c \log 2 - 1)5^{a-1}$, so the two block-diagonal masks together contribute at most

$$N_c L, \quad N_c = 2 + \frac{5}{4(50c \log 2 - 1)}. \quad (132)$$

The far remainder

Write A for the lower far mask, $A_{ij} = Y_{ij} \mathbf{1}\{\ell_j + 1 < \ell_i\}$, and $w_{ij} = \mathbb{E} A_{ij}^2$. If $A_{ik} \neq 0$ and $\ell_j \leq \ell_k$, then $i \notin S_{\ell_k+1}$ while $j \in S_{\ell_k}$, so the screening property gives $w_{ij} \leq M^2/(b_{\ell_k}^3 + 1)$. Summing over the at most b_r^2 column pairs with $\max(\ell_j, \ell_k) = r$,

$$\sum_{j,k} \sum_i w_{ij} w_{ik} \leq M^4 \sum_{r \geq 0} \frac{b_r^2}{b_r^3 + 1} \leq 2M^4.$$

The centered Gram $G = A^\top A - \text{diag}(\sum_i w_{ij})$ has independent centered summands in each entry, and regularity at order two gives $\mathbb{E} Y_{ij}^4 \leq \alpha^4 v_{ij}^2$, so $\mathbb{E} \|G\|_F^2$ is at most α^4 times this triple sum. Since $\|A\|^2 \leq M^2 + \|G\|_F$ deterministically, two applications of Cauchy–Schwarz give $\mathbb{E} \|A\| \leq M \sqrt{1 + \sqrt{2} \alpha^2}$; the transpose handles the other far mask. No independence among the entries of G is used, and no general matrix inequality.

Assembly and symmetrization

Adding the five contributions gives

$$\begin{aligned} \mathbb{E}\|Y\| &\leq A_\alpha M + B_\alpha D, & A_\alpha &= \sqrt{31} + 2\sqrt{1 + \sqrt{2}\alpha^2} + N_c \ell_c \sqrt{\alpha} \left(1 + \frac{\chi_0 \sqrt{P_2 - 1}}{2} \alpha^{2+\nu_c}\right), \\ & & B_\alpha &= N_c \ell_c \chi_0 K_c \alpha^{5/2+\tau_c} =: C_c \alpha^{\zeta_c}. \end{aligned} \tag{133}$$

Every power of α in A_α is below $\zeta_c = \frac{5}{2} + \tau_c$, and $A_\alpha \leq A_1 \alpha^{\zeta_c}$ with $A_1 < C_c$ for each admitted c (Table 3), so $\mathbb{E}\|Y\| \leq C_c \alpha^{\zeta_c} (M + D)$, which is (124); this takes the larger of the two coefficients and does not add them. The two-coefficient form is the informative one: the variance-first allocation is what keeps the coefficient of M small, and the deletion parameter enters only through the blocks $a \geq 3$.

c	P_1, P_2, P_3	ν_c	τ_c	$\tau_c - m_c$	K_c	tail prefactor	N_c	C_c	ζ_c
2	70, 348, 1734	7.5287	9.1272	1.127	27.388	25.10	2.0183	454.96	11.6272
1	36, 174, 868	6.5287	8.1286	1.129	19.372	17.75	2.0371	417.05	10.6286
$\frac{1}{2}$	18, 88, 434	5.5455	7.1286	1.129	13.690	12.55	2.0766	495.33	9.6286
$\frac{1}{4}$	10, 44, 218	4.5455	6.1340	1.134	9.692	8.88	2.1631	992.90	8.6340

Table 3: The assembly constants for the four admitted moment policies (126), rounded; the exact expressions are (130)–(132) with $\ell_c = 2.31e^{1/(2c)}$ and $\chi_0 = \frac{7}{3} 2^{1/4}$. The column “tail prefactor” is the bound $(\frac{33}{16})^{2/17} (\frac{4250c+1}{16})^{1/2}$ on the blocks $a \geq 4$, which must be below K_c ; $\tau_c - m_c$ must exceed 1.094. The corresponding values of A_1 are 169.4, 158.0, 190.5 and 381.5. The centered constants $2C_c 2^{\zeta_c}$ are $2.88 \cdot 10^6$, $1.32 \cdot 10^6$, $7.85 \cdot 10^5$ and $7.89 \cdot 10^5$ at $\alpha = 1$. Smaller c trades a larger dimension factor for fewer doublings of the weak-moment order; the exponent of α is a property of this choice, not of the theorem. The printed symmetric bounds cross at $\alpha \approx 1.19$ ($c = 1$ to $c = \frac{1}{2}$) and at $\alpha \approx 2.01$ ($c = \frac{1}{2}$ to $c = \frac{1}{4}$).

For general centered X , the symmetrization $Y = X - X'$ by an independent copy has 2α -regular symmetric entries, $M(Y) = \sqrt{2}M(X)$, and $\mathbb{E}\|X\| \leq \mathbb{E}\|Y\|$ by Jensen. For every deterministic I ,

$$R_{Y,I}(p) \leq \begin{cases} 2^{1/p} R_{X,I}(p), & 0 < p < 1, \\ 2R_{X,I}(p), & p \geq 1, \end{cases}$$

by subadditivity of $x \mapsto x^p$ in the first case and Minkowski’s inequality in the second. The budgets $k \geq 2$ have orders $\log(k+1) \geq \log 3 > 1$ and are charged the factor 2; the budget $k = 1$ has the subunit order $\log 2$ and is charged $2^{1/\log 2} = e$. This alone gives $D(Y) \leq eD(X)$ and the prefactor e of the formalized version. At budget one a second bound is available: with $u_1 = \min_{|I| \leq 1} R_{Y,I}(\log 2)$, monotonicity in the order and the independent centered entries give $u_1 \leq R_{Y,\emptyset}(2) \leq \min\{\sigma_r(Y), \sigma_c(Y)\} \leq M(Y)/2 = M(X)/\sqrt{2}$, since $\sum_{ij} \mathbb{E}Y_{ij}^2 s_i^2 t_j^2 \leq \sigma_r(Y)^2$ for unit vectors s, t . Combining the two bounds on u_1 with the weights $2/e$ and $1 - 2/e$ gives $u_1 \leq 2D(X) + (1 - 2/e)M(X)/\sqrt{2}$, while every other budget contributes at most $2D(X)$. Hence

$$M(Y) + D(Y) \leq \frac{3 - 2/e}{\sqrt{2}} M(X) + 2D(X) \leq 2\{M(X) + D(X)\},$$

and $C_\alpha = 2C_{2\alpha}^{\text{sym}} = 2C_c(2\alpha)^{\zeta_c}$. Every budget keeps its exact order $\log(k+1)$, including $\log 2 < 1$ at $k = 1$, and the same original set is deleted from both axes throughout.

The analytic reference constant and the formalized version

The earlier version of this assembly used a $\frac{1}{2}$ -net in the thin-axis estimate, the crude contraction $2B_\alpha = 4 \cdot 5^{1/4} H_\alpha$ in the local theorem, the uniform doubling (78) ten times per block and three times in the local mean, and the bound $3L$ for the maximum over the blocks of each mask; with the strong/weak constant $2H_\alpha^2$ it gives the analytic reference constant

$$C_\alpha^{\text{sym,ref}} = \sqrt{31} + 2\sqrt{1 + \sqrt{2}\alpha^2} + 6L_\alpha^{\text{ref}}(1 + 2H_\alpha^2 J_\alpha^{10}), \quad L_\alpha^{\text{ref}} = 4e \cdot 5^{1/4} H_\alpha, \quad J_\alpha = H_\alpha^7, \quad (134)$$

about $5 \cdot 10^{19}$ at $\alpha = 1$. The formal development identified in Section 11 formalizes that earlier version as an existence theorem, with the three deliberate differences recorded in Section 11: $\sqrt{3}\alpha^2$ in place of H_α in the cone comparison, $(2n)^{1/(2q)}$ in place of $n^{1/(2q)}$, and the prefactor e in place of 2 in the symmetrization. The literal constant it certifies is fixed by the definitions in that development and is not identified with (134) here. The refinements of the present version, the spherical thin-axis estimate, the parity majorant, the order-adapted doubling, the variance-first treatment of the two lowest blocks and the moment policy (126), are analytic and not formalized; they change constants, not the structure of the argument.

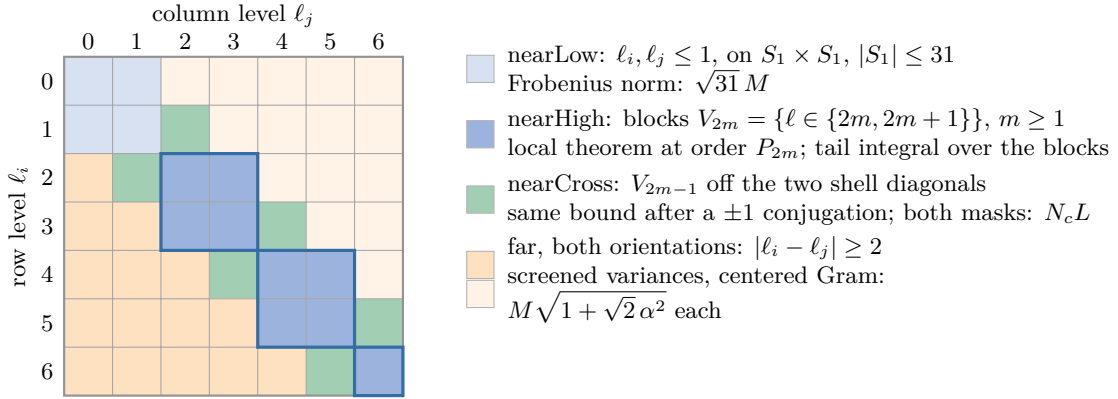


Figure 10: The five deterministic masks of the deletion argument, drawn in the plane of row and column levels (ℓ_i, ℓ_j) . Exactly one condition holds at each pair of levels. The near band $|\ell_i - \ell_j| \leq 1$ is covered by three masks: the small nearLow block, and two block-diagonal masks whose blocks (heavy outlines for nearHigh) are principal submatrices on consecutive shells, each disjoint from a deleted set two stages earlier. Only these blocks use the graded method, through Theorem 4.11 at a moment order that grows with the block index. The far masks are controlled by the screening built into the nested family and an elementary Gram estimate; no moment method enters there.

Remark 5.8 (Two routes to the same estimate). The local estimate (125) is the contribution of the method, and it is the same in both versions of the proof; what changed is how the two analytic inputs around it are obtained. The earlier version quoted the strong/weak comparison of Latała–Strzelecka [45, Theorem 1.1] in Lemma 3.31, which supplied the weak input at the same order $4q$ as the strong moment, and it ran the deletion on the matrix-symmetric dilation with a paired deletion budget, estimating the remainder by the variance-envelope theorem of Latała–van Handel–Youssef [49, Theorem 4.4] through a stochastic domination by exponential-type variables. That theorem assumes two-sided polynomial moment growth, which α -regular entries need not have, so the adaptation carried its own hypotheses. The present version proves the comparison it needs (Theorem 3.24) by an independent copy, which also lowers the weak order to $2q$; runs the deletion on the original rectangular matrix with one shared index family, which is precisely what D quantifies over, so that every entry family is

genuinely independent and no dilation budget arises; and raises the screening threshold from b_k^2 to b_k^3 , at which point the far remainder becomes summable by the elementary Gram second moment above (Lemma G.5) rather than by a general fourth-moment matrix inequality. The upper bound then uses, from the literature, only the Boolean hypercontractivity theorem for the signs (in its L_4 case proved in Lemma 3.18), all its constants are explicit, and it is in a form that a proof assistant can check.

The lower comparison The lower half of (123) is [48, Theorem 4.1], an external result used with attribution; it is not part of what the method proves, and it is not formalized. Its structure is recorded because it uses a genuinely different input. Paley–Zygmund on each row and column sum of squares gives $M(X) \leq 8\sqrt{2}\alpha^4 \mathbb{E}\|X\|$. For the deletion parameter, a large weak moment at order $P = \log(k+1)$ is shown to have a sparse witness, supported on at most $\sqrt{k}$ indices, using the scalar moment-growth estimate $\|S\|_{L_p} \leq K_\alpha(p/r)^{\max\{1/2, \log_2 \alpha\}} \|S\|_{L_r}$ of [45, Corollary 1.4]; deleting successive witnesses produces $\lfloor \sqrt{k} \rfloor$ independent bilinear sums on disjoint entries, each exceeding a fixed fraction of its L_P moment with probability at least $1/(4\sqrt{k}+1)$, and the maximum of these independent events gives $D(X) \leq C_\alpha \mathbb{E}\|X\|$. Proposition 3.25 is not a substitute for that input: it compares $2p$ with p , whereas sparsification needs the growth rate in p itself.

5.5 Which estimate supplies which factor

Table 4 records the structural estimate used in each application and the scale it supplies. The applications use different contractions of the same exact representation. Lower bounds, where asserted, require their own same-input arguments. The last four rows point forward to the analytic refinements proved in Section 10.

Result	Structural estimate used	Factor supplied	Verification scope
SparseStack, Theorem 5.1	Shared-factor Gram, Lemma 3.1; hollow reference law	$d + \ell - 1$ instead of $d\ell$; hence the row scale $d + q$	Direct proof here (Appendix B); original coupled core in Lean 4
Graph upper bound, Theorem 5.3	Connected creation, Proposition 3.9; separator factorization	$\sqrt{K+1}$ per attached floating component; hence $(\log n)^{b_*/2}$	Proof here; reported fixed- p Lean theorem; uniform strong moments are analytic
Graph lower bound, Theorem 5.3	Separate excursion and projection arguments, Section F.7	Matching expectation and fixed- p scale $T_{H,n}$	Proof here; reported fixed- p classification in Lean 4 (Section 11)
Walsh sampling, Theorem 5.5	Positive-effects selector estimate; original row-channel and normal-ordering bounds	Allowance $d + 27q^3$; fixed-size constant 8192	Corollary of Theorem 10.1 here; original real Walsh core in Lean 4
Regular entries, Theorem 5.6	Parity counts and polynomial-cone contraction, Corollary 4.14; deterministic deletion; known lower bound	Local $M + R(\log(n+1))$ at logarithmic order, then the mean-norm scale $M(X) + D(X)$	New constants: analytic proof here. Earlier upper theorem: explicit Lean term in Section 11. Lower half: [48]
Coherent sampling, Theorem 10.1	Collective coefficients; commuting-transport kernel and reflection; positive effects	Allowance $\beta d + (8\beta + 4\alpha)q^2$; fixed-size constant 1024	Analytic proof here
Direct SparseStack, Theorem 10.6	Shared-factor Gram; block parity; returning-path majorant	$s < 26.5q/\epsilon$, $m < 179(d+q)/\epsilon^2$	Analytic proof and rational certificate here
Triangle, regular entries, Theorem 10.7	Local moment theorem on the pinned-row blocks; Hölder for the pinned magnitudes	$\sqrt{n} \ X\ _{L_2[\log n]}$ with explicit constants	Analytic proof here; exact finite checks
Triangle, sparse entries, Theorem 10.9	Exact support-conditioned block representation; neighborhood degrees and degeneracy	$\rho^{-3/2}$ times the largest neighborhood norm; critical factor $\sqrt{\log n / \log \log n}$	Analytic proof here; exact finite checks; Bandeira–van Handel for $\beta < \frac{1}{2}$

Table 4: The origin of the main factors and the scope of formal verification. The abstract frame and positive-effects lemmas as stated here are analytic extensions, even where an original formalized proof uses their special cases.

6 Where the losses are

6.1 Coefficient loss: improve the estimate on the same space

The first example is the two-round allowance in Table 2: $d + 27q^3$ becomes $d + 12q^2$. The random model, multiplication operator, and moment cutoff stay fixed. The cubic estimate bounds row channels separately and replaces coefficient maps by occupation counts too early. The collective contractions of Section 3.5 keep those maps until their Parseval and Bessel identities have been used. For flat transforms, combining these contractions with the grouped reflection in the double-annihilation calculation gives the quadratic allowance; Theorem 10.1 and Appendix A give the complete derivation. The test therefore locates this loss in a coefficient estimate, before any restriction to a smaller state space is attempted.

A second instance is the direct SparseStack heavy swap, where the missing structure is visible in the external coefficient space.

In direct SparseStack a site is vacuum, light, or heavy. Let L_g, H_g count light and heavy sites in block g , and write $L = \sum_g L_g$, $H = \sum_g H_g$. The light states e_a are orthonormal, and the heavy simplex states satisfy $\langle f_a, f_c \rangle = (b\delta_{ac} - 1)/(b - 1)$. The leg $R_a = |f_a\rangle\langle e_a|$ converts light to heavy. The heavy-swap family is

$$H_0 = \sum_{g,a,i \neq j} u_i u_j^* \otimes (R_{a,gi} R_{a,gj}^* + R_{a,gi}^* R_{a,gj}).$$

Appendix B defines the local space and all eight families.

Every coefficient family has the exact external compression $F(U) = (U^* \otimes I)F(I_n)(U \otimes I)$. For $U = I_n$, a *marked* site is the site named by the external coordinate index; it is retained along with the occupation state. The first heavy-swap orientation has its marked external site heavy in source and target; the second has it light. These supports are orthogonal. Each local exchange has norm one, because $(e_a \otimes f_a)_a$ and $(f_a \otimes e_a)_a$ are orthonormal. The two neighbor counts are L_g and H_g . Taking their maximum before external compression gives

$$\|H_0\| \leq \sum_{g: L_g H_g > 0} \max(L_g, H_g). \quad (135)$$

If every L_g is even, this is at most

$$C(L, H) = \max\{L, H, L + H - 3\}, \quad L \geq 2, H \geq 1,$$

and the family vanishes when $LH = 0$. If every contributing block is light-dominated the sum is at most L ; if every one is heavy-dominated it is at most H . Otherwise the minimum count is at least one in a strictly light-dominated block and at least two in a strictly heavy-dominated block. Subtracting these minima gives $L + H - 3$. The full proof and vectors attaining the one-block bound $\max(L, H)$ are in Propositions C.3 and C.4.

6.2 Symmetry-sector loss: an excluded representation can raise the norm

The direct operator preserves the light parity of each block, since each term acts twice in that block. Its vacuum orbit lies in the sector where all L_g are even. Appendix B uses this symmetry before bounding the heavy Grams. There is a further exact symmetry: simultaneously relabel the buckets of a block and flip all column signs in any chosen bucket. Locally,

$$\Omega \mapsto \Omega, \quad e_a \mapsto \epsilon_a e_{\pi(a)}, \quad f_a \mapsto f_{\pi(a)}, \quad \epsilon_a \in \{\pm 1\}, \quad \pi \in S_b.$$

Every family commutes with this action, since each term uses two legs with the same bucket label. We call its fixed space the signed-bucket invariant space.

Take one block, three columns, and $U = (1, 1, 1)^*/\sqrt{3}$. On source counts $(L, H) = (2, 1)$ the heavy-pair map is

$$H_+ = \frac{2}{3} \sum_{i < j} \sum_a R_{a,i} R_{a,j}.$$

Its signed-bucket invariant source has the orthonormal basis

$$w_k = \frac{1}{\sqrt{b}} \sum_a f_a^{(k)} \bigotimes_{i \neq k} e_a^{(i)}, \quad k = 1, 2, 3.$$

All three map to $2/(3\sqrt{b})$ times $F_3 = \sum_a f_a^{\otimes 3}$. The simplex Gram gives $\|F_3\|^2 = b[1 - (b-1)^{-2}]$, so

$$\|H_+|_{\text{inv}}\| = \frac{2}{\sqrt{3}} \frac{\sqrt{b(b-2)}}{b-1}. \quad (136)$$

At $b = 4$ the full block-even sector has strictly larger norm:

$$\|H_+\|_{\text{parity}} = \sqrt{\frac{8(9 + \sqrt{33})}{81}} > \sqrt{\frac{32}{27}} = \|H_+\|_{\text{inv}}. \quad (137)$$

Propositions C.1 and C.2 prove the invariant-space description and the complete tetrahedral block calculation. Thus an excluded bucket representation causes a genuine channel-norm loss.

6.3 Spectral-weight loss survives reduction to the cyclic space

Return to Example 2.1. Its two-column scalar error is $Z = \sigma_1 \sigma_2 \mathbf{1}_{\{h_1=h_2\}}$. It equals 1, -1 with probabilities $1/(2b)$ each and is otherwise zero. The three cyclic states have grades 0, 2, 4, and the cyclic matrix has spectrum $\{0, 1, -1\}$. The signed-bucket action has exactly three orbits on the sample space: different buckets, equal bucket with sign product $+1$, and equal bucket with sign product -1 . Its entire fixed space is therefore this three-dimensional cyclic space.

Proposition 6.1 (An invariant norm cannot recover the vacuum weight). *Every invariant subspace of the error multiplication operator containing Ω contains the three-state cyclic space and has restricted norm one. For $q \geq 2$, when the grade cutoff contains those states,*

$$\langle \Omega, M_{\text{cut}}^{2q} \Omega \rangle = \frac{1}{b}, \quad \frac{\|M_{\text{cut}}\|}{\|M_{\text{cut}}^q \Omega\|^{1/q}} = b^{1/(2q)}. \quad (138)$$

Proof. The functions $1, Z, Z^2$ are linearly independent on the three values of Z , each having positive probability. Thus $\Omega, M\Omega, M^2\Omega$ span the cyclic space, and every invariant subspace containing Ω contains it. Multiplication by Z has norm one both there and on the full probability space. Finally $\mathbb{E}Z^{2q} = 1/b$. For $q \geq 2$ the grade cutoff contains the cyclic space, proving the formulas. $\square$

The large eigenvalues persist in the smallest invariant space, but their total vacuum mass is $1/b$. Recovering this factor requires moment or path information. Figure 11 displays the distinction. For an orthonormal eigenbasis (v_j) of a finite self-adjoint matrix, its vacuum spectral measure places mass $|\langle v_j, \Omega \rangle|^2$ at the corresponding eigenvalue λ_j ; hence its $2q$ -th moment is $\sum_j |\langle v_j, \Omega \rangle|^2 \lambda_j^{2q}$. The Gaussian quadrature construction in Section 2 uses this same finite Jacobi spectral measure for moment matching. In this example, panel (a) is the two-node Gauss quadrature rule for the measure in panel (b). Its moments through degree three agree: reaching the omitted third state of the Jacobi chain and returning requires at

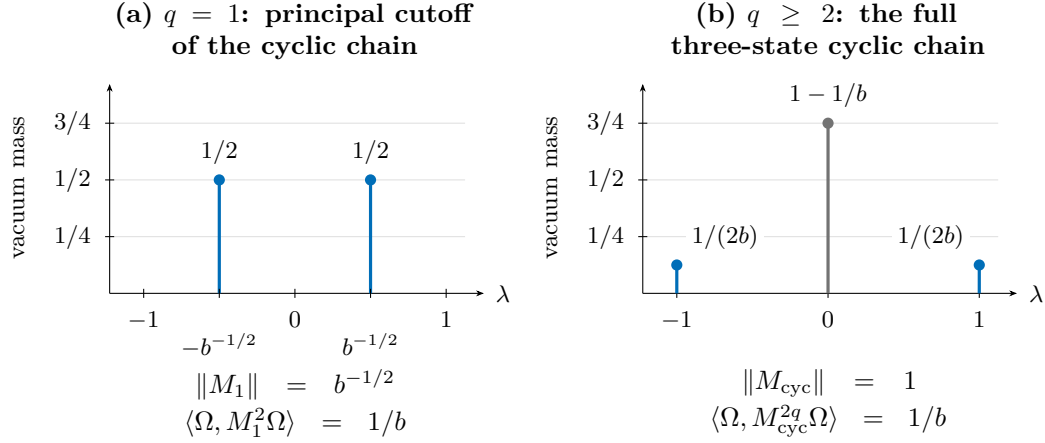


Figure 11: Vacuum spectral measures for the two-site example. Locations and heights are drawn at $b = 4$; atom labels give the formulas for arbitrary $b > 1$. In panel (a), M_1 is the upper-left 2×2 principal cutoff of the cyclic matrix in (3). It is also the signed-bucket invariant grade-two cutoff; it is not the full block-even occupation cutoff, whose norm is $\sqrt{2/b}$. Panel (a) is the two-node Gauss rule for the measure in (b). Its nonzero atoms have total mass one; in panel (b), that mass is $1/b$. The norm bound replaces this mass by one, while every positive even moment of the full cyclic matrix remains $1/b$.

least four multiplications. The full three-node rule in panel (b) is exact for every polynomial, since it is the law of Z itself.

At $q = 1$ the bucket-invariant grade-two cutoff has norm $1/\sqrt{b}$, whereas the block-even cutoff has norm $\sqrt{2/b}$. Its unscaled map from the vacuum and the two single-heavy spaces into the equal-bucket light-pair space has Gram

$$\mathbf{11}^* + 2(b-1)(\langle f_a, f_c \rangle)_{a,c} = 2bI - \mathbf{11}^*.$$

The constant bucket direction has eigenvalue b and its orthogonal complement has eigenvalue $2b$. Division by $m = b$ gives the two norms. Thus the first-cutoff gain is a symmetry effect, while (138) is a spectral-weight effect.

Occupation projections require a separate distinction. In the three-site uniform scalar model put

$$\psi = \sum_{j=1}^3 \sum_a f_a^{(j)} \bigotimes_{i \neq j} e_a^{(i)}.$$

Let $P_{2,1}$ project onto the sector with two light sites and one heavy site. Pair creation followed by a mixed transition gives

$$P_{2,1} M^2 \Omega = \frac{8\sqrt{b-1}}{9b^2} \psi.$$

There are two ordered ways to create each specified heavy site, each with coefficient $(2/3)^2 \sqrt{b-1}$ before division by b^2 . This is an occupation-projected component, but not a scalar cyclic vector. Indeed the three-site scalar error is

$$Z_3 = \frac{2}{3} \sum_{i < j} \sigma_i \sigma_j \mathbf{1}_{\{h_i = h_j\}}.$$

A negative-sign pair collision with the third column in another bucket and a three-column collision with mixed signs both have $Z_3 = -2/3$. On these events ψ equals respectively

$b/\sqrt{b-1}$ and $-b\sqrt{b-1}$. It is not a function of Z_3 , whereas every cyclic vector is such a function.

The continuation examples give two further checks on the coefficient estimate. The Parseval insertion in Section 3.4.1 admits the positive budget $C_{\text{graph}}(RR^* \otimes I)C_{\text{graph}}^* \preceq C_{\text{graph}}C_{\text{graph}}^*$ before pinching, but its norm can decrease by a factor m after output-count pinching. The two-creation example in Section 3.3.4 has a primitive Gram that vanishes on one input while the continued Gram is positive there. The first needs an unpinched budget; the second needs the ordered child product. The pinching example rules out a dimension-independent repair uniformly in m . The continued-child example, whose primitive Gram vanishes on the selected input, rules out every scalar multiple of that primitive descriptor for the stated continuation. Proposition 7.1 states this as a general test: a descriptor that identifies two states which a later interaction distinguishes fails a kernel condition that no constant can repair, whereas a descriptor that passes it is priced by a Loewner inequality.

For variable magnitudes, a separate domain check is needed. The actual walk prefixes lie in $\mathcal{P}_j^+$, but their full linear span contains signed coefficient combinations. Proposition 3.33 and Lemma 4.8 bound the preserved family directly; a full-span operator norm would impose a stronger requirement.

6.4 A diagnostic procedure

For a new model, record the exact moment identity and the first inequality that loses the desired parameter. Keep the operator fixed while testing sharper coefficient contractions, exact symmetry sectors, and vacuum moments. A coefficient improvement locates a loss in the estimate. A strict symmetry-sector norm gap locates a loss in the chosen domain. A large eigenvalue with small vacuum mass inside the cyclic space calls for a path or moment estimate. Exact small matrices can distinguish these mechanisms; each supplies a different next step for the proof. When the next step is a further interaction, Section 7 supplies the admissibility test for the retained description, and Sections 8.4 and 8.9 supply exact original-law values against which the candidate estimates can be checked.

7 Retained state under continuation

Each structural estimate of Section 3 ends by naming what it retains and which further maps may act on it: a positive budget that survives congruence, a walk cone preserved by the actual multiplication, an occupied coefficient span, an ordered child Gram. Section 6 exhibited continuations that a primitive descriptor cannot support. This section isolates the common question. A proof retains a *description* of the current state, a further interaction is applied, and one asks whether the retained description determines the continued state and at what quantitative price. Three answers are given, in decreasing order of exactness. An *exact descent* test decides whether a proposed update is even defined on the retained state, and a Loewner inequality prices it. A *live register* keeps a label or an independent name through repeated visits, so that a specified family of further insertions closes on it exactly. A *signed conditional core* keeps a retained group of sign characters and charges the information that a product returns to that group, giving an approximate closure with a certified remainder. An exact reduction for supplied coefficient algebras and a set of obstructions complete the section. Table 5 summarizes the four mechanisms. Letters are local to each subsection, as the notation guide records; in particular T , P , G and H do not carry their meanings from Sections 2 and 4 here.

Mechanism	Retained object	Continuations supported	Price
Exact descent (§7.1)	An encoding Φ of descriptions into states	Updates T with $\ker \Phi \subseteq \ker(\Phi'T)$	The Loewner constant L in $T^*\Gamma'T \preceq L^2\Gamma$
Live register (§7.2)	The label-indexed state, or one open register per independent name	Block insertions with the same live label; arbitrary interleaving of independent names	Endpoint second moments; interior norms
Signed core (§7.3)	$\Pi_H F$ and an L_P radius for $F - \Pi_H F$	Sums, adjoints, ordered products; retained-character insertions act exactly	$e_{FG} \leq e_F n_G + a_F e_G$; returning modes are charged
Cyclic seed space (§7.4)	The active space of a supplied algebra and finitely many seeds	Every product in the algebra and the seed outer products	$\dim W \leq r \sum_\lambda d_\lambda^2$; multiplicity weights

Table 5: Four ways to retain a state through a further interaction. The first is a test, the second and fourth are exact closure rules for specified families of operations, and the third is an approximate closure with a certified remainder. None asserts that an arbitrary polynomial matrix admits a small retained state.

7.1 Exact descent through an encoding

Takeaway. A proposed update of a compressed description is realized on the retained state precisely when it kills the kernel of the encoding; its cost is a Loewner inequality between the two description Grams. When the kernel test fails, no constant repairs it.

Let E be a finite-dimensional space of descriptions, such as coefficient arrays indexed by labels, and let $\Phi : E \rightarrow \mathcal{H}$ be the linear *encoding* that sends a description to the state it represents. Two descriptions with the same encoding are the same state. A further interaction is proposed as an update $T : E \rightarrow E'$ of descriptions, followed by the encoding $\Phi' : E' \rightarrow \mathcal{H}'$ of the continued state. The update is *realized on the retained state* if there is a linear map $\bar{T}$ on $\text{range } \Phi$ with $\Phi'T = \bar{T}\Phi$. Write $\Gamma = \Phi^*\Phi$ and $\Gamma' = \Phi'^*\Phi'$ for the description Grams.

Proposition 7.1 (Descent test). *A map $\bar{T}$ with $\Phi'T = \bar{T}\Phi$ exists if and only if*

$$\ker \Gamma \subseteq \ker(\Phi'T). \quad (139)$$

Such a map with $\|\bar{T}\| \leq L$ exists if and only if

$$T^*\Gamma'T \preceq L^2\Gamma. \quad (140)$$

Proof. Since $\langle x, \Gamma x \rangle = \|\Phi x\|^2$, one has $\ker \Gamma = \ker \Phi$. The value $\Phi'Tx$ is a function of Φx alone precisely when $\Phi x = 0$ forces $\Phi'Tx = 0$, which is (139); the map is then well defined and linear on $\text{range } \Phi$. The inequality (140) states that $\|\Phi'Tx\|^2 \leq L^2\|\Phi x\|^2$ for every x , which is the norm bound for $\bar{T}$ on $\text{range } \Phi$; extending by zero on the orthogonal complement changes nothing. $\square$

After taking adjoints, the proposition is the finite-dimensional case of Douglas's lemma on majorization, factorization and range inclusion [53]: $(\Phi'T)^* = \Phi^*\bar{T}^*$ has a solution precisely when $\text{range}(\Phi'T)^* \subseteq \text{range } \Phi^*$, and a solution of norm at most L precisely when $(\Phi'T)^*(\Phi'T) \preceq L^2\Phi^*\Phi$. It is stated here as an interface for retained descriptions, not as a new factorization theorem; what is specific to the method is the identification of the encoding, the update and the price in each continuation.

The two tests have different characters. The kernel test is qualitative: if it fails, there are two descriptions of the same state whose continued states differ, and no enlargement of a constant can restore a map. The Loewner test is quantitative and is the one-step inequality of Lemma 4.9: if the state at stage j is described by $x \in E_j$ with encoding Φ_j , the descriptions are updated by T_j with $\Phi_{j+1}T_j = \mathcal{T}\Phi_j$, and the energy is a quadratic form $\mathcal{E}_j(f) = \langle f, W_j f \rangle$, then with the weighted Grams $\Gamma_j = \Phi_j^* W_j \Phi_j$ the inequality $\mathcal{E}_{j+1}(\mathcal{T}f) \leq a_j^2 \mathcal{E}_j(f)$ reads $T_j^* \Gamma_{j+1} T_j \preceq a_j^2 \Gamma_j$, which is (140) for the weighted encodings $W_j^{1/2} \Phi_j$. The two requirements are distinct: an energy written as $\langle x, \Gamma_j x \rangle = \|W_j^{1/2} \Phi_j x\|^2$ is automatically a function of the encoded state, whereas it is the kernel test that makes the *update* T_j well defined on retained states.

Three qualifications keep the test honest. First, it is a test of a *proposed* update: a compressed transport need not remain normal or commute, and the test must be applied to the actual composed map, as already noted after Lemma 3.3. Second, on a cone the test is restricted: for the walk family of Section 3.6 the encoding is the identity on $\mathcal{H}$, descent is automatic, and the quantitative condition is the inequality $\|\Phi' T x\| \leq L \|\Phi x\|$ for x in the cone only. This is weaker than (140) on the full space, and the full-space form is not implied. Third, the test reads the continuation examples of Section 6. In the two-creation residual of Section 3.3.4, the description is the primitive pair Gram, the update is the next creation consuming its rows, and the primitive Gram vanishes on an input where the continued Gram is positive. That input lies in $\ker \Gamma$ but not in $\ker(\Phi' T)$, so (139) fails: the primitive descriptor has identified two states that the child products distinguish, which is why no scalar multiple of it dominates the continued Gram. The pinching example of Section 3.4.1 is a loss of the second kind: the unpinched positive budget survives the insertion, and a descriptor that pinches by output counts can only recover it at the price $L^2 = m$ in (140).

The following example shows the failure on actual SparseStack coefficients with a summed label.

Example 7.2 (A cancelled coherent sum can revive). Use the SparseStack notation of Section 5.1, with rows u_i^* of the isometry U , edge coefficients $A_{ij} = u_i u_j^* + u_j u_i^*$, and, within one block, the sign-collision variables $z_{ij} = \sigma_i \sigma_j \mathbf{1}_{\{h_i = h_j\}}$. A word of the error expansion visits an internal label v between boundary labels; summing over v produces coherent sums such as $\sum_v z_{av} A_{av}$. Consider the encoding that retains only such a sum, $(C_v)_v \mapsto \sum_v C_v$, and the update that right-multiplies each summand by its incident coefficient, $(C_v)_v \mapsto (C_v A_{vb})_v$.

Take $d = 2$ and the eight rows

$$(1, 0), (0, 1), (1, 1), (-1, -1), (1, -1), (1, -1), (2, 0), (0, 2),$$

all divided by 3; their outer products sum to I_2 . Name the first four rows a, b, v, w . Then

$$A_{av} = \frac{1}{9} \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}, \quad A_{vb} = \frac{1}{9} \begin{pmatrix} 0 & 1 \\ 1 & 2 \end{pmatrix}, \quad A_{aw} = -A_{av}, \quad A_{wb} = -A_{vb},$$

and the coefficients do not commute. Hence

$$A_{av} + A_{aw} = 0, \quad A_{av} A_{vb} + A_{aw} A_{wb} = \frac{2}{81} \begin{pmatrix} 1 & 4 \\ 0 & 1 \end{pmatrix} \neq 0. \quad (141)$$

The pair (A_{av}, A_{aw}) lies in the kernel of the coherent-sum encoding but not in the kernel of its updated encoding, so the kernel test (139) fails for this update. The failure is realized on the original hash and sign inputs. Condition on the event that the four hashes agree and $\sigma_v = \sigma_w$, which has probability $1/(2b^3)$. On this event the path sum $z_{av} A_{av} + z_{aw} A_{aw}$ vanishes pointwise, while its incident extension is $\sigma_a \sigma_b$ times the nonzero matrix in (141),

with conditional squared Frobenius norm $8/729$. On this event the scalar words coincide, $z_{av} = z_{aw}$ and $z_{av}z_{vb} = z_{aw}z_{wb}$; what the coherent sum discards is the association of each summand with its future coefficient. This is an obstruction to that encoding, not a lower bound for the sketch.

The remedy is to keep the label as a register, which is the subject of the next subsection. A useful sufficient condition for exact elimination is *last incidence*: if a primitive X_0 is to be discarded, a primitive Y_0 is retained, and every later factor depends only on Y_0 and on fresh primitives independent of (X_0, Y_0) , then

$$\mathbb{E}[L(Y_0) F(X_0, Y_0) R(Y_0) \mid Y_0] = L(Y_0) \mathbb{E}[F \mid Y_0] R(Y_0),$$

so integrating X_0 out at its last incidence preserves every later ordered expectation. For a Frobenius energy this rule must be applied to the complete paired expression F^*F , not to one prematurely compressed half. A label that returns with the same original index after a long absence is not fresh; it uses the same primitive, and the register must still be open.

7.2 Live registers

Takeaway. Keeping one label, or one open register per independent name, through every visit gives an all-length composition rule whose price is a second moment at the two endpoints and a norm at each interior visit.

7.2.1 One live label through repeated visits

Continue with the SparseStack coefficients $A_{ij} = u_i u_j^* + u_j u_i^*$ of Example 7.2. Fix an internal label set I and put

$$H_v = u_v u_v^*, \quad F_I = \sum_{v \in I} H_v \preceq I_d, \quad \mu_I = \max_{v \in I} \ell_v, \quad \ell_v = \|u_v\|^2,$$

with empty sums and maxima equal to zero. For a boundary label $a \notin I$ define the insertion and the block-diagonal transport

$$\begin{aligned} V_a : \mathbb{C}^d &\rightarrow \bigoplus_{v \in I} \mathbb{C}^d, & (V_a x)_v &= A_{av} x, & D_a &= \bigoplus_{v \in I} A_{av}, \\ g_a &= \left\| \sum_{v \in I} A_{av}^2 \right\|, & \theta_a &= \max_{v \in I} \|A_{av}\|. \end{aligned} \tag{142}$$

Then $V_a^* V_a = \sum_v A_{av}^2$, so $\|V_a\| = \sqrt{g_a}$, and $\|D_a\| = \theta_a$. The state on the register is the label-indexed vector $(x_v)_{v \in I}$; its encoding into $\bigoplus_v \mathbb{C}^d$ is the identity, so every update of the form below passes the descent test.

Theorem 7.3 (Live-label contraction). *For every sequence $a_1, \dots, a_r$ of boundary labels outside I , $r \geq 2$,*

$$\sum_{v \in I} A_{a_1 v} A_{a_2 v} \cdots A_{a_r v} = V_{a_1}^* D_{a_2} \cdots D_{a_{r-1}} V_{a_r}, \tag{143}$$

$$\left\| \sum_{v \in I} A_{a_1 v} \cdots A_{a_r v} \right\| \leq \sqrt{g_{a_1} g_{a_r}} \prod_{j=2}^{r-1} \theta_{a_j}. \tag{144}$$

Moreover

$$\sum_{v \in I} A_{av}^2 = (\text{tr } F_I) H_a + H_a F_I + F_I H_a + \ell_a F_I, \quad g_a \leq (d+3)\ell_a, \quad \theta_a \leq \min\{1, 2\sqrt{\ell_a \mu_I}\}, \tag{145}$$

so that at even length $r = 2t$

$$\left\| \sum_{v \in I} A_{a_1 v} \cdots A_{a_{2t} v} \right\| \leq (d+3) 4^{t-1} \mu_I^{t-1} \prod_{j=1}^{2t} \|u_{a_j}\|. \quad (146)$$

Proof. The block-diagonal maps keep the label v fixed at every visit, and the final adjoint insertion sums over it; this is (143), and taking norms gives (144). Expanding $A_{av}^2 = \langle u_v, u_a \rangle u_a u_v^* + \ell_v H_a + \ell_a H_v + \langle u_a, u_v \rangle u_v u_a^*$ and summing over $v \in I$ gives the identity in (145), since the frame is real and $\sum_v \langle u_v, u_a \rangle u_a u_v^* = H_a F_I$. Each summand A_{av}^2 is positive, so the sum over I is dominated by the sum over all labels including the algebraic loop $A_{aa} = 2H_a$; by Parseval that full sum is $(d+2)H_a + \ell_a I$, of norm $(d+3)\ell_a$. The two possibly nonzero eigenvalues of A_{av} are $\langle u_a, u_v \rangle \pm \sqrt{\ell_a \ell_v}$, which gives $\theta_a \leq 2\sqrt{\ell_a \mu_I}$; for distinct labels $A_{av} = U^*(e_a e_v^* + e_v e_a^*)U$ is a compression of a norm-one matrix, which gives $\theta_a \leq 1$. Substitution proves (146). $\square$

The theorem is a restricted family of future operations closing on an explicitly retained state. The retained state is the label-indexed register, the allowed operations are the block transports D_a and the terminal contraction, and the price is one θ_a per interior visit. It is not a recurrence on the norm of a previously summed join: Example 7.2 shows that such a norm can be zero while the continued state is not. For arbitrary conformable matrices C_v the positive update

$$\sum_{v \in I} C_v A_{av}^2 C_v^* \preceq \theta_a^2 \sum_{v \in I} C_v C_v^*$$

survives every later congruence, including noncommuting coefficient congruences, but no stability under one-sided multiplication in Loewner order is asserted, for the reason given in Example 4.7. In the occupation language of Section 3, the register is a marked site carried as an orthogonal label; Lemma 3.1 already uses this device when it retains auxiliary labels until the coefficient contraction has been estimated.

7.2.2 Interleaved independent names

The same device handles several independent random matrices whose occurrences interleave arbitrarily in an ordered word: open a register at the first occurrence of a name, reuse the same random variable at every interior occurrence, and close the register at the last one.

Proposition 7.4 (Register contraction for interleaved names). *Let $Z_1, \dots, Z_k$ be mutually independent Hermitian $n \times n$ random matrices with $\|Z_i\| \leq L_i$ almost surely and $\mathbb{E} Z_i^2 \preceq v_i I_n$. Let $(i_1, \dots, i_N)$ be a word in which each name i occurs $t_i \geq 2$ times, and let $D_0, \dots, D_N$ be deterministic $n \times n$ matrices. Then*

$$\|\mathbb{E}[D_0 Z_{i_1} D_1 Z_{i_2} \cdots Z_{i_N} D_N]\| \leq \prod_{j=0}^N \|D_j\| \prod_{i=1}^k v_i L_i^{t_i-2}. \quad (147)$$

The same bound holds after a common tensor amplification of all matrices by an identity, with deterministic insertions on the amplified space. Centering is not required, and a name occurring exactly twice needs only its second-moment bound.

Proof. Put the randomness of Z_i on a probability space $\mathbf{X}_i$. For a set A of currently open names, the register space is $\mathcal{H}_A = L^2(\prod_{i \in A} \mathbf{X}_i; \mathbb{C}^n)$ with the product law. Process the factors from right to left, in the order in which they act on a vector, and maintain the invariant that after processing a suffix the current function equals that suffix applied to the initial vector, with exactly the closed names integrated out; Fubini and mutual independence give this

invariant. The rightmost occurrence of i opens its register by $(C_i f)(\omega_A, \omega_i) = Z_i(\omega_i) f(\omega_A)$, whose squared norm is $\int \langle f, \mathbb{E}[Z_i^2] f \rangle \leq v_i \|f\|^2$. An interior occurrence multiplies pointwise by the same retained $Z_i(\omega_i)$, at cost L_i . The leftmost occurrence closes the register by $C_i^* g = \int Z_i(\omega_i) g(\cdot, \omega_i) d\mu_i(\omega_i)$, again of norm at most $\sqrt{v_i}$. Other open registers only amplify these maps, and deterministic matrices act pointwise in their original positions. The final composition is the original expectation, and multiplying the map norms proves (147). $\square$

Interleaving is free in this certificate; identifying names is not. Merging two identically distributed independent names of multiplicities t and u turns $v^2 L^{t+u-4}$ into $v L^{t+u-2}$, a ratio L^2/v that scalar Bernoulli variables attain. Nor can mutual independence be weakened to pairwise independence: for independent signs $\varepsilon_1, \varepsilon_2$, the variables $A = (1 + \varepsilon_1)/2$, $B = (1 + \varepsilon_2)/2$ and $C = (1 + \varepsilon_1 \varepsilon_2)/2$ have the one- and two-variable laws of independent Bernoulli variables, but $\mathbb{E} A^2 B^2 C^2 = \mathbb{P}\{\varepsilon_1 = \varepsilon_2 = 1\} = 1/4$ instead of $1/8$. The proposition needs almost-sure bounds at the interior visits, so it does not cover the regular entries of Section 2.7; there, the repeated magnitudes are retained inside the coefficient functions of the walk cone, and Proposition 3.33 prices them through moments.

7.3 Signed conditional cores

Takeaway. Projecting a sign polynomial onto a retained group of characters is a conditional expectation. Insertions supported on the group act exactly, a general product returns to the group only through two discarded frequencies, and this returning term is an off-diagonal block of a retained positive Gram. An explicit error rule carries the discarded information through a circuit of sums and ordered products.

Let $\varepsilon = (\varepsilon_e)_{e \in \mathcal{E}}$ be independent uniform signs, as in Section 2.7 or in any of the sign registers of Section 5, and write $\varepsilon_S = \prod_{e \in S} \varepsilon_e$. A matrix polynomial in the signs is $F = \sum_{S \subseteq \mathcal{E}} C_S \varepsilon_S$ with matrix coefficients C_S of a common size. A *retained character group* is a family $H \subseteq 2^{\mathcal{E}}$ containing $\emptyset$ and closed under symmetric difference. Define the core and the remainder

$$\Pi_H F = \sum_{S \in H} C_S \varepsilon_S, \quad R = F - \Pi_H F. \quad (148)$$

In this subsection A, B denote cores, R, R' remainders, and C_S a coefficient matrix, not the operators of earlier sections.

Lemma 7.5 (The core is a conditional expectation). *Π_H is the conditional expectation onto the σ -algebra generated by the retained characters $\{\varepsilon_S : S \in H\}$. Consequently Π_H maps pointwise positive semidefinite matrix functions to positive semidefinite ones; for polynomials L, B supported on H of compatible sizes,*

$$\Pi_H(LFB) = L(\Pi_H F)B, \quad \Pi_H(LRB) = 0; \quad (149)$$

and the discarded energies are

$$\mathbb{E} R^* R = \sum_{S \notin H} C_S^* C_S, \quad \mathbb{E} R R^* = \sum_{S \notin H} C_S C_S^*, \quad \mathbb{E} \|R\|_F^2 = \sum_{S \notin H} \|C_S\|_F^2. \quad (150)$$

Proof. Since $\varepsilon_S \varepsilon_{S'} = \varepsilon_{S \Delta S'}$, the span $\mathcal{A}_H$ of the retained characters is closed under products, contains the constants, and consists of real functions. A subalgebra of the functions on the finite set $\{\pm 1\}^{\mathcal{E}}$ with these properties is spanned by its minimal idempotents, which are the indicators of the blocks of a partition; so $\mathcal{A}_H$ is the space of functions measurable with respect to that partition, and its orthogonal projection in L_2 , applied entrywise, is the conditional expectation. Conditional expectation is positive and commutes with multiplication

by measurable factors; in operator-algebra language it is a completely positive projection onto a subalgebra that is a bimodule map over that subalgebra [54], and these standard properties give the first assertion and (149); the second identity in (149) holds because R is orthogonal to $\mathcal{A}_H$ and LRB has coefficients only at frequencies $S \triangle S' \triangle S''$ with $S, S'' \in H$ and $S' \notin H$, which lie outside H . Parseval in the characters gives (150). $\square$

Lemma 7.6 (Returning modes). *Let $F = A + R$ and $G = B + R'$ with $A = \Pi_H F$ and $B = \Pi_H G$. Then*

$$\Pi_H(FG) = AB + \Pi_H(RR'), \quad (151)$$

and the returning term is an off-diagonal block of a positive semidefinite matrix function:

$$\begin{pmatrix} \Pi_H(RR^*) & \Pi_H(RR') \\ \Pi_H(R'^*R^*) & \Pi_H(R'^*R') \end{pmatrix} = \Pi_H \left[\begin{pmatrix} R \\ R'^* \end{pmatrix} \begin{pmatrix} R^* & R' \end{pmatrix} \right] \succeq 0. \quad (152)$$

Proof. Expand $FG = AB + AR' + RB + RR'$. The product AB is supported on H because H is closed under symmetric difference. A frequency of AR' or RB is $S_1 \triangle S_2$ with exactly one of S_1, S_2 in H , hence outside H , so these terms are killed by Π_H . This proves (151). The matrix in (152) is the conditional expectation of a pointwise Gram, hence positive by Lemma 7.5. $\square$

Two frequencies that were discarded individually can combine into a retained one: with H the characters supported in $\{1, 3\}$, the products $\varepsilon_1 \varepsilon_2$ and $\varepsilon_2 \varepsilon_3$ are both discarded, yet their product $\varepsilon_1 \varepsilon_3$ is retained. Projecting each factor and multiplying the projections is therefore not the same as projecting the product; this is the concrete form of the warning after Lemma 3.3 that compression does not justify replacing a product by the product of its compressed factors. The two energies in (150) bound the returning term but cannot reconstruct it: for $R = \varepsilon_1 \varepsilon_2 X$ and $R' = \pm \varepsilon_2 \varepsilon_3 Y$ with fixed matrices X, Y , the discarded energies agree while $\Pi_H(RR') = \pm \varepsilon_1 \varepsilon_3 XY$. The block Gram (152) retains the sign and the orientation of the returning term; whether a continuation rule keeps this cross block exactly or only bounds it must be part of its stated interface. In the parity–amplitude representation of Section 2.7, Π_H discards ordinary monomials and therefore preserves the cone $\mathcal{P}_d^+$, but it does not commute with $\mathcal{T}$; the cone argument never projects between steps, which is why it carries no remainder.

At confidence-scale moment orders, the L_2 energies become useful through the sign hypercontractivity already used in Section 3.

Lemma 7.7 (Weighted remainder energy at high order). *For $R = \sum_S C_S \varepsilon_S$ and every real $u \geq 2$,*

$$(\mathbb{E} \|R\|_F^u)^{1/u} \leq \left(\sum_S (u-1)^{|S|} \|C_S\|_F^2 \right)^{1/2}. \quad (153)$$

In particular, if every discarded frequency has $|S| \leq a$, the L_u radius of the remainder is at most $(u-1)^{a/2} (\mathbb{E} \|R\|_F^2)^{1/2}$.

Proof. Let N_ρ be the noise operator of Section 3.6.1, applied entrywise, and take $\rho = (u-1)^{-1/2}$. Since N_ρ averages over independent resamplings of the signs, Jensen gives $\|N_\rho f\|_F \leq N_\rho \|f\|_F$ pointwise for every matrix function f , and scalar Boolean hypercontractivity [47, Theorem 1.1] applied to the scalar function $\|f\|_F$ gives $\|N_\rho \|f\|_F\|_{L_u} \leq \|\|f\|_F\|_{L_2}$. Apply this to $f = \sum_S \rho^{-|S|} C_S \varepsilon_S$, so that $N_\rho f = R$, and use Parseval for $\|f\|_{L_2}$. Lemma 3.18 is the case $u = 4$. $\square$

If H contains all characters supported on a set of retained coordinates, one may condition on those coordinates, count $|S|$ only over the remaining free coordinates, and bound the conditional coefficient matrices by their suprema over the retained signs; this charges high degree only where it is actually discarded.

The following statement is the approximate closure rule. It evaluates a supplied circuit once on the cores and once, in principle, on the original polynomials, and bounds the difference. No primitive is resampled when it is reused: the identities below hold pointwise on the same sample.

Theorem 7.8 (Continuation with a certified remainder). *Let $F_1, \dots, F_k$ be matrix polynomials in the signs, and let $A_i = \Pi_H F_i$ with the same retained group H . Let a circuit be built from the leaves F_i and deterministic constant matrices by sums, adjoints, and ordered products with deterministic insertions. Assign weight one to a leaf, zero to a constant, the maximum of the child weights to a sum or an adjoint, and the sum of the child weights to a product; let $\ell \geq 1$ be the weight of the output, fix a target exponent $p \geq 2$, and set $P = p\ell$. Suppose that at this exponent*

$$\| \|F_i - A_i\|_F \|_{L_P} \leq e_i, \quad \| \|A_i\| \|_{L_P} \leq a_i, \quad \| \|F_i\| \|_{L_P} \leq n_i, \quad (154)$$

where one may always take $n_i = a_i + e_i$. Evaluate the circuit on the cores to obtain A_{out} and on the leaves to obtain F_{out} . Propagate radii by $e_{F+G} \leq e_F + e_G$, $a_{F+G} \leq a_F + a_G$, $n_{F+G} \leq n_F + n_G$, unchanged radii at an adjoint, and at an ordered product

$$a_{FG} \leq a_F a_G, \quad n_{FG} \leq n_F n_G, \quad e_{FG} \leq e_F n_G + a_F e_G, \quad (155)$$

where a gate of weight $t > 0$ is certified at exponent P/t and a constant contributes its norm exactly. Then

$$\| \|F_{\text{out}} - A_{\text{out}}\|_F \|_{L_p} \leq e_{\text{out}}. \quad (156)$$

If the circuit is a sum of words of length at most ℓ with total norm of coefficient products at most C and common leaf radii (a, e) , then

$$e_{\text{out}} \leq C \ell e \max\{1, a + e\}^{\ell-1}. \quad (157)$$

Consequently, with $a_p = \| \|A_{\text{out}}\| \|_{L_p}$,

$$\max\{0, a_p - e_{\text{out}}\} \leq \| \|F_{\text{out}}\| \|_{L_p} \leq a_p + e_{\text{out}}. \quad (158)$$

Proof. For a product gate with child weights $t_F, t_G > 0$, the exponents satisfy $t_F/P + t_G/P = (t_F + t_G)/P$. Apply Hölder's inequality to the pointwise identity

$$FG - A_F A_G = (F - A_F)G + A_F(G - A_G),$$

with $\|UV\|_F \leq \|U\|_F \|V\|_F$ in the first term and $\|UV\|_F \leq \|U\| \|V\|_F$ in the second; this proves (155) at exponent $P/(t_F + t_G)$. If one child is a constant, use its exact norm and omit its reciprocal exponent. A sum of weight $\max\{t_F, t_G\}$ is certified at the smaller of the two child exponents, $P/\max\{t_F, t_G\}$, where both child bounds remain valid by monotonicity of L_p norms, and Minkowski's inequality applies; adjoints preserve all norms. Induction over the circuit proves (156), since the output is certified at exponent $P/\ell = p$. For a word $D_0 F_1 D_1 \cdots F_j D_j$ with $j \leq \ell$, telescoping $F_1 \cdots F_j - A_1 \cdots A_j = \sum_{i=1}^j A_1 \cdots A_{i-1} (F_i - A_i) F_{i+1} \cdots F_j$ and bounding the i -th term by $a^{i-1} e(a+e)^{j-i}$ gives $j e(a+e)^{j-1} \leq \ell e \max\{1, a+e\}^{\ell-1}$ per unit of coefficient mass, which is (157). Finally (158) is Minkowski's inequality together with $\|F_{\text{out}} - A_{\text{out}}\| \leq \|F_{\text{out}} - A_{\text{out}}\|_F$. $\square$

The lower estimate in (158) is for the same matrix; no different saturating model supplies it. It compares moments, not quantiles: a quantile statement needs a separate coupling of the tail event of $\|F_{\text{out}} - A_{\text{out}}\|$ with the quantiles of $\|A_{\text{out}}\|$. Reused leaves and reused primitives cost nothing extra because every identity is pointwise.

Example 7.9 (The core tracks a cancellation that word norms discard). Let $A = \varepsilon_1(E_{12} + E_{21})$ on $\mathbb{C}^2$, so that $A^2 = I$, and let $F = A + \lambda \varepsilon_2 E_{11}$ with $0 < \lambda \leq 1$. Retain $H = \{\emptyset, \{1\}\}$, so $\Pi_H F = A$ and the remainder $R = \lambda \varepsilon_2 E_{11}$ has $e = \lambda$ at every exponent, with $a = 1$ and $n = 1 + \lambda$. For the circuit $F^2 - I$ the core output is $A^2 - I = 0$, and (155) certifies $\|F^2 - I\| \leq 2\lambda + \lambda^2$; the exact value is $F^2 - I = \lambda \varepsilon_1 \varepsilon_2 (E_{12} + E_{21}) + \lambda^2 E_{11}$, of norm between λ and $\lambda + \lambda^2$. A triangle inequality over the words of $F^2 - I$ gives a certificate of at least 2 for every λ . The core is exact on the retained algebra, where the cancellation $A^2 = I$ lives, and the remainder is charged once.

Remark 7.10 (Affine sign constraints and the size of the core). Conditioning the signs on an event $\{\varepsilon_{S_1} = \eta_1, \dots, \varepsilon_{S_s} = \eta_s\}$ with $\mathbb{F}_2$ -independent index sets, which has probability 2^{-s} , identifies ε_S with $\eta_i \varepsilon_{S \Delta S_i}$. The coefficients then merge in signed sums over the cosets of the constraint group, before any norm is taken, and a retained group is a subgroup of the quotient. A fresh constraint whose frequency lies in H keeps the core an exact projection, and the elementary inequality $(U + \eta V)^*(U + \eta V) \preceq 2(U^*U + V^*V)$ shows that s such constraints multiply the discarded energies by at most 2^s ; on the conditioned law an L_u radius grows by at most $2^{s/u}$, a single factor compiled at the end rather than at every gate. A constraint whose frequency lies outside H can merge a discarded coset with a retained one; restoring an exact projection then requires the discarded coefficient from the source polynomial, which the energies alone do not contain.

The rule is a core plus a certified remainder, not a theorem that every input has a small accurate core. A large discarded energy, a high discarded degree in (153), or a large coefficient mass C in (157) can make the certificate useless, and the bound (157) grows with the word length ℓ . In particular, a bounded-degree continuation is not the same as iterating through q moment steps: a useful theorem at logarithmic order needs the returning modes controlled at each step, for instance through the block Gram (152), rather than a polynomial Lipschitz estimate applied to M^q . Nor does the rule extend the cone argument of Section 3.6: original magnitude powers and cone preservation need their own proof.

7.4 Exact closure on a cyclic seed space

Takeaway. When the coefficients lie in a supplied algebra with known sector structure and a further interaction is generated by finitely many vectors, the space they generate is an exact reducing subspace whose dimension is charged through the seeds; the untouched multiplicity copies keep their trace weights.

Proposition 2.4 restricts to an invariant sector. A symmetry-breaking insertion, such as a low-rank coefficient perturbation that is not invariant, leaves that sector. The following reduction handles finitely many such insertions exactly. Suppose the base coefficients lie in an algebra given in its sector form

$$\mathcal{H} = \bigoplus_{\lambda \in \Lambda} \mathbb{C}^{d_\lambda} \otimes \mathbb{C}^{\mu_\lambda}, \quad \mathcal{A} = \bigoplus_{\lambda \in \Lambda} M_{d_\lambda}(\mathbb{C}) \otimes I_{\mu_\lambda}, \quad (159)$$

as the commutant of a unitary group action is by Schur's lemma, with d_λ the multiplicity and μ_λ the dimension of the irreducible representation λ , and that the new interactions are generated by outer products $v_a v_b^*$ of seed vectors $v_1, \dots, v_r \in \mathcal{H}$; a rank- t insertion is listed through bases of its range and of its adjoint range, using at most $2t$ seeds. Write $v_a = \sum_\lambda \sum_{i=1}^{d_\lambda} e_i \otimes r_{a,\lambda,i}$ and $M_\lambda = \text{span}\{r_{a,\lambda,i} : a \leq r, i \leq d_\lambda\} \subseteq \mathbb{C}^{\mu_\lambda}$, with $t_\lambda = \dim M_\lambda$.

Theorem 7.11 (Cyclic seed reduction). *The active space*

$$W = \mathcal{A}\text{span}\{v_1, \dots, v_r\} = \bigoplus_{\lambda} \mathbb{C}^{d_{\lambda}} \otimes M_{\lambda} \quad (160)$$

reduces every element of $\mathcal{A}$ and every seed outer product, and

$$t_{\lambda} \leq \min\{\mu_{\lambda}, rd_{\lambda}\}, \quad \dim W = \sum_{\lambda} d_{\lambda} t_{\lambda} \leq r \sum_{\lambda} d_{\lambda}^2. \quad (161)$$

For every ordered polynomial P in elements of $\mathcal{A}$, the seed outer products, and scalar random coefficients,

$$P = P|_W \oplus \bigoplus_{\lambda: \mu_{\lambda} > t_{\lambda}} P_{\lambda}^{(0)} \otimes I_{\mu_{\lambda} - t_{\lambda}}, \quad (162)$$

where $P_{\lambda}^{(0)}$ is obtained by evaluating the same polynomial in sector λ with every seed operator set to zero. The decomposition is preserved by further sums, ordered products, adjoints, and later insertions from $\mathcal{A}$.

The proof is in Appendix E.

The state can be built from the coefficient Gram entries $\langle v_a, E_{\lambda, ij} v_b \rangle$ without choosing coordinates in a large multiplicity space, and all cross-sector couplings and phases are retained. Two bookkeeping rules matter when this reduction is used inside the moment framework. First, one copy of each untouched sector preserves the operator norm but not the trace: $\text{tr}(X \otimes I_{\mu}) = \mu \text{tr} X$, so the trace of (162) weights $\text{tr} P_{\lambda}^{(0)}$ by $\mu_{\lambda} - t_{\lambda}$. An unweighted reduced trace is a norm proxy, not the original trace moment. Second, for a moment computation the vacuum must be among the seeds, or its components in the untouched copies must be weighted in the same way; the vacuum weight of Section 6 is not preserved by an unweighted restriction. The benefit depends on the base algebra: on $\mathbb{C}^n \otimes \mathbb{C}^n$ with $\mathcal{A} = M_n \otimes I_n$, the single seed $\sum_i e_i \otimes e_i$ already generates the whole n^2 -dimensional space, so a low-rank input does not imply a low-dimensional active space. Conversely a sector of exponential multiplicity is activated in at most rd_{λ} of its multiplicity directions, and the untouched copy is what keeps the exact representation economical. The reduction is a rule for supplied symmetric coefficient classes with finitely many symmetry-breaking insertions; it is not a requirement imposed on the general framework.

7.5 What a retained descriptor cannot discard

The positive rules above have three different kinds of closure: exact descent closes prescribed updates on an encoding, a live register closes label-dependent extensions, and a certified core closes an inequality while possibly losing exact spectral information. The following obstructions keep these apart. They are tests that every proposed simplification must pass; they rule out particular descriptors, not richer signed descriptors or a more restricted class of future operations.

A cancelled live-label sum can revive. Example 7.2 shows that an encoding retaining only a coherent sum $\sum_v C_v$ can fail the kernel test for a label-dependent extension, even on actual SparseStack inputs. Any descriptor that merges label-indexed summands must certify that no later coefficient distinguishes them.

Pairwise positive data can miss a signed cycle. The next example keeps fixed dimension, fixed polynomial degree and three primitive signs, and separates exact overlap information from signed coefficient alignment.

Theorem 7.12 (A fixed-size polynomial separation). *There are two triples of real symmetric 8×8 matrices $A_i^\pm$ with the following identical retained data: the law of the three independent signs, every individual coefficient spectrum, $\sum_i (A_i^\pm)^2 = I_8$, each positive operator $[A_i^\pm, A_j^\pm]^* [A_i^\pm, A_j^\pm]$, the law of the norm, indeed of the spectrum, of every real-weighted one- or two-term signed sum, and the normalized second and fourth trace moments of $X_\pm = \sum_{i=1}^3 \varepsilon_i A_i^\pm$. Nevertheless, for the polynomial $p_\tau(x) = (x^2 - 2/3)(x^2 - 2) + \tau$ with $0 < \tau \leq 2/9$,*

$$\|p_\tau(X_+)\| = \tau, \quad \|p_\tau(X_-)\| \geq 2/9 \quad \text{on every sign input.} \quad (163)$$

Consequently no constant-factor estimate of these polynomial outputs can depend only on the listed retained data and the polynomial.

The two triples are $A_i^\pm = \Gamma(v_i^\pm)/\sqrt{3}$, where $\Gamma(v) = \sum_{j=1}^4 v_j \Gamma_j$ for four anticommuting symmetric 8×8 matrices Γ_j squaring to I_8 , and $v_i^\pm$ are unit vectors in $\mathbb{R}^4$ whose pairwise inner products are all $+1/2$ in one triple and all $-1/2$ in the other. Every pairwise datum listed above depends only on the individual inner products up to sign, whereas $X_\pm^2 = (1 \pm T/3)I_8$ with $T = \varepsilon_1\varepsilon_2 + \varepsilon_1\varepsilon_3 + \varepsilon_2\varepsilon_3$ sees the product of the three, a signed cycle. The verification is in Appendix E.

The L_p norm ratios and every quantile ratio in this example are at least $2/(9\tau)$. The complete signed coefficient array, the actual commutator matrices, or the vectorized covariance do distinguish the two models; the theorem rules out the specified positive pairwise state, not these richer descriptors.

Positive return kernels do not give a Loewner recurrence. Example 4.7 shows that the trace comparison of Theorem 4.3 cannot be upgraded to an operator recurrence even when every first return is positive. Ordered noncommuting products require more than positivity of their ingredients.

Remark 7.13 (A complete Gram closes an algebra only in a named representation). For the Pauli matrices, the triples (X, Y, Z) and $(X, Y, -Z)$ have the same anticommutator Gram I_3 , but $I + iXYZ = 0$ while $I + iXY(-Z) = 2I$. A retained joint Gram of anticommuting generators determines every ordered word in a faithful representation that contains both orientations of the volume element, as a doubled representation does, and fails after an unrecorded change to an irreducible representation of odd Clifford dimension, which fixes one orientation. Whenever a descriptor is claimed to be sufficient, the claim must name the representation and the allowed operations.

These tests refine the continuation rules recorded in Section 9.1. The next substantive step they call for is to take a coefficient family of Section 5 whose descriptor fails after a specified further interaction, to retain the missing label or ordered child information, and to prove that the revised encoding passes the descent test with a useful Loewner constant for that family of insertions, rather than repackaging a product of norms.

8 Calibration and comparison

8.1 The parameters supplied by the structural estimates

For the three motivating models, the method gives the conjectured Nelson–Nguyen row and sparsity orders, the exact logarithmic exponent for every fixed dense sign graph shape, and the quadratic confidence allowance $d + \log^2(1/\delta)$ for two-round sampling. The corresponding proof steps are the additive shared Gram, connected commutator cancellation, and collective

row contractions followed by a positive selector estimate. Each retains an algebraic relation that would be lost by bounding its component maps separately.

These conclusions sharpen specific previous estimates. Cohen's trace-inequality analysis has an additional logarithmic row factor [24]. Chenakkod, Dereziński, Dong, and Rudelson obtain optimal embedding dimension [25]; subsequent results improve the sparsity [26] and approach both conjectured scales up to sub-polylogarithmic factors [27]. The shared Gram supplies $d + q$ directly in the SparseStack occupation estimate. For dense graph matrices, the trace and matrix-chaos frameworks give separator-based powers of n and logarithmic upper bounds [28, 29]; tracking attached floating components through the commutator identifies the precise logarithmic exponent. In two-round sampling, the quadratic bound improves the cubic allowance of the formalized original proof (Theorem 5.5) on the same sign occupation spaces. These are comparisons of established bounds; the mechanisms provide concrete targets for refinements of other methods.

8.2 Matrix Chernoff as a calibration

If $A_j \succeq 0$, $\sum_j A_j = I$, and $\|A_j\| \leq R$, then $L_q \leq qR$. The proof in Section 3.4 gives $m \gtrsim nR \log(d/\delta)/\epsilon^2$ for independent row sampling, recovering the matrix Chernoff order [10]. Its more specific parameter is the largest norm of an actual q -element subset sum. It improves the singleton relaxation precisely when the available subset bound improves qR .

The Laplace-transform proof organizes positivity through matrix exponential inequalities and independence. Our proof organizes it through a row factorization and the Gram of selector creation. Both retain a positive sum; the displayed proof explains why that structure also admits an occupation-space treatment.

8.3 The same frame identity in Gaussian covariance

Gaussian trace comparisons use coefficient information through variance and tensor covariance. We can compute both exactly for a circular Gaussian series generated by a Parseval frame. This gives a second use of the same instruction: contract the frame before replacing its terms by separate norms.

For Hermitian matrices H_i and independent standard real Gaussians g_i , write

$$Y = \sum_i g_i H_i, \quad V = \mathbb{E}Y^2 = \sum_i H_i^2, \quad \mathcal{A}_Y = \left(\sum_i H_i \otimes H_i \right) S,$$

where $S(x \otimes y) = y \otimes x$. We use the ordinary complex Hilbert tensor product and inner products conjugate-linear in the first variable. Every $H_i \otimes H_i$ is self-adjoint and commutes with S , so $\mathcal{A}_Y$ is self-adjoint. The estimate

$$\|\mathcal{A}_Y\| \leq \|V\| \tag{164}$$

follows by factoring $\sum_i H_i \otimes H_i$ as the row $[H_i \otimes I]_i$ times the column $[I \otimes H_i]_i^\top$. The squared norms of these factors are $\|V \otimes I\| = \|I \otimes V\| = \|V\|$, and S is unitary. Positivity of $\mathcal{A}_Y$ is a further structural property of the specific series below.

Let $u_1, \dots, u_n \in \mathbb{C}^d$, $d \geq 1$, satisfy $\sum_p u_p u_p^* = I$. Put $P_p = u_p u_p^*$ and $\ell_p = \|u_p\|^2$. For $p < r$, define

$$H_{pr,1} = \frac{u_p u_r^* + u_r u_p^*}{\sqrt{2}}, \quad H_{pr,2} = \frac{i(u_p u_r^* - u_r u_p^*)}{\sqrt{2}}. \tag{165}$$

With independent standard real $g_{pr,1}, g_{pr,2}$, the resulting series is

$$Y_0 = \sum_{p < r} \left(z_{pr} u_p u_r^* + \overline{z_{pr}} u_r u_p^* \right), \quad z_{pr} = \frac{g_{pr,1} + i g_{pr,2}}{\sqrt{2}}.$$

Thus $\mathbb{E}|z_{pr}|^2 = 1$ and $\mathbb{E}z_{pr}^2 = 0$. This normalization specifies the circular complex law completely.

Proposition 8.1 (Circular-frame covariance). *For the series (165),*

$$V_0 = dI - \sum_p \ell_p P_p, \quad \mathcal{A}_0 = I - \sum_p P_p \otimes P_p \succeq 0. \quad (166)$$

Writing $\kappa_0 = \|V_0\|$, we have

$$\|\mathcal{A}_0\| = \min\{\kappa_0, 1\}. \quad (167)$$

The associated covariance map has the exact formula

$$\mathcal{K}_0(X) := \mathbb{E}[Y_0 X Y_0] = \text{tr}(X)I - \sum_p \text{tr}(P_p X) P_p \quad (168)$$

for every complex $d \times d$ matrix X .

Proof. Put $A = u_p u_r^*$. The squares of the two directions for this pair sum to $AA^* + A^*A = \ell_r P_p + \ell_p P_r$. Since $\sum_p \ell_p = d$ and $\sum_p P_p = I$, summing over pairs proves the first identity in (166).

Their tensor squares sum to $A \otimes A^* + A^* \otimes A$. On an elementary tensor,

$$(A \otimes A^*)S(x \otimes y) = (u_p \otimes u_r)\langle u_p, x \rangle \langle u_r, y \rangle = (P_p \otimes P_r)(x \otimes y).$$

Consequently

$$\mathcal{A}_0 = \sum_{p \neq r} P_p \otimes P_r = I - \sum_p P_p \otimes P_p.$$

Every term in the first sum is positive, and the last sum is positive, so $0 \preceq \mathcal{A}_0 \preceq I$.

For $d \geq 2$, choose a nonzero antisymmetric tensor. Its inner product with $u_p \otimes u_p$ vanishes, so each $P_p \otimes P_p$ annihilates it. Hence $\mathcal{A}_0$ has eigenvalue one and norm one. The Parseval identity gives $P_p \preceq I$ and $\ell_p \leq 1$, so $V_0 \succeq (d-1)I$ and $\kappa_0 \geq 1$. For $d = 1$, both covariances are the scalar $1 - \sum_p \ell_p^2 \in [0, 1]$. This proves (167), including the zero-series case.

Finally the contribution of a pair to $\mathcal{K}_0(X)$ is $AXA^* + A^*XA = \text{tr}(P_r X)P_p + \text{tr}(P_p X)P_r$. Summing over pairs gives $\sum_{p \neq r} \text{tr}(P_r X)P_p$. Add and subtract its diagonal terms and use Parseval to obtain (168). $\square$

For $Y = Y_0/\sqrt{b}$, $b > 0$, the three covariance quantities become V_0/b , $\mathcal{A}_0/b$, and $\mathcal{K}_0/b$. In particular, their two norm parameters are exactly κ_0/b and $\min\{\kappa_0, 1\}/b$.

For completeness, the standard Bogoliubov–Kubo–Mori (BKM) quadratic form at a Hermitian background B is the unnormalized canonical-correlation form associated with e^B ; see Petz [60]. It is the Hessian of the trace exponential, by the standard derivative formula for operator exponentials [59]:

$$Q_B(H) = \frac{d^2}{dt^2} \text{tr} e^{B+tH} \Big|_{t=0} = \int_0^1 \text{tr}(e^{(1-u)B} H e^{uB} H) du.$$

The derivative identity follows by differentiating the exponential series: the coefficient of $B^j H B^k$ in $\int_0^1 e^{(1-u)B} H e^{uB} du$ is $\int_0^1 (1-u)^j u^k du / (j!k!) = 1/(j+k+1)!$. Differentiating $\text{tr} e^{B+tH}$ first gives $\text{tr}(e^{B+tH} H)$ by cyclicity, and a second derivative gives the displayed formula. In an eigenbasis of B , with eigenvalues λ_i ,

$$Q_B(H) = \sum_{i,j} |H_{ij}|^2 \int_0^1 e^{(1-u)\lambda_i + u\lambda_j} du \geq 0.$$

Convexity of the scalar exponential bounds this integral by $(e^{\lambda_i} + e^{\lambda_j})/2$. Therefore a Gaussian series $Y = \sum_i g_i H_i$ satisfies the elementary estimate

$$\mathbb{E} Q_B(Y) = \sum_i Q_B(H_i) \leq \text{tr}(e^B V) \leq \|V\| \text{tr} e^B.$$

The variance and tensor covariance above also enter comparisons with the fourth directional derivative $(d^4/dt^4) \text{tr} e^{B+tY} \Big|_{t=0}$. Our exact identities isolate the coefficient information supplied by the frame, while the occupation estimates use it to control finite moments. The same contraction thus supplies inputs to two analytic representations of a random-matrix problem.

The two representations read out their information differently, and the difference is not cosmetic. A Gaussian trace comparison is a statement about the full trace $\text{tr} e^{B+tY}$, whose convexity properties come from cyclicity; the moment identity (6) instead takes the trace over the external index only and reads the occupation side through the vacuum vector. A vector state is a positive functional, but the convexity in t of $t \mapsto \text{tr} e^{B+tH}$, which comes from cyclicity, is not inherited by $t \mapsto \langle v, e^{B+tH} v \rangle$ for noncommuting B, H : for $B = \text{diag}(0, 1)$, $H = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ and $v = (3, -2)/\sqrt{13}$,

$$\frac{d^2}{dt^2} \langle v, e^{B+tH} v \rangle \Big|_{t=0} = \frac{5-2e}{13} < 0, \quad \text{whereas} \quad \frac{d^2}{dt^2} \text{tr} e^{B+tH} \Big|_{t=0} = Q_B(H) = 3e - 1.$$

The example rules out an unrestricted substitution of the vacuum functional for the trace in a trace-exponential comparison. One sufficient route into the occupation representation is a reducing subspace: compression by a projection commuting with the background and with every coefficient is again an ordinary trace. Other weighted or specially structured comparisons are not excluded by the example, and none is asserted here; taking the full trace over the occupation space instead would reintroduce the spurious dimension factor discussed after Theorem 2.3. The positivity in (166) is likewise a property of the circular law and not of the frame alone. With a single real quadrature $H_{pr,1}$ per pair, the terms $(u_p \otimes u_p)(u_r \otimes u_r)^*$ that the second quadrature cancels survive, and for the standard frame $u_p = e_p$ the compression of $\mathcal{A}_Y$ to the span of $e_p \otimes e_p$ and $e_r \otimes e_r$ is $\begin{pmatrix} 0 & 1/2 \\ 1/2 & 0 \end{pmatrix}$, which has a negative eigenvalue. The same two-by-two computation shows that no nonzero family of real symmetric coefficients with zero diagonal in a common basis has $\mathcal{A}_Y \succeq 0$: its entry between $e_a \otimes e_a$ and $e_b \otimes e_b$ is $\sum_i (H_i)_{ab}^2$, and the two diagonal entries vanish.

8.4 Fourth-order energies of the original law and a Gaussian comparison

The frame identities above describe a Gaussian model. The next identity says exactly how far the original law of a sum of independent components is from that model at fourth order, for every ordered quadratic expression. This is an ordered-matrix version of the elementary fourth-moment decomposition into covariance pairings and a one-component correction. Gaussian pairings are classical [62]; the point here is to retain deterministic insertions and the actual reused-component contribution.

Lemma 8.2 (Quadratic energy identity with component reuse). *Let $Y_1, \dots, Y_k$ be independent centered random matrix tuples with finite fourth moments, and let $G_1, \dots, G_k$ be independent centered jointly Gaussian tuples, G_a having the complete entry covariance of Y_a : for real tuples the covariance of all entries, and for complex tuples the joint covariance of the real and imaginary parts of all entries, equivalently both $\mathbb{E}[X_{ab}\overline{X_{cd}}]$ and $\mathbb{E}[X_{ab}X_{cd}]$. Put $S = \sum_a Y_a$ and $G = \sum_a G_a$. For every homogeneous ordered quadratic matrix polynomial Q , with arbitrary deterministic insertions and rectangular output,*

$$\mathbb{E} Q(S)^* Q(S) = \mathbb{E} Q(G)^* Q(G) - \sum_a \mathbb{E} Q(G_a)^* Q(G_a) + \sum_a \mathbb{E} Q(Y_a)^* Q(Y_a). \quad (169)$$

The left-energy identity with $Q(\cdot)Q(\cdot)^*$ holds as well. If the Y_a are identically distributed and $S = \sum_a w_a Y_a$ with real weights, put $v = \sum_a w_a^2 > 0$ and $\rho = \sum_a w_a^4/v^2$; if G_0 has the covariance of one component, then

$$\mathbb{E} Q(S/\sqrt{v})^* Q(S/\sqrt{v}) = (1 - \rho) \mathbb{E} Q(G_0)^* Q(G_0) + \rho \mathbb{E} Q(Y_1)^* Q(Y_1). \quad (170)$$

Proof. The expression $Q(S)^* Q(S)$ has four random factors in a fixed order. Expand each in components. A component name appearing exactly once integrates to zero by centering and independence. The surviving patterns are two components appearing twice each, and one component appearing four times. A pattern of the first kind depends only on the two entry covariances, so it has the same value for Y and for G . Subtracting the one-component Gaussian terms and adding the actual one-component terms gives (169). Under identical distribution and weights, G has the law of $\sqrt{v} G_0$, quadratic homogeneity scales the energies by v^2 , and the one-component terms carry $\sum_a w_a^4$; dividing by v^2 proves (170). $\square$

The identity separates the covariance-controlled contribution from the repeated-component correction, while every insertion stays in its written position and the sum is formed before any trace or norm. The full second-order matching is necessary for complex tuples: for independent real signs regarded as complex scalars and proper complex Gaussians with the same $\mathbb{E}|G_a|^2 = 1$, the quadratic $Q(z) = z^2$ gives $\mathbb{E}|Y_1 + Y_2|^4 = 8$, whereas the right side of (169) would evaluate to $8 - 4 + 2 = 6$; the missing datum is $\mathbb{E}Y_a^2 = 1$ against $\mathbb{E}G_a^2 = 0$. For heterogeneous components the difference of the two Gaussian and actual one-component sums need not be positive; the identically distributed form (170) is a positive mixture. Three uses are immediate.

Example 8.3 (SparseStack at fourth order). The SparseStack error (114) is $E = s^{-1} \sum_{g=1}^s E_g$ with independent identically distributed blocks $E_g = \sum_{i < j} z_{g,ij} A_{ij}$, $A_{ij} = u_i u_j^* + u_j u_i^*$ and $z_{g,ij} = \sigma_{gi} \sigma_{gj} \mathbf{1}_{\{h_{gi}=h_{gj}\}}$. Since $\mathbb{E} z_{g,ij} z_{g,kl} = b^{-1} \mathbf{1}_{\{i,j\}=\{k,l\}}$, one block has the covariance of the real Gaussian series $G_0 = b^{-1/2} \sum_{i < j} g_{ij} A_{ij}$ with independent standard g_{ij} , whose $\mathbb{E}G_0^2 = b^{-1}[(d+1)I - 2 \sum_i \ell_i u_i u_i^*]$ agrees with (117) at $m = b$. With $w_g = s^{-1}$ one has $v = s^{-1}$ and $\rho = s^{-1}$, so for every homogeneous ordered quadratic Q

$$\mathbb{E} Q(\sqrt{s} E)^* Q(\sqrt{s} E) = \left(1 - \frac{1}{s}\right) \mathbb{E} Q(G_0)^* Q(G_0) + \frac{1}{s} \mathbb{E} Q(E_1)^* Q(E_1).$$

With $Q(X) = X^2$ this is an exact formula for $\mathbb{E}E^4$, in which the Gaussian term is a Wick sum determined by the frame, while the one-block term retains the actual collision law: an edge used four times in one block has weight $\mathbb{E}z^4 = b^{-1}$ rather than the Gaussian $3b^{-2}$, and a four-cycle of collisions, which the Gaussian model does not see at all, has weight b^{-3} ; every such difference lives in $\mathbb{E}E_1^4$. This is the fourth-order companion of the second-moment check (117) and a benchmark for the diagnostic procedure of Section 6.

The second use is a reuse test. Two occurrences of the same observation in a layered construction must have their weights combined before ρ is computed; treating them as independent copies changes ρ and hence the answer. The third use is calibration of a Gaussian comparison for a noncommuting insertion $Q(X) = XDX$: the exact fourth-order comparison error is the last two terms of (169), and for a weighted layer it is ρ times the difference of the two one-component energies. The identity is a fourth-order statement and is not to be extrapolated to logarithmic moment orders, where the returning-path and cone estimates of Section 4 are the tools.

8.5 Noncommutative Khintchine and matrix chaos

The noncommutative Khintchine inequalities of Lust-Piquard [61] and Lust-Piquard–Pisier [37] provide a basic comparison point. For a Hermitian Gaussian series $X = \sum_i g_i A_i$, they control the $2q$ trace-moment root by a constant times

$$\sqrt{q} D^{1/(2q)} \left\| \sum_i A_i^2 \right\|^{1/2}.$$

The $\sqrt{q}$ scale is attained already by a scalar Gaussian. In families with a smaller spectral scale, finer coefficient information can be decisive. The free-probability comparison of Bandeira, Boedihardjo, and van Handel [11] and its two-sided extension [12] capture such information through a free model. Our estimates retain it through cross Grams and grading relations in the exact moment.

Bandeira, Lucca, Nizić-Nikolac, and van Handel develop a systematic framework for higher-order matrix chaos [29]. Its parameters are coefficient flattenings, and its strong versions remove logarithmic losses when the relevant flattenings are small. This is the closest structural comparison for graph matrices: flattenings expose different ways to share coefficient indices, while creation commutators expose cancellation between overlapping embeddings. We use the latter to count exactly which floating components contribute a moment factor.

8.6 Weak moments, coefficient domains, and independent entries

The regular-entry estimate uses the same three stages: exact original multiplication, occupation localization, and a contraction of retained coefficient data. Its retained data include all magnitude powers and the signs of the original monomial coefficients. The thin-axis factor is $\rho_q(|I|) < 1.37$, with $|I| \leq q$ and $p = 4q$; the positive-polynomial factor is $H_\alpha = \sqrt{3}\alpha$. Thus the local moment estimate loses only an α -dependent constant at orders comparable to $\log n$. No enumeration of the occupation space is part of this estimate.

The upper bound quotes from the literature only the Boolean hypercontractivity theorem for the signs: its fourth-moment case, the magnitude-product bound, the spherical thin-axis estimate and the independent-coordinate strong/weak comparison are proved in Section 3.6, and the deletion remainder is estimated by an elementary centered Gram in Section 5.4. The earlier route through the strong/weak theorem of Latała–Strzelecka and the variance-envelope theorem of Latała–van Handel–Youssef is recorded in Remark 5.8. The deletion architecture and the lower comparison are due to Latała–Świątkowski; the contraction on the preserved walk cone is the local mechanism that the present method adds.

A calibration shows what the deletion functional can and cannot capture. For weighted nested triangular supports $A(b)_{ij} = b_j \mathbf{1}_{\{i \leq j\}}$ with $b_1 \geq \dots \geq b_n \geq 0$ and independent signs, the weak moment $W_p(A) = \sup_{\|s\|_2, \|t\|_2 \leq 1} \left\| \sum a_{ij} \varepsilon_{ij} s_i t_j \right\|_{L_p}$ and the sparse-mask maximum $H_p(A) = \max_{|E| \leq |p|} \|(|a_{ij}| \mathbf{1}_E)\|$ of Latała–Świątkowski [48, Proposition 1.5] satisfy $\frac{1}{2}H_p \leq W_p \leq 3\|A_r\| \leq 3H_p$ for $p \geq 1$, where A_r is the leading $r \times r$ block with $r = \min\{n, \lfloor \sqrt{p} \rfloor\}$: the block is a mask of at most p entries, the sum over its columns is bounded pointwise by $\|A_r\|$, the remaining columns have coefficients at most b_{r+1} and are bounded by Khintchine by $\sqrt{p}b_{r+1} \leq (r+1)b_{r+1} \leq 2\|A_r\|$, the last step by testing A_r against the normalized all-ones vector. On this family the logarithmic gap of that proposition closes by prefix truncation alone; moreover the deletion $I_k = \{1, \dots, k\}$ leaves coefficients at most $b_{k+1} \leq \|b\|_2 / \sqrt{k+1}$, so $D(X) \leq \|b\|_2 / \sqrt{2} \leq M(X) / \sqrt{2}$, and $\mathbb{E}\|X\| \asymp \|b\|_2$ already follows from the first row and the fourth-moment bound of Latała [32]. The family is a calibration, not a test, of the deletion mechanism.

This mechanism complements the collective row and connected-creation estimates. It depends on a coefficient cone preserved by the particular walk, whereas those estimates

can control complete graded spaces. An arbitrary deterministic coefficient insertion must be checked against the relevant domain. Efficient evaluation of $D(X)$, discovery of hidden coefficient structure, and matching quantiles are separate tasks.

8.7 Universality and extreme eigenvalues

Universality compares sums of independent random matrices with Gaussian models having matching mean and covariance, with quantitative error terms [13]. It handles broad classes without a separate local occupation inventory. The sketching and fixed-graph applications instead have explicit small local factors and useful coefficient identities, making a direct finite-moment calculation effective.

Tropp’s extreme-eigenvalue comparison theorem treats the maximum eigenvalue, with minimum-eigenvalue and spectral-norm consequences [14]. Its complex sparse-sketch application provides a lower-edge injectivity result, as recorded in the Simons workshop problem list [19]. Our SparseStack theorems control the two-sided Gram error for real signs. These application statements specify the relevant models and events, while the comparison theory itself encompasses both spectral edges. Since the two results overlap exactly at the lower edge, the constants can be compared there.

Remark 8.4 (Lower-edge constants for SparseStack). Theorem 3.6 of [14] concerns the SparseStack matrix with independent complex Steinhaus signs and uniform hashes. For a fixed isometry $U \in \mathbb{C}^{n \times d}$, lower distortion $\alpha \in (0, 1)$ and failure probability $p \in (0, 1]$, the parameters

$$\zeta \geq 6\alpha^{-1} \log(d/p), \quad k \geq 16\alpha^{-2}(d \vee \log(d/p))$$

give $\mathbb{P}\{\sigma_{\min}^2(\Phi U) > 1 - \alpha\} \geq 1 - p$ together with $\mathbb{E}\sigma_{\min}^2(\Phi U) \geq 1 - \alpha$, where ζ is the column sparsity and $k = \zeta b$ the embedding dimension. Nothing is asserted about the upper edge. In our notation $\zeta = s$, $k = m$, and the real-sign two-sided event $\|U^* S^* S U - I_d\| \leq \epsilon$ is the intersection of $\sigma_{\min}^2(SU) \geq 1 - \epsilon$ and $\sigma_{\max}^2(SU) \leq 1 + \epsilon$. It implies the non-strict lower-edge event; it is not an event containing all lower-edge successes. The prescriptions compare as follows, with q the moment order of the respective theorem and $\Lambda = 90(1 - e^{-1})^{-2}$.

Result	Signs; edges	Sparsity	Dimension	Leading constants
[14, Thm. 3.6]	complex; lower	$6\alpha^{-1} \log(d/p)$	$16\alpha^{-2}(d \vee \log(d/p))$	6; 16
Theorem 10.6	real; both	$\lceil \frac{17}{2}(2q+1)/\epsilon \rceil$, $q = \lceil \log_4(d/\delta) \rceil$	$< 179(d+q)/\epsilon^2$	$17/\log 4 \approx 12.3$; 81
Theorem 5.1	real; both; Lean 4	$\lceil \Lambda(2q+1)/\epsilon \rceil$, $q = \lceil \log_2(d/\delta) \rceil$	$< 306456(d+q)/\epsilon^2$	$2\Lambda/\log 2 \approx 650$; $\Lambda^2 \approx 50732$
Steinhaus local-factor comparison	complex	matching local Grams; full charge-graded assembly not checked here	no separate prescription asserted	see Remark 8.5

The leading constants are the coefficients of $\epsilon^{-1} \log(d/\delta)$ in the sparsity and of $\epsilon^{-2}d$ in the dimension; for the dimension of Theorem 10.6 the coefficient 81 is the one in the choice of b , before the ceiling in s raises the displayed product bound to 179. At $d = 1000$, $\epsilon = \alpha = 1/2$ and $\delta = p = 10^{-2}$ the three prescriptions are $(\zeta, k) = (139, 64079)$, $(s, m) = (323, 330429)$ and $(s, m) = (15767, 2.1 \times 10^8)$. Thus the comparison theorem is sharper by a factor of about two in sparsity and about five in dimension than our analytic direct theorem, and by two to three orders of magnitude than the formalized coupled theorem, on the event the two share.

Several differences must be separated before attributing this gap to an individual estimate. The comparison theorem controls one edge: it compares the minimum eigenvalue of an

independent sum of self-adjoint matrices with a Gaussian proxy of the same mean and covariance ([14, Corollary 1.3]), under a one-sided bound on the summands, and the positive semidefinite Gram summands $U^* S_i^* S_i U$ satisfy that bound automatically; our two-sided theorem also controls the upper edge. This difference in target event does not, by itself, quantify its cost. The laws differ as well: complex phase averaging and real-sign averaging retain different charge words. Remark 8.5 records the exact relation and the limits of what the local Gram identities establish. And our constants are all-order certificates in a fixed moment convention ($q = \lceil \log_4(d/\delta) \rceil$ for the direct theorem, $\lceil \log_2(d/\delta) \rceil$ for the coupled one), so an extra factor enters through the moment-to-tail conversion of Section 4; the coupled theorem additionally pays the transfer cost $c_*^2 = (1 - e^{-1})^{-2}$ in the moment root. On the other side, our theorems give the upper edge with the same parameters, which the comparison theorem does not address, and the direct constants are certified for every moment order by the rational inequality of Section 10. Where the lower edge alone is the target, the comparison theorem should be preferred.

Remark 8.5 (Sign-model comparison: identities and limitations). The charge-to-parity relation can be written exactly. With Steinhaus signs the local factor at a site is $L^2(S^1 \times [b]) = \bigoplus_{c \in \mathbb{Z}} \sigma^c L^2([b])$, graded by the charge c . In the bucket basis $e_a^{(c)} = \sigma^c \sqrt{b} \mathbf{1}_{\{b=a\}}$, which is orthonormal in (c, a) , multiplication by e_a and by $\bar{e}_a$ are charge shifts with a bucket-diagonal amplitude:

$$e_a e_{a'}^{(c)} = \delta_{aa'} \sqrt{b} e_a^{(c+1)}, \quad \bar{e}_a e_{a'}^{(c)} = \delta_{aa'} \sqrt{b} e_a^{(c-1)},$$

and the vacuum is $\Omega = b^{-1/2} \sum_a e_a^{(0)}$. Each term $\bar{J}_{a,gi} J_{a,gj}$ of the error multiplier lowers the charge at (g, i) and raises it at (g, j) , so charge is conserved in every block. The real factor of Section B.1 is the quotient by $\sigma^2 = 1$: the reduction $\pi(\sigma^c \phi) = \sigma^{c \bmod 2} \phi$ is a ring homomorphism, hence intertwines the two multiplications, and therefore $\mathcal{M}_{\mathbb{R}}^q \Omega = \pi \mathcal{M}_{\mathbb{C}}^q \Omega$ on the tensor product over sites. To recover the trace, use all external vacuum vectors: $x = \bigoplus_{a=1}^d \mathcal{M}_{\mathbb{C}}^q(e_a \otimes \Omega)$. For a charge pattern $\mathbf{c}$, remove its character and denote its coefficient vector by $\tilde{x}_{\mathbf{c}}$, a vector in the common external/bucket amplitude space. Distinct charge sectors are orthogonal before folding; the following inner products are between these *amplitudes after that identification*, not between vectors in distinct charge subspaces. Then

$$\mathbb{E} \operatorname{tr} E_{\mathbb{R}}^{2q} = \mathbb{E} \operatorname{tr} E_{\mathbb{C}}^{2q} + \sum_{\substack{\mathbf{c} \neq \mathbf{c}' \\ \mathbf{c} \equiv \mathbf{c}' \pmod{2}}} \langle \tilde{x}_{\mathbf{c}}, \tilde{x}_{\mathbf{c}'} \rangle. \quad (171)$$

Indeed, the real norm squares the sum of amplitudes in each common parity class, whereas the complex norm sums their squared norms separately. The ordered cross terms have a real total because conjugate pairs occur together. This is an identity for the *difference* of the moments; without another argument it does not give a sign to that difference. It records words that real parity can retain although complex charge averaging removes them.

At $q = 1$ the only interfering pairs are the two orientations of a single term, and (171) evaluates to

$$\mathbb{E} \operatorname{tr} E_{\mathbb{R}}^2 = \frac{d(d+1) - 2 \sum_i \ell_i^2}{m}, \quad \mathbb{E} \operatorname{tr} E_{\mathbb{C}}^2 = \frac{d^2 - \sum_i \ell_i^2}{m}, \quad \ell_i = \|u_i\|^2,$$

when the complex second moment is positive, their ratio is

$$1 + \frac{d - \sum_i \ell_i^2}{d^2 - \sum_i \ell_i^2} \leq 1 + \frac{1}{d}.$$

If the denominator vanishes, both second moments vanish and no ratio is asserted.

Some local Gram identities are unchanged. The charge-raising leg on a charge- ± 1 site has Gram bI , as does the combined real leg $e_a \mapsto \Omega + \sqrt{h}f_a$; the vacuum-creation and heavy-frame identities also agree. These are useful algebraic comparisons, but they are not a completed verification of every charge sector and transition in Appendix B. That assembly has not been rerun here, so no separate complex two-sided prescription is claimed.

The scalar sector majorants used by this proof discard the relative amplitudes responsible for the folding terms. This is a limitation of those estimates, not of every possible norm estimate or of the exact vacuum norm. Even if the completed charge-graded argument produces the same certificate, equality of sufficient bounds would not establish equality of optimal thresholds or sharpness of the coefficient bounds.

The following table records the reported Monte Carlo comparison for a fixed Gaussian-orthonormalized frame. The second-moment column uses the exact formula; the other columns are sample estimates. The underlying simulation runs were not independently replayed in this audit revision.

(n, d, s, b)	N	$\mathbb{E} \operatorname{tr} E^2$	$(\mathbb{E} \operatorname{tr} E^6)^{1/6}$	$\mathbb{E} \ E\ $	$\mathbb{E} \lambda_{\min}(E) $
(128, 8, 4, 16)	20000	1.117	1.103	1.069	1.038
(256, 16, 8, 32)	8000	1.058	1.047	1.039	1.027
(256, 16, 4, 64)	8000	1.058	1.049	1.041	1.028
(512, 32, 8, 128)	3000	1.029	1.022	1.023	1.018

For these particular frames and sizes, the reported norm and edge statistics differ modestly. Such finite comparisons do not establish a uniform bound on the real/complex difference, an optimal-threshold comparison, or a causal allocation of the gap between sufficient prescriptions. Returning-path estimates, coefficient estimates, domain restrictions, and moment-to-tail conversion remain distinct places for controlled comparison. A same-model improvement should be assessed by replacing one justified inequality at a time.

Sparsity also has genuine structural constraints. Huang, Rudelson, and Tikhomirov prove obstructions to constant-sparsity oblivious subspace injections under stated support-overlap hypotheses [36]. Those hypotheses and their one-sided event specify a different scope from a two-sided, distortion-dependent SparseStack theorem. Their result provides a useful benchmark for any attempt to decrease sparsity by changing the support law.

8.8 Chaining, resampling, and testing geometry

Chaining retains metric information in a supremum of correlated tests. Krahmer, Mendelson, and Rauhut apply it to chaos processes and restricted isometry [39]; Dirksen develops general tail bounds in this framework [16]. Our fixed-subspace moment estimates can be combined with finite-family union bounds. Sharper uniform statements require additional control of the testing family, which is the information chaining is designed to preserve.

Matrix Efron–Stein inequalities [38] express fluctuations through resampling. Rajendran and Tulsiani develop this route for polynomial random matrices [17]. A resampling argument seeks contractions in derivative or difference Grams; an occupation argument seeks them in maps between orthogonal grades. For a new polynomial model, the useful comparison is which of these Grams makes its coefficient identities most transparent.

The compressed matrix can still be enormous. The aim of the method is an analytic contraction that avoids constructing it. Small exact matrices, such as Example 2.1, diagnose which information a proposed estimate loses.

8.9 Exact moments for polynomials in a few aggregate statistics

Takeaway. When the random matrix is a polynomial of degree R in a short vector of sums of local sign statistics, its $2p$ -th trace moment is an exact linear functional of the joint moments of that vector through degree $2Rp$, and those moments can be computed by eliminating the signs along a supplied schedule, with a message of $\binom{w+2Rp}{w}$ scalars per boundary state. The joint law of the vector may have exponentially many atoms and is never enumerated.

Exact small matrices are the diagnostic instrument of Section 6. Building the multiplication operator is not the only way to obtain exact moments. Let $\varepsilon_1, \dots, \varepsilon_M$ be independent uniform signs. For $j = 1, \dots, L$ supply an index set $\mathcal{V}_j \subseteq [M]$ with $|\mathcal{V}_j| \leq a$, a rational table $\phi_j : \{\pm 1\}^{\mathcal{V}_j} \rightarrow \mathbb{Q}^w$, and a zero-one constraint table η_j on the same variables. Put

$$T = \sum_{j=1}^L \phi_j(\varepsilon_{\mathcal{V}_j}) \in \mathbb{Q}^w, \quad \Xi = \left\{ \prod_{j=1}^L \eta_j(\varepsilon_{\mathcal{V}_j}) = 1 \right\}, \quad \rho = \mathbb{P}(\Xi) > 0, \quad (172)$$

and condition on Ξ . Repeated occurrences of a sign are the same named variable, so reused equalities, overlapping parity conditions and implied constraints remain actual shared events. The random matrix is

$$Z = Q(T), \quad Q(t) = \sum_{|\alpha| \leq R} C_\alpha t^\alpha, \quad C_\alpha \in M_d(\mathbb{Q}(i)), \quad (173)$$

with commuting scalar coordinates and noncommuting matrix coefficients; a written ordered product of affine matrix pencils with inserted fixed matrices is included by multiplying its coefficients in their original order and collecting equal monomials. For the supplied factor order define the live boundaries and their width

$$\partial_j = \left(\bigcup_{i \leq j} \mathcal{V}_i \right) \cap \left(\bigcup_{i > j} \mathcal{V}_i \right), \quad \beta = \max_j |\partial_j|. \quad (174)$$

Fix $p \geq 1$, set $N = 2Rp$, and work in the truncated polynomial ring $\mathcal{R}_N = \mathbb{Q}[z_1, \dots, z_w] / \langle z^\alpha : |\alpha| > N \rangle$, which has dimension $\binom{w+N}{w}$. Define the local truncated exponential $\kappa_j(u; z) = \eta_j(u) \sum_{|\alpha| \leq N} \phi_j(u)^\alpha z^\alpha / \alpha!$, where $\alpha! = \prod_k \alpha_k!$, and the boundary messages, starting from $m_0(\emptyset; z) = 1$,

$$m_j(v; z) = 2^{-|\mathcal{V}_j^+ \setminus \partial_j|} \sum_{\substack{u \in \{\pm 1\}^{\mathcal{V}_j^+} \\ u|_{\partial_j} = v}} m_{j-1}(u|_{\partial_{j-1}}; z) \kappa_j(u|_{\mathcal{V}_j}; z) \quad \text{in } \mathcal{R}_N, \quad \mathcal{V}_j^+ = \partial_{j-1} \cup \mathcal{V}_j. \quad (175)$$

Lemma 8.6 (Exact aggregate moments with constraint reuse). *For every multi-index α with $|\alpha| \leq N$,*

$$[z^\alpha] m_L(\emptyset; z) = \frac{\mathbb{E}[\mathbf{1}_\Xi T^\alpha]}{\alpha!}. \quad (176)$$

In particular the constant term is ρ , and $\alpha!/\rho$ times the coefficient is the conditional moment $\mathbb{E}[T^\alpha \mid \Xi]$. The transfer has no approximation error.

Proof. In $\mathcal{R}_N$ the product of two truncated exponentials of scalar linear forms is the truncated exponential of their sum, because terms of degree above N cannot affect a coefficient of degree at most N . For a complete sign assignment, the product of the κ_j is therefore $\mathbf{1}_\Xi \exp(\langle z, T \rangle)$ through degree N . After factor j , every sign omitted from ∂_j has no later occurrence, so averaging it in (175) is exact under the product law. Induction identifies m_j with the corresponding partial expectation, and at the empty terminal boundary the exponential coefficient gives (176). Since $\eta_j^2 = \eta_j$ and logical implications hold pointwise before averaging, a repeated or implied constraint acquires no fresh probability factor. $\square$

Proposition 8.7 (Exact conditional norm moments). *Write $(Q(t)^*Q(t))^p = \sum_{|\alpha| \leq N} C_{p,\alpha} t^\alpha$ with the matrix products taken in their written order, and put*

$$M_p = \sum_{|\alpha| \leq N} \text{tr}(C_{p,\alpha}) \mathbb{E}[T^\alpha \mid \Xi], \quad \mathbf{m}_{2p} = M_p^{1/(2p)}.$$

*Then $M_p = \mathbb{E}[\text{tr}((Z^*Z)^p) \mid \Xi]$ and*

$$d^{-1/(2p)} \mathbf{m}_{2p} \leq (\mathbb{E}[\|Z\|^{2p} \mid \Xi])^{1/(2p)} \leq \mathbf{m}_{2p}, \quad (177)$$

with ratio at most $\sqrt{e}$ when $p \geq \log d$. Individual ordered trace terms may be negative or nonreal; they are summed before any absolute value is taken, and the total is real and nonnegative.

Proof. Lemma 8.6 evaluates the expectation of every scalar monomial in the ordered expansion, and for the singular values of the same realization $\|Z\|^{2p} \leq \text{tr}((Z^*Z)^p) \leq d\|Z\|^{2p}$. $\square$

Appendix E records the operations under which the descriptor is closed, the arithmetic costs, and three model classes with explicit parameters; in particular an aggregate with 2^{80} distinct values needs only 13 message coefficients for its moments through degree 12.

What the transfer should and should not do is also clear. It supplies exact conditional moments for hand-specified examples with overlapping variables, noncommuting coefficients and reused constraints, and these serve as an independent check on an explicit construction of the multiplication operator. It does not compute count-sector block norms: a moment evaluator and an operator-block evaluator answer different questions. Nor is a terminal moment table an extension-stable descriptor: a later interaction involving a sign that was eliminated forces recomputation with the larger boundary, so the source tables must be retained. And the theorem requires all random entries to factor through a small supplied aggregate vector with a low-width schedule; an arbitrary dense SparseStack or Walsh matrix need not admit such a representation, and none is inferred here. Matching moments do not imply matching quantiles; a separate strict lower certificate follows from Paley–Zygmund after running the transfer to degree $4Rp$, with a success probability that is part of the conclusion and may be small.

9 Scope: proved obstructions and missing estimates

The following examples isolate five structural issues: independence, simultaneous testing, tensor order, reuse of randomness, and exact column normalization. Each has a concrete obstruction or a precise additional estimate to supply.

9.1 Recovery and the domain of a continuation

Each structural estimate specifies both its retained information and the maps that may subsequently act on it.

SparseStack. The shared-factor identity of Lemma 3.1 remains the starting coefficient contraction. Projection variance controls a whole input superposition; exceptional labels and their occupied coefficient spans sharpen the residual boundary. The explicitly supplied marked rung family realizes the same collective inclusion operator as the graph interface. Arbitrary nonlinear changes of coefficients are not included in these continuation rules.

Two-round SRHT. The row and selector estimates in Section 10.1 and Appendix A retain their original sampling conclusions. The residual estimates consume those same row inputs, retain the ordered child-pair Grams, and quantify a further creation. A common row-input equality is an isometry on the actual full boundary; it does not equate distinct original pairs. These interface estimates alone assert no improved complete sampling rate or arbitrary sequence of residual insertions.

Graph matrices. The connected-creation and separator arguments in the original applications retain their stated support hypotheses. The weighted edge creator has an arbitrary-support count contraction and an input variance Gram valid under actual congruence. A separator-dependent coefficient map still needs its own compatibility with the retained occupations and coefficient products; independent-entry norm bounds do not themselves prove a new general graph-separator theorem.

Independent entries. The weighted-sign theorem is a full operator estimate. The regular-entry proof in Section 5.4 uses a different, smaller admissible family: vector polynomials with nonnegative coefficients in the original entry monomials. Parity supplies orthogonality while all magnitude multiplicities remain in the coefficients. The actual matrix walk preserves that family. An arbitrary coefficient insertion may not. This distinction explains how a sharp independent-entry application can be obtained without claiming a uniform bound on every vector of a raw Jacobi polynomial span.

All the positive block budgets in Section 3 survive their actual input congruences. A squared bound on K directly controls KL ; for a rebuilt cross $L_1^*KL_2$ it must also charge $\|L_1\|^2$ unless L_1 is a contraction. A common left output map changes the shared Gram. Identifying original random variables changes the local multiplication algebra and must be analyzed as that specified operation. These are domains of the proved rules, rather than a proposed program of further research. Section 7 adds the general test behind these rules, Proposition 7.1, together with two closure rules that pass it exactly, the live register of Theorem 7.3 and the cyclic seed space of Theorem 7.11, one that passes it approximately with a certified remainder, Theorem 7.8, and the obstructions of Section 7.5 that every proposed simplification must respect. None of the analytic extensions here is presented as a new formal proof-assistant certificate or a matching-quantile theorem.

9.2 Scope and the information retained by the proof

The regular-entry estimate of Theorem 5.6 concerns the expected norm of the same original matrix throughout. Its constants depend only on the entry regularity parameter. The deletion sets are deterministic for the law, use the same original indices on both axes, and retain the exact moment order $\log(k+1)$. The lower comparison and the deletion construction have their stated public origins; the new local argument is the parity-count contraction on positive original-coefficient walk polynomials.

Three features of that local argument prevent unsupported reductions. First, the magnitudes and their repeated powers are never replaced by static entry heights or fresh copies. Second, sector orthogonality comes from the original signs and current vertices, not from magnitude monomials on disjoint rows. Third, the multiplier estimate is used only on a cone preserved by the actual walk prefixes. It does not bound all signed linear combinations in a polynomial span. The Hilbert Boolean estimate, the scalar product-moment comparison, and the independent-coordinate strong/weak comparison are proved explicitly, and the deletion remainder uses only an elementary Gram second moment.

The upper half of the regular-entry theorem is formalized in Lean 4, with the two stated differences recorded in Section 11; the lower half is the known result and is not

formalized. No efficient algorithm for evaluating $D(X)$ or $R_X(p)$ is claimed. The moment estimates can supply upper-tail bounds by Markov's inequality, but do not by themselves identify matching quantiles. Likewise, arbitrary noncommuting coefficient insertions can destroy the nonnegative monomial cone; applications with such insertions need their own preserved-coefficient argument.

9.3 Correct support marginals do not control dependence

Proposition 9.1 (A shared-support rank obstruction). *There are random matrices whose individual columns have exactly the uniform signed s -subset law, but which fail to embed every fixed d -dimensional coordinate subspace when $d > s$.*

Proof. Choose one uniform s -subset T of $[m]$ and use it as the support of every column, with independent signs on its entries and normalization $s^{-1/2}$. Each column has the required marginal law. The whole matrix has rank at most s , so its restriction to a d -dimensional coordinate subspace is noninjective whenever $d > s$. $\square$

Thus exact marginal sparsity alone is insufficient. A specified dependent law requires a replacement for product orthogonality or a valid transfer comparison. Sparse graph matrices already have powerful model-specific analyses: Kothari, Potechin, and Xu obtain constant-factor estimates in an ultra-sparse setting [34], while Xu develops switching arguments for regular graphs [35]. These supply useful benchmarks for any extension of the present contraction lemmas to dependent laws.

9.4 Fixed-subspace control is not a simultaneous theorem

For a finite family of N predetermined d -subspaces, substituting δ/N in a fixed-subspace theorem and taking a union bound is valid. For RIP of order k , one may take the $\binom{n}{k}$ coordinate subspaces. The resulting moment order contains $\log(k \binom{n}{k} / \delta)$, and therefore includes the usual support entropy $k \log(en/k)$, since $\binom{n}{k} \leq (en/k)^k$.

The logical gap is real: every linear map with fewer rows than columns has a nonzero kernel vector, so no realization can preserve all vectors in the ambient space. A theorem for each fixed vector has different quantifiers from a theorem for all vectors simultaneously. Improved uniformity on a restricted family needs information about that family; no invariant-sector reduction by itself reverses these quantifiers.

9.5 Tensor order can create a genuine rare-event cost

Proposition 9.2 (Ordinary TensorSketch at small failure probability). *For the ordinary one-shot TensorSketch formed by cyclically convolving t independent two-coordinate CountSketch outputs, take $x = ((1, 1)/\sqrt{2})^{\otimes t}$. If its output has m coordinates, then*

$$m < \frac{2^t}{1 + \epsilon} \implies \mathbb{P}\{\|Sx\|^2 > 1 + \epsilon\} \geq 2^{-t}.$$

Proof. With probability 2^{-t} the two signs agree in every mode. On that event each mode's output has coordinate sum of magnitude $\sqrt{2}$, independently of its hashes. Convolution multiplies coordinate sums, so $|\sum_r (Sx)_r| = 2^{t/2}$. Cauchy-Schwarz yields $\|Sx\|^2 \geq 2^t/m$. $\square$

Thus for $\delta < 2^{-t}$ the row requirement is exponential in t for this sketch family, even for one fixed vector. The threshold $\delta < 2^{-t}$ specifies the confidence regime of the obstruction. Recursive constructions use a different architecture; see [18]. The proof identifies an event that any occupation estimate for this one-shot model must retain.

9.6 Adaptivity requires a conditional hypothesis

Proposition 9.3 (Reuse versus fresh randomness). *If a realized sketch $S \in \mathbb{R}^{m \times n}$ with $m < n$ is revealed before the input is chosen, an input depending on S can fail any nontrivial lower norm guarantee with probability one. In contrast, if at round j the input is measurable with respect to the past and the new sketch has, conditionally on that past, the prescribed independent law with failure at most δ_j for every fixed input, then all rounds succeed with probability at least $1 - \sum_j \delta_j$.*

Proof. For the first assertion choose a unit vector in $\ker S$. A measurable choice can be made by a deterministic elimination and normalization rule. For the second, condition on the past so that the current input is fixed. Apply the conditional guarantee, average, and use a union bound. $\square$

The first assertion concerns full revelation and arbitrary sketch-aware input choice. The second shows exactly how a fixed-input theorem can be reused with fresh randomness and a conditional hypothesis.

9.7 An independent-entry diagonal changes the sparsity scale

Replacing exact-column supports by independent entries creates an additional scalar fluctuation. For the coordinate vector e_i under three-point entries with density $1/b$ and scaling $s^{-1/2}$, take integers $b \geq 2$, $s \geq 1$, and distortion $\epsilon > 0$. Then

$$\|Se_i\|^2 = K/s, \quad K \sim \text{Bin}(bs, 1/b), \quad \text{Var}(K/s) = \frac{1 - 1/b}{s}.$$

An elementary atom bound makes the obstruction quantitative:

$$\mathbb{P}\{|K/s - 1| \leq \epsilon\} \leq \min \left\{ 1, \frac{2\epsilon s + 1}{\sqrt{s(1 - 1/b)}} \right\}. \quad (178)$$

To prove it, let $N = bs$ and $p = 1/b$. Fourier inversion for an integer-valued variable gives, for each integer k ,

$$\mathbb{P}\{K = k\} \leq \frac{1}{2\pi} \int_{-\pi}^{\pi} |1 - p + pe^{it}|^N dt.$$

Since $|1 - p + pe^{it}|^2 = 1 - 4p(1 - p)\sin^2(t/2)$ and $|\sin(t/2)| \geq |t|/\pi$ on this interval, the integrand is at most $\exp[-2Np(1 - p)t^2/\pi^2]$. Extending the integral to the real line bounds each atom by $\sqrt{\pi}/(2\sqrt{2Np(1 - p)}) \leq 1/\sqrt{Np(1 - p)}$. The interval $[s - \epsilon s, s + \epsilon s]$ contains at most $2\epsilon s + 1$ integers, proving (178). Fourier inversion here follows directly by integrating the finite expansion $(1 - p + pe^{it})^N e^{-ikt}$ term by term.

For fixed $b \geq 2$, the right side tends to zero when $s \rightarrow \infty$ and $\epsilon\sqrt{s} \rightarrow 0$. Thus an $s = O(1/\epsilon)$ prescription in this regime cannot give a fixed positive success probability. Exact-column sketches eliminate precisely this diagonal fluctuation.

Together these examples give a useful research boundary: preserving finite moments is broadly possible, but dependence, testing geometry, tensor order, conditioning, and diagonal fluctuation each require appropriate estimates.

10 Extensions of the operator and transfer principles

10.1 Coherent unitaries and quadratic confidence dependence

We extend the Walsh argument by controlling the actual coefficient kernels of two arbitrary complex unitaries. Let $A, B \in \mathbb{C}^{n \times n}$ be unitary, let $U \in \mathbb{C}^{n \times d}$ satisfy $U^*U = I_d$, and define

$$\alpha = n \max_{j,a} |A_{ja}|^2, \quad \beta = n \max_{a,i} |B_{ai}|^2, \quad \kappa = \max(\alpha, \beta).$$

For two independent families of independent unbiased signs, set $W = AD_yBD_xU$ and $w_j^* = e_j^*W$. The coefficient vectors are

$$(g_{ji})_a = \sqrt{n} \overline{A_{ja}B_{ai}}, \quad c_i = e_i^*U. \quad (179)$$

Expansion of the matrix products gives the row map $R_j = n^{-1/2} \sum_i c_i \otimes X_i \otimes Y(g_{ji})$ of Section 3.5.

Theorem 10.1 (Sampling after two coherent unitary transforms). *For independent Bernoulli(ρ) selectors, $0 < \rho < 1$, independent of the signs, put*

$$E_\rho = \rho^{-1}W^* \text{diag}(\eta)W - I_d, \quad m = \rho n, \quad D_q = \beta d + (8\beta + 4\alpha)q^2 \leq \beta d + 12\kappa q^2.$$

For every integer $q \geq 1$,

$$\mathbb{E} \text{tr}(E_\rho^{2q}) \leq d \left[6\sqrt{\frac{D_q}{m}} + 9\frac{D_q}{m} \right]^{2q}. \quad (180)$$

For $0 < \epsilon < 1$, $0 < \delta < 1/2$, choose

$$q = \lceil \log_4(4d/\delta) \rceil, \quad M = \min\{n, \lceil 1024\epsilon^{-2}D_q \rceil\}. \quad (181)$$

Let T be a uniform M -element subset independent of the signs. Define the coordinate restriction $R_T : \mathbb{C}^n \rightarrow \ell^2(T)$ by $(R_T z)_j = z_j$ for $j \in T$, and let $S = \sqrt{n/M} R_T A D_y B D_x$. Then

$$\mathbb{P}\{\|U^*S^*SU - I_d\| > \epsilon\} \leq \delta.$$

Both statements allow $\min(n, 4q)$ -wise independent unbiased signs in each family. The Bernoulli statement also allows selectors with $\min(n, 2q)$ -wise independent Bernoulli marginals, provided the three families remain mutually independent.

Proof. We first work with full independence. For a fixed j , synthesis in the vectors $(g_{ji})_i$ has matrix $\sqrt{n} \text{diag}(\overline{A_{j,:}})\overline{B}$, whose norm is at most $\sqrt{\alpha}$. For fixed i , row synthesis has matrix $\sqrt{n} \text{diag}(\overline{B_{:,i}})A^*$, whose norm is at most $\sqrt{\beta}$. Taking adjoints gives the corresponding analysis bound. Thus both Bessel hypotheses of Section 3.5 hold.

Let $K_T = \sum_{j \in T} R_j^* R_j$, and fix input grades (r, s) . The four channels have distinct output grade pairs, so the diagonal block $\Pi_{r,s} K_T \Pi_{r,s}$ is the sum of their positive Grams. Lemmas 3.16, 3.17, and A.2 give, with $p = (r-1)_+$, $v = (s-1)_+$,

$$n\|\Pi_{r,s} K_T \Pi_{r,s}\| \leq \beta[d + p + pv + (r+1)s] + \alpha|T|(v + s + 1). \quad (182)$$

The last lemma, including its complex coefficient factorizations, is proved in Appendix A. The right side of (182) is nondecreasing in $r, s, |T|$. For $r, s \leq 2q$ and $|T| \leq q$, it is at most

$$\beta[d + 8q^2] + 4\alpha q^2 = D_q \leq \beta d + 12\kappa q^2. \quad (183)$$

Indeed, at $r = s = 2q$ and $|T| = q$, one has $p + pv + (r + 1)s = 8q^2$ and $q(v + s + 1) = 4q^2$ exactly.

Let P_{sig} retain both sign grades at most $\min(2q, n)$. The row effects

$$A_j^{\text{eff}} = P_{\text{sig}} M_{w_j w_j^*} P_{\text{sig}}$$

are positive and sum to the identity on this space because $W^*W = I_d$ pointwise. The sign-grade support of a row Gram changes each grade by 0 or ± 2 , so every grade pair has at most nine neighbors. For a positive operator C , its block form satisfies

$$|\langle z_\gamma, C z_\lambda \rangle| \leq \langle z_\gamma, C z_\gamma \rangle^{1/2} \langle z_\lambda, C z_\lambda \rangle^{1/2};$$

apply Cauchy–Schwarz to $C^{1/2} z_\gamma$ and $C^{1/2} z_\lambda$. Using the diagonal bound D_q/n and $2ab \leq a^2 + b^2$, and summing over the at most nine neighbors, gives

$$L_q := \max_{|T| \leq q} \left\| \sum_{j \in T} A_j^{\text{eff}} \right\| \leq \min\{1, 9D_q/n\}. \quad (184)$$

Label the orthonormal basis $1, (\eta - \rho)/\sqrt{\rho(1 - \rho)}$ by $|0\rangle, |1\rangle$. With $a = |0\rangle\langle 1|$ and $N = |1\rangle\langle 1|$, centered selector multiplication is $\sigma(a + a^*) + \tau N$, where

$$\sigma = \sqrt{(1 - \rho)/\rho}, \quad \tau = (1 - 2\rho)/\rho.$$

Compress selector grade to at most q as well. By Lemma 3.12 and (184),

$$\|M_{\text{cut}}\| \leq 2\sigma\sqrt{L_q} + |\tau|L_q \leq 6\sqrt{D_q/m} + 9D_q/m.$$

Each multiplication raises either sign grade by at most two and the selector grade by at most one. The cutoff therefore retains the first q iterates from the vacuum exactly. Applying the multiplication identity of Theorem 2.3,

$$\mathbb{E} \text{tr}(E_\rho^{2q}) = \sum_{a=1}^d \|M_{\text{cut}}^q(e_a \otimes \Omega)\|^2,$$

which proves (180).

For fixed-size sampling, the case $M = n$ is exact by unitarity, including $n = 1$. Otherwise apply Lemma A.3, conditioned on the signs, to the Hermitian coefficients $w_j w_j^*$ and rescale both centered sums by ρ^{-1} , with $\rho = M/n$. This gives

$$\mathbb{E} \text{tr}[(U^* S^* S U - I_d)^{2q}] \leq d \left[2 \left(6\sqrt{D_q/M} + 9D_q/M \right) \right]^{2q}.$$

The choice $M \geq 1024D_q/\epsilon^2$ bounds this root by

$$\frac{3}{8}\epsilon + \frac{9}{512}\epsilon^2 \leq \frac{201}{512}\epsilon < \epsilon/2.$$

The spectral theorem and Markov give failure below $d4^{-q} \leq \delta/4$.

Finally, each error entry has degree at most two in each sign family and one in the selectors. In a trace product of length $2q$, at most $4q, 4q, 2q$ distinct coordinates from the respective families occur. Matching their joint marginals therefore matches the even trace moment exactly. For fixed-size sampling this degree argument changes only the signs; the row subset keeps its uniform law. $\square$

For flat pairs, $\alpha = \beta = 1$, so the allowance in (183) is exactly $d + 12q^2$. We obtain $O(\epsilon^{-2}[d + \log^2(1/\delta)])$ rows, capped at n , since $\log^2 d = O(d)$. Taking $A = B$ equal to the normalized discrete Fourier matrix gives this conclusion at every integer dimension.

Remark 10.2 (The catalog form of the two-round question, and its constants). The open-problem catalog of Townsend [43] states the two-round question in a fixed form: n a power of two, $1 \leq r \leq n$, $0 < \varepsilon < 1/2$, $\Omega = \sqrt{n/k} D_1 F D_2 F S$ with F the normalized Walsh matrix, D_1, D_2 independent Rademacher diagonals, S a uniformly random k -subset of coordinates sampled without replacement, and the width prescribed as $k = \min\{n, \lceil Cr/\varepsilon^2 \rceil\}$; it asks for a universal C such that Ω is a $(1 \pm \varepsilon)$ subspace embedding of every fixed r -dimensional subspace with probability at least 0.99. This is the fixed-size assertion of Theorem 10.1 with $A = B = H$, $\alpha = \beta = 1$, $d = r$, $\delta = 0.01$, and $R_T H D_y H D_x = S^\top F D_2 F D_1$: the sampling model, the normalization, the order of the two rounds and the event coincide. At $\delta = 0.01$ the moment order is $q = \lceil \log_4(400d) \rceil$, and

$$D_q = d + 12q^2 \leq 301d \quad (d \geq 1),$$

the ratio being largest at $d = 1$, where $q = 5$. The completion of the proof uses only $M \geq 1024D_q/\epsilon^2$, so it applies at the prescribed width, and the catalog question is answered with

$$C = 1024 \cdot 301 = 308\,224.$$

The machine-checked cubic statement, Theorem 5.5, gives the same conclusion with $D_q^{\text{old}} = d + 27q^3 \leq 3376d$ and $C = 8192 \cdot 3376 = 27\,656\,192$. Beyond the constant, both theorems carry the dependence on the failure probability explicitly, $M = O(\epsilon^{-2}[d + \log^2(d/\delta)])$ and $O(\epsilon^{-2}[d + \log^3(d/\delta)])$ respectively, allow $\min(n, 4q)$ -wise independent signs, and, in the quadratic form, arbitrary coherent unitary pairs.

Another proof of the catalog statement is given by Yang [44], by a high-order trace argument for large ranks and a Frobenius second-moment bound below a rank cutoff R_0 . Its explicit constant is $C = 200R_0 + 8196K_0$ with $R_0 = 2^{20000}$ and $K_0 = 576 \cdot 4^{24}$, that is $C \approx 2^{20008}$, which the author describes as uniform but not a practical prescription; the success probability is fixed at 0.99 and the range is $0 < \varepsilon < 1$. That development also reports a Lean 4 formalization of its prescribed-width theorem. The two arguments are of different kinds: the trace argument there counts walk contributions directly and absorbs a polynomial dependence on the rank into R_0 , whereas the sector contractions here keep the row coefficients collected, which is what produces a constant of the order 10^5 and the additive $\log^2(d/\delta)$ term. Neither proof has had external peer review.

Remark 10.3 (The identity inner transform). Coherence provides a sufficient bound on the kernels. Its limitations are visible by setting the inner transform $B = I$: then $\beta = n$ and $W = A D_y D_x U$, a single transform with an unbiased sign diagonal. If A is flat and $U = e_i$, every squared row magnitude is exactly $1/n$. Every nonempty fixed-size sample is therefore exact after rescaling, although the coherence prescription reaches the all-row endpoint. This identifies slack in the coefficient bound in a regime where the underlying multiplication representation remains exact.

10.2 Common-density strata and an exact signed barycenter

The transfer layer can change the support law while keeping the reference hollow operator unchanged. We make this precise for common-density stratifications.

Fix $m = bs$, with integers $b \geq 2$ and $s \geq 1$. For each input column i , partition $[m]$ into sets I_{ig} of sizes bs_{ig} , where the positive integers s_{ig} sum to s . The partitions may differ by column. In each stratum choose exactly s_{ig} positions uniformly and give them independent

signs. All strata and columns are independent. Write $X_{ri} \in \{0, \pm 1\}$ for this unnormalized array and $S = s^{-1/2}X$. Every column of S has squared norm one.

Lemma 10.4 (Signed stratum barycenter). *For a stratum of size bk there is a coupling of a uniform signed k -subset X and an independent-entry array ζ with $\mathbb{P}\{\zeta_r = 1\} = \mathbb{P}\{\zeta_r = -1\} = 1/(2b)$ such that*

$$\mathbb{E}[\zeta \mid X] = a_k X, \quad a_k = \frac{\mathbb{E} \min(K, k)}{k}, \quad K \sim \text{Bin}(bk, 1/b).$$

Moreover $a_k \geq a_0 := 1 - e^{-1}$. One can replace ζ by an array with the same independent-entry law whose conditional mean is $a_0 X$.

Proof. Draw the independent-entry array first, with support B and $K = |B|$. If $K \geq k$, choose a uniform k -subset T of B ; if $K < k$, adjoin a uniform $(k - K)$ -subset of the complement. Retain old signs on $B \cap T$ and assign independent signs to the new positions. For every fixed K , permutation symmetry makes T uniform among all k -subsets. Its signs are independent and unbiased, also conditionally on K , so K is independent of the final signed array. The intersection has size $\min(K, k)$. Given X , each position of its support is consequently retained from B with probability a_k . Its old sign, when retained, equals its sign in X . An old sign outside the final support still has mean zero, even conditionally on X : flipping that sign leaves the construction and X unchanged. This proves the conditional mean formula.

To bound a_k , independently assign each active position a uniform label in $[k]$. The number of distinct labels is at most $\min(K, k)$. A particular label is absent with probability $(1 - 1/(bk))^{bk}$. Thus

$$\mathbb{E} \min(K, k) \geq k[1 - (1 - 1/(bk))^{bk}] \geq k(1 - e^{-1}).$$

Finally multiply the entire proxy array by an independent sign τ with mean a_0/a_k . This changes its conditional mean to $a_0 X$. Its marginal distribution is unchanged: the symmetric product law of ζ is invariant under simultaneous negation. In particular its coordinates remain independent in that marginal law. $\square$

Theorem 10.5 (Transfer to common-density stratifications). *Let U be a real isometry with d columns. Let $E_X = U^* S^* S U - I_d$ for the stratified sketch above. Independently sample a proxy array ζ_{ri} with the three-point law in Lemma 10.4, and let*

$$Z_\zeta = \frac{1}{s} \sum_{r=1}^m \sum_{i \neq j} \zeta_{ri} \zeta_{rj} u_i u_j^*.$$

For every integer $q \geq 1$,

$$\mathbb{E} \text{tr} E_X^{2q} \leq a_0^{-4q} \mathbb{E} \text{tr} Z_\zeta^{2q}. \quad (185)$$

Consequently the parameter prescription in Theorem 5.1 also gives an oblivious subspace embedding for this stratified model.

Proof. Construct the equalized coupling independently in every stratum and column. Each proxy stratum still has a product distribution, and the strata are independent, so the full proxy array has the specified independent-entry law. Its conditional coordinate means are all $a_0 X_{ri}$. The column couplings are independent; conditionally on the entire target array they remain independent across columns. Hence, for $i \neq j$,

$$\mathbb{E}[\zeta_{ri} \zeta_{rj} \mid X] = a_0^2 X_{ri} X_{rj}.$$

Since the target diagonal is deterministic, $\mathbb{E}[Z_\zeta \mid X] = a_0^2 E_X$. Conditional Jensen for the convex even trace power, as in Lemma 2.6, proves (185).

The one external operator input of this paper enters here. The independent-entry hollow theorem of the SparseStack paper [1], proved there by the coupled operator argument that is formalized, is

$$\mathbb{E} \operatorname{tr} Z_\zeta^{2q} \leq d(3\beta_q)^{2q}, \quad \beta_q = (3 + \sqrt{2}) \left[\sqrt{\frac{d+2q+1}{m}} + \frac{d+2q+1}{m} + \frac{2q+1}{s} \right].$$

Thus the transferred moment root is at most $3a_0^{-2}\beta_q$. For $\Lambda = 90a_0^{-2}$ and the reference parameter prescription, this root divided by ϵ is at most

$$3a_0^{-2}(3 + \sqrt{2})(2/\Lambda + 1/\Lambda^2) < \frac{1}{3} + \frac{1}{540} < \frac{1}{2}.$$

Here we used $3 + \sqrt{2} < 5$ and $a_0^{-2} \geq 1$. The reference moment choice and Markov give the desired failure bound. $\square$

The transfer inequality (185) is proved here in full. Its consequence for the parameter prescription is the only statement of this paper whose proof invokes an operator estimate proved elsewhere, namely the moment bound for the independent three-point proxy; for SparseStack itself, the case $sig = 1$ with a common partition, Appendix B gives a direct proof that needs no proxy.

One stratum gives a uniform exact- s support. Taking s strata of size b gives SparseStack. The common density is what permits a single independent-entry proxy and a single conditional-mean factor across all coordinates, even when the partitions vary by column.

10.3 Direct SparseStack and an all-order constant certificate

Theorem 10.6 (Direct SparseStack constants). *For independent real SparseStack, let $d \geq 1$, $0 < \epsilon \leq 1$, $0 < \delta \leq 1$, and define*

$$q = \max\{1, \lceil \log_4(d/\delta) \rceil\}, \quad s = \left\lceil \frac{(17/2)(2q+1)}{\epsilon} \right\rceil, \\ b = \left\lceil \frac{81(d+2q+1)}{s\epsilon^2} \right\rceil, \quad m = sb.$$

For every fixed real isometry U ,

$$\mathbb{P}\{\|U^* S^* S U - I_d\| > \epsilon\} \leq \delta, \quad s < \frac{26.5q}{\epsilon}, \quad m < \frac{179(d+q)}{\epsilon^2}.$$

Proof. We use the direct local multiplier (212). Appendix B proves its complete channel inventory and all required bounds. The shared-factor Gram controls light creation; coordinate-frame compression controls the local heavy exchanges; even light parity in each block sharpens the marked heavy Grams. These give the grade envelopes (224)–(225) and the scalar vacuum comparison (226).

For the displayed parameters, the row-sum envelope is the positive increasing concave function (228). Its vacuum edge is at most $\epsilon/9$. The exact midpoint product (229) bounds $\exp \int_0^1 \log \Psi < 1/2$, and the displayed rational boundary inequality verifies Lemma 4.2. This proves $\mathbb{E} \operatorname{tr} E^{2q} < d(\epsilon/2)^{2q}$ at every $q \geq 2$. The vacuum edge handles $q = 1$. Markov and the ceiling estimates at the end of that appendix complete the proof. $\square$

The direct theorem has a base-four moment choice; the original SparseStack statement in Theorem 5.1 uses base two. The direct law eliminates the conditional-mean cost, while the parity and returning-path estimates remove additional losses inside its operator proof. The explicit constants depend on all three steps. The heavy-swap estimate in Appendix C can also be inserted into a comparison matrix retaining (L, H) separately; the uniform theorem above already follows from its simpler bound $\|H_0\| \leq L + H$.

10.4 One shape beyond signs: regular and sparse entries

The classification of Theorem 5.3 concerns uniform signs. This subsection follows one shape into two other entry laws with tools already proved: the local moment theorem (Theorem 4.11) for symmetric entries of regular moment growth, and the exact original-law representation of Section 2 for sparse signed Bernoulli entries. The shape is the triangle rx, ry, xy with boundaries $U = (r, x)$ and $V = (r, y)$. Its minimum separators are $\{r, x\}$ and $\{r, y\}$, each containing one edge, and it has no attached floating component, so Theorem 5.3 gives the order $\sqrt{n}$ for signs. The first theorem shows that for regular symmetric laws the same shape has the order $\sqrt{n}$ times the entry's moment at the logarithmic order, with explicit constants; the second shows that for sparse laws the polynomial exponent changes at $\rho = n^{-1/2}$, where a critical correction $\sqrt{\log n / \log \log n}$ appears, although the ambient graph still has degree of order $\sqrt{n}$. The regular-law proof uses the existing local moment theorem after conditioning. The sparse proof uses elementary support arguments at and below the critical internal degree, and the explicitly cited Bandeira–van Handel estimate in its denser regime. Neither requires the graph classification, a new cone estimate, or a general calculus.

10.4.1 The block structure

Assume $n \geq 3$; for $n \leq 2$ the triangle matrix is zero. Let $X_{ij} = X_{ji}$, $1 \leq i < j \leq n$, be independent copies of a real symmetric variable X with $\mathbb{E}X^2 = 1$ and all moments finite, and set $X_{ii} = 0$. By (118), with rows and columns indexed by ordered pairs of distinct labels,

$$\mathcal{T}_n[(i, j), (i', k)] = \mathbf{1}_{\{i=i'\}} \mathbf{1}_{\{|\{i, j, k\}|=3\}} X_{ij} X_{ik} X_{jk}. \quad (186)$$

With $N = n - 1$, $d^{(i)} = (X_{ij})_{j \neq i} \in \mathbb{R}^N$, $D_i = \text{diag}(d^{(i)})$ and $X^{(i)}$ the principal $N \times N$ minor of X without row and column i ,

$$\mathcal{T}_n = \bigoplus_{i=1}^n B_i, \quad B_i = D_i X^{(i)} D_i, \quad \|\mathcal{T}_n\| = \max_i \|B_i\|. \quad (187)$$

For fixed i the vector $d^{(i)}$ is independent of $X^{(i)}$, since they use disjoint edges; different blocks share entries and are not independent, and the proofs below never assume that they are.

Proof guide. For regular laws, first condition on one pinned row. The remaining minor has independent entries, so the local cone theorem applies after a triangular split. We then remove the conditioning, bound the maximum over roots without assuming block independence, and construct a lower witness using a large edge and independent reserved vertices. For sparse laws, the better route is to condition on support instead: this exposes signed neighborhood graphs, whose maximum degree and degeneracy control the norm. The two proofs use different information from the same exact block identity.

10.4.2 Regular symmetric entries

Write $\mu_r = \|X\|_{L_r}$ and assume the regularity (7), $\mu_{2r} \leq \alpha \mu_r$ for $r \geq 1$. The relevant scale is the entry's moment at the logarithmic order,

$$m_n = \mu_{2^{\lceil \log n \rceil}} = \|X\|_{L_{2^{\lceil \log n \rceil}}} :$$

$m_n = 1$ for signs, $m_n \asymp \sqrt{\log n}$ for standard Gaussians, $m_n \asymp \log n$ for the variance-one Laplace law, and in general $m_n \asymp_\alpha \mathbb{E} \max_{i < j \leq n} |X_{ij}|$, the maximum-entry scale of the matrix.

Theorem 10.7 (The triangle with regular symmetric entries). *Let $\hat{\Theta}_q < 2.31$ and $\chi_\alpha = \frac{7}{3} 2^{1/4} \alpha^2$ be the constants of Theorem 4.11.*

(i) *For every integer $q \geq 1$,*

$$\begin{aligned} (\mathbb{E} \|\mathcal{T}_n\|^{2q})^{1/(2q)} &\leq 2 \hat{\Theta}_q \alpha^{(q-1)/(2q)} (nN)^{1/(2q)} \\ &\times \left[N^{1/(4q)} \mu_{4q} (\sqrt{N} + 2(2q-1)^{1/4} N^{1/4} \mu_{4q}) + \chi_\alpha \sqrt{2q-1} \mu_{2q} N^{1/(2q)} \mu_{4q}^2 \right]. \end{aligned} \quad (188)$$

(ii) *Consequently, at $q = \lceil \log n \rceil$,*

$$(\mathbb{E} \|\mathcal{T}_n\|^{2q})^{1/(2q)} \leq 4.62 e^{5/4} \alpha^{3/2} \sqrt{n} m_n \left[1 + \frac{2(2q)^{1/4} \alpha m_n}{n^{1/4}} + \frac{\chi_0 e^{1/4} \alpha^3 \sqrt{2q} m_n^2}{\sqrt{n}} \right], \quad (189)$$

with $\chi_0 = \frac{7}{3} 2^{1/4}$, and for fixed $p > 0$ and $n \geq e^{p/2}$ the left side dominates $(\mathbb{E} \|\mathcal{T}_n\|^p)^{1/p}$. Since $m_n \leq \alpha \lceil \log n \rceil^{\log_2 \alpha}$ by (7), the bracket tends to 1.

(iii) *Let $\alpha > 1$ and $p > 0$, and let $n \geq n_0(\alpha)$, where $n_0(\alpha) = \max\{2\alpha^8, 512\alpha^4, 68\}$. With $k_\alpha = \max\{0, \lceil \log_2(10 \log \alpha) \rceil\}$,*

$$(\mathbb{E} \|\mathcal{T}_n\|^p)^{1/p} \geq \frac{(1 - e^{-1})^{1/p}}{4 \cdot 8^{1/p}} \alpha^{-k_\alpha} \sqrt{n} m_n. \quad (190)$$

For $\alpha = 1$ the entries are signs, $m_n = 1$, and (190) holds with $k_\alpha = 0$.

In particular $(\mathbb{E} \|\mathcal{T}_n\|^p)^{1/p} \asymp_{\alpha,p} \sqrt{n} m_n$ for every symmetric α -regular law and every fixed $p > 0$.

Proof. We first prove the all-order upper bound, then specialize its moment order. The lower bound is a separate construction: one edge supplies a large magnitude, and fresh edges supply a row with Euclidean norm of order $\sqrt{n}$. Throughout the upper-bound argument, q is fixed and $\|Z\|_{L_r}$ denotes a scalar or Euclidean-valued random-variable norm, whereas $\|B_i\|$ denotes an operator norm.

Step 1. Condition on the pinned row, not on the other blocks.

Fix a root i . Let

$$\mathcal{F}_i = \sigma(X_{ij} : j \neq i), \quad d = d^{(i)}, \quad D = D_i, \quad W = X^{(i)}.$$

The vector d is $\mathcal{F}_i$ -measurable. The minor W uses no edge incident to i , so it is independent of $\mathcal{F}_i$. This is the conditional independence that we need; independence between different root blocks is neither true nor required.

Write $W = L + L^\top$, where L is strictly upper triangular, and set $Y = DLD$. Once d is fixed, the nonzero entries

$$Y_{jk} = d_j d_k L_{jk}, \quad j < k,$$

are independent scalar multiples of the original entries. They are symmetric and α -regular, including when the multiplier is zero. We can therefore apply Theorem 4.11 conditionally. Write $\sigma_d(Y)$ and $R_d(Y; 2q)$ for the variance scale and weak moment calculated over L with d held fixed. The theorem gives

$$(\mathbb{E}[\|Y\|^{2q} \mid \mathcal{F}_i])^{1/(2q)} \leq N^{1/(2q)} \widehat{\Theta}_q \alpha^{(q-1)/(2q)} \{\sigma_d(Y) + \chi_\alpha R_d(Y; 2q)\}.$$

Splitting the minor was necessary: W itself has repeated entries across its diagonal, while L has genuinely independent entries.

Step 2. Compute the two conditional parameters.

For every row j ,

$$\sum_{k>j} \mathbb{E}[Y_{jk}^2 \mid \mathcal{F}_i] = d_j^2 \sum_{k>j} d_k^2 \leq \|d\|_\infty^2 \|d\|_2^2.$$

The same bound holds for each column. Thus

$$\sigma_d(Y) \leq \|d\|_\infty \|d\|_2.$$

This is the favorable variance geometry: one maximum magnitude and one Euclidean norm, rather than two maximum magnitudes.

For the weak moment, fix deterministic unit vectors s, t and put $c_{jk} = d_j d_k s_j t_k$. Represent L_{jk} in law as $\varepsilon_{jk} |L_{jk}|$, with independent signs independent of the magnitudes. Conditional Khintchine at order $2q$, followed by Minkowski in L_q , gives

$$\begin{aligned} \left\| \sum_{j<k} c_{jk} L_{jk} \right\|_{L_{2q}(W)} &\leq \sqrt{2q-1} \left\| \sum_{j<k} c_{jk}^2 |L_{jk}|^2 \right\|_{L_q(W)}^{1/2} \\ &\leq \sqrt{2q-1} \mu_{2q} \left(\sum_{j<k} c_{jk}^2 \right)^{1/2} \\ &\leq \sqrt{2q-1} \mu_{2q} \|d\|_\infty^2. \end{aligned}$$

The last step uses $\sum_{j<k} s_j^2 t_k^2 \leq 1$. Taking the supremum over s, t proves the same bound for $R_d(Y; 2q)$.

Finally, $B_i = DWD = Y + Y^\top$ and $\|B_i\| \leq 2\|Y\|$. We have proved

$$\begin{aligned} (\mathbb{E}[\|B_i\|^{2q} \mid \mathcal{F}_i])^{1/(2q)} &\leq 2N^{1/(2q)} \widehat{\Theta}_q \alpha^{(q-1)/(2q)} \Lambda(d), \\ \Lambda(d) &= \|d\|_\infty \|d\|_2 + \chi_\alpha \sqrt{2q-1} \mu_{2q} \|d\|_\infty^2. \end{aligned} \tag{191}$$

Here the dependence of Λ on the fixed order q is suppressed. The crude inequality $\|DWD\| \leq \|d\|_\infty^2 \|W\|$ would discard the distinction between the two terms in Λ .

Step 3. Separate the two random weights by Hölder.

Raise (191) to the power $2q$, average over the pinned row, and then take a $2q$ -th root. Define

$$A_q = \|\|d\|_\infty\|_{L_{4q}}, \quad B_q = \|\|d\|_2\|_{L_{4q}}.$$

Minkowski and Hölder on the law of d give

$$\|\Lambda(d)\|_{L_{2q}} \leq A_q B_q + \chi_\alpha \sqrt{2q-1} \mu_{2q} A_q^2.$$

No independence between $\|d\|_\infty$ and $\|d\|_2$ is asserted. There are N coordinates, so the maximum is controlled directly:

$$A_q^{4q} = \mathbb{E} \max_{j \neq i} |d_j|^{4q} \leq \sum_{j \neq i} \mathbb{E} |d_j|^{4q} = N \mu_{4q}^{4q}.$$

Therefore $A_q \leq N^{1/(4q)} \mu_{4q}$.

Step 4. Estimate the Euclidean norm by centering its square.

The coordinates of d have variance one. Set

$$Z_j = d_j^2 - 1, \quad \mathbb{E} Z_j = 0.$$

Then

$$B_q^2 = \left\| \sum_{j \neq i} d_j^2 \right\|_{L_{2q}} \leq N + \left\| \sum_{j \neq i} Z_j \right\|_{L_{2q}}.$$

To bound the centered sum, let Z'_j be independent copies and let ε_j be auxiliary independent signs. Conditional Jensen, symmetry of $Z_j - Z'_j$, and Minkowski imply

$$\begin{aligned} \left\| \sum_j Z_j \right\|_{L_{2q}} &\leq \left\| \sum_j (Z_j - Z'_j) \right\|_{L_{2q}} \\ &= \left\| \sum_j \varepsilon_j (Z_j - Z'_j) \right\|_{L_{2q}} \\ &\leq 2 \left\| \sum_j \varepsilon_j Z_j \right\|_{L_{2q}}. \end{aligned}$$

Khintchine conditional on the Z_j , followed by Minkowski in L_q , now gives

$$\left\| \sum_j \varepsilon_j Z_j \right\|_{L_{2q}} \leq \sqrt{2q-1} \left(\sum_j \|Z_j\|_{L_{2q}}^2 \right)^{1/2}.$$

Each term satisfies

$$\|Z_j\|_{L_{2q}} \leq \|d_j^2\|_{L_{2q}} + 1 = \mu_{4q}^2 + 1 \leq 2\mu_{4q}^2,$$

because $\mu_{4q} \geq \mu_2 = 1$. Combining the preceding inequalities,

$$B_q^2 \leq N + 4\sqrt{2q-1}\sqrt{N} \mu_{4q}^2.$$

Taking a square root and using $\sqrt{x+y} \leq \sqrt{x} + \sqrt{y}$ yields

$$B_q \leq \sqrt{N} + 2(2q-1)^{1/4} N^{1/4} \mu_{4q}.$$

This completes the estimate of the random pinned row.

Step 5. Take the maximum without assuming block independence.

The block identity (187) and exchangeability give

$$\mathbb{E} \|\mathcal{T}_n\|^{2q} = \mathbb{E} \max_{1 \leq i \leq n} \|B_i\|^{2q} \leq \sum_{i=1}^n \mathbb{E} \|B_i\|^{2q} = n \mathbb{E} \|B_1\|^{2q}.$$

This is an inequality for nonnegative numbers followed by expectation; it does not use any relation between different blocks. Substitution of the estimates for A_q and B_q into Steps 1–3 proves (188).

Step 6. Choose the logarithmic moment order.

Set $q = \lceil \log n \rceil$. Since $N < n$ and $q \geq \log n$,

$$(nN)^{1/(2q)} \leq e, \quad N^{1/(4q)} \leq e^{1/4}, \quad N^{1/(2q)} \leq e^{1/2}.$$

Regularity gives $\mu_{2q} = m_n$ and $\mu_{4q} \leq \alpha m_n$; also $\alpha^{(q-1)/(2q)} \leq \sqrt{\alpha}$ and $2\hat{\Theta}_q < 4.62$. Applying these bounds term by term to (188), and factoring out $4.62e^{5/4}\alpha^{3/2}\sqrt{n}m_n$, gives (189).

The asymptotic assertion needs one more observation. With $a = \lceil \log_2 q \rceil$, monotonicity and a applications of regularity starting at order two give

$$m_n = \mu_{2q} \leq \mu_{2^{a+1}} \leq \alpha^a \mu_2 \leq \alpha q^{\log_2 \alpha}.$$

Thus both remainder terms in the bracket of (189) are a fixed power of $\log n$ divided by a positive power of n , and tend to zero for fixed α . For a fixed $p > 0$, the additional condition $p \leq 2q$ permits $\|\mathcal{T}_n\|_{L_p} \leq \|\mathcal{T}_n\|_{L_{2q}}$. This proves part (ii).

Step 7. Obtain a usable maximum from independent candidate edges.

We now turn to the lower bound. First consider iid copies $\xi_1, \dots, \xi_m$ of X . Suppose $\alpha > 1$ and $r \geq 1$ satisfy $m \geq 4\alpha^{2r}$. Apply Paley–Zygmund to $Z = |\xi|^r$:

$$\begin{aligned} \mathbb{P}\{|\xi| \geq \mu_r/2\} &\geq \mathbb{P}\{|\xi|^r \geq \mu_r^r/2\} \\ &\geq \frac{1}{4} \frac{(\mathbb{E}|\xi|^r)^2}{\mathbb{E}|\xi|^{2r}} \\ &\geq \frac{1}{4} \alpha^{-2r}. \end{aligned}$$

The first inclusion uses $r \geq 1$. Independence across the candidates then yields

$$\mathbb{P}\left\{\max_{\ell \leq m} |\xi_\ell| < \mu_r/2\right\} \leq \left(1 - \frac{1}{4}\alpha^{-2r}\right)^m \leq e^{-m\alpha^{-2r}/4} \leq e^{-1}.$$

Consequently, for every $p > 0$,

$$\left\|\max_{\ell \leq m} |\xi_\ell|\right\|_{L_p} \geq \frac{1}{2}(1 - e^{-1})^{1/p} \mu_r.$$

This probability argument is valid even when $p < 1$.

Step 8. Pair the selected maximum with a fresh row energy.

Take $m = \lfloor n/4 \rfloor$ disjoint candidate edges

$$(i_\ell, j_\ell) = (2\ell - 1, 2\ell), \quad 1 \leq \ell \leq m,$$

and reserve the labels

$$K = \{2m + 1, \dots, n\}, \quad N_0 = |K| \geq n/2.$$

Let $\mathcal{F}_{\text{cand}}$ be generated by the candidate entries. Let J select the largest $|X_{i_\ell j_\ell}|$, with the smallest index chosen in case of a tie. The index J is $\mathcal{F}_{\text{cand}}$ -measurable.

For each candidate define

$$R_\ell^2 = \sum_{k \in K} X_{i_\ell k}^2 X_{j_\ell k}^2.$$

A restriction of column j_J in block B_{i_J} gives the deterministic lower bound

$$\|\mathcal{T}_n\| \geq \|B_{i_J} e_{j_J}\|_2 \geq |X_{i_J j_J}| R_J.$$

Every primitive edge used in the R_ℓ is different from every candidate edge. Hence, conditional on $\mathcal{F}_{\text{cand}}$, R_J has the same law as R_1 . Selecting a large candidate therefore does not bias the fresh row energy.

For $A_k = X_{i_1 k}^2 X_{j_1 k}^2$, independence and normalization imply

$$\mathbb{E}A_k = 1, \quad \mathbb{E}A_k^2 = (\mathbb{E}X^4)^2 = \mu_4^8 \leq \alpha^8.$$

The variables A_k , $k \in K$, are independent, so

$$\mathbb{E}R_1^2 = N_0, \quad \mathbb{E}R_1^4 = N_0\mu_4^8 + N_0(N_0 - 1).$$

A second application of Paley–Zygmund gives

$$\mathbb{P}\{R_1^2 \geq N_0/2\} \geq \frac{1}{4} \frac{N_0}{N_0 - 1 + \mu_4^8} \geq \frac{1}{8}, \quad N_0 \geq \alpha^8.$$

The hypothesis $n \geq 2\alpha^8$ ensures the last condition. On the event in this display, $R_J \geq \sqrt{n}/2$. Taking the conditional expectation of the deterministic lower bound therefore gives

$$\mathbb{E}[\|\mathcal{T}_n\|^p \mid \mathcal{F}_{\text{cand}}] \geq \frac{1}{8} \left(\frac{\sqrt{n}}{2} \right)^p \max_{\ell \leq m} |X_{i_\ell j_\ell}|^p.$$

Average once more. The lower bound has now been reduced to the independent maximum from Step 7.

Step 9. Compare the available maximum order with m_n .

Define the admissible integer order

$$r_* = \left\lfloor \frac{\log(m/4)}{2 \log \alpha} \right\rfloor.$$

It satisfies $m \geq 4\alpha^{2r_*}$. We verify that $r_* \geq 1$ and that only a bounded number of doublings separates it from $2\lceil \log n \rceil$.

For $n \geq 68$,

$$m/4 \geq (n - 4)/16 \geq n/17.$$

For $n \geq 289\alpha^4$ this implies

$$r_* \geq \frac{\log n - \log 17}{2 \log \alpha} - 1 \geq \frac{\log n}{4 \log \alpha}.$$

The threshold in the theorem implies both conditions, and also makes the final lower bound at least one. Moreover,

$$\frac{2\lceil \log n \rceil}{r_*} \leq 8 \log \alpha \left(1 + \frac{1}{\log n} \right) \leq 10 \log \alpha.$$

If the ratio is at most one, moment monotonicity suffices. Otherwise, iterating regularity from r_* shows

$$m_n \leq \alpha^{k_\alpha} \mu_{r_*}, \quad k_\alpha = \max\{0, \lceil \log_2(10 \log \alpha) \rceil\}.$$

Insert this comparison into Step 7 and then into Step 8. Taking a p -th root gives precisely (190).

When $\alpha = 1$, equality between the second and fourth moments of $|X|$ forces $|X| = 1$ almost surely; symmetry makes X a uniform sign. Every row of every root block then has Euclidean norm $\sqrt{n-2}$. This deterministic bound implies the stated weaker lower bound with $k_\alpha = 0$ and completes the proof. $\square$

For laws of exact power type $\|X\|_{L_q} \asymp q^{\theta/2}$ with $0 \leq \theta \leq 2$, the scale $\sqrt{n}(\log n)^{\theta/2}$ is also given by [30, Theorem 5.2] for the typed factor model, in which the three edges carry independent arrays (separator $\{r, x\}$ or $\{r, y\}$ with one factor supported on it and no active component). The coloring identity behind that paper's upper transfer is pointwise in the entries, so its upper bound carries over to the shared array; its stated lower transfer is for signs. This distinction is not by itself a novelty claim for the overlapping fixed-law scales: extensions of the sign-Fourier reduction must also be taken into account. Theorem 10.7 proves both halves directly for every α -regular law, with constants depending only on α and p and with the law's own moment m_n in place of a power of the logarithm. Its methodological content is the use of Theorem 4.11 inside a dependent polynomial matrix: condition on the pinned variables, split the symmetric minor into two independent-entry halves, keep the true variance geometry $\|d\|_\infty \|d\|_2$ of each half, and charge the pinned magnitudes through Hölder.

10.4.3 Sparse signed Bernoulli entries

Now let $X_{ij} = \rho^{-1/2} \varepsilon_{ij} B_{ij}$ for $i \neq j$, where $B_{ij} = B_{ji}$ are independent Bernoulli variables of parameter $\rho \in (0, 1]$ and $\varepsilon_{ij} = \varepsilon_{ji}$ are independent uniform signs independent of them; the support is a graph $G \sim G(n, \rho)$, and $\mathbb{E} X_{ij}^2 = 1$. This law has $\|X\|_{L_r} = \rho^{1/r-1/2}$ and $\|X\|_{L_{2r}} / \|X\|_{L_r} = \rho^{-1/(2r)}$, so its smallest regularity constant is $\alpha_\rho = \rho^{-1/2}$, which diverges as $\rho \rightarrow 0$; Theorem 10.7 applies at each fixed ρ but does not describe $\rho = \rho(n) \rightarrow 0$. The original-law representation does.

For a root i let $N(i) = \{j : B_{ij} = 1\}$, let A_i be the signed adjacency matrix of the graph induced by G on $N(i)$, $(A_i)_{jk} = \varepsilon_{jk} B_{jk}$ for $j, k \in N(i)$, and let $E_i = \text{diag}(\varepsilon_{ij} : j \in N(i))$. Since $X_{ij} X_{ik} \neq 0$ forces $j, k \in N(i)$, the nonzero part of the block B_i of (187) is $\rho^{-3/2} E_i A_i E_i$, and E_i is orthogonal, so

$$\|\mathcal{T}_n\| = \rho^{-3/2} S_n, \quad S_n = \max_i \|A_i\|. \quad (192)$$

Conditionally on the root's incident support, A_i is the signed adjacency matrix of $G(|N(i)|, \rho)$; the matrices for different roots are dependent. Define the largest number of common neighbors of an active edge,

$$\Delta_n = \max_{i < j} B_{ij} \sum_{k \notin \{i, j\}} B_{ik} B_{jk}, \quad (193)$$

which is exactly the largest degree among all neighborhood graphs $G[N(i)]$. The column norm and the largest absolute row sum give, for every support and every sign assignment,

$$\sqrt{\Delta_n} \leq S_n \leq \Delta_n, \quad (194)$$

with $S_n = \Delta_n = 0$ if G has no triangle. The key parameter is now visible: a neighborhood has about $n\rho$ vertices but conditional mean degree $\lambda_n = n\rho^2$, so the neighborhood graphs reach constant mean degree at $\rho = n^{-1/2}$, while the ambient graph still has degree of order $\sqrt{n}$.

Lemma 10.8 (Degeneracy converts degrees into a norm). *If a finite simple graph has maximum degree Δ and is k -degenerate (every subgraph has a vertex of degree at most k), then every signed adjacency matrix A on that support satisfies $\|A\| \leq 2\sqrt{k\Delta}$.*

Proof. Remove a minimum-degree vertex successively and orient every incident remaining edge away from it. The oriented signed matrix Q has absolute row sums at most k and absolute column sums at most Δ , so $\|Q\| \leq \sqrt{\|Q\|_{1 \rightarrow 1} \|Q\|_{\infty \rightarrow \infty}} \leq \sqrt{k\Delta}$, and $A = Q + Q^\top$. $\square$

Theorem 10.9 (The triangle with sparse entries: a one-shape phase diagram). *Let $\rho = n^{-\beta}$ with fixed $\beta \geq 0$, and let $p > 0$ be fixed. Then*

$$(\mathbb{E}\|\mathcal{T}_n\|^p)^{1/p} \asymp_{\beta,p} \begin{cases} n^{(1+\beta)/2}, & 0 \leq \beta < \frac{1}{2}, \\ n^{3/4} \sqrt{\log n / \log \log n}, & \beta = \frac{1}{2}, \\ n^{3\beta/2}, & \frac{1}{2} < \beta < 1, \end{cases} \quad (195)$$

and in each of these regimes the stated order also holds with probability tending to one. At $\beta = 1$, $(\mathbb{E}\|\mathcal{T}_n\|^p)^{1/p} \sim (1 - e^{-1/6})^{1/p} n^{3/2}$, and for $\beta > 1$, $(\mathbb{E}\|\mathcal{T}_n\|^p)^{1/p} \sim 6^{-1/p} n^{3\beta/2 + 3(1-\beta)/p}$; in these two regimes the matrix vanishes with probability bounded away from zero, respectively tending to one.

Proof. The exact identity (192) separates the deterministic normalization $\rho^{-3/2}$ from the random quantity

$$S_n = \max_i \|A_i\|.$$

We estimate S_n first and restore the normalization at the end of each regime. The argument has two different lower witnesses. Near and below the neighborhood transition it uses an edge with many common neighbors. When triangles are rare, it uses the event that at least one triangle exists.

Step 1. A tail bound for the maximum active codegree.

For a fixed unordered pair ij , the common-neighbor count is

$$C_{ij} = \sum_{k \notin \{i,j\}} B_{ik} B_{jk} \sim \text{Bin}(n-2, \rho^2).$$

It is independent of B_{ij} , because its primitive edges are all different from ij . The binomial tail bound $\mathbb{P}\{\text{Bin}(N, \pi) \geq t\} \leq (eN\pi/t)^t$ and a union bound therefore give

$$\mathbb{P}\{\Delta_n \geq t\} \leq n^2 \rho \left(\frac{en\rho^2}{t} \right)^t, \quad t \in \{1, 2, \dots\}. \quad (196)$$

The quantities C_{ij} need not be independent across different pairs. A union bound does not require them to be.

Step 2. At $\beta = 1/2$, identify the maximum degree scale.

Here $\rho = n^{-1/2}$ and $n\rho^2 = 1$. Put

$$\ell_n = \frac{\log n}{\log \log n},$$

for sufficiently large n . With $t = \lceil C\ell_n \rceil$, the logarithm of the right side of (196) is

$$\frac{3}{2} \log n + t(1 - \log t) = \left(\frac{3}{2} - C + o(1) \right) \log n.$$

Thus, for each prescribed $A > 0$, there is a constant C_A such that

$$\mathbb{P}\{\Delta_n > C_A \ell_n\} \leq n^{-A}$$

for all sufficiently large n .

For a lower bound, take $m = \lfloor n/4 \rfloor$ disjoint candidate edges (i_a, j_a) and reserve the other $N_0 = n - 2m \geq n/2$ vertices as a set K . Count common neighbors only in K :

$$C_a = \sum_{k \in K} B_{i_a k} B_{j_a k}.$$

The pairs $(B_{i_a j_a}, C_a)$ are independent across a . Although two candidates use the same reserved vertex labels, they use different primitive edges. Also $C_a \sim \text{Bin}(N_0, 1/n)$.

For $1 \leq t \leq N_0$, retaining the event $C_a = t$ gives

$$\begin{aligned} \mathbb{P}\{B_{i_a j_a} = 1, C_a \geq t\} &\geq n^{-1/2} \binom{N_0}{t} n^{-t} (1 - 1/n)^{N_0} \\ &\geq \frac{1}{4\sqrt{n}} (2t)^{-t}. \end{aligned}$$

We used $\binom{N_0}{t} \geq (N_0/t)^t$, $N_0 \geq n/2$, and $(1 - 1/n)^n \geq 1/4$. Let $t = \lfloor \ell_n/4 \rfloor$. The sum of the success probabilities over the candidates is at least

$$\frac{\sqrt{n}}{32} (2t)^{-t} = n^{1/4+o(1)},$$

which tends to infinity. Independence of the candidate experiments bounds the probability of no success by the exponential of minus this sum. A success supplies an active edge with at least t common neighbors. We have therefore proved

$$\Delta_n \asymp \ell_n \quad \text{with probability tending to one,}$$

with the polynomial upper-tail control stated above.

Step 3. At criticality, no neighborhood has a dense core.

The degree estimate alone would only give $S_n \leq \Delta_n$. To obtain a square root, we verify the degeneracy hypothesis of Lemma 10.8.

Let $\mathcal{E}_{\text{rmsize}}$ be the event that every root has at most $2\sqrt{n}$ neighbors. A binomial Chernoff bound gives

$$\mathbb{P}(\mathcal{E}_{\text{size}}^c) \leq ne^{-c\sqrt{n}}.$$

For a fixed root, condition on its full incident support. Its neighborhood graph then has law $G(N, n^{-1/2})$. When $N \leq 2\sqrt{n}$, fix an integer $k \geq 12$. A subgraph of minimum degree at least k would contain a vertex set of size $t \geq k+1$ spanning at least $kt/2$ edges. Counting vertex sets and then using the binomial tail, uniformly over this conditional neighborhood,

$$\mathbb{P}\{\text{such a set of size } t\} \leq \left[\frac{2e\sqrt{n}}{t} \left(\frac{et}{k\sqrt{n}} \right)^{k/2} \right]^t.$$

We now remove the conditioning for each root and take a union bound. With $a = k/2 - 1$ and $C_k = 2e(e/k)^{k/2}$,

$$\begin{aligned} &\mathbb{P}\{\mathcal{E}_{\text{size}} \text{ and some neighborhood has such a set}\} \\ &\leq n \sum_{t=k+1}^{\lfloor 2\sqrt{n} \rfloor} [C_k (t/\sqrt{n})^a]^t. \end{aligned}$$

This is still a union bound over dependent neighborhoods.

Split the sum into two ranges. For $t \leq n^{1/4}$, its base is at most $C_k n^{-a/4}$, so the sum over this range is $O_k(n^{-a(k+1)/4})$. For $t > n^{1/4}$, the base is at most

$$C_k 2^a = e(2e/k)^{k/2} < \frac{1}{2},$$

and the corresponding sum is at most $2 \cdot 2^{-n^{1/4}}$. Choose k so that $a(k+1)/4 > A+1$. Including the leading factor n and the event $\mathcal{E}_{\text{size}}^c$, we conclude that every neighborhood is $(k-1)$ -degenerate outside an event of probability at most n^{-A} , after increasing the lower threshold on n .

Step 4. Convert critical support geometry into norm moments.

By the orientation lemma and Steps 2–3,

$$S_n \leq 2\sqrt{(k-1)\Delta_n} \leq C_A\sqrt{\ell_n}$$

outside an event of probability at most n^{-A} . The deterministic lower bound $S_n \geq \sqrt{\Delta_n}$ gives the matching lower scale with probability tending to one.

For fixed $p > 0$, choose $A > p + 1$. Since $S_n \leq n$ for every realization,

$$\mathbb{E}S_n^p \leq (C_A\sqrt{\ell_n})^p + n^p n^{-A} \lesssim_p \ell_n^{p/2}.$$

The high-probability lower event gives the reverse inequality. This reasoning also applies when $p < 1$: it integrates nonnegative random variables and uses no subunit triangle inequality. Multiplication by $\rho^{-3/2} = n^{3/4}$ proves the critical line of (195).

Both the row-norm lower bound and the orientation upper bound hold for every sign assignment. The critical conclusion therefore remains valid even if the signs are chosen after the support is observed.

Step 5. Denser neighborhoods: $0 \leq \beta < 1/2$.

Now $\lambda_n = n\rho^2 = n^{1-2\beta}$ grows faster than $\log n$. With probability at least $1 - ne^{-cnp}$, all neighborhoods have size at most $2n\rho$. Conditional on a root support of size $N \leq 2n\rho$, A_i has independent symmetric upper-triangular entries, bounded by one, and row variance scale at most $\sqrt{2\lambda_n}$. The bounded-entry tail estimate in [31, Corollary 3.12] gives

$$\mathbb{P}\{\|A_i\| \geq 3\sqrt{2\lambda_n} + t \mid N(i)\} \leq Ne^{-t^2/C}$$

for an absolute constant C ; empty neighborhoods cause no difficulty. Choose $t = C_A\sqrt{\log n}$ and take a union bound over the roots. It follows that

$$S_n \leq C_A(\sqrt{\lambda_n} + \sqrt{\log n}) \lesssim_{\beta,A} \sqrt{\lambda_n}$$

with polynomial upper tails of any prescribed order.

For the lower bound, reuse the disjoint candidates and reserve from Step 2. A candidate is active with probability ρ , and $m\rho \rightarrow \infty$, so at least one is active with probability tending to one. Select the first active candidate using only the candidate indicators. Conditional on that selection, its reserved common-neighbor count is still $\text{Bin}(N_0, \rho^2)$, with mean at least $\lambda_n/2$. A binomial lower-tail bound makes this count at least $\lambda_n/4$ with probability tending to one. Therefore $S_n \geq \sqrt{\lambda_n}/2$ on that event.

The same truncation by $S_n \leq n$ as in Step 4 proves the fixed-moment comparison. Finally,

$$\rho^{-3/2}\sqrt{\lambda_n} = \sqrt{n/\rho} = n^{(1+\beta)/2}.$$

Step 6. Thinner neighborhoods, but many triangles: $1/2 < \beta < 1$.

For each $A > 0$, choose an integer

$$t > \frac{A + 2 - \beta}{2\beta - 1}.$$

Equation (196) is then $O_{\beta,t}(n^{-A})$, because its power of n is $2 - \beta - (2\beta - 1)t$. Hence $S_n \leq \Delta_n = O_{\beta,A}(1)$ with polynomial upper tails.

Let Z be the number of support triangles. Its mean and variance are

$$\begin{aligned}\mathbb{E}Z &= \binom{n}{3}\rho^3, \\ \text{Var } Z &= \binom{n}{3}\rho^3(1 - \rho^3) + 2\binom{n}{2}\binom{n-2}{2}(\rho^5 - \rho^6).\end{aligned}$$

Only pairs of triangles sharing an edge contribute a covariance; pairs sharing at most a vertex use disjoint primitive edges. For $\beta < 1$,

$$\frac{\text{Var } Z}{(\mathbb{E}Z)^2} = O\left(\frac{1}{n^3\rho^3} + \frac{1}{n^2\rho}\right) \rightarrow 0.$$

Thus a triangle exists with probability tending to one. Any one triangle gives $S_n \geq 1$. Together with the upper tail this proves $\|S_n\|_{L_p} \asymp_{\beta,p} 1$, and multiplication by $\rho^{-3/2}$ proves the third line of (195).

Step 7. The endpoint $\beta = 1$: a nonvanishing probability of zero.

Two deterministic facts connect the norm to Z :

$$\mathbf{1}_{\{Z>0\}} \leq S_n \leq \Delta_n \leq Z.$$

If no two triangles share an edge, every neighborhood graph is a matching. A nonempty signed matching has operator norm one, so on this event $S_n = \mathbf{1}_{\{Z>0\}}$.

At $\rho = 1/n$, we can verify the required Poisson calculation by factorial moments. For each fixed integer r , the ordered, vertex-disjoint r -tuples of triangles contribute

$$\frac{(n)_{3r}}{6^r} n^{-3r} \rightarrow (1/6)^r$$

to $\mathbb{E}(Z)_r$. Every other tuple of distinct triangles has a connected component containing at least two triangles and therefore more edges than vertices in that component. Its contribution is $O(n^{-1})$; there are only finitely many intersection patterns for fixed r . Hence $\mathbb{E}(Z)_r \rightarrow (1/6)^r$. Bonferroni's inequalities applied to $\mathbb{P}\{Z = 0\}$, followed by increasing the truncation order, yield

$$\mathbb{P}\{Z = 0\} \rightarrow e^{-1/6}.$$

The same factorial-moment calculation bounds every fixed ordinary moment of Z uniformly in n .

Let $\mathcal{E}_{\text{share}}$ be the event that two triangles share an edge. A union bound gives

$$\mathbb{P}(\mathcal{E}_{\text{share}}) \leq \binom{n}{2}\binom{n-2}{2}n^{-5} = O(n^{-1}).$$

For fixed $p > 0$, Hölder and the bounded higher moments show that $\mathbb{E}[Z^p \mathbf{1}_{\mathcal{E}_{\text{share}}}] \rightarrow 0$. Outside this event, $S_n^p = \mathbf{1}_{\{Z>0\}}$. Therefore

$$\mathbb{E}S_n^p \rightarrow 1 - e^{-1/6}.$$

Since $\rho^{-3/2} = n^{3/2}$, this proves the endpoint assertion.

Step 8. Beyond the endpoint: $\beta > 1$.

Now $\lambda_T = \binom{n}{3}\rho^3 \rightarrow 0$. For any fixed integer $r \geq 1$, expand Z^r as a sum over ordered r -tuples of triangles. Tuples using only one distinct triangle contribute exactly λ_T . A union of two or more distinct triangles has $v \geq 4$ vertices and at least v edges, because every vertex has degree at least two. Its contribution is at most

$$O(n^v \rho^v) = O((n\rho)^v) = o((n\rho)^3) = o(\lambda_T).$$

Only finitely many patterns occur for fixed r , so $\mathbb{E}Z^r = \lambda_T(1 + o(1))$.

Cauchy–Schwarz and Markov give

$$\frac{(\mathbb{E}Z)^2}{\mathbb{E}Z^2} \leq \mathbb{P}\{Z > 0\} \leq \mathbb{E}Z,$$

whence $\mathbb{P}\{Z > 0\} \sim \lambda_T$. Choose an integer $r \geq p$. Because Z is integer-valued,

$$\mathbb{P}\{Z > 0\} \leq \mathbb{E}S_n^p \leq \mathbb{E}Z^p \leq \mathbb{E}Z^r = \lambda_T(1 + o(1)).$$

Thus $\mathbb{E}S_n^p \sim \lambda_T$. Restoring the normalization gives

$$(\mathbb{E}\|\mathcal{T}_n\|^p)^{1/p} \sim \rho^{-3/2} \lambda_T^{1/p} = 6^{-1/p} n^{3\beta/2+3(1-\beta)/p}.$$

Here the norm is zero with probability tending to one; the displayed moment comes from rare occupied triangles, not from a typical nonzero matrix. This completes the proof. $\square$

Remark 10.10 (The support is a projector). For $0 < \rho < 1$, the original law of $X = \rho^{-1/2}\varepsilon B$ is supported on three points. In the orthonormal basis $\Omega = 1$, $e_1 = X$, $e_2 = (B - \rho)/\sqrt{\rho(1 - \rho)}$ of its L_2 space, multiplication by X is

$$J_\rho = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & \sqrt{(1 - \rho)/\rho} \\ 0 & \sqrt{(1 - \rho)/\rho} & 0 \end{pmatrix},$$

which is $\rho^{-1/2}$ times the matrix (3) of Example 2.1 at $b = 1/\rho$. Its eigenvalues are $0, \pm\rho^{-1/2}$, the support indicator is the projector $P_B = \rho J_\rho^2$, and $\langle \Omega, J_\rho^{2q} \Omega \rangle = \rho^{1-q}$, whereas $\|J_\rho\|^{2q} = \rho^{-q}$: a reused edge pays its support probability once, not once per visit. At $\rho = 1$, the support is deterministic and the appropriate basis is the two-state sign basis $(1, X)$, with support projector I . For several edges, let $m_e > 0$ denote the multiplicities of the edges that actually occur in a monomial. Its expectation is $\prod_{e: m_e > 0} \rho^{1-m_e/2}$ when all those multiplicities are even, and is zero otherwise; unused edges contribute the factor one. In the proof above, conditioning on the root's support projections selects $N(i)$ exactly, the root signs are removed by the orthogonal gauge E_i , and the remaining norm is a question about the conditional support graph: its row energies are the common-neighbor counts (193), and at criticality the orientation lemma converts them into a norm. This is the original-law viewpoint of Section 2 applied where a global regularity constant would be uninformative.

Remark 10.11 (What the two triangles say about other laws). For the triangle, a minimum separator contains one edge, so the natural extrapolation of Theorem 5.3 to a symmetric law with moment m_n at the logarithmic order is $\sqrt{n} m_n$. Theorem 10.7 proves this for every α -regular law. For the sparse law $m_n \asymp_\beta \rho^{-1/2}$, and $\sqrt{n} m_n \asymp \sqrt{n/\rho}$ is the truth only for $0 \leq \beta < \frac{1}{2}$: for $\frac{1}{2} < \beta < 1$ the full separator $\{r, x, y\}$, which is not of minimum cardinality but contains all three edges, supplies the larger typical scale $\rho^{-3/2}$. At and above $\beta = 1$, occupancy of the whole triangle must also be retained: the fixed moments have the separate rare-event asymptotics in Theorem 10.9. At the tie $\beta = \frac{1}{2}$ the critical factor $\sqrt{\log n / \log \log n}$

appears, which no assignment of logarithms to attached floating components can produce because there are none. In the moment-weighted cut cost $|S| - \beta|E(S)|$ of [30, Proposition 5.4] the two candidates cost $2 - \beta$ and $3 - 3\beta$; they tie exactly at $\beta = \frac{1}{2}$, where that proposition's hypothesis that every minimizing cut has minimum cardinality fails, so the critical correction is consistent with it, and the polynomial scales on both sides agree with the sparse bounds of Kothari, Potechin and Xu [34], which maximize over all separators. The single-matrix critical correction at $\rho = 1/n$ is classical and is recorded as a calibration in [30, Proposition 5.7]; here it arises inside a degree-three graph matrix at a different ambient density, because the matrices exposed by conditioning on the shared boundary are sparser than the ambient graph.

10.4.4 The trace words of the triangle and limited independence

The block structure (187) has an exact counterpart at the level of trace moments. Since $B_i = D_i X^{(i)} D_i$ and the trace is cyclic, $\text{tr } B_i^{2q} = \text{tr}(D_i^2 X^{(i)})^{2q}$: the pinned row enters every moment only through the squares X_{ij}^2 . Written out, with $j_{2q+1} = j_1$,

$$\text{tr } \mathcal{T}_n^{2q} = \sum_{i=1}^n \sum_{\substack{j_1, \dots, j_{2q} \neq i \\ j_\ell \neq j_{\ell+1}}} \left(\prod_{\ell=1}^{2q} X_{ij_\ell}^2 \right) \left(\prod_{\ell=1}^{2q} X_{j_\ell j_{\ell+1}} \right). \quad (197)$$

A word is a closed walk of length $2q$ in the neighborhood of the root i , avoiding i and without immediate repetition, and each visited vertex charges its root edge twice. For the sparse law this is the trace form of Remark 10.10: a root edge reused by the walk pays its support once, $B_{ij}^2 = B_{ij}$, and the $2q$ root factors of a word involve only as many support variables as the walk has distinct vertices.

The words also show how much independence the moments use. Proposition 4.15 transfers a trace moment to every law with the product marginals on the sets of coordinates that a single word can occupy.

Corollary 10.12 (Limited independence for the triangle). *Let $q \geq 1$ be an integer, let $\hat{X} = (\hat{X}_{jk})$ be a symmetric random array with zero diagonal, let $\hat{\mathcal{T}}_n$ be the matrix (186) built from it, and put $N_e = \binom{n}{2}$.*

- (i) *If the entries $\hat{X}_{jk}$, $j < k$, have a common marginal law with finite moments and are $\min\{N_e, 4q\}$ -wise independent, then $\mathbb{E} \text{tr } \hat{\mathcal{T}}_n^{2q} = \mathbb{E} \text{tr } \mathcal{T}_n^{2q}$, the value for independent entries of that law. If the law is the symmetric α -regular law of Section 10.4, then (188) holds for $\hat{\mathcal{T}}_n$ with an additional factor $N^{1/(2q)}$ on the right, and (189) holds with $4.62 e^{5/4}$ replaced by $4.62 e^{7/4}$.*
- (ii) *For signs, $\min\{N_e, 2q\}$ -wise independence suffices in (i); at $q = 1$ none is needed, since $\text{tr } \mathcal{T}_n^2 = n(n-1)(n-2)$ identically.*
- (iii) *Let $\hat{X}_{jk} = \rho^{-1/2} \hat{\varepsilon}_{jk} \hat{B}_{jk}$ with a sign family independent of a support family, the signs uniform and $\min\{N_e, 2q\}$ -wise independent, and the supports $\min\{N_e, 2q+1\}$ -wise independent with Bernoulli(ρ) marginals. Then $\mathbb{E} \text{tr } \hat{\mathcal{T}}_n^{2q}$ equals its value for the independent law of Theorem 10.9.*

The generic count for a degree-three polynomial matrix is $6q$.

Proof. A word in (197) uses at most $2q$ distinct root edges and at most $2q$ distinct internal edges, hence at most $4q$ entries of $\hat{X}$, with their powers. Its expectation is the same under every law that agrees with the product law on each set of $4q$ entries, and Proposition 4.15

gives the first claim of (i). For the second claim, the proof of Theorem 10.7 bounds the block moments under the product law: by (191) and $\text{tr } B_i^{2q} \leq N \|B_i\|^{2q}$,

$$\mathbb{E} \text{tr } \mathcal{T}_n^{2q} = \sum_{i=1}^n \mathbb{E} \text{tr } B_i^{2q} \leq nN(2N^{1/(2q)} \hat{\Theta}_q \alpha^{(q-1)/(2q)})^{2q} \mathbb{E} \Lambda(d)^{2q},$$

and $\|\Lambda(d)\|_{L_{2q}}$ is at most the bracket of (188), as shown there. Since $\|\hat{\mathcal{T}}_n\|^{2q} \leq \text{tr } \hat{\mathcal{T}}_n^{2q}$, the transferred moment gives the stated bound, and $N^{1/(2q)} \leq e^{1/2}$ at $q = \lceil \log n \rceil$. For (ii) the root squares are one, so a word is a monomial in at most $2q$ internal signs. For (iii), average the signs first: under $2q$ -wise independence a word has sign expectation one if every internal edge is used an even number of times and zero otherwise, exactly as under the product law. A surviving walk uses at most q distinct internal edges; its graph is connected, so it visits at most $q + 1$ vertices, and the word is ρ^{-3q} times a product of at most $q + (q + 1)$ distinct support indicators, whose expectation is fixed by the $(2q + 1)$ -wise marginals. $\square$

At $n = 5$ and $q = 2$ the counts are exact in the following sense. The product law of signs gives $\mathbb{E} \text{tr } \mathcal{T}_5^4 = 300$, and so does every 4-wise independent sign law, whereas conditioning the four signs of one 4-cycle to have product one, a 3-wise independent law, gives 308; after averaging the signs, unnormalized Bernoulli($\frac{1}{2}$) supports give 15, and a 4-wise independent law that fixes the parity of the five supports of one word gives $\frac{59}{4}$ or $\frac{61}{4}$. What the corollary transfers is the unconditioned trace moment and the upper bounds derived from it. The lower bound (190) and the lower events of Theorem 10.9 use full independence, and the upper half of Theorem 10.9 is proved through the exact representation (192) and neighborhood degrees rather than through trace moments, so neither is transferred by this route; nor is a short seed for the support family, which is a separate sampling problem. Under the sparse law the transferred moments are those of Remark 10.10, and a bound on them at $q \asymp \log n$ sharp enough to reproduce the critical factor of (195) is not claimed.

11 Formalization

11.1 Formalized cores and analytic extensions

We have formalized the three original proof cores in Lean 4, and the upper half of the regular-entry theorem in its earlier version. Each record below names the repository, the checked source commit, the toolchain, the literal declarations and the mathematical statement they are reported to certify. Build and axiom-audit instructions are supplied by the pinned companion records. No new Lean replay was performed for this editorial audit. The cited records report transitive axiom audits of the named declarations using only `propext`, `Classical.choice` and `Quot.sound`, and the developments contain no `sorry` outside the intentional Challenge placeholders noted. Formal verification here means that the stated formal propositions have complete proof terms accepted by the Lean kernel; it does not mean that a proof assistant has checked the prose of this paper or the correspondence between the formal statements and the informal ones, which is a human comparison.

SparseStack. Repository <https://github.com/DiarHaidary/nelson-nguyen-sparse-fock>, commit `b9da2b5d96e5fd70252e9c6551aefc92643e7d0b`, Lean 4.33.0, Mathlib commit `db584cd6d46c92f209a44c0f1c829460d327499d`; the artifact builds with `cd formal && lake build`. The declaration `SparseFock.MainTheorem.fully_independent_sparseStack_matrix` (`formal/SparseFockFormal/MainTheorem.lean`) has the hypotheses $d \geq 1$, $0 < \epsilon \leq 1$, $0 < \delta \leq 1$, a real matrix U and $U^\top U = I_d$, and asserts, for the fully independent real SparseStack law with the prescription of Theorem 5.1, the failure bound $\mathbb{P}\{\|U^* S^* S U - I_d\| > \epsilon\} \leq \delta$, the two-sided embedding event, exact column sparsity, $b \geq 2$,

$m = sb$ and the displayed size bounds. The formalized proof is the coupled argument recorded in Section 5.1, not the direct proof of Appendix B.

Two-round Walsh sampling. Repository <https://github.com/DiarHaidary/Rerandomized-Subsampled-Trigonometric-Transforms>, source commit f813b5c3c14ddda0cd0dc429e657b0dbe6f94079, verification run 34373116757 of 9 September 2026, Lean 4.33.0, Mathlib commit db584cd6. The declarations `SRHT.two_round_srht_ose` and `SRHT.two_round_srht_norm_failure` state the fixed-size assertion of Theorem 5.5, with the cubic allowance, the constant 8192, and sign families matching independent unbiased signs on at most $\min(n, 4q)$ coordinates; `SRHT.BernoulliGeneral.moment_bound` and `SRHT.BernoulliGeneral.moment_bound_limited` state its Bernoulli moment bound and the limited-independence form with the orders $\min(n, 4q)$, $\min(n, 4q)$, $\min(n, 2q)$; `SRHT.two_round_srht_ose_log_confidence` states the combined row bound. The selected theorem `RerandomizedSTT.problem_5_6`, with fully independent signs and the prescribed width, additionally passed Comparator against an independent Challenge statement and an independent NanoDa replay; it is registered as Palomar entry PALOMAR-2026-09-09-000003 at commit 929653a019eb4910fc6efb9cb34cc3c3c2f949d4. The quadratic and coherent statements of Theorem 10.1 are not formalized.

Graph matrices. Repository <https://github.com/DiarHaidary/Sharp-Bounds-for-Graph-Matrices>, source commit 01994362f8543743d556e7544d2538e93c62c02b, verification workflow 7, run 34267577829 of 8 September 2026, Lean 4.33.0 with pinned Mathlib; Palomar registry entry PALOMAR-2026-09-08-000005. The declarations `GraphMatrices.Classification.full_dense_sign_classification` (Challenge.lean and Solution.lean, with the proof in `GraphMatrices/FullClassification.lean`), `GraphMatrices.Classification.expected_dense_sign_classification` and `GraphMatrices.Classification.classification_logarithmic_exponent` state the first assertion of Theorem 5.3: for every fixed shape and every fixed $p > 0$, the L_p size of $\|M_{H,n}\|$ is within shape-dependent constants of $n^{(h+i_0-s)/2}(\log(2n))^{b_*/2}$, together with its expected-norm specialization and the half-integrality of the logarithmic exponent. The run included Comparator’s comparison of the independent Challenge and Solution statements with their definition dependencies, an independent NanoDa proof replay and a kernel replay. The strong-moment statement with its uniform logarithmic range, and the limited-independence transfer, are not formalized in full. The analytic proof of both halves is Appendix F.

Regular entries. Repository <https://github.com/DiarHaidary/Spectral-norms-of-independent-entries-with-regular-moment-growth>, commit 762bd5ec5050a96f5e6ba3926b6cda4816fcd4b0, Lean 4.33.0 (leanprover/lean4:v4.33.0), Mathlib commit db584cd6d46c92f209a44c0f1c829460d327499d; Palomar registry entry PALOMAR-2026-09-14-000006, version 1, checked on 14 September 2026 (workflow run 34852526384 of PalomarRegistry/PalomarSubmission). The declaration `MI32.main_upper`, stated in Challenge.lean with the intentional placeholder and proved in Solution.lean through `exists_upperBoundAt` of `MI32/SymmetricDeletion.lean`, asserts: for every $\alpha \geq 1$ there is $C > 0$ such that $\mathbb{E}\|X\| \leq C\{M(X) + D(X)\}$ for every $n \geq 1$ and every real $n \times n$ matrix X with independent centered entries with finite absolute moments satisfying (7). The proof supplies the explicit Lean witness

$$\text{SymmetricDeletion.deletionConstant } (2\alpha) \cdot \exp(1).$$

Its axiom report is `[propext, Classical.choice, Quot.sound]`. This identifies a term defined in the pinned source, not the newer analytic decimal constant in this manuscript.

The extensions proved in this paper are analytic: the quadratic and coherent two-round bounds, the abstract frame and positive-effects lemmas, the commuting-transport bridge and reflection budget, the direct SparseStack constants, and the symmetry and path refinements.

Their proofs terminate in explicit finite identities and, for the uniform constants, rational inequalities. The complete analytic argument accompanies those certificates here.

The occupied-span, cyclic-child and weighted-sign arguments are analytic as well, and so are the two one-shape theorems of Section 10.4 with their trace-word corollary, the ordered renewal comparison of Section 4.3, the descent, live-register, signed-core and cyclic-seed statements of Section 7, and the quadratic energy identity and aggregate-moment transfer of Sections 8.4 and 8.9. Their internal proof reviews, and the finite checks used to test their algebra, are not a formal verification of these results.

The formal statement of the regular-entry development fixes the moment functional, the operator norm, and M and D as ordinary definitions over an arbitrary probability space, with one deterministic set deleted from both axes and the order $\log(k + 1)$ unchanged; it quantifies over every dimension and every matrix of independent centered entries with finite absolute moments satisfying (7), and asserts the existence of a constant. Its transitive axiom audit reports only `propxt`, `Classical.choice` and `Quot.sound`. Three deliberate differences from the analytic argument are recorded there: the certified cone comparison carries $H_\alpha = \sqrt{3} \alpha^2$ in place of the $\sqrt{3} \alpha$ that Lemma 3.21 gives; the dimension factor in Theorem 4.11 is $(2n)^{1/(2q)}$ rather than $n^{1/(2q)}$; and the symmetrization prefactor is e rather than the 2 obtained at the end of Section 5.4 by the second bound at budget one. None changes the structure of the argument. The literal certified witness is the Lean term displayed above. Equation (134) is an analytic reference formula; it should not be read as an extraction of that Lean term by implicit substitutions. Matching the definitions and constants is a separate statement-level comparison. The analytic constants displayed in Sections 3.6, 4.4.1 and 5.4, in particular the spherical thin-axis estimate, the parity majorant, the order-adapted doubling and the variance-first assembly with its moment policy, are not formalized. The lower half of the theorem is the known result and is not formalized, and the formal development has not been independently re-verified.

11.2 Reusable interfaces

The proof separates into five interfaces, with the admissible domain made explicit at each stage. A probability interface constructs the local weighted inner products and multiplication matrices. A grading interface proves the vacuum identity and exact cutoff. A coefficient interface proves insertion-Gram, commutator, positive-map or polynomial-cone bounds, and states the domain preserved by further interactions. For the cone argument, the actual iterates and localization masks must retain nonnegative original monomial coefficients, with ordinary degree tracked separately from parity. A moment interface converts a nonnegative path majorant or a sequence of cone contractions into a moment bound; a tail claim then requires its own distributional step. A transfer interface uses conditional barycenters or the stated symmetrization comparison to change the probability law. Arbitrary signed coefficient maps need not preserve the cone domain.

This separation identifies the mathematical work in each extension. Coherent transforms change the complex coefficient contractions while preserving the selector and moment interfaces. Direct SparseStack changes the local factor to the exact $2b$ -point law and adds parity restrictions before the scalar path estimate. The Gaussian variant uses Hermite recurrences and finite spectral measures for each fixed moment. A formal proof can therefore state each interface once and instantiate its local hypotheses in the application.

Supplementary material. The four Lean developments are public at the repositories named above, at the commits named there. All analytic arguments used here, including the four application proofs (Appendices A, B, F and G), the coefficient calculations and the rational

constant certificates, are contained in this manuscript and its appendices. External inputs are attributed where used: these include the general Boolean hypercontractive comparison, the known lower regular-entry comparison, the dense-regime independent-entry norm estimate for the sparse triangle, and the operator input in Section 10.2. The author release should include the checks named in the text: `regular_constants_check.py`, `triangle_checks.py`, `sparse_triangle_checks.py` and `trace_word_checks.py`. Those four files were not supplied with the source audited here. This revision instead includes a separate, explicitly scoped `v11_audit_checks.py` and its output; these finite checks are not a Lean replay or a substitute for the general proofs.

Acknowledgments

I used ChatGPT and Claude in developing and organizing the proofs, exploring examples, preparing calculations, and revising the exposition. The two one-shape theorems of Section 10.4, their trace-word corollary, and the readout example of Section 8.3 were proposed and audited in that exchange. The separate finite-check bundle and its limitations are identified immediately above. I directed this work and take responsibility for the mathematical statements and the distinction between formalized results and analytic extensions.

A Coefficient kernels and finite sampling for two-unitary rows

We prove the double-annihilation and sampling estimates used in Theorem 10.1. All Hilbert spaces below are finite dimensional. The occupation operators and complex convention are those of Section 3.5. In particular, $a_y(g) = \sum_a \bar{g}_a a_{y,a}$.

A.1 Lifting a coefficient kernel

Lemma A.1 (A retained-coordinate lift). *Let $\mathcal{H}$ be an auxiliary Hilbert space, and let $K = (K_{li})_{l,i \in [n]}$ be an operator on $\mathcal{H} \otimes \mathbb{C}^n$, written as a matrix of operators on $\mathcal{H}$. On occupation grade p , define*

$$\mathcal{L}_p(K) = \sum_{l,i} K_{li} \otimes a_l^* a_i.$$

Then $\|\mathcal{L}_p(K)\| \leq p\|K\|$. If a kernel acts on $\mathcal{H} \otimes \mathbb{C}_x^n \otimes \mathbb{C}_y^n$, its corresponding two-species lift has norm at most $pv\|K\|$ on grades (p, v) .

Proof. Carry $\mathcal{H}$ along in the extraction map E_p of Lemma 3.15. Expanding on occupation basis vectors gives

$$\mathcal{L}_p(K) = E_p^*(K \otimes I)E_p, \quad E_p^*E_p = pI,$$

with the tensor factors placed in the displayed order. The first bound follows. For two species, extract one coordinate from each. The resulting map $E_{p,v}$ satisfies $E_{p,v}^*E_{p,v} = pvI$, and the double lift is $E_{p,v}^*(K \otimes I)E_{p,v}$. This also proves the claim when one grade is zero. $\square$

A.2 The complex double-annihilation estimate

Let $A, B, U, g_{ji}, \alpha, \beta$ be as in Section 10.1, and set

$$P = UU^*, \quad c_i = e_i^*U, \quad \ell_i = P_{ii}.$$

Thus P and $P^\top = \bar{P}$ are positive Hermitian projections, $0 \leq \ell_i \leq 1$, and $\sum_i \ell_i = d$. For a row set T , collect

$$R_j^{--} = n^{-1/2} \sum_i c_i \otimes a_{x,i} \otimes a_y(g_{ji}) \quad (j \in T)$$

as an operator from $\mathbb{C}^d \otimes \mathcal{X}_r \otimes \mathcal{Y}_s$ to $\ell^2(T) \otimes \mathcal{X}_{r-1} \otimes \mathcal{Y}_{s-1}$.

Lemma A.2 (Collective double annihilation). *For $r, s \geq 1$, put $p = r - 1$, $v = s - 1$, and $t = |T|$. Then*

$$n\|R_T^-\|^2 \leq \beta(d + p + pv) + \alpha tv. \quad (198)$$

If either input grade is zero, the map is zero. Replacing p, v by $(r - 1)_+, (s - 1)_+$ gives a nonnegative bound valid for every input grade.

Proof. We give the output Gram and its exact normal ordering, retaining the one- x term together with its occupation correction. The output Gram $nR_T^-(R_T^-)^*$ has row blocks

$$\mathcal{G}_{jk} = \sum_{i,l} P_{il} a_{x,i} a_{x,l}^* \otimes a_y(g_{ji}) a_y(g_{kl})^*, \quad j, k \in T, \quad (199)$$

restricted to grades (p, v) . This identity follows by multiplying the row maps and using $c_i c_l^* = P_{il}$.

Write $N_{x,i} = a_{x,i}^* a_{x,i}$, $S_i = I - 2N_{x,i}$, and similarly $N_{y,a}$, S_a^y . On one site, $aa^* = I - N$ and $a^*a = N$; operators at distinct sites commute. Hence

$$a_{x,i} a_{x,l}^* = a_{x,l}^* a_{x,i} + \mathbf{1}_{i=l} S_i, \quad (200)$$

$$a_y(g) a_y(h)^* = a_y(h)^* a_y(g) + \langle g, h \rangle I - 2 \sum_a \bar{g}_a h_a N_{y,a}. \quad (201)$$

Multiply these two identities in (199). The result is the exact five-term expansion

$$\mathcal{G} = \mathcal{E} + \mathcal{X} - \mathcal{F} + \mathcal{Y} + \mathcal{Z}, \quad (202)$$

whose blocks are

$$\begin{aligned} \mathcal{E}_{jk} &= \sum_{i,a} \ell_i \overline{(g_{ji})_a} (g_{ki})_a S_i \otimes S_a^y, \\ \mathcal{X}_{jk} &= \sum_{i,l} P_{il} \langle g_{ji}, g_{kl} \rangle a_{x,l}^* a_{x,i} \otimes I, \\ \mathcal{F}_{jk} &= 2 \sum_{i,l,a} P_{il} \overline{(g_{ji})_a} (g_{kl})_a a_{x,l}^* a_{x,i} \otimes N_{y,a}, \\ \mathcal{Y}_{jk} &= \sum_i \ell_i S_i \otimes a_y(g_{ki})^* a_y(g_{ji}), \\ \mathcal{Z}_{jk} &= \sum_{i,l} P_{il} a_{x,l}^* a_{x,i} \otimes a_y(g_{kl})^* a_y(g_{ji}). \end{aligned}$$

We now bound these terms on the output grades, grouping $\mathcal{X} - \mathcal{F}$ before taking its norm.

The double commutator. On the full row register put $v_a = Ae_a$. These vectors are orthonormal. Substitution of (179) gives

$$\mathcal{E} = \sum_a |v_a\rangle \langle v_a| \otimes \left(\sum_i n \ell_i |B_{ai}|^2 S_i \right) \otimes S_a^y,$$

followed by compression of the row register to T . Before compression, this is a direct sum over a . The middle factor has norm at most $\sum_i n \ell_i |B_{ai}|^2 \leq \beta d$, and $\|S_a^y\| \leq 1$. Compression is contractive, so

$$\|\mathcal{E}\| \leq \beta d. \quad (203)$$

The one- x term and its occupation correction. Define one-particle matrices

$$D_a = \text{diag}_i(\sqrt{n}B_{ai}), \quad C_a = D_a^* P^\top D_a.$$

Each C_a is positive semidefinite, with $\|C_a\| \leq \beta$. Its entry with output l and input i is $n\overline{B_{al}}P_{il}B_{ai}$. These are precisely the auxiliary-mode blocks in Lemma 3.3, for the diagonal transports displayed in Section 3.2; this also explains the transpose of P . Since

$$\langle g_{ji}, g_{kl} \rangle = n \sum_a A_{ja} \overline{A_{ka}} B_{ai} \overline{B_{al}},$$

the one-particle kernel of $\mathcal{X}$ on the full row register is $\sum_a |v_a\rangle\langle v_a| \otimes C_a$. Therefore, before row compression,

$$\mathcal{X} = \sum_a |v_a\rangle\langle v_a| \otimes \mathcal{L}_p(C_a) \otimes I, \quad \mathcal{F} = 2 \sum_a |v_a\rangle\langle v_a| \otimes \mathcal{L}_p(C_a) \otimes N_{y,a}.$$

Keep the common positive coefficient block through the correction:

$$\mathcal{X} - \mathcal{F} = \sum_a |v_a\rangle\langle v_a| \otimes \mathcal{L}_p(C_a) \otimes (I - 2N_{y,a}). \quad (204)$$

Each $I - 2N_{y,a}$ is a self-adjoint unitary preserving the y grade. The row modes are orthogonal, so Lemma A.1 gives the bound before compression; common row compression to T then preserves it:

$$\|\mathcal{X} - \mathcal{F}\| \leq \beta p. \quad (205)$$

Taking the norms of the two summands separately would give $3\beta p$. The saving uses their common kernel and reflection, rather than the negative sign alone.

This grouping also retains an order budget for further interactions. On the full row register, $\mathcal{X} \succeq 0$ commutes with

$$\mathcal{S} = \sum_a |v_a\rangle\langle v_a| \otimes I \otimes (I - 2N_{y,a}), \quad \mathcal{X} - \mathcal{F} = \mathcal{X}\mathcal{S}.$$

Let J_T include the rows in T , with identity on the occupation factors, and put $B_T = J_T^* \mathcal{X} J_T$, $H_T = J_T^* (\mathcal{X} - \mathcal{F}) J_T$. Lemma 3.4 gives

$$-B_T \preceq H_T \preceq B_T, \quad \|LH_T R\| \leq \|LB_T^{1/2}\| \|B_T^{1/2}R\|. \quad (206)$$

Thus a continuation can retain the actual boundary Grams even when row compression mixes the auxiliary modes. No involution identity for the compressed reflection is required.

The one- y term. This term is Hermitian, as is seen by exchanging j, k and taking an adjoint. For $\psi = (\psi_j)_{j \in T}$, S_i commutes with every y operator and has norm one. Cauchy–Schwarz and $2ab \leq a^2 + b^2$ give

$$\begin{aligned} |\langle \psi, \mathcal{Y}\psi \rangle| &\leq \sum_{i,j,k} \ell_i \|a_y(g_{ki})\psi_j\| \|a_y(g_{ji})\psi_k\| \\ &\leq \sum_{i,j,k} \ell_i \|a_y(g_{ki})\psi_j\|^2. \end{aligned}$$

The second squared-norm sum produced by $2ab \leq a^2 + b^2$ equals the first after interchanging j, k . Set

$$G_T = \sum_{k \in T} \sum_i \ell_i |g_{ki}\rangle \langle g_{ki}|.$$

At each fixed k , the synthesis bound and $0 \leq \ell_i \leq 1$ imply $\sum_i \ell_i |g_{ki}\rangle\langle g_{ki}| \preceq \alpha I$. Thus $G_T \preceq \alpha t I$. Since $a_y(g)^* a_y(g) = \mathcal{L}_v(|g\rangle\langle g|)$, the preceding quadratic form is at most

$$\sum_j \langle \psi_j, (I \otimes \mathcal{L}_v(G_T)) \psi_j \rangle \leq \alpha t v \|\psi\|^2.$$

Hermiticity now gives

$$\|\mathcal{Y}\| \leq \alpha t v. \quad (207)$$

The fully ordered term. Its one- x /one- y coefficient kernel, with the row register retained, is

$$K_{jk} = \sum_{i,l} P_{il} |l\rangle\langle i| \otimes |g_{kl}\rangle\langle g_{ji}|.$$

Write an input as $z_{k,i} \in \mathbb{C}_y^n$. The following three maps compose to K :

$$z'_{k,i,j} = \langle g_{ji}, z_{k,i} \rangle, \quad z''_{j,l,k} = \sum_i P_{il} z'_{k,i,j}, \quad z'''_{j,l} = \sum_{k \in T} g_{kl} z''_{j,l,k}.$$

The first map has norm at most $\sqrt{\beta}$ by the fixed- i row Bessel bound. The second applies $P^\top$ in the coordinate register and permutes the two row registers, so its norm is at most one. The last map has norm at most $\sqrt{\beta}$ by the fixed- l row synthesis bound. Therefore $\|K\| \leq \beta$. Its two-species occupation lift is exactly $\mathcal{Z}$, so

$$\|\mathcal{Z}\| \leq \beta p v. \quad (208)$$

The triangle inequality in (202), using (203)–(208), gives $\|\mathcal{G}\| \leq \beta(d + p + pv) + \alpha t v$. Since $\mathcal{G}$ is the output Gram, this proves the lemma. A removal from grade zero is zero, which proves the remaining cases. $\square$

A.3 A centered finite sampling comparison

Lemma A.3 (Bernoulli to fixed-size sampling). *Let $1 \leq M < n$, $\rho = M/n$, and let $H_1, \dots, H_n$ be arbitrary Hermitian matrices of the same size. For a Bernoulli subset $\mathcal{B}$ with independent inclusion probability ρ and a uniform M -element subset $\mathcal{T}$, define*

$$Z_{\mathcal{B}} = \sum_j (\mathbf{1}_{j \in \mathcal{B}} - \rho) H_j, \quad Z_{\mathcal{T}} = \sum_j (\mathbf{1}_{j \in \mathcal{T}} - \rho) H_j.$$

Then, for every integer $q \geq 1$,

$$\mathbb{E} \operatorname{tr}(Z_{\mathcal{T}}^{2q}) \leq 2^{2q} \mathbb{E} \operatorname{tr}(Z_{\mathcal{B}}^{2q}). \quad (209)$$

Proof. Draw $\mathcal{B}$ first. If its cardinality K exceeds M , choose $\mathcal{T}$ uniformly among its M -subsets; if $K < M$, add uniformly chosen missing elements until its size is M . At equality take $\mathcal{T} = \mathcal{B}$. Conditional on $K = k$, permutation symmetry makes $\mathcal{T}$ uniform. Thus K and $\mathcal{T}$ are independent, and both required marginal laws are preserved.

Put $a = \mathbb{E}(K - M)_+ = \mathbb{E}(M - K)_+$; the equality follows from $\mathbb{E}K = M$. Conditioning on $\mathcal{T}$, nestedness and symmetry give inclusion probabilities

$$p_{\text{in}} = 1 - a/M, \quad p_{\text{out}} = a/(n - M)$$

inside and outside $\mathcal{T}$. Set $v = n\rho(1 - \rho)$ and $c = p_{\text{in}} - p_{\text{out}} = 1 - a/v$. Since $p_{\text{out}} = (1 - c)\rho$, these probabilities imply the matrix identity

$$\mathbb{E}[Z_{\mathcal{B}} \mid \mathcal{T}] = c Z_{\mathcal{T}}. \quad (210)$$

We verify $c \geq 1/2$ directly. Let Y be binomial with parameters $n-1, \rho$. The elementary identity $k \binom{n}{k} = n \binom{n-1}{k-1}$ gives $\mathbb{E}[K \mathbf{1}_{K \geq M+1}] = M \mathbb{P}(Y \geq M)$. Also K has the same law as Y plus an independent Bernoulli(ρ) variable. Hence

$$a = M[\mathbb{P}(Y \geq M) - \mathbb{P}(K \geq M+1)] = M(1-\rho)\mathbb{P}(Y = M) = v\mathbb{P}(Y = M).$$

The two neighboring binomial atoms are positive and obey

$$\frac{\mathbb{P}(Y = M)}{\mathbb{P}(Y = M-1)} = \frac{n-M}{M} \frac{\rho}{1-\rho} = 1.$$

They are disjoint events, so $\mathbb{P}(Y = M) \leq 1/2$, and therefore $c = 1 - \mathbb{P}(Y = M) \geq 1/2$.

Apply Lemma 2.6 to (210), with $a = c$ and conditioning on $\mathcal{T}$. Its trace-convexity proof gives

$$\mathbb{E} \operatorname{tr}(Z_{\mathcal{T}}^{2q}) \leq c^{-2q} \mathbb{E} \operatorname{tr}(Z_{\mathcal{B}}^{2q}).$$

Since $c \geq 1/2$, this is (209). $\square$

A.4 A confidence obstruction for finitely many rounds

Fix an integer number of rounds $t \geq 1$, fully independent sign diagonals $D_1, \dots, D_t$, and a uniform sample T of M distinct rows; let $L \geq 2$ be a power of two, $n = L^2$, and $H = H_L \otimes H_L$ the normalized Walsh matrix of order n . Put $u = L^{-1/2} \mathbf{1}_L \otimes e_0$ and $v = e_0 \otimes L^{-1/2} \mathbf{1}_L$, unit vectors with L nonzero coordinates; character orthogonality gives $Hu = v$ and $Hv = u$. Require the signs of round j to be +1 on the current support, that of u at odd rounds and of v at even rounds. These events use different independent sign layers, each has probability 2^{-L} , and on their intersection the state alternates between u and v ; if the sample then misses the final L -point support, the sketched vector is zero. Hence, for $S_t = \sqrt{n/M} R_T H D_t \cdots H D_1$, $1 \leq M \leq n - L$ and $0 < \epsilon < 1$,

$$\mathbb{P}\{\|S_t u\|^2 < 1 - \epsilon\} \geq 2^{-tL} \frac{\binom{n-L}{M}}{\binom{n}{M}} = 2^{-tL} \prod_{j=0}^{L-1} \left(1 - \frac{M}{n-j}\right) \geq 2^{-tL} \left(1 - \frac{M}{n-L+1}\right)^L.$$

A guarantee of failure probability at most $\delta < 2^{-tL}$, already for the one-dimensional subspace spanned by u , therefore requires $M \geq (n-L+1)(1-2^t \delta^{1/L})$, which is automatic when $M \geq n-L+1$. For two rounds and $\delta = 2^{-4L}$ this reads $M \geq \frac{3}{4}(n-L+1)$. Consequently no bound of the form $M \leq C\epsilon^{-2}(d + \log(1/\delta))$ with a fixed C can hold at every confidence level for a fixed number of these rounds: along $\delta_L = e^{-aL}$ with $a > t \log 2$ it would give $M = O(L)$, whereas a positive fraction of L^2 is necessary. This is consistent with Theorems 5.5 and 10.1, whose confidence terms reach the all-row endpoint $M = n$ once $\log(1/\delta)$ is of order $\sqrt{n}$. A stronger family of alignment events, with probability $4L^2 2^{-2L}$ for two rounds, is obtained by requiring each sign diagonal on the current support to equal one of the $2L$ signed Walsh characters of order L .

B A direct SparseStack proof with explicit constants

We give the operator estimates and scalar certificate used in Theorem 10.6. We use the actual signed bucket variables, so no distributional comparison enters this proof.

B.1 Local space, counts, and the eight families

At site (g, i) use $L^2(\{\pm 1\} \times [b])$ with uniform probability, where $b \geq 2$. Write $h = b - 1$, $r_b = b/(b - 1)$, and

$$\Omega = 1, \quad e_a = \sqrt{b} \sigma \mathbf{1}_{\{\mathfrak{h}=a\}}, \quad f_a = \frac{b \mathbf{1}_{\{\mathfrak{h}=a\}} - 1}{\sqrt{b - 1}},$$

where $\mathfrak{h}$ is the local hash. The local space splits orthogonally into the vacuum, light, and heavy spaces

$$\mathcal{H}_0 = \mathbb{R}\Omega, \quad \mathcal{H}_1 = \text{span}\{e_a : a \in [b]\}, \quad \mathcal{H}_2 = L_0^2([b]).$$

Their dimensions are $1, b, b - 1$. Direct averaging gives

$$\langle e_a, e_c \rangle = \delta_{ac}, \quad \langle f_a, f_c \rangle = \frac{b\delta_{ac} - 1}{b - 1}, \quad \sum_a f_a = 0, \quad \sum_a |f_a\rangle\langle f_a| = r_b I_{\mathcal{H}_2}. \quad (211)$$

For the last identity, a mean-zero hash function v satisfies $\langle f_a, v \rangle = v(a)/\sqrt{b - 1}$; summing its squared values proves the frame identity. Define

$$P_a^* = |e_a\rangle\langle\Omega|, \quad P_a = |\Omega\rangle\langle e_a|, \quad R_a = |f_a\rangle\langle e_a|, \quad R_a^* = |e_a\rangle\langle f_a|.$$

Multiplication by e_a is exactly

$$J_a = P_a + P_a^* + \sqrt{h}(R_a + R_a^*). \quad (212)$$

Indeed it sends Ω to e_a , sends e_c to $\delta_{ac}(\Omega + \sqrt{h}f_a)$, and sends a heavy vector v to $\sqrt{h}\langle f_a, v \rangle e_a$. These vectors span the local space.

Tensor the local spaces over $[s] \times [n]$ and then with $\mathbb{R}^d$. For a type pattern let L_g, H_g count light and heavy sites in block g , and put $L = \sum_g L_g$, $H = \sum_g H_g$, $N = L + H$, $\nu = L + 2H$. The subspace with fixed totals is $\mathcal{K}_{L,H}$, and $\mathcal{K}_\nu = \bigoplus_{L+2H=\nu} \mathcal{K}_{L,H}$. Write $U_{ij} = u_i u_j^*$, with $\sum_i u_i u_i^* = I_d$ and $m = sb$. The exact error multiplication operator is

$$\mathcal{M} = \frac{1}{m} \sum_{g,a} \sum_{i \neq j} U_{ij} \otimes J_{a,gi} J_{a,gj}, \quad (213)$$

because $\sum_a e_{a,gi} e_{a,gj} = b \sigma_{gi} \sigma_{gj} \mathbf{1}_{\{h_{gi}=h_{gj}\}}$.

In the next definitions every sum is over g, a and ordered $i \neq j$; subscripts on the legs are suppressed:

$$\begin{aligned} L_+ &= \sum U_{ij} \otimes P_i^* P_j^*, & L_0 &= \sum U_{ij} \otimes (P_i^* P_j + P_i P_j^*), \\ X_+ &= \sum U_{ij} \otimes P_i^* R_j, & Y_+ &= \sum U_{ij} \otimes R_i P_j^*, \\ X_0 &= \sum U_{ij} \otimes P_i^* R_j^*, & Y_0 &= \sum U_{ij} \otimes R_i^* P_j^*, \\ H_+ &= \sum U_{ij} \otimes R_i R_j, & H_0 &= \sum U_{ij} \otimes (R_i R_j^* + R_i^* R_j). \end{aligned} \quad (214)$$

Expanding the local multiplier gives the bands of shifts $2, 0, -2$:

$$\begin{aligned} mB^+ &= L_+ + \sqrt{h}(X_+ + Y_+) + hH_+, & B^- &= (B^+)^*, \\ mB^0 &= L_0 + \sqrt{h}(X_0 + X_0^* + Y_0 + Y_0^*) + hH_0. \end{aligned} \quad (215)$$

Figure 8 in Section 5.1 displays these local legs and the count changes that assemble into the three bands.

The output counts of L_+, X_+, Y_+, H_+ are respectively $(L + 2, H)$, $(L, H + 1)$, $(L, H + 1)$, $(L - 2, H + 2)$. Those of X_0, Y_0 are $(L + 2, H - 1)$, while L_0, H_0 preserve both totals. Invalid transitions are zero. Every leg changes light parity, and each term acts twice in one block. Thus every family preserves

$$\mathcal{K}^{\text{ev}} = \text{span}\{\text{patterns with every } L_g \text{ even}\}, \quad (216)$$

which contains the vacuum.

B.2 Coordinate-frame estimates for the heavy legs

We use two elementary reductions. A block matrix with at most a nonzero blocks per column, at most c per row, and block norms at most k has norm at most $k\sqrt{ac}$. To prove this, apply Cauchy–Schwarz to each output sum and then sum: each input squared norm is counted at most a times and each output sum has at most c terms.

Every family with coefficients $u_i u_j^*$, including sums with $i = j$, has the exact external compression

$$F(U) = (U^* \otimes I)F(I_n)(U \otimes I). \quad (217)$$

Occupation projections commute with these isometries. Thus a count-sector bound for the coordinate frame proves the same bound for every U .

For $U = I_n$, decompose into blocks specified by the external index and the type pattern, retaining the light and heavy vector spaces inside each block. The following local maps have the indicated norms:

$$\sum_a |\Omega, e_a\rangle\langle e_a, \Omega| : 1, \quad \sum_a |f_a, e_a\rangle\langle e_a, \Omega| : 1, \quad \sum_a |e_a, e_a\rangle\langle f_a, \Omega| : \sqrt{r_b}.$$

The first two are isometries on their initial spaces; the third has Gram $r_b I$. The exchange $\sum_a |f_a, e_a\rangle\langle e_a, f_a|$ has norm one, since both families indexed by a are orthonormal.

Put $H' = \sum U_{ij} \otimes P_i P_j^*$. For H' and Y_+ the external index is initially at a free site j , and the other site i is light. There are at most L possible output type blocks. Reading an output backward also selects one of its L light sites. For Y_0 there are at most H choices of an input heavy site and at most $L + 2$ choices of a light site when reading backward. The block inequality therefore proves

$$\|H'\| \leq L, \quad \|Y_+\| \leq L, \quad \|Y_0\| \leq \sqrt{r_b H(L+2)}. \quad (218)$$

These counts are totals over all sketch blocks.

For a single block the two coordinate-frame orientations of H_0 are

$$A_g = \sum_{a, i \neq j} |i\rangle\langle j| \otimes R_{a,gi} R_{a,gj}^*, \quad B_g = \sum_{a, i \neq j} |i\rangle\langle j| \otimes R_{a,gi}^* R_{a,gj}.$$

The marked external site is heavy on both sides of A_g , and light on both sides of B_g . Each orientation is self-adjoint. The exchange kernel and neighbor counts give

$$\|A_g\| \leq L_g, \quad \|B_g\| \leq H_g. \quad (219)$$

They vanish unless both counts are positive, and their marked supports are orthogonal. Summing over blocks and compressing gives $\|H_0\| \leq N$, which suffices below. The sharper maximum bound is proved in Appendix C.

B.3 Collected Grams and parity

For a leg Z define

$$S_{ga}(Z) = \sum_i u_i \otimes Z_{a,gi} : \mathcal{F} \longrightarrow \mathbb{R}^d \otimes \mathcal{F}, \quad G(Z) = \sum_{g,a} S_{ga}(Z) S_{ga}(Z)^*.$$

Here $\mathcal{F}$ is the occupation tensor product without its external factor; thus every $G(Z)$ acts on $\mathbb{R}^d \otimes \mathcal{F}$. These positive operators preserve each block's light and heavy counts. Let $\Pi_{k,gi}$ project onto the local level $\mathcal{H}_k$ at site (g, i) , and write $G_g(Z) = \sum_a S_{ga}(Z) S_{ga}(Z)^*$ for the

contribution from block g . In the coordinate frame put $Q_{g,L} = \sum_i |i\rangle\langle i| \otimes \Pi_{1,gi}$ and define $Q_{g,H}$ similarly. Expanding the diagonal terms gives

$$G_g(R^*) = Q_{g,L} + B_g, \quad G_g(R) = r_b Q_{g,H} + A_g.$$

Together with (219), these identities prove

$$\begin{aligned} G(R^*) &\preceq \left[\sum_{g:L_g>0} (H_g + 1) \right] I, \\ G(R) &\preceq r_b \left[\sum_{g:H_g>0} (L_g + 1) \right] I. \end{aligned} \tag{220}$$

External compression proves them for general U . On $\mathcal{K}_{L,H}^{\text{ev}}$ the first bound is at most $H + L/2$; the second is at most $r_b N$. If at least two heavy sites occur in one block, the second improves to $r_b(N - 1)$, since the number of heavy-active blocks is then at most $H - 1$. The span of those patterns is invariant under $G(R)$.

For the light Gram $G_P = G(P^*)$ we retain the actual frame. Freeze a heavy assignment, label light particles by (g, a, i) , and map an unordered configuration isometrically to the sum over its $L!$ orderings, divided by $\sqrt{L!}$. The light move on this labelled tensor product is the sum of L shared-factor contractions in Lemma 3.1, with (g, a) a passive register. Compression forbids repeated physical sites and the frozen heavy sites. The resulting operator is exactly G_P : it replaces (g, a, j) by an allowed (g, a, i) with coefficient $u_i u_j^*$. Therefore

$$0 \preceq G_P \preceq (d + L - 1)I \quad (L \geq 1), \quad G_P = 0 \quad (L = 0). \tag{221}$$

Define $D_0 = b \sum_{g,i} u_i u_i^* \otimes \Pi_{0,gi}$ and $D_1 = \sum_{g,i} u_i u_i^* \otimes \Pi_{1,gi}$. Expansion gives $G(P) = D_0 + H'$ and $L_0 = G_P - D_1 + H'$. Here $D_0 \preceq mI$ and $D_1 \preceq LI$. The inequalities $-D_1 \preceq G_P - D_1 \preceq G_P$ give $\|G_P - D_1\| \leq \max\{L, d + L - 1\} = d + L - 1$ when $L > 0$. Combining this with (218),

$$\|G(P)\| \leq m + L, \quad \|L_0\| \leq d + 2L - 1 \quad (L > 0), \quad L_0 = 0 \quad (L = 0). \tag{222}$$

B.4 All channel bounds used in the direct theorem

Block Cauchy–Schwarz gives

$$\left\| \sum_k B_k A_k \right\|^2 \leq \left\| \sum_k A_k^* A_k \right\| \left\| \sum_k B_k B_k^* \right\|.$$

This is the norm bound for the product of a block row and block column; orthogonal endpoint projections may be inserted before forming the Grams. The local products $(P_a^*)^2, R_a^2, P_a^* R_a, P_a^* R_a^*$ vanish, so

$$\begin{aligned} L_+ &= \sum_{g,a} S_{ga}(P^*) S_{ga}(P)^*, & X_+ &= \sum_{g,a} S_{ga}(P^*) S_{ga}(R^*)^*, \\ H_+ &= \sum_{g,a} S_{ga}(R) S_{ga}(R^*)^*, & X_0 &= \sum_{g,a} S_{ga}(P^*) S_{ga}(R)^*. \end{aligned}$$

Use the preceding Grams on the indicated source and target spaces. For H_+ the target has two newly heavy sites in one block, so the $N - 1$ bound applies. This gives the full list on the block-even sector:

$$\begin{aligned} \|L_+\| &\leq \sqrt{(m + L)(d + L + 1)}, & \|X_+\| &\leq \sqrt{(H + L/2)(d + L - 1)}, \\ \|Y_+\| &\leq L, & \|H_+\| &\leq \sqrt{r_b(N - 1)(H + L/2)}, \\ \|X_0\| &\leq \sqrt{r_b N(d + L + 1)}, & \|Y_0\| &\leq \sqrt{r_b H(L + 2)}, \\ \|L_0\| &\leq d + 2L - 1, & \|H_0\| &\leq N. \end{aligned} \tag{223}$$

The X_+, H_+ bounds are used when $L \geq 2$; these families vanish at $L = 0$. The X_0, Y_0 families vanish at $H = 0$, L_0 at $L = 0$, and H_0 when either count is zero. These conventions cover all endpoints.

Each family maps different (L, H) sectors to different output sectors, so its norm on a grade is their maximum. On positive even grades put

$$m\alpha_\nu = \sqrt{(m+\nu)(d+\nu+1)} + \sqrt{h\nu(d+\nu-1)/2} + \nu\sqrt{h} + \sqrt{bh\nu(\nu-1)/2}, \quad (224)$$

$$m\chi_\nu = d + 2\nu - 1 + 2\sqrt{b(\nu-1)(d+\nu-1)} + (\nu+1)\sqrt{b} + h\nu. \quad (225)$$

Then $\|B^+|_{\mathcal{K}_\nu^{\text{ev}}}\| \leq \alpha_\nu$ and $\|B^0|_{\mathcal{K}_\nu^{\text{ev}}}\| \leq \chi_\nu$. For X_0 use $H \geq 1$, giving $N \leq \nu - 1$ and $d + L + 1 \leq d + \nu - 1$. For Y_0 use

$$H(L+2) \leq \frac{(\nu+2)^2}{8} \leq \frac{(\nu+1)^2}{4} \quad (\nu \geq 2).$$

The latter inequality is equivalent to $\nu^2 \geq 2$. The other terms follow from $L \leq \nu$, $H + L/2 = \nu/2$, and $N - 1 \leq \nu - 1$. At the vacuum $\alpha_0 = \sqrt{(d+1)/m}$ and $\chi_0 = 0$. Let T_q have diagonal χ_{2k} and edges $(T_q)_{k,k+1} = \alpha_{2k}$, $0 \leq k < q$. The edge bounds increase. The exact first- q -iterate compression and (97) give

$$\mathbb{E} \text{tr } E^{2q} \leq d(T_q^{2q})_{00}. \quad (226)$$

B.5 A rational certificate for every moment order

Set $B = 17/2$, $D = d + 2q + 1$, and choose

$$s = \left\lceil \frac{B(2q+1)}{\epsilon} \right\rceil, \quad M_0 = \frac{81D}{\epsilon^2}, \quad b = \lceil M_0/s \rceil, \quad m = sb. \quad (227)$$

Assume $d, q \geq 1$ and $0 < \epsilon \leq 1$. Since $Bd > \epsilon$ and $s < [B(2q+1) + \epsilon]/\epsilon$, $M_0/s > 81/(B\epsilon) \geq 162/17$, so $b \geq 10$. For $u = (\nu+1)/(2q+1)$ define

$$\begin{aligned} \Psi(u) &= \frac{2}{9} \sqrt{1 + \frac{u}{85}} + \frac{2 + \sqrt{2}}{9} \sqrt{\frac{2u}{17}} + \frac{1}{81} \\ &\quad + \frac{2}{17} (1 + \sqrt{2} + G_{10})u, \quad G_{10} = \frac{2\sqrt{9/10} + 1}{\sqrt{10}}. \end{aligned} \quad (228)$$

The row sums of T_q/ϵ are bounded by $\Psi(u_k)$, $u_k = (2k+1)/(2q+1)$, as follows. For $\nu = 2k > 0$ the row sum is at most $2\alpha_\nu + \chi_\nu$. The light-creation terms give the first term of (228), using $\nu/m \leq u/(10B)$ and $(d+\nu+1)/m \leq \epsilon^2/81$. The two X_+ and two X_0 terms give $(\sqrt{2}+2)\sqrt{u/B}/9$. The two heavy-creation terms are at most $\sqrt{2}u/B$. The nonradical loop terms combine as

$$d + 2\nu - 1 + h\nu = d + \nu + 1 + b\nu - 2$$

and contribute at most $1/81 + u/B$. The remaining terms contribute at most

$$\frac{2\sqrt{1-1/b} + 1}{\sqrt{b}} \frac{u}{B} \leq G_{10} \frac{u}{B}.$$

Here $y \mapsto 2y\sqrt{1-y^2} + y$ increases on $[0, 1/\sqrt{2}]$, since its derivative is $2(1-2y^2)/\sqrt{1-y^2} + 1 \geq 1$. We used $\epsilon \leq 1$ when removing a remaining $\sqrt{\epsilon}$ or ϵ factor. At $k = 0$ the row sum is at most $1/9$.

The function Ψ is positive, increasing, and concave, and the vacuum edge is at most $\epsilon/9$. For an exact midpoint certificate use

$$\sqrt{2} \leq \frac{1414214}{10^6}, \quad \sqrt{9/10} \leq \frac{948684}{10^6}, \quad \sqrt{1/10} \leq \frac{316228}{10^6}.$$

The table gives rational upper bounds; its middle columns are numerators over 10^6 , and its last column over 10^4 :

u	$\sqrt{1+u/85}$	$\sqrt{2u/17}$	$\Psi(u)$
1/8	1000736	121268	3298
3/8	1002204	210043	4617
5/8	1003670	271164	5832
7/8	1005134	320845	7003

Squaring the radical bounds and substituting in (228) verifies every entry by rational arithmetic. The final comparison is

$$3298 \cdot 4617 \cdot 5832 \cdot 7003 = 621887986831536 < 625000000000000 = \frac{10^{16}}{16}. \quad (229)$$

Thus (99) gives $J = \exp \int_0^1 \log \Psi < 1/2$. The boundary condition, with $a = 1/9$, is

$$a^2 J < \frac{1}{162} \leq \left(\frac{19}{81}\right)^3 = \Psi(0)^3.$$

Lemma 4.2 and (226) give $\mathbb{E} \operatorname{tr} E^{2q} < d(\epsilon/2)^{2q}$ for every $q \geq 2$. For $q = 1$ the vacuum edge gives the stronger root bound $\epsilon/9$. Taking $q = \max\{1, \lceil \log_4(d/\delta) \rceil\}$ proves the failure probability bound by Markov. Finally the ceiling inequalities give

$$s < \frac{17q + 19/2}{\epsilon} \leq \frac{(53/2)q}{\epsilon},$$

and

$$m < M_0 + s < \epsilon^{-2} \left(81d + 179q + \frac{181}{2} \right) < \frac{179(d+q)}{\epsilon^2},$$

where the last step uses $98d > 181/2$.

B.6 The monotone form and the reference prescription

Corollary B.1 (Monotone form of the direct bound). *Let $d, q \geq 1$, $0 < \epsilon \leq 1$, and let s, b be integers with $s \geq \frac{17}{2}(2q+1)/\epsilon$, $b \geq 10$ and $sb \geq 81(d+2q+1)/\epsilon^2$. Then the fully independent real SparseStack sketch with these parameters satisfies $\mathbb{E} \operatorname{tr} E^{2q} < d(\epsilon/2)^{2q}$ if $q \geq 2$ and $\mathbb{E} \operatorname{tr} E^2 \leq d(\epsilon/9)^2$; hence $\mathbb{P}\{\|E\| > \epsilon\} \leq d4^{-q}$ for $q \geq 2$ and $\mathbb{P}\{\|E\| > \epsilon\} \leq d/81$ for $q = 1$. In particular the prescription of Theorem 5.1 gives $\mathbb{P}\{\|E\| > \epsilon\} \leq \delta$.*

Proof. The family bounds (223) and the envelopes (224)–(225) hold for every $m = sb$ with $b \geq 2$. In bounding the row sums of T_q/ϵ by Ψ only the inequalities $1/s \leq \epsilon/(B(2q+1))$, $1/m \leq \epsilon^2/(81D)$, $b \geq 10$ and $\epsilon \leq 1$ were used, and every term decreases when s or b increases: the terms in ν/m and $(d+\nu+1)/m$, the bounds $h\nu/m \leq \nu/s$ and $\sqrt{bh\nu(\nu-1)/2}/m \leq \sqrt{\nu(\nu-1)/2}/s$, the terms $(\nu+1)\sqrt{b}/m = (\nu+1)/(s\sqrt{b})$ and $2\sqrt{b(\nu-1)(d+\nu-1)}/m = 2\sqrt{(\nu-1)(d+\nu-1)/(s^2b)}$, the ratio $r_b = b/(b-1)$, and $(2\sqrt{1-1/b+1})/\sqrt{b}$, which decreases in b for $b \geq 2$ as shown above. The certificate (229) and Lemma 4.2 therefore apply verbatim, and Markov's inequality gives the failure bounds. For the prescription of Theorem 5.1 put

$q_0 = \max\{1, \lceil \log_4(d/\delta) \rceil\} \leq q = \max\{1, \lceil \log_2(d/\delta) \rceil\}$ and note $\Lambda = 90c_*^2 > 225$. Then $s = \lceil \Lambda(2q+1)/\epsilon \rceil \geq \frac{17}{2}(2q_0+1)/\epsilon$; $m = sb \geq \Lambda^2 D_q/\epsilon^2 \geq 81(d+2q_0+1)/\epsilon^2$; and, since $s \leq 2\Lambda(2q+1)/\epsilon$ and $D_q \geq 2q+1$, $b \geq \Lambda^2 D_q/(s\epsilon^2) \geq \Lambda D_q/(2(2q+1)\epsilon) \geq \Lambda/2 > 10$. Applying the first part at the order q_0 gives $\mathbb{P}\{\|E\| > \epsilon\} \leq d4^{-q_0} \leq \delta$ when $q_0 \geq 2$, and $\leq d/81 < \delta$ when $q_0 = 1$, since then $d/\delta \leq 4$. $\square$

C Exact bucket-sector calculations

We give the complete finite calculations behind the two bucket examples and the improved heavy-swap estimate. Throughout $b \geq 2$. The local light vectors e_a are orthonormal, the heavy vectors satisfy

$$\langle f_a, f_c \rangle = \begin{cases} 1, & a = c, \\ -1/(b-1), & a \neq c, \end{cases} \quad \sum_a f_a = 0, \quad R_a = |f_a\rangle\langle e_a|. \quad (230)$$

These are the local spaces and legs of (5). Bucket permutations send e_a, f_a to $e_{\pi(a)}, f_{\pi(a)}$. An independent sign on bucket a multiplies e_a by that sign and fixes the heavy space. We call invariance under both actions *signed-bucket invariance*.

C.1 A heavy-pair norm on the invariant sector

Take one block, three sites, one external dimension, and $U = (1, 1, 1)^*/\sqrt{3}$. From the sector with two light sites and one heavy site to the all-heavy sector, the unscaled heavy-pair family is

$$H_+ = \frac{2}{3} \sum_{1 \leq i < j \leq 3} \sum_{a=1}^b R_{a,i} R_{a,j}. \quad (231)$$

The factor $2/3$ includes the two orders of the selected pair. The local coefficient $(b-1)/m$ in the error operator is separate.

Proposition C.1 (Exact invariant heavy-pair norm). *The signed-bucket invariant source has dimension three, with orthonormal basis*

$$w_k = \frac{1}{\sqrt{b}} \sum_a f_a^{(k)} \bigotimes_{i \neq k} e_a^{(i)}, \quad k = 1, 2, 3.$$

The norm of (231) on this source is

$$\|H_+\|_{\text{inv}} = \frac{2}{\sqrt{3}} \frac{\sqrt{b(b-2)}}{b-1}. \quad (232)$$

The same norm is obtained using bucket permutations alone.

Proof. Fix the heavy position. Invariance under independent bucket signs forces the two light labels to coincide. Every such source vector is therefore $\sum_a e_a \otimes e_a \otimes v_a$, with v_a heavy, up to the indicated ordering of sites. Permutation invariance requires $v_{\pi(a)} = T_\pi v_a$. Identify the heavy representation with $\{x \in \mathbb{C}^b : \sum_a x_a = 0\}$. The stabilizer of label a fixes exactly the one-dimensional subspace of vectors constant on the other $b-1$ coordinates, namely the span of f_a . Thus $v_a = c f_a$ with the same scalar c for all a . This proves that the fixed space at each heavy position is one-dimensional. The three positions are orthogonal occupation patterns, and the light factors make different summands in each w_k orthogonal. Hence the displayed vectors are an orthonormal basis. The argument includes $b = 2$.

Put $F_3 = \sum_a f_a^{\otimes 3}$. Direct application of the two light-to-heavy legs gives

$$H_+ w_k = \frac{2}{3\sqrt{b}} F_3, \quad \|F_3\|^2 = b + b(b-1) \left(-\frac{1}{b-1} \right)^3 = \frac{b^2(b-2)}{(b-1)^2}.$$

The three orthonormal input vectors have the same image. The norm is therefore $\sqrt{3}$ times the norm of that image, proving (232). At $b = 2$ the image is zero.

Every bucket-sign action is the identity on the all-heavy target. Because H_+ intertwines these actions, $H_+ = H_+ \Pi_{\text{sign}}$, where Π_{sign} averages the independent signs. This projection commutes with bucket averaging: permutations merely permute the signs being averaged. Hence every permutation-invariant input has the same image as its signed-bucket invariant projection, whose norm is no larger. This proves equality of the two restricted operator norms. $\square$

C.2 The full twenty-seven-dimensional calculation at four buckets

Proposition C.2 (A strict bucket-sector reduction). *For (231) with $b = 4$, the full block-even source and the bucket-invariant source have norms*

$$\|H_+\|_{\text{parity}} = \sqrt{\frac{8(9 + \sqrt{33})}{81}}, \quad \|H_+\|_{\text{inv}} = \sqrt{\frac{32}{27}}. \quad (233)$$

Their squared-norm difference is $8(\sqrt{33} - 3)/81 > 0$.

Proof. For a fixed heavy position k , the only active light inputs are the equal-bucket pairs. Identify their space with $\mathbb{C}^b$, and define $T_k : \mathbb{C}^b \otimes \mathbb{H} \rightarrow \mathbb{H}^{\otimes 3}$ by

$$T_k(e_a \otimes v) = v^{(k)} \bigotimes_{i \neq k} f_a^{(i)}.$$

The source patterns for different k are orthogonal. Consequently

$$H_+ = \frac{2}{3} [T_1 \ T_2 \ T_3], \quad H_+ H_+^* = \frac{4}{9} \mathcal{K}, \quad \mathcal{K} = \sum_{i < j} \mathcal{S}_{ij}, \quad (234)$$

where $\mathcal{S} = \sum_a |f_a \otimes f_a\rangle \langle f_a \otimes f_a|$ acts on the indicated pair, with identity on the remaining heavy factor. Inputs with unequal light buckets are in the kernel, so (234) computes the norm on the entire source. Its two light sites already satisfy block-even parity.

Choose the tetrahedral simplex in the heavy space $\mathbb{R}^3$:

$$f_1 = (1, 1, 1)/\sqrt{3}, \quad f_2 = (1, -1, -1)/\sqrt{3}, \quad f_3 = (-1, 1, -1)/\sqrt{3}, \quad f_4 = (-1, -1, 1)/\sqrt{3}.$$

Let $|ijk\rangle$ denote the tensor of the three standard heavy coordinate vectors. A matrix entry of $\mathcal{S}$ is $4/9$ when each coordinate occurs an even number of times among its four indices, and zero otherwise. Indeed the four sign triples above have product one, and their average annihilates a product of two distinct coordinate signs. It follows that $L = (9/4)\mathcal{S}$ acts by

$$L|ii\rangle = \sum_{j=1}^3 |jj\rangle, \quad L|ij\rangle = |ij\rangle + |ji\rangle \quad (i \neq j). \quad (235)$$

Set $A = \sum_{i < j} L_{ij}$, so $H_+ H_+^* = (16/81)A$.

The parity of each heavy coordinate's multiplicity is preserved by A . Since there are three slots, either all three multiplicities are odd, or exactly one is odd. The all-odd block consists of the six permutations of $|123\rangle$. On it, $A = 3I + F_{12} + F_{13} + F_{23}$, where F_{ij} exchanges slots.

The sum of the exchanges is the adjacency matrix of $K_{3,3}$, with the two vertex classes given by even and odd permutations. Its spectrum is $3, -3, 0, 0, 0, 0$: constant vectors on the two classes give the first two eigenvalues, and vectors summing to zero within each class give zero. Thus this block of A has eigenvalues $6, 0, 3, 3, 3, 3$.

Fix the unique odd coordinate i , and let j, k be the other two. This block has dimension seven. Write $x = |iii\rangle$ and, for $r = 1, 2, 3$, let $p_r^{(j)}$ have i in slot r and j in the other slots; define $p_r^{(k)}$ similarly. Equation (235) gives

$$\begin{aligned} Ax &= 3x + \sum_r p_r^{(j)} + \sum_r p_r^{(k)}, \\ Ap_r^{(j)} &= x + 3p_r^{(j)} + \sum_{s \neq r} p_s^{(j)} + p_r^{(k)}, \end{aligned}$$

with the analogous formula interchanging j, k . On the normalized symmetric vectors $x, 3^{-1/2} \sum_r p_r^{(j)}, 3^{-1/2} \sum_r p_r^{(k)}$, the matrix is

$$B = \begin{pmatrix} 3 & \sqrt{3} & \sqrt{3} \\ \sqrt{3} & 5 & 1 \\ \sqrt{3} & 1 & 5 \end{pmatrix}. \quad (236)$$

The difference of the last two vectors has eigenvalue four. On its orthogonal complement, B becomes $\begin{pmatrix} 3 & \sqrt{6} \\ \sqrt{6} & 6 \end{pmatrix}$, whose eigenvalues are $(9 \pm \sqrt{33})/2$.

For completeness, the remaining four-dimensional subspace consists of coefficient pairs $z, w \in \mathbb{C}^3$ with $\sum_r z_r = \sum_r w_r = 0$ on the two triples $p_r^{(j)}, p_r^{(k)}$. The preceding action sends (z, w) to $(2z + w, z + 2w)$. It therefore has eigenvalues three and one, each with multiplicity two. The three seven-dimensional blocks and the six-dimensional block exhaust all 27 dimensions. Their largest eigenvalue is $(9 + \sqrt{33})/2$, proving the first formula in (233).

Proposition C.1 gives the second formula. We can also locate its eigenspace explicitly:

$$F_3 = \frac{4}{3\sqrt{3}} \sum_{\pi \in S_3} |\pi(1)\pi(2)\pi(3)\rangle.$$

It belongs to the eigenvalue-six block of A . Thus the larger singular value comes from a bucket representation excluded by the vacuum's symmetry. Subtracting the two exact squared norms gives the stated positive difference. $\square$

C.3 Heavy swaps: separate the marked external supports

Let $U : \mathbb{C}^d \rightarrow \mathbb{C}^n$ be an isometry with rows u_i^* , and let L_g, H_g count the light and heavy sites in block $g \in [s]$. Define

$$H_0(U) = \sum_{g,a,i \neq j} u_i u_j^* \otimes (R_{a,gi} R_{a,gj}^* + R_{a,gi}^* R_{a,gj}). \quad (237)$$

The operator preserves each block's two counts.

Proposition C.3 (Marked-support heavy-swap estimate). *On a sector with specified counts in each block,*

$$\|H_0(U)\| \leq \sum_{g: L_g H_g > 0} \max(L_g, H_g). \quad (238)$$

If every L_g is even, then on total counts (L, H) with $L \geq 2, H \geq 1$,

$$\|H_0(U)\| \leq C(L, H) := \max\{L, H, L + H - 3\}. \quad (239)$$

The operator vanishes when $LH = 0$. On even-light total grade $\nu = L + 2H \geq 4$, it has norm at most $\nu - 2$.

Proof. Each summand has the exact external compression

$$(U^* \otimes I)(|i\rangle\langle j| \otimes T)(U \otimes I) = u_i u_j^* \otimes T.$$

Occupation projections act on the other factor and commute with this compression. We first prove the bound in the coordinate frame I_n . For one block write the two orientations as

$$A_g = \sum_{a, i \neq j} |i\rangle\langle j| \otimes R_{a,gi} R_{a,gj}^*, \quad B_g = \sum_{a, i \neq j} |i\rangle\langle j| \otimes R_{a,gi}^* R_{a,gj}.$$

Both are self-adjoint, by interchanging i, j after taking the adjoint. Let $\Pi_{H,gi}$ and $\Pi_{L,gi}$ be the local heavy and light projections, and set

$$Q_{g,H} = \sum_i |i\rangle\langle i| \otimes \Pi_{H,gi}, \quad Q_{g,L} = \sum_i |i\rangle\langle i| \otimes \Pi_{L,gi}.$$

The input marked index j and output marked index i are both heavy for A_g , and both light for B_g . Thus

$$A_g = Q_{g,H} A_g Q_{g,H}, \quad B_g = Q_{g,L} B_g Q_{g,L}, \quad Q_{g,H} Q_{g,L} = 0.$$

Their joint norm is consequently the maximum of their two norms.

The local exchange from light-heavy to heavy-light is

$$K = \sum_a |f_a \otimes e_a\rangle\langle e_a \otimes f_a|.$$

Both indexed families in this formula are orthonormal: the light factor makes cross inner products zero, while each heavy vector is unit. Hence K is a partial isometry and $\|K\| = 1$.

Decompose a marked-heavy sector by its physical occupation pattern and its marked external site. Each such block has exactly L_g possible exchange neighbors under A_g , each of norm at most one. The reverse neighbor count is also L_g . For a vector of these components, Cauchy-Schwarz at each output and then summation give

$$\|A_g x\|^2 \leq \sum_{\beta} L_g \sum_{\alpha \sim \beta} \|x_{\alpha}\|^2 \leq L_g^2 \sum_{\alpha} \|x_{\alpha}\|^2.$$

Thus $\|A_g\| \leq L_g$. The marked-light argument gives $\|B_g\| \leq H_g$. When either count is zero both orientations vanish. Taking the maximum first and then externally compressing proves the one-block bound for U . Triangle inequality over blocks proves (238).

If every contributing block satisfies $L_g \geq H_g$, the sum in (238) is at most L . If every one satisfies $H_g \geq L_g$, it is at most H . Otherwise there is a strictly light-dominated contributing block and a strictly heavy-dominated one. Their respective minima $\min(L_g, H_g)$ are at least one and two: the latter uses the positive even light count. Subtracting these minima from the sum of all counts gives at most $L + H - 3$. This proves (239), including mixtures with ties. Finally each of $L, H, L + H - 3$ is at most $L + 2H - 2$ when $L \geq 2, H \geq 1$. These are precisely the possible nonzero even-light sectors, whose minimum grade is four. $\square$

Proposition C.4 (Sharpness within the signed-bucket sector). *For one block and any $L, H \geq 1$, the bound $\max(L, H)$ is attained for the coordinate frame $U = I_{L+H}$. When L is even, it is attained by signed-bucket invariant vectors.*

Proof. Put $N = L + H$. For a heavy set $T \subseteq [N]$ of size H , define

$$z_T = \frac{1}{\sqrt{b}} \sum_a \bigotimes_{i \in T} f_a^{(i)} \bigotimes_{i \notin T} e_a^{(i)}.$$

At least one factor is light, so different bucket summands are orthogonal and $\|z_T\| = 1$. Permuting buckets fixes their sum. If L is even, each bucket-sign action multiplies each summand by the factor $\epsilon_a^L = 1$, so z_T is signed-bucket invariant.

Fix $H - 1$ heavy sites T_0 , and let j range over the other $L + 1$ sites. The vectors $|j\rangle \otimes z_{T_0 \cup \{j\}}$ are orthonormal. A heavy marked exchange moves j to any of the L light sites, with coefficient one: the local action sends $f_a \otimes e_a$ to $e_a \otimes f_a$. Thus A_g acts on this subspace as the adjacency matrix of the complete graph on $L + 1$ vertices, and B_g vanishes there. The uniform sum is an eigenvector with eigenvalue L .

Similarly, fix $L - 1$ light sites and vary the remaining marked light site over $H + 1$ choices. The corresponding vectors have H heavy sites and form a complete-graph subspace for B_g , with eigenvalue H , while A_g vanishes. The upper bound from Proposition C.3 therefore attains $\max(L, H)$. Both constructions have the asserted symmetry when L is even. $\square$

D Numerical certificates for the regular-entry constants

The constants of Sections 3.6, 4.4.1 and 5.4 are closed forms; the decimal bounds asserted for them are certified by evaluating the following finite expressions with outward-rounded interval arithmetic at forty decimal digits.

The parity constant. With $s_\infty = \frac{1}{2} + \frac{\log 17}{8}$ and $f(u) = \log \cosh(s_\infty u)$, convexity of f gives, for $N = 256$,

$$\frac{1}{N} \sum_{i=0}^{N-1} f\left(\frac{i+1/2}{N}\right) \leq \int_0^1 f(u) du \leq \frac{1}{N} \left[\frac{f(0) + f(1)}{2} + \sum_{i=1}^{N-1} f\left(\frac{i}{N}\right) \right],$$

and $\frac{\sqrt{17}}{2} \exp(\cdot)$ of the two sides brackets C_{sph} between 2.30997833 and 2.30998095; hence $\hat{\Theta}_q < 2.31$ for every q by Theorem 4.13. The same construction with $s = \frac{1}{2} \log 3 + \frac{1}{8} \log 17$ in place of s_∞ , which corresponds to the sign factor $3^{k/(2q)}$ of the L_4 interpolation instead of β_q^k , gives a constant below 2.34.

The assembly constants. For each admitted c in (126), with $\ell_c = 2.31e^{1/(2c)}$, $\chi_0 = \frac{7}{3}2^{1/4}$, $N_c = 2 + \frac{5}{4(50c \log 2 - 1)}$, $\delta = \frac{8}{5 \log 2} - 2$, $P_3 = 2[625c \log 2]$, $\tau_c = \delta + \eta(P_3/4)$ and $K_c = (\frac{7}{3})^{\delta/2} \sqrt{(P_3 - 1)/3}$, the certified finite comparisons are: $\tau_c - m_c > 1.094$ and $(\frac{33}{16})^{2/17} (\frac{4250c+1}{16})^{1/2} < K_c$ (the tail of the blocks $a \geq 4$), $A_1 < C_c$, and the roundings of C_c , ζ_c and $2C_c 2^{\zeta_c}$ in Table 3 and (124). At $c = 1$: $K_1 = 19.3720\dots$, $\tau_1 = 8.12858\dots$, $N_1 = 2.03713\dots$, $C_1 = 417.0507\dots < 418$, $\zeta_1 = 10.62858\dots < 10.629$, $A_1 = 158.015\dots$ and $2C_1 2^{\zeta_1} = 1320514\dots < 1.33 \cdot 10^6$; at $c = 2$: $K_2 = 27.3882\dots$, $\tau_2 = 9.12722\dots$, $C_2 = 454.956\dots$, $\zeta_2 = 11.62722\dots$. The bound $\frac{19}{17}(2 - \frac{256}{250+8/17}) = 1.0930\dots < 1.094$ is the common tail exponent margin. The script `regular_constants_check.py`, named for the author release, evaluates every expression of this appendix in interval arithmetic and prints the brackets; it uses only `mpmath`.

The local mean. The values quoted in the proof of Corollary 4.14 are $3^{2/\log 2 - 1} \sqrt{3} = 13.744\dots$, $4/\log 2 - 1 = 4.7707\dots$, $3^{2/\log 3 - 1} \sqrt{5} = 5.507\dots$, $4/\log 3 - \frac{1}{3} = 3.3076\dots$, $\sqrt{23/3} = 2.769\dots$, $\eta(6) = 8/3$, $\sqrt{31/5} = 2.489\dots$, $\eta(16/3) = 5/2$, $\sqrt{33/5} = 2.569\dots$ and $\eta(17/3) = 2.588\dots$.

E Constructions for the continuation rules and the aggregate transfer

This appendix contains the proofs and construction details removed from Sections 7 and 8.9 so that the statements and their limitations could be read without interruption.

E.1 Proof of the cyclic seed reduction

Proof of Theorem 7.11. A matrix unit E_{ij} of sector λ maps v_a to $e_i \otimes r_{a,\lambda,j}$ and annihilates the other sectors, and these vectors span the right side of (160); this proves the equality and (161). The space W is invariant under the self-adjoint algebra $\mathcal{A}$, hence reducing. Every seed lies in W , so $v_a v_b^*$ maps into W and vanishes on $W^\perp$. The complement is $\bigoplus_\lambda \mathbb{C}^{d_\lambda} \otimes M_\lambda^\perp$, on which every seed operator is zero and $\mathcal{A}$ acts as $M_{d_\lambda} \otimes I_{\mu_\lambda - t_\lambda}$; this gives (162). Restriction to a common reducing subspace is a $*$ -homomorphism, which proves the closure under later operations. $\square$

E.2 Proof of the signed-cycle separation

Proof of Theorem 7.12. Let $J = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and $\Delta = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, and put

$$\Gamma_1 = \Delta \otimes I \otimes I, \quad \Gamma_2 = J \otimes \Delta \otimes I, \quad \Gamma_3 = J \otimes J \otimes \Delta, \quad \Gamma_4 = J \otimes J \otimes J.$$

These are traceless symmetric matrices that square to I_8 and anticommute in distinct indices, because $J\Delta = -\Delta J$ at the first differing tensor position. Hence $\Gamma(v) = \sum_j v_j \Gamma_j$ satisfies $\Gamma(v)\Gamma(w) + \Gamma(w)\Gamma(v) = 2\langle v, w \rangle I_8$. Choose

$$\begin{aligned} v_1^+ &= (1, 0, 0, 0), & v_2^+ &= \frac{1}{2}(1, 1, 1, 1), & v_3^+ &= \frac{1}{2}(1, 1, 1, -1), \\ v_1^- &= (1, 0, 0, 0), & v_2^- &= \frac{1}{2}(-1, 1, 1, 1), & v_3^- &= \frac{1}{2}(-1, -1, -1, -1), \end{aligned}$$

and $A_i^\pm = \Gamma(v_i^\pm)/\sqrt{3}$. The vectors are unit vectors with all pairwise inner products $+1/2$ in the first triple and $-1/2$ in the second. Therefore $(A_i^\pm)^2 = I_8/3$, every spectrum is $\pm 1/\sqrt{3}$ with multiplicity four, and for unit v, w with inner product c the Clifford relation gives $[\Gamma(v), \Gamma(w)]^*[\Gamma(v), \Gamma(w)] = 4(1 - c^2)I_8$, so every retained pairwise positive operator is $I_8/3$. For real weights α, β , $\|\alpha \varepsilon_i A_i^\pm + \beta \varepsilon_j A_j^\pm\|^2 = (\alpha^2 + \beta^2 \pm \alpha\beta \varepsilon_i \varepsilon_j)/3$, whose law does not depend on the sign of the alignment. With $T = \varepsilon_1 \varepsilon_2 + \varepsilon_1 \varepsilon_3 + \varepsilon_2 \varepsilon_3$, which is 3 with probability $1/4$ and -1 otherwise,

$$X_\pm^2 = (1 \pm T/3) I_8.$$

Both normalized second moments are 1 and both normalized fourth moments are $4/3$, while the sixth moments are $20/9$ and $16/9$. The missing signed invariant is the product of the three pairwise inner products, $+1/8$ against $-1/8$; a sign change of one coefficient alters two edges of this triangle and cannot change the product. Finally $T^2 = 3 + 2T$ gives $p_0(X_+) = 0$ and $p_0(X_-) = (4T/9)I_8$, and adding $\tau \in (0, 2/9]$ proves (163); since the output is scalar, one coordinate vector certifies its norm on every input. $\square$

E.3 Closure operations, costs and examples for the aggregate transfer

Continue with the notation of Section 8.9. The descriptor is closed, exactly, under the operations a calibration needs: products and word extension by ordered coefficient convolution $C_\nu = \sum_{\alpha+\beta=\nu} C'_\alpha C''_\beta$; addition before norms; the adjoint, which conjugates and transposes coefficients and reverses every ordered product; concatenation of aggregate lists with every original sign identification retained; and the imposition of an actual equality or parity

relation by identifying named variables or adjoining its zero–one indicator, followed by recomputation of the schedule and of ρ . With naive truncated convolution the transfer costs $O(L2^{\beta+a}(w+1)^{\binom{w+N}{w}^2})$ rational operations and the matrix readout $O(d^3 \binom{w+N}{w}^2 \log(p+1))$, with $O(2^\beta \binom{w+N}{w})$ scalar and $O(d^2 \binom{w+N}{w})$ matrix coefficients stored; at fixed w, a, β, R and $p = O(\log(d/\delta))$ these are polynomial in d and L . The magnitude of the rational data is charged separately: an aggregate with a very large range does not enlarge the state, although its bit length does enter the arithmetic. The elimination itself is the standard variable-elimination scheme of probabilistic inference [52]; the point here is the exact truncated-exponential message, which carries all moments through degree N at once.

The compression is real. For independent weighted rows $T = \sum_j a_j \varepsilon_j$ one has $a = 1$ and $\beta = 0$; the aggregate $T = \sum_{j=0}^{79} 2^j \varepsilon_j$ takes 2^{80} distinct values, yet every moment through degree 12 is carried by 13 rational message coefficients. Overlapping path statistics such as $T_1 = \sum_j a_j \varepsilon_j$, $T_2 = \sum_j b_j \varepsilon_j \varepsilon_{j+1}$, $T_3 = \sum_j c_j \varepsilon_j \varepsilon_{j+2}$ have $w = 3$, $a \leq 3$ and a schedule with $\beta \leq 2$, and their dependence is retained rather than replaced by independent aggregate coordinates. A bucket in 2^h locations can be encoded by its h bits with the vertex sign beside them, so that weighted collision statistics on a graph of frontier width t have boundary at most $(h+1)t$ and local arity at most $2(h+1)$; reused edges and closed triangles are then actual constraints, and a polynomial in a fixed number of these statistics can carry dense, irreducible, noncommuting coefficient matrices.

F The classification of dense sign graph matrices

This appendix gives the operator-based proof of Theorem 5.3 for the model (118). The expected-norm classification also has the proof of Xu, Fu, Qiao, and Tao [30, Theorem 1.3]. The common separator and role-partitioning background is described in Section 5.2 and [28, Sections 5–6]. Section 5.2 introduced the objects, the moment identity (120) and the factorization (121); here every step is proved. From the body of the paper we use three results: the connected creation estimate, Proposition 3.9; the vector-valued Boolean fourth-moment inequality, Lemma 3.18; and the moment compression of Theorem 2.3, whose graph instance is proved again below in the form in which it is used. The remaining estimates are proved below; neither of the cited classification proofs is invoked as a black box.

Throughout, the shape $H = (\mathcal{V}, E)$ with its boundary lists U, V is fixed, $h = |\mathcal{V}|$, $e = |E|$, and $s, \mathcal{S}_{\min}, b_{\text{att}}(S), b_*, i_0$ and $T_{H,n}$ are as in (119). Constants $c_H, C_H > 0$ depend only on the shape, $C_{H,A}$ also on a displayed parameter A , and both may change from line to line; “eventually” means for $n \geq n_0(H)$, or $n \geq n_0(H, A)$ when A is present. For a real random variable Z and $p > 0$ we write $\|Z\|_{L_p} = (\mathbb{E}|Z|^p)^{1/p}$. We write $(m)_q = m(m-1) \cdots (m-q+1)$, with $(m)_q = 0$ when $q > m$, and $d_q = (2q-1)!!$, $d_0 = 1$, the number of perfect matchings of $2q$ points. All random spaces are finite sign spaces, so every expectation is a finite sum.

Route through the proof. Coloring separates vertex roles without changing the norm scale by more than shape-dependent constants. Connected scalar fields supply both upper moments and simultaneous lower excursions. The upper matrix bound then distinguishes subcritical signatures, handled by a Schur count, from critical signatures, factored at a minimum separator. For the lower bound, normalized side columns extract the separator contribution without replacing maxima of compatible factors by unrelated maxima. Boundary-free components are reattached last. The fixed-moment conclusion and the limited-independence upper bound have different probability inputs, which are stated separately.

F.1 Three elementary tools

Lemma F.1 (Schur test). *If a finite real matrix $A = (a_{ij})$ has absolute row sums at most r and absolute column sums at most c , then $\|A\| \leq \sqrt{rc}$.*

Proof. For every vector x , the Cauchy–Schwarz inequality with weights $|a_{ij}|$ gives $\sum_i |\sum_j a_{ij}x_j|^2 \leq \sum_i (\sum_j |a_{ij}|)(\sum_j |a_{ij}||x_j|^2) \leq r \sum_j |x_j|^2 \sum_i |a_{ij}| \leq rc \|x\|^2$. $\square$

The occupation operators below are sums of partial permutations with nonnegative integer coefficients, and every application of the Schur test counts, for a fixed source state, the target states with multiplicity, and conversely.

Lemma F.2 (Paley–Zygmund). *If $Y \geq 0$, $0 < \mathbb{E}Y^2 < \infty$, and $0 < \theta < 1$, then $\mathbb{P}\{Y \geq \theta \mathbb{E}Y\} \geq (1 - \theta)^2 (\mathbb{E}Y)^2 / \mathbb{E}Y^2$. The case $Y = 0$ almost surely is treated separately and needs no quotient by its second moment.*

Proof. The event $\{Y < \theta \mathbb{E}Y\}$ contributes at most $\theta \mathbb{E}Y$ to $\mathbb{E}Y$, so its complement contributes at least $(1 - \theta)\mathbb{E}Y$, and by Cauchy–Schwarz that contribution is at most $(\mathbb{E}Y^2)^{1/2} \mathbb{P}\{Y \geq \theta \mathbb{E}Y\}^{1/2}$. $\square$

Lemma F.3 (Fixed-degree sign polynomials). *Let X be a finite real vector whose coordinates are multilinear polynomials of degree at most d in independent uniform signs. Then*

$$(\mathbb{E}\|X\|_2^4)^{1/2} \leq 3^d \mathbb{E}\|X\|_2^2, \quad \mathbb{E}\|X\|_2 \geq c_d (\mathbb{E}\|X\|_2^2)^{1/2}, \quad c_d = (4\sqrt{2} 3^{2d})^{-1}. \quad (240)$$

Proof. Lemma 3.18 applied to the Euclidean-valued polynomial X gives $(\mathbb{E}\|X\|_2^4)^{1/4} \leq 3^{d/2} (\mathbb{E}\|X\|_2^2)^{1/2}$, which is the first inequality. Apply Lemma F.2 to $Y = \|X\|_2^2$ with $\theta = \frac{1}{2}$: the probability that $\|X\|_2^2 \geq \frac{1}{2} \mathbb{E}\|X\|_2^2$ is at least $\frac{1}{4} (\mathbb{E}Y)^2 / \mathbb{E}Y^2 \geq \frac{1}{4} 3^{-2d}$, and on that event $\|X\|_2 \geq (\frac{1}{2} \mathbb{E}\|X\|_2^2)^{1/2}$. $\square$

F.2 Reductions and the colored model

Lemma F.4 (Reductions). (i) *Deleting all edges internal to U or internal to V preserves the family of separators, the number s , and every count $b_{\text{att}}(S)$ at a minimum separator. It changes $M_{H,n}$ only by multiplication on the left and on the right by diagonal sign matrices.*

(ii) *No minimum separator contains a role of an original connected component that misses both boundaries. Deleting all such components preserves the minimum-separator family and every count $b_{\text{att}}(S)$.*

(iii) *If H' is obtained by deleting the i_0 isolated interior roles, then for $n \geq h$*

$$M_{H,n} = (n - |V(H')|)_{i_0} M_{H',n}, \quad (241)$$

and s and the counts $b_{\text{att}}(S)$ are unchanged.

Proof. (i) If a U -to- V path avoids a proposed separator, truncate it at its first visit to V and then at its last preceding visit to U . The result is a U -to- V path with no boundary role in its interior, hence using no edge internal to a boundary; a path of length zero needs no change. So a set separates U from V before the deletion if and only if it does so after it. A component of the edge-deleted graph that misses both boundaries contains no endpoint of a deleted edge, so it was a component before the deletion, and conversely; its edges to a separator are unaffected. Each deleted edge internal to U is a sign depending on the row assignment alone, and each remaining deleted edge is a sign depending on the column assignment alone (an

edge internal to both boundaries is assigned to one factor); these are diagonal matrices with entries ± 1 . No interior role becomes isolated.

(ii) A role of an original boundary-free component lies on no boundary-to-boundary path, so removing it from a separator leaves a separator; a minimum separator therefore avoids such components. Deleting them leaves the same minimum separators, and the deleted components have no edge to any separator, so they are never counted in b_{att} .

(iii) Every embedding of H' extends to the deleted roles in exactly $(n - |V(H')|)_{i_0}$ ordered ways, and no monomial changes because an isolated role has no edge. The separator assertion follows from (ii). $\square$

The colored model. Fix pairwise disjoint pools $P_x \subseteq [n]$, one for each role x . The colored matrix M^{col} retains the terms of (118) with $\phi(x) \in P_x$ for every role; it is viewed as a zero-padded matrix on the original boundary spaces. Coloring makes global injectivity automatic. Distinct edge roles use disjoint blocks of primitive signs,

$$\mathcal{B}_{xy} = \{\{i, j\} : i \in P_x, j \in P_y\}, \quad xy \in E,$$

so a primitive edge determines the edge role it represents, and the primitive edge support of a colored monomial determines the labels of all nonisolated roles. Different sets of edge roles use independent signs.

Lemma F.5 (Color average). *For $h > 0$, give each ground label an independent uniform color from the h roles and let P_x be the class of color x . Then*

$$M_{H,n} = h^h \mathbb{E}_{\text{colors}} M^{\text{col}}, \quad (242)$$

an identity between polynomials in the signs. Consequently a bound on $\|M^{\text{col}}\|_{L_p}$, $p \geq 1$, that holds uniformly over all pools of sizes at most n transfers to $M_{H,n}$ at the cost of the factor h^h .

Proof. An injective embedding survives the coloring exactly when each of its h labels receives the color of its role, an event of probability h^{-h} ; average (118) term by term. The transfer is the triangle inequality in L_p over the convex combination. $\square$

Lemma F.6 (Fourier projection to one coloring). *Suppose H has no isolated interior role, fix arbitrary disjoint pools, and let $a_H = |\text{Aut}(H; U \cup V)|$ be the number of graph automorphisms fixing every boundary role. For $p \geq 1$,*

$$\|M_{H,n}\|_{L_p} \geq a_H \|M^{\text{col}}\|_{L_p}. \quad (243)$$

Proof. Compress the row and column spaces to the boundary pools; call the result N . Average N over every primitive sign outside the blocks $\mathcal{B}_{xy}$; call the result N_0 . A monomial uses distinct signs, so it survives this conditional expectation exactly when all its primitive edges lie in allowed blocks. For $\sigma \in \{\pm 1\}^E$ let W^σ multiply every sign of the block $\mathcal{B}_{xy}$ by σ_{xy} , and put

$$\mathcal{Q}N_0(W) = 2^{-e} \sum_{\sigma \in \{\pm 1\}^E} \left(\prod_{xy \in E} \sigma_{xy} \right) N_0(W^\sigma). \quad (244)$$

A monomial with k_{xy} primitive edges in $\mathcal{B}_{xy}$ is multiplied by $\prod_{xy} \sigma_{xy}^{k_{xy}+1}$, whose average is one if every k_{xy} is odd and zero otherwise. Since the total degree is e , $\mathcal{Q}$ selects the monomials using each of the e blocks exactly once.

Consider such a monomial, coming from an uncolored embedding ϕ . Every nonisolated pool is an endpoint of a used block and is occupied by some $\phi(x)$; the pools of isolated

boundary roles are occupied by the boundary compression; there are no other isolated roles. Hence the h roles occupy all h pools, and the map sending x to the role of the pool containing $\phi(x)$ is a bijection of $\mathcal{V}$ which maps E onto E (each edge role lands in exactly one block) and fixes the boundary roles: a boundary-fixing automorphism. Conversely, a colored embedding reindexed by any such automorphism survives both averages and contributes the same monomial at the same boundary indices. Hence

$$\mathcal{Q}N_0 = a_H M^{\text{col}}. \quad (245)$$

Compression contracts the operator norm, conditional expectation contracts L_p sizes by Jensen, and each block flip preserves the law of the independent signs, so the triangle inequality in (244) gives $|||\mathcal{Q}N_0|||_{L_p} \leq |||N_0|||_{L_p} \leq |||M_{H,n}|||_{L_p}$. If $e = 0$ all roles are boundary roles and the sign averages are empty. $\square$

The inequality (F.6) compares moments, not realizations: full independence enters through the conditional expectation and the invariance under block flips. This is why the lower half of Theorem 5.3 is stated under full independence.

F.3 Connected scalar fields

Let J be a fixed simple graph with root roles R and a nonempty interior C , $c = |C|$, such that $J[C]$ is connected and every interior role has positive degree; remove root-internal edges. Give all roles disjoint pools, fix a root assignment A , and define the colored scalar field

$$F_A = \sum_{\phi: \phi|_R=A, \phi(x) \in P_x \ (x \in C)} \prod_{xy \in E(J)} W_{\phi(x)\phi(y)}. \quad (246)$$

Unused roots cause no difficulty, and a rootless connected graph with an edge is allowed. Put $B_c = 6c + 8$.

Proposition F.7 (Finite scalar moments). *If all interior pools have size at most n and $\rho = (2cq)^{B_c}/n < 1$, then*

$$\mathbb{E}F_A^{2q} \leq \frac{d_q}{1-\rho} n^{cq}. \quad (247)$$

If all interior pools have the same size m , then also

$$\mathbb{E}F_A^{2q} \geq d_q ((m)_q)^c. \quad (248)$$

Consequently, for every fixed $B > 0$, uniformly for $1 \leq q \leq (\log m)^B$ and equal pools, $\mathbb{E}F_A^{2q} = (1 + o(1)) d_q m^{cq}$.

Proof. Expand the moment over $2q$ ordered colored embeddings. All coefficients are nonnegative, since a product of independent signs has expectation zero or one, and a term survives only if every primitive sign occurs an even number of times.

Multiplicities

Every interior label occurring in a surviving term occurs in at least two copies: otherwise an edge incident to its unique occurrence would be a singleton primitive edge, since role colors prevent a primitive edge from representing two different edge roles. Let v be the number of distinct interior labels and $d = cq - v \geq 0$ the *defect*; the $2cq$ interior occurrences satisfy $\sum_{\lambda} (k_{\lambda} - 2) = 2d$, where k_{λ} is the number of copies containing the label λ . Mark every copy that contains a label of multiplicity at least three. Since $k \leq 3(k-2)$ for $k \geq 3$, the number r of marked copies satisfies $r \leq \sum_{k_{\lambda} \geq 3} k_{\lambda} \leq 6d$.

Unmarked copies pair up as wholes

An interior label x of an unmarked copy occurs in exactly one other copy, and every edge of the first copy at x must recur in that copy, because the primitive edge occurs an even number of times and the only other occurrence of x is there. Propagating along the connected graph $J[C]$ (using a root attachment in the case $c = 1$), all interior labels of the first copy occur in that same copy, which therefore has the same interior image set and is unmarked; role colors identify roles, so the two embeddings agree role by role, and no third copy uses any of their labels. The unmarked copies are thus partitioned into *clean pairs*.

Defect zero

Count equality patterns canonically, scanning copy positions and then the roles in a fixed order and naming each label at its first appearance. At defect zero every copy is in a clean pair, so there are exactly d_q patterns, one per pairing of the $2q$ positions, and on equal pools each has $((m)_q)^c$ labelings (each role receives q distinct labels, one per pair). Since all terms are nonnegative, this proves (248).

Positive defect

Encode a pattern of defect $d \geq 1$ by the set of marked positions, a pairing of the unmarked positions, and an arbitrary set partition of the cr marked role occurrences; the clean pairs introduce their labels at the first copy of each pair. This reconstructs every pattern, so the number $N_{q,d}$ of patterns of defect d satisfies

$$\begin{aligned} N_{q,d} &\leq d_q \sum_{0 \leq r \leq \min(2q, 6d)} \binom{2q}{r} (cr)^{cr} \\ &\leq d_q (2q+1) (2cq)^{6(c+1)d} \\ &\leq d_q (2cq)^{(6c+8)d}. \end{aligned}$$

The factors encode, respectively, the clean-pair matching, the marked positions, and an overcount of the marked-label partition. Here the pairing count of a smaller even set is at most d_q , $\binom{2q}{r} (cr)^{cr} \leq (2cq)^{(c+1)r}$ and $2q+1 \leq (2cq)^2$. A pattern with $cq - d$ distinct labels has at most n^{cq-d} realizations on pools of sizes at most n ; summing over $d \geq 0$ gives the geometric series (247).

Asymptotics

Finally $(m)_q/m^q = \prod_{j < q} (1 - j/m) = 1 + O(q^2/m)$ uniformly when $q^2/m \rightarrow 0$, and $\rho \rightarrow 0$ throughout a fixed polylogarithmic range of q . $\square$

Corollary F.8 (Scalar upper moments). *For arbitrary pools of sizes at most n , every fixed $A_0 > 0$, and all sufficiently large n ,*

$$\|F_A\|_{L_p} \leq C_{J, A_0} n^{c/2} \sqrt{p}, \quad 1 \leq p \leq A_0 \log(2n),$$

uniformly in the root labels; a rootless field obeys the same bound.

Proof. Choose an integer q with $p \leq 2q \leq p+2$. Then $\|F_A\|_{L_p} \leq \|F_A\|_{L_{2q}} \leq (d_q/(1-\rho))^{1/(2q)} n^{c/2} \leq C \sqrt{2q} n^{c/2}$ by $d_q \leq (2q)^q$, and ρ is bounded away from one in the stated range since $q = O(\log n)$. $\square$

Proposition F.9 (Conditional moment concentration). *Suppose J has at least one root-to-interior edge and all interior pools have size m . Split the signs of (246) into the interior*

signs ω and the root-to-interior signs x , and put $Z_q(\omega) = \mathbb{E}_x F_A^{2q}$, $\mu_q = \mathbb{E}_\omega Z_q$. For every fixed $B > 0$, uniformly for $1 \leq q \leq (\log m)^B$ and all large m ,

$$\mu_q = (1 + o(1)) d_q m^{cq}, \quad \frac{\text{Var } Z_q}{\mu_q^2} \leq C_c \frac{(4cq)^{B_c}}{m}. \quad (249)$$

After identifying the root-to-interior arrays by their interior role labels, $Z_q(\omega)$ is the same function of the fixed interior signs for every root assignment A . In particular its law is independent of A .

Proof. 1. Separate the conditioned arrays from their labels.

Once the interior signs ω are fixed, changing the root assignment only renames the root-to-interior sign variables over which $\mathbb{E}_x$ is taken. These variables have the same product law and are independent of ω . Thus the conditional moment is the *same function* of ω for every root assignment, not merely a random variable with the same distribution. Its mean is

$$\mu_q = \mathbb{E} F_A^{2q} = (1 + o(1)) d_q m^{cq}$$

by Proposition F.7.

2. Express the second moment using two replicas.

To compute $\mathbb{E}_\omega Z_q^2$, use two groups of $2q$ copies of the field. Both groups use the same interior signs, but the root-to-interior arrays are replicated independently between the two groups. A term survives only when each interior primitive sign has even multiplicity and each root-edge sign has even multiplicity *within its own replica group*.

There are $4cq$ interior occurrences. If v distinct labels occur, write $d = 2cq - v$ for the defect. The marking argument in the preceding proof leaves at most $6d$ marked copies. Each unmarked copy has a unique partner with the same interior labels. Because the component has a root attachment, equality of the attachment sign together with its replica index forces this partner to belong to the same group.

3. Count the zero-defect and positive-defect terms.

At defect zero, pair the positions in each group separately. There are d_q^2 choices. All $2q$ pair labels at each role are distinct across both groups, giving the exact contribution

$$d_q^2 ((m)_{2q})^c.$$

For $d \geq 1$, the same marked-copy encoding as before gives at most $d_q^2 (4cq)^{B_c d}$ patterns. Each has at most m^{2cq-d} labelings. With $\rho_2 = (4cq)^{B_c}/m < 1$, summing these upper bounds yields

$$\mathbb{E} Z_q^2 \leq \frac{d_q^2 m^{2cq}}{1 - \rho_2}.$$

This is an upper bound; it need not preserve the exact zero-defect labeling factor.

4. Normalize and subtract the square of the mean.

The lower bound $\mu_q \geq d_q ((m)_q)^c$ implies

$$\frac{\text{Var } Z_q}{\mu_q^2} \leq \frac{1}{1 - \rho_2} \left(\frac{m^q}{(m)_q} \right)^{2c} - 1.$$

In the stated polylogarithmic range, both ρ_2 and q^2/m tend to zero. The product formula for $(m)_q$ therefore gives

$$\frac{\text{Var } Z_q}{\mu_q^2} \leq C_c \left(\rho_2 + \frac{q^2}{m} \right) \leq C_c \rho_2.$$

This proves (249) uniformly in that range. $\square$

Lemma F.10 (Simultaneous component excursions). *Let $C_1, \dots, C_b$, $b \geq 1$, be nonempty connected interiors of sizes $c_1, \dots, c_b$, each attached by at least one edge to a common root set S , all roles with disjoint pools of size m , and let $F_{j,A}$ be the field of C_j at the root tuple A . Then*

$$\mathbb{P}\left\{\max_A \prod_{j=1}^b |F_{j,A}| \geq c_H m^{(c_1+\dots+c_b)/2} (\log m)^{b/2}\right\} \longrightarrow 1. \quad (250)$$

Proof. The components share the root tuple, so their individual maxima cannot simply be multiplied. We instead construct many independent trials at which *all* components are large simultaneously.

1. Obtain typical conditional moments of every component.

Condition on all interior signs. Choose $q = \lfloor \delta \log m \rfloor$, with the positive constant δ to be fixed below. Apply Proposition F.9 at q and $2q$ to each of the finitely many components. Chebyshev's inequality gives an interior event of probability tending to one on which, for a component of size c ,

$$Z_q \geq \frac{1}{4} d_q m^{cq}, \quad Z_{2q} \leq 4 d_{2q} m^{2cq}.$$

These bounds hold for every component simultaneously. They also hold for every root tuple on the same event: as proved in Proposition F.9, its conditional moments are the same functions of the fixed interior arrays after relabeling the independent root arrays. Equality in distribution alone would not suffice here.

2. Produce a conditional excursion at one tuple.

Fix such an interior realization. Conditional Paley–Zygmund applied to $|F|^{2q}$ gives

$$\mathbb{P}_x \left\{ |F| \geq (Z_q/2)^{1/(2q)} \mid \omega \right\} \geq \frac{Z_q^2}{4Z_{2q}} \geq \frac{d_q^2}{256d_{2q}} \geq e^{-C_0 q}.$$

The last inequality follows from $d_q \geq q! \geq (q/e)^q$ and $d_{2q} \leq (4q)^{2q}$, after choosing an absolute C_0 . On this excursion,

$$|F| \geq (d_q/8)^{1/(2q)} m^{c/2} \geq c_0 m^{c/2} \sqrt{q}$$

for an absolute $c_0 > 0$.

3. Use disjoint root-array trials.

Enumerate the pool of each root z as $\lambda_z(1), \dots, \lambda_z(m)$, and consider only the m tuples

$$A^{(i)} = (\lambda_z(i))_{z \in S}, \quad 1 \leq i \leq m.$$

Given the interior signs, the root-to-interior primitive arrays are disjoint between components and between these tuples. They are therefore independent under the conditional law. At each tuple all b excursions occur with conditional probability at least $e^{-C_0 b q}$.

Choose δ so that $C_0 b \delta < 1/2$. The probability, conditional on the typical interiors, that every trial fails is at most

$$(1 - e^{-C_0 b q})^m \leq \exp\{-m e^{-C_0 b q}\} \leq \exp\{-m^{1/2}\}$$

for all sufficiently large m .

4. Multiply the excursions at the successful tuple.

At a successful trial the factors refer to the *same* root tuple, and

$$\prod_{j=1}^b |F_{j,A}| \geq c_0^b m^{(c_1+\dots+c_b)/2} q^{b/2}.$$

Since $q \asymp \log m$ and the typical interior event has probability tending to one, this proves (250). $\square$

The constant c_H in (250) depends on δ , hence on b ; only the order $(\log m)^{b/2}$ is identified.

F.4 Occupation states, patterns and signatures

For the operator argument all roles have disjoint pools of sizes at most n . Let $\mathcal{E}$ be the set of all unordered pairs of ground labels, including pairs unused by the shape. The Walsh characters $\chi_\omega = \prod_{a \in \omega} W_a$, $\omega \subseteq \mathcal{E}$, form an orthonormal basis of the sign space, with $\chi_\omega \chi_{\omega'} = \chi_{\omega \Delta \omega'}$. We identify χ_ω with the occupation state $|\omega\rangle$ of grade $|\omega|$; $P_{\leq K}$ projects onto grades at most K ; the *vertex support* of ω is the set of endpoints of its edges, of size at most $2|\omega|$. Multiplying by a primitive sign toggles it in and out of ω , so ω records the signs used an odd number of times, and the expectation of a product of signs is the coefficient of the vacuum $|\emptyset\rangle$.

For a colored embedding ϕ write $E_\phi = \{\{\phi(x), \phi(y)\} : xy \in E\}$. Let $\mathcal{H}_U, \mathcal{H}_V$ be the spaces of colored boundary assignments. Multiplication by the self-adjoint dilation of M^{col} has the Walsh representation $\mathcal{W}$ on $(\mathcal{H}_U \oplus \mathcal{H}_V) \otimes \ell^2(2^\mathcal{E})$ whose *forward* block, from the U -register to the V -register, is the adjoint direction of the matrix,

$$|a, \omega\rangle \mapsto \sum_{\phi: \phi|_U = a} |\phi|_V, \omega \Delta E_\phi\rangle, \quad (251)$$

and whose reverse block is its adjoint. The coefficients are nonnegative integers: all cancellation is absorbed into the geometry of the basis.

Lemma F.11 (Vacuum compression). *For every integer $q \geq 1$, with $K = eq$,*

$$\mathbb{E} \text{tr}[(M^{\text{col}}(M^{\text{col}})^*)^q] = \sum_a \langle a, \emptyset | (P_{\leq K} \mathcal{W} P_{\leq K})^{2q} | a, \emptyset \rangle \leq n^{|U|} \|P_{\leq K} \mathcal{W} P_{\leq K}\|^{2q}. \quad (252)$$

Proof. The U -diagonal block of $\mathcal{W}^{2q}$ represents multiplication by $(M^{\text{col}}(M^{\text{col}})^*)^q$, and expectation is the vacuum coefficient, so the expected trace is the sum over the boundary assignments a of the vacuum-to-vacuum matrix elements of $\mathcal{W}^{2q}$. Every step toggles e distinct primitive edges, so it changes the grade by at most e ; a contributing walk of $2q$ steps from the vacuum to the vacuum therefore has grade at most $e \min(j, 2q - j) \leq eq$ after j steps, and the compression retains it entirely. There are at most $n^{|U|}$ legal assignments a , and each compressed matrix element is at most the operator norm to the power $2q$. $\square$

This is the graph instance of Theorem 2.3, with the external trace dimension $n^{|U|}$ and no dimension factor from the occupation space. It remains to bound the compressed norm uniformly over all states of grade at most K .

Patterns and signatures. For $F \subseteq E$ let $\mathcal{A}_F$ be the forward block (251) restricted to embeddings with $E_\phi \cap \omega = \phi(F)$, so that $\sum_F \mathcal{A}_F$ is the full forward block, $\mathcal{A}_F$ annihilates the edges of F , creates the others, and changes the grade by $e - 2|F|$. Assume there are no isolated interior roles, and let R_F, Z_F, S_F be the signature defined in Section 5.2: with $I = \mathcal{V} \setminus (U \cup V)$, $B = U \cap V$ and $\delta_H(x)$ the edges at x , $R_F = \{x \in (U \setminus B) \cup I : \delta_H(x) \subseteq F\}$,

$Z_F = \{x \in (V \setminus B) \cup I : \delta_H(x) \cap F = \emptyset\}$, $S_F = \mathcal{V} \setminus (R_F \cup Z_F)$. An isolated exclusive boundary role belongs vacuously to its free set; common boundary roles stay in S_F . A role of Z_F is *fresh*: in the forward direction its label ranges over its pool. A role of R_F is *released*: in the reverse direction, where created and annihilated edges are exchanged, its label ranges over its pool. Every other noninput role is *pinned*: it has an edge in F , whose primitive image lies in ω , so its label lies in the vertex support of ω .

Lemma F.12 (Signature and Schur count). *The sets R_F, Z_F are disjoint, no edge joins them, and S_F is a separator; hence $|R_F| + |Z_F| \leq h - s$. At grades at most K , the forward column sums and the reverse row sums of $\mathcal{A}_F$, with multiplicities, are at most*

$$n^{|Z_F|}(1 + 2K)^{h-|U|-|Z_F|} \quad \text{and} \quad n^{|R_F|}(1 + 2K)^{h-|V|-|R_F|}. \quad (253)$$

If $|R_F| + |Z_F| < h - s$, then

$$\|P_{\leq K} \mathcal{A}_F P_{\leq K}\| \leq n^{(h-s-1)/2} (1 + 2K)^h. \quad (254)$$

Proof. Only an isolated interior role could lie in both free sets. An edge between them would belong to F , by its endpoint in R_F , and not belong to F , by its endpoint in Z_F . Every U -role outside S_F lies in R_F and every V -role outside S_F lies in Z_F , so a U -to- V path either starts or ends in S_F , or starts in R_F and ends in Z_F and crosses through S_F because no edge joins the two sets; thus S_F separates and $|S_F| \geq s$.

Fix a forward source (a, ω) . Every noninput role incident to F receives a label in the vertex support of ω , of size at most $2K$, and the only unrestricted noninput roles are those of Z_F ; counting unused or inconsistent choices as well only overcounts. This proves the first bound in (253); in the reverse direction the occupied embedding edges are $E \setminus F$ and the free roles are R_F . Lemma F.1 gives a norm at most $n^{(|R_F|+|Z_F|)/2} (1 + 2K)^{h-(|U|+|V|+|R_F|+|Z_F|)/2} \leq n^{(|R_F|+|Z_F|)/2} (1 + 2K)^h$, and strict inequality in $|R_F| + |Z_F| \leq h - s$ saves $n^{1/2}$. $\square$

A pattern is *critical* when $|R_F| + |Z_F| = h - s$, that is, when S_F is a minimum separator. The Schur count charges a factor $2K$ for every pinned label, which at $K \asymp \log n$ would give $(\log n)^h$ instead of $(\log n)^{b_{\text{att}}(S_F)/2}$; the critical patterns are therefore factored at their separator and each piece is estimated with a sharper tool.

F.5 The two creation estimates

Both estimates concern *pure creation*: adding the edges of an embedding to a state none of whose edges is already present. The first is Proposition 3.9: for a rooted shape J with connected interior of size c , fixed root labels, $r = |E(J)| \geq 1$ and $A_c = 2^c + c$, the pure creation operator $\mathcal{C}$ of (41) satisfies

$$\|\mathcal{C}P_{\leq K}\|, \|\mathcal{C}^*P_{\leq K}\| \leq \sqrt{K+1} \left[n^{c/2} + \sqrt{A_c} n^{(c-1)/2} (1 + 2K)^{(c+1)/2} \right], \quad (255)$$

on states that may carry arbitrary edges outside the blocks of J . It is the source of the logarithm. The second concerns a shape whose boundary labels change and whose input boundary is a minimum separator; it contributes no logarithm.

Let L be a graph with an input boundary S and an output boundary O , possibly overlapping, such that S is a minimum S -to- O separator in L , every component of $L - S$ meets O , edges internal to S have been omitted, and every role outside O has an incident edge; isolated roles in O , including common boundary roles, are allowed. Put $v_L = |V(L)|$, $\ell = v_L - |S|$, and define the side creation operator

$$\mathcal{C}_L |a, \omega\rangle = \sum_{\substack{\phi|_{S=a} \\ E_\phi \cap \omega = \emptyset}} |\phi|_O, \omega \cup E_\phi\rangle. \quad (256)$$

Lemma F.13 (Side creation). *If $\ell \geq 1$, then*

$$\|\mathcal{C}_L P_{\leq K}\|, \|\mathcal{C}_L^* P_{\leq K}\| \leq n^{\ell/2} + 2^{v_L/2} (1 + 2K)^{v_L} n^{(\ell-1)/2}. \quad (257)$$

If $\ell = 0$, then $O = S = V(L)$, there are no edges, and $\mathcal{C}_L$ is the identity on the common register.

Proof. If $\ell = 0$, minimality implies $|O| \geq |S|$, while $O \subseteq V(L) = S$. Thus $O = S = V(L)$; all edges were internal to S and have been omitted. The operator is the identity.

Assume henceforth that $\ell \geq 1$. We estimate the Gram matrix $\mathcal{C}_L^* \mathcal{C}_L$ on one source grade k .

1. The diagonal counts extensions of the input labels.

Fix a source (a, ω) . There are at most n^ℓ embeddings extending a . If two such embeddings give the same output assignment and target state, they have the same added edge support. Every role outside O is then determined by an incident edge, while every role in O is determined by the output assignment. Role coloring prevents an edge from representing a different edge role. Hence the embeddings coincide, and the Gram diagonal is at most n^ℓ .

2. An off-diagonal coincidence pins changed labels.

An off-diagonal contribution consists of embeddings ϕ, ψ from different sources with the same output assignment and target state. Let

$$T = \{x : \phi(x) \neq \psi(x)\}$$

be their nonempty set of changed roles. It is disjoint from O . Every ψ -edge incident to T differs from the corresponding ϕ -edge and, by role coloring, from every other edge of ϕ . It nevertheless belongs to the common target. Therefore it was already in the old source state ω .

Every changed role has an incident edge, so every changed ψ -label lies in the vertex support of ω , of size at most $2k$. This restricts the second embedding, but we still need to save one free label in the first embedding.

3. Find an unchanged frontier role outside the input.

We claim there is

$$y \in (N_L(T) \setminus T) \setminus S.$$

Suppose otherwise. Then every neighbor of T outside T lies in S . If $T \cap S = \emptyset$, the set T is a union of components of $L - S$ missing O , contradicting the component hypothesis. If $T \cap S \neq \emptyset$, the smaller set $S \setminus T$ separates S from O : paths starting in $S \setminus T$ are intercepted there, and paths starting in $S \cap T$ cannot leave T without visiting $S \setminus T$. Since $T \cap O = \emptyset$, this contradicts minimality of S .

Choose such a y canonically from T . It is unchanged, and its edge toward T in ψ lies in ω . Thus $\phi(y) = \psi(y)$ has at most $2k$ possible labels. Because $y \notin S$, this pins one of the ℓ otherwise free labels of ϕ .

4. Count the coincidences and apply Schur's bound.

For fixed T , there are at most

$$n^{\ell-1} (2k)^{|T|+1}$$

pairs: choose the pinned first-embedding label and its remaining free labels, then the changed second-embedding labels in ω . The second source is determined by the pair, namely

$$(\psi|_S, (\omega \cup E_\phi) \setminus E_\psi).$$

Summing over at most 2^{v_L} changed sets gives the convenient uniform bound

$$2^{v_L}(1+2k)^{2v_L}n^{\ell-1}$$

on each off-diagonal row sum. Symmetry of the Gram matrix gives the same column bound. Lemma F.1 now yields

$$\|\mathcal{C}_L^* \mathcal{C}_L\| \leq n^\ell + 2^{v_L}(1+2k)^{2v_L}n^{\ell-1}$$

on grade k . Take square roots and use $\sqrt{x+y} \leq \sqrt{x} + \sqrt{y}$.

Distinct source grades have distinct target grades. Maximizing over $k \leq K$ proves the creation estimate. On an output grade at most K , the adjoint is the transpose of a creation block whose source grade is no larger; the same bound proves the adjoint estimate. $\square$

The hypotheses on L are used once, to produce the frontier role y . If a component of $L - S$ missed O , the changed set could be that component, no ϕ -label would be pinned, and the off-diagonal sum would be comparable to n^ℓ times a polynomial in the grade; that is the situation of a floating component, where the norm grows with the grade. In the application both hypotheses follow from the minimality of the separator.

F.6 Critical patterns and the core upper bound

In this subsection H is a core shape, without edges internal to either boundary by Lemma F.4(i), and $f_0 = (h-s)/2$. Fix a critical pattern F and put $S = S_F$, a minimum separator by Lemma F.12. Every component of $H - S$ lies wholly in R_F or wholly in Z_F , since no edge joins the two sets; a component meeting U lies in R_F and one meeting V lies in Z_F . Let $C_1, \dots, C_b$ be the remaining components; each has an edge to S because H is a core, so $b = b_{\text{att}}(S)$. Let L be the subgraph on S and the components meeting U , with input S and output U , and R its counterpart for V ; include all edges at their roles outside S and omit edges internal to S . With $\ell = |V(L) \setminus S|$, $r = |V(R) \setminus S|$ and $c_j = |C_j|$,

$$\ell + r + \sum_{j=1}^b c_j = h - s. \quad (258)$$

Lemma F.14 (Side hypotheses). *S is a minimum input-to-output separator in both side graphs, every component of a side outside S meets its output, and every role of a side outside its output has positive degree.*

Proof. The component statement holds by construction. If a smaller set separated U from S in L , it would separate U from V in H : a boundary-to-boundary path has a prefix ending at its first visit to S (of length zero when the path starts at a common role), and that prefix lies in L . This contradicts the minimality of S ; the right side is the same with the path reversed. A role of $S \setminus U$ with no left edge would be redundant in a separator between S and U , contradicting the minimality just proved, and a role outside $S \cup U$ lies in a component meeting U and is not isolated. $\square$

For each C_j take the full register S as its roots and include its internal and attachment edges; let $\mathcal{C}_j$ be its pure scalar creation operator, block diagonal in the separator assignment, and put $\mathcal{B}_j = \mathcal{C}_j^*$ if $C_j \subseteq R_F$ and $\mathcal{B}_j = \mathcal{C}_j$ if $C_j \subseteq Z_F$. For a separator assignment A the edges internal to S have prescribed primitive images; let $\mathcal{D}_S$ project onto the states in which exactly those of them prescribed by F are occupied and then toggle all of them. It preserves A and has norm at most one. Write $\mathcal{C}_L, \mathcal{C}_R$ for the side operators (256).

Lemma F.15 (Exact critical product). *The forward block of a critical pattern factors as*

$$\mathcal{A}_F = \mathcal{C}_R \mathcal{D}_S \left(\prod_{j=1}^b \mathcal{B}_j \right) \mathcal{C}_L^*, \quad (259)$$

the floating factors in any order. If the source and target grades are at most K , every intermediate grade is at most $K + e$.

Proof. Read from the right. Every edge at a left component belongs to F , so $\mathcal{C}_L^*$ annihilates exactly the left edges; it chooses the left labels and the separator labels extending the input assignment, and stores the latter in the register. Each floating factor chooses its component's labels and creates or annihilates its edges, as its signature prescribes. The separator factor supplies the remaining tests and toggles. Every edge at a right component lies outside F , so $\mathcal{C}_R$ creates exactly those edges and chooses the right labels and the output assignment. The edge-role families of the factors are disjoint, hence, by role coloring, so are their primitive edge families; no factor can alter the occupancy test of another, and distinct pools make all chosen labels globally injective. The procedure is therefore a bijection, with multiplicities, between contributions to the product and embeddings with $E_\phi \cap \omega = \phi(F)$, and the same disjointness permits reordering the floating factors. At most the e edges of one embedding change during the product, so an intermediate grade exceeds the source grade by at most e ; inserting the projection $P_{\leq K+e}$ at every stage changes no term with endpoint grades at most K . $\square$

Proposition F.16 (Critical norm bound). *There are constants C_H and $D_H = 6h + 4$ such that every critical pattern with separator S satisfies*

$$\|P_{\leq K} \mathcal{A}_F P_{\leq K}\| \leq C_H n^{f_0} (K+1)^{b_{\text{att}}(S)/2} + C_H n^{f_0-1/2} (K+1)^{D_H}, \quad (260)$$

uniformly over pools, boundary assignments and occupied states.

Proof. Apply Lemma F.15 with the cutoff $K + e$ at every stage, Lemma F.13 to the two sides, (255) to each floating factor, and the norm-one bound to $\mathcal{D}_S$. The product of the leading terms is at most $n^{\ell/2} n^{r/2} \prod_j \sqrt{K+e+1} n^{c_j/2} \leq C_H n^{f_0} (K+1)^{b/2}$ by (258). Every other term of the expanded product contains an error term and therefore loses at least $n^{1/2}$, with a remaining polynomial dependence on $K+1$ whose degree is bounded by $v_L + v_R + \sum_j (c_j + 2) \leq 5h$; the shift by e changes only the shape-dependent constant. The degree $D_H = 6h + 4$ also exceeds the degree h of (254). $\square$

Proposition F.17 (Core upper bound). *For every fixed core shape, every fixed $A > 0$, and arbitrary disjoint pools of sizes at most n ,*

$$\|M^{\text{col}}\|_{L_p} \leq C_{H,A} n^{(h-s)/2} (\log(2n))^{b_*/2}, \quad 1 \leq p \leq A \log(2n), \quad (261)$$

and the same bound holds for $M_{H,n}$.

Proof. The forward block is the sum of the 2^e patterns and the dilation has the norm of its forward block, so (254) and (260) give, since the number of patterns is fixed and the largest floating count is b_* ,

$$\|P_{\leq K} \mathcal{W} P_{\leq K}\| \leq C_H n^{f_0} (K+1)^{b_*/2} + C_H n^{f_0-1/2} (K+1)^{D_H}.$$

With $K = eq$ and e absorbed into C_H , Lemma F.11 gives

$$\mathbb{E} \text{tr}[(M^{\text{col}}(M^{\text{col}})^*)^q] \leq n^{|U|} \left[C_H n^{f_0} (q+1)^{b_*/2} + C_H n^{f_0-1/2} (q+1)^{D_H} \right]^{2q}. \quad (262)$$

Choose an integer q with $\max\{p/2, \log(2n)\} \leq q \leq C_A \log(2n)$. For any matrix $\|M\|^{2q} \leq \text{tr}[(MM^*)^q]$; the prefactor has $n^{|U|/(2q)} \leq e^{|U|/2}$; and $n^{-1/2}(q+1)^{D_H-b_*/2} \rightarrow 0$, so the second term in the bracket is eventually smaller than the first. Taking $2q$ -th roots and using $p \leq 2q$ proves the colored bound, and Lemma F.5 transfers it to $M_{H,n}$; the boundary-internal edges removed at the start are restored by Lemma F.4(i). Empty pools give the zero matrix unless the role set is empty, and a shape without edges has only boundary roles, no floating factors, and side operators that are identities or all-ones output factors, to which the same argument applies at grade zero. $\square$

The choice $q \asymp \log n$ balances the estimates in this proof. When $|U| > 0$, it makes the trace prefactor $n^{|U|/(2q)}$ bounded, and the critical creation factor becomes $(q+1)^{b_*/2} \asymp (\log n)^{b_*/2}$, matching the lower-bound scale. This is not a necessity claim for every shape or every proof: for example, a scalar exterior space has no growing trace-dimension factor.

F.7 The core lower bound

Let H be a nonempty core shape, choose a minimum separator S with $b_{\text{att}}(S) = b_*$, and give every role a pool of size $m = \lfloor n/h \rfloor$. Let L and R be the side graphs of Section F.6, now as random matrices: $L[a, A]$ is the sum over colored embeddings of the left side with $\phi|_U = a$ and $\phi|_S = A$ of the product of the left-side signs, and $R[A, b]$ is defined in the same way with $\phi|_V = b$; edges internal to S are excluded from both. Let $C_1, \dots, C_b$ be the floating components at S and $F_{j,A}$ their fields (246).

Lemma F.18 (Separator product). *There is the exact factorization*

$$M^{\text{col}} = LDR, \quad D_{A,A} = \chi_{H[S]}(A) \prod_{j=1}^b F_{j,A}, \quad (263)$$

where $\chi_{H[S]}(A)$ is the product of the signs on the separator-internal edges, and L, D, R are independent.

Proof. For fixed separator labels the free roles of the two sides and of the floating components are disjoint and their pools make injectivity automatic, so an embedding of H is the same thing as a separator assignment A together with embeddings of L , of R and of each C_j extending A , and its monomial is the product of the monomials of the pieces; common boundary roles impose their constraints through the same register. The three factors use disjoint edge-role blocks, hence independent signs. $\square$

This is the decomposition (259) for the random matrix itself rather than for one pattern of its occupation operator. For a separator assignment A put $u_A = Le_A / \|Le_A\|$ and $v_A = R^*e_A / \|R^*e_A\|$, either vector being zero when its denominator vanishes.

Lemma F.19 (Normalized-column symmetry). *For each fixed A ,*

$$\mathbb{E}_L[L^*u_A] = e_A \mathbb{E}_L\|Le_A\|, \quad \mathbb{E}_R[Rv_A] = e_A \mathbb{E}_R\|R^*e_A\|. \quad (264)$$

Consequently, with $\alpha_L = \mathbb{E}\|Le_A\|$ and $\alpha_R = \mathbb{E}\|R^*e_A\|$, which do not depend on A ,

$$\mathbb{E}\|LDR\| \geq \alpha_L \alpha_R \mathbb{E} \max_A |D_{A,A}|. \quad (265)$$

Proof. The test columns depend on the side matrices. We first show that their expected action is diagonal in the separator labels, then select the largest diagonal entry of the independent middle factor.

1. Separate distinct separator assignments.

Fix A and $B \neq A$, and choose a separator role z with $B_z \neq A_z$. If $z \in U$, the columns Le_A and Le_B have disjoint row supports. Thus $\langle Le_B, u_A \rangle = 0$ pointwise in this case.

Suppose $z \notin U$. Minimality of S gives a U -to- V path that avoids $S \setminus \{z\}$. Its prefix up to its first separator visit reaches z through some edge zx , with x in a left component.

2. Use a law-preserving flip on one role block.

Flip every sign in the block $\mathcal{B}_{zx}$ whose z -label is B_z . Each monomial of Le_B contains exactly one such sign, so this column changes sign. The column Le_A , its norm, and u_A do not change because $A_z \neq B_z$. Distinct role blocks cannot coincide. The flip preserves the independent sign law, and therefore

$$\mathbb{E}_L \langle Le_B, u_A \rangle = 0 \quad (B \neq A).$$

For $B = A$, the inner product is $\|Le_A\|$, also when that column vanishes by the definition of u_A . This proves

$$\mathbb{E}_L[L^*u_A] = e_A \mathbb{E}_L\|Le_A\|.$$

The right-side identity is proved in the same way with paths reversed. Relabeling within pools shows that the two expected column norms do not depend on A .

3. Select the middle diagonal before averaging the sides.

Let $A = A(D)$ maximize $|D_{A,A}|$, using a fixed tie-breaking rule. Conditionally on D , this is a fixed separator assignment; L and R retain their independent original laws. The two diagonal identities then give

$$\begin{aligned} \mathbb{E}_{L,R}[u_A^* LDR v_A \mid D] &= \sum_B D_{B,B} \mathbb{E}_L \langle u_A, Le_B \rangle \mathbb{E}_R \langle R^* e_B, v_A \rangle \\ &= D_{A,A} \alpha_L \alpha_R. \end{aligned}$$

The sums are finite, so exchanging expectation and summation needs no additional integrability argument.

4. Compare the selected scalar test with the operator norm.

Both test vectors have norm at most one. Hence, pointwise,

$$\text{sgn}(D_{A,A}) u_A^* LDR v_A \leq \|LDR\|.$$

Conditional expectation gives $\mathbb{E}_{L,R}[\|LDR\| \mid D] \geq \alpha_L \alpha_R \max_A |D_{A,A}|$. Average over D to obtain (265). The selection uses only D ; it does not maximize independent side columns at incompatible separator assignments. $\square$

Lemma F.20 (Column scale). *With $\ell = |V(L) \setminus S|$ and $r = |V(R) \setminus S|$, $\mathbb{E}\|Le_A\|^2 = m^\ell$ and $\mathbb{E}\|R^*e_A\|^2 = m^r$, and*

$$\alpha_L \geq c_H m^{\ell/2}, \quad \alpha_R \geq c_H m^{r/2}. \quad (266)$$

Proof. In a fixed output coordinate, a pair of monomials of Le_A survives the expectation only if their primitive edge supports agree, both being square-free products of signs; every summed role has an incident edge and is identified by its color, so the supports together with the output labels identify the embedding. Isolated output roles duplicate entries over their free coordinates. There are exactly m^ℓ free assignments, each contributing one, which proves the second moment. The column is a vector of sign polynomials of degree at most e , so Lemma F.3 gives (266). A side with no role outside S has an identity register and column norm one. $\square$

Proposition F.21 (Core lower bound). *For every fixed core shape, eventually $\mathbb{E}\|M_{H,n}\| \geq c_H n^{(h-s)/2} (\log(2n))^{b_*/2}$.*

Proof. The empty shape is immediate. Otherwise, every floating component at S is attached, so Lemma F.10 gives $\mathbb{E} \max_A |D_{A,A}| \geq c_H m^{(\sum_j c_j)/2} (\log m)^{b_*/2}$, the event in (250) having probability at least $\frac{1}{2}$ eventually and $|\chi_{H[S]}(A)| = 1$; for $b_* = 0$ the maximum is one. Lemmas F.19 and F.20 with (258) give $\mathbb{E}\|M^{\text{col}}\| \geq c_H m^{(h-s)/2} (\log m)^{b_*/2}$, and Lemma F.6 applies because a core has no isolated interior role; $m \asymp_H n$. $\square$

F.8 Completion of the proof of Theorem 5.3

Original boundary-free components

Assume first that H has no isolated interior role, and remove its original boundary-free components $G_1, \dots, G_t$ to obtain the possibly empty core H_0 ; each G_ν has an edge, $c_\nu = |V(G_\nu)|$, and s, b_* are those of the core by Lemma F.4(ii). For every fixed coloring there is the exact identity

$$M^{\text{col}} = M_{H_0}^{\text{col}} \prod_{\nu=1}^t X_{G_\nu}^{\text{col}}, \quad (267)$$

where $X_{G_\nu}^{\text{col}}$ is the rootless colored scalar field of G_ν : a colored embedding of H is an embedding of H_0 together with one of each G_ν , and its monomial is the product of theirs. Under full independence the factors are independent, since their edge-role blocks are disjoint. For pools of sizes at most n , Corollary F.8 gives $\|X_{G_\nu}^{\text{col}}\|_{L_p} \leq C_{H,A} n^{c_\nu/2} \sqrt{p}$ for $1 \leq p \leq A \log(2n)$; with Proposition F.17 and independence,

$$\| \| M^{\text{col}} \| \|_{L_p} \leq C_{H,A} n^{(h-s)/2} (\log(2n))^{b_*/2} p^{t/2}, \quad 1 \leq p \leq A \log(2n), \quad (268)$$

and Lemma F.5 gives the same bound for $M_{H,n}$. For the lower bound use equal pools of size $m = \lfloor n/h \rfloor$. Distinct colored embeddings of G_ν have distinct sign supports, so $\mathbb{E}|X_{G_\nu}^{\text{col}}|^2 = m^{c_\nu}$, and Lemma F.3 gives $\mathbb{E}|X_{G_\nu}^{\text{col}}| \geq c_H m^{c_\nu/2}$. Independence in (267) and the colored estimate of Proposition F.21 give $\mathbb{E}\|M^{\text{col}}\| \geq c_H m^{(h-s)/2} (\log m)^{b_*/2}$, and Lemma F.6 transfers this to $M_{H,n}$. An original boundary-free component thus adds to the polynomial exponent and to the factor $p^{t/2}$, but not to the logarithm: it is a single scalar, not indexed by separator labels, and its typical size is what the expectation sees.

Isolated interior roles

Delete the i_0 isolated interior roles to obtain H' . By (241) the two matrices differ by the exact factor $(n - |V(H')|)_{i_0} \asymp n^{i_0}$, and s, b_* are unchanged, so the polynomial exponent becomes $(|V(H')| - s)/2 + i_0 = (h + i_0 - s)/2$. This proves the second statement of Theorem 5.3 and the expectation lower bound $\mathbb{E}\|M_{H,n}\| \geq c_H T_{H,n}$ for every shape.

Fixed positive moments

Let $Z = \|M_{H,n}\|$ and $T = T_{H,n}$. The bounds at $p = 1$ and $p = 2$ give $\mathbb{E}Z \geq c_H T$ and $\mathbb{E}Z^2 \leq C_H T^2$, so Lemma F.2 gives shape-dependent $c, d > 0$ with $\mathbb{P}\{Z \geq cT\} \geq d$ eventually, hence $\|Z\|_{L_p} \geq cd^{1/p}T$ for every fixed $p > 0$. For fixed $p \geq 1$ the upper bound is the second statement, and for $0 < p < 1$ monotonicity gives $\|Z\|_{L_p} \leq \mathbb{E}Z$. This completes the proof of Theorem 5.3. $\square$

F.9 Upper tails and limited independence

The upper half was proved at moments of logarithmic order and uses only expectations of polynomials of logarithmic degree in the signs. Two consequences, stated in Section 5.2, follow.

Polynomial tails

Fix $D > 0$ and take $p = a \log(2n)$ with $a > D$. The second statement of Theorem 5.3 gives $\|Z\|_{L_p} \leq C_{H,a} T_{H,n} (\log(2n))^{t/2}$, and Markov's inequality at e times this scale gives $\mathbb{P}\{Z > C_{H,D} T_{H,n} (\log(2n))^{t/2}\} \leq e^{-p} = (2n)^{-a} \leq n^{-D}$ eventually. For a core shape, $t = 0$ and the threshold is the expected scale itself; for $t > 0$ the extra power of the logarithm is necessary, since already the rootless single-edge scalar $2 \sum_{i < j} W_{ij}$ exceeds any fixed multiple of its standard deviation with probability bounded away from zero.

Limited independence

Unbiased k -wise independent signs have the same expectation as independent signs for every polynomial of total degree at most k : reduce powers by $W_a^2 = 1$ and apply independence to each square-free monomial. The trace polynomial $\text{tr}[(M^{\text{col}}(M^{\text{col}})^*)^q]$ has degree at most $2eq$, so if $2eq \leq k_n$ its expectation equals the fully independent one and (262) transfers exactly; with q a sufficiently large fixed multiple of $\log(2n)$ this gives the core bound (261) for $p \leq A \log(2n)$ whenever $k_n \geq K_{H,A} \log(2n)$. For a shape with boundary-free components the factors of (267) may now be dependent; Hölder gives $\|M^{\text{col}}\|_{L_p} \leq \|M_{H_0}^{\text{col}}\|_{L_{(t+1)p}} \prod_{\nu} \|X_{G_{\nu}}^{\text{col}}\|_{L_{(t+1)p}}$, each scalar factor is transferred at an even order $2r$ with $(t+1)p \leq 2r < (t+1)p + 2$ through the polynomial $(X_{G_{\nu}}^{\text{col}})^{2r}$ of degree $2r|E(G_{\nu})|$, and the color average (242) is an identity for every sign realization. Hence the second statement of Theorem 5.3 holds for unbiased k_n -wise independent signs with $k_n \geq K_{H,A} \log(2n)$. The lower bound does not transfer by this argument: the excursion lemma conditions on some signs and uses the independence of the others, and the Fourier projection uses the invariance of the law under block flips.

G The deletion assembly for regular entries: detailed verifications

Section 5.4 proves the upper half of Theorem 5.6 from the local theorem (Theorem 4.11), the scalar comparisons of Section 3.6, and a deterministic deletion argument. The argument there is complete except for six elementary verifications, which are collected here so that the proof can be read without any external source: the size and screening properties of the schedule, the partition into five masks, the monotonicity of weak moments under deletion and under principal restriction, the maximum of finitely many variables with moment control, the centered Gram estimate for the far remainder, and the symmetrization step. Throughout, Y is a real $n \times n$ matrix with independent entries symmetric in law, satisfying (7), $M = M(Y)$, $D = D(Y)$, $v_{ij} = \mathbb{E}Y_{ij}^2$, $\sigma_r, \sigma_c, \sigma$ and $R_{Y,I}$ are as in (8)–(9), and $M > 0$.

G.1 The schedule

Recall from Section 5.4 that $b_k = 2^{5^k} - 1$, $\pi_k = \log(b_k + 1) = 5^k \log 2$, $t_k = b_k^3$, that I_k is a deterministic minimizer of $I \mapsto R_{Y,I}(\log(k+1))$ over $|I| \leq k$ for $k < n$, with $I_k = [n]$ for $k \geq n$, and that $S_0 = I_1$ and S_{k+1} is obtained from S_k by adding every row i with $(t_k + 1)v_{ij} > M^2$ for some column $j \in S_k$, every column j with $(t_k + 1)v_{ij} > M^2$ for some row $i \in S_k$, and the set I_{b_k} . Since one and the same index set is deleted from both axes, “row i ” and “column i ” refer to the same element $i \in [n]$.

Lemma G.1 (Sizes and screening). (i) $b_{k+1}+1 = (b_k+1)^5$, and $|S_k| \leq b_k$ for every $k \geq 0$; moreover $S_k = [n]$ as soon as $b_{k-1} \geq n$, in particular $S_{n+1} = [n]$.

(ii) If $j \in S_r$ and $i \notin S_{r+1}$, then $v_{ij} \leq M^2/(b_r^3 + 1)$ and $v_{ji} \leq M^2/(b_r^3 + 1)$.

(iii) $R_{Y, S_{a-1}}(\pi_{(a-2)+}) \leq D$ for every $a \geq 1$.

Proof. **1. Size and eventual saturation**

The first identity is the definition. For a fixed j , $\sum_i v_{ij} \leq \sigma_c^2 \leq M^2$, so fewer than $t_k + 1$ rows i satisfy $(t_k + 1)v_{ij} > M^2$; the same holds for columns. Hence $|S_{k+1}| \leq |S_k| + 2t_k|S_k| + |I_{b_k}| \leq 2(1 + t_k)|S_k| + b_k$. By induction, $|S_0| \leq 1 = b_0$, and if $|S_k| \leq b_k$ then $|S_{k+1}| \leq 2b_k + 2b_k^4 + b_k \leq b_{k+1}$, because $b_{k+1} = (b_k + 1)^5 - 1 \geq 5b_k^4 + 5b_k \geq 2b_k^4 + 3b_k$. If $b_{k-1} \geq n$ then $I_{b_{k-1}} = [n] \subseteq S_k$.

2. Variance screening

Row i was not added at stage r although $j \in S_r$, so $(t_r + 1)v_{ij} \leq M^2$; the transposed statement is the column rule.

3. The available deletion moment

For $a = 1$, $S_0 = I_1$ and $\pi_0 = \log 2$, so $R_{Y, S_0}(\log 2) = \min_{|I| \leq 1} R_{Y, I}(\log 2) \leq D$ by (10). For $a \geq 2$, $S_{a-1} \supseteq I_{b_{a-2}}$ and $\pi_{a-2} = \log(b_{a-2} + 1)$; if $b_{a-2} < n$ then $R_{Y, I_{b_{a-2}}}(\pi_{a-2}) \leq D$ by (10), and if $b_{a-2} \geq n$ then $I_{b_{a-2}} = [n]$ and the weak moment vanishes. Enlarging the deleted set can only decrease $R_{Y, I}(p)$, since the supremum in (9) is then taken over fewer coordinates (Lemma G.3(i)). $\square$

G.2 The five masks

Let ℓ_i be the first r with $i \in S_r$, so that $\{i : \ell_i \leq r\} = S_r$ has at most b_r elements, and consider, for a pair $(\ell_i, \ell_j) = (a, b)$ of levels, the five conditions

$$\begin{aligned} \text{nearLow: } a, b \leq 1; \quad \text{nearHigh: } \lfloor a/2 \rfloor = \lfloor b/2 \rfloor \geq 1; \\ \text{nearCross: } \lfloor (a+1)/2 \rfloor = \lfloor (b+1)/2 \rfloor, a \neq b; \quad \text{far: } b+1 < a; \quad \text{far}^\top: a+1 < b. \end{aligned}$$

Lemma G.2 (The mask partition). *Exactly one of the five conditions holds for every pair of levels. The nearHigh mask is block diagonal with blocks $V_a = \{i : \ell_i \in \{a, a+1\}\}$, $a = 2m$, $m \geq 1$; the nearCross mask is block diagonal with blocks V_a , $a = 2m-1$, $m \geq 1$, restricted to the pairs with $\ell_i \neq \ell_j$; and $|V_a| \leq b_{a+1}$. Moreover, for the symmetric sign matrix $\Sigma_a = \text{diag}(\pm 1)$ with $+1$ on the shell $\{\ell_i = a\}$ and -1 on the shell $\{\ell_i = a+1\}$,*

$$Y[V_a, V_a] \mathbf{1}\{\ell_i \neq \ell_j\} = \frac{1}{2}(Y[V_a, V_a] - \Sigma_a Y[V_a, V_a] \Sigma_a), \quad (269)$$

so its norm is at most $\|Y[V_a, V_a]\|$ for every realization.

Proof. If $|a-b| \geq 2$ exactly one far condition holds, and no near condition does, since each near condition forces $|a-b| \leq 1$. If $a = b$, then nearLow holds when $a \leq 1$ and nearHigh when $a \geq 2$, and nearCross never. If $b = a+1$: for $a = 0$ only nearLow holds; for even $a \geq 2$, nearHigh holds and nearCross fails since $\lfloor (a+1)/2 \rfloor = a/2 < a/2 + 1 = \lfloor (a+2)/2 \rfloor$; for odd $a \geq 1$, nearHigh fails since $\lfloor a/2 \rfloor < \lfloor (a+1)/2 \rfloor$ and nearCross holds since $\lfloor (a+1)/2 \rfloor = (a+1)/2 = \lfloor (a+2)/2 \rfloor$; in each case the far conditions fail. The case $a = b+1$ is symmetric. The block descriptions restate the two near conditions, and $V_a \subseteq S_{a+1}$ gives $|V_a| \leq b_{a+1}$. Conjugation by Σ_a preserves the entries with $\ell_i = \ell_j$ and reverses the sign of the others, which proves (269); the norm bound follows from $\|\Sigma_a Y \Sigma_a\| = \|Y\|$. $\square$

G.3 Weak moments of deleted and restricted matrices

- Lemma G.3** (Monotonicity). (i) If $I \subseteq I'$ then $R_{Y,I'}(p) \leq R_{Y,I}(p)$ for every $p > 0$.
(ii) If $V \subseteq [n]$ and $V \cap I = \emptyset$, then $R_{Y[V,V]}(p) \leq R_{Y,I}(p)$ for every $p > 0$, where the left side is the weak moment (9) of the principal submatrix.
(iii) For every deterministic I , $R_{Y,I}(2)^2 = \max_{i,j \notin I} v_{ij} \leq \min\{\sigma_r, \sigma_c\}^2 \leq (M/2)^2$, where the maximum over an empty set is defined to be zero.

Proof. (i) and (ii): a bilinear form $\sum_{i,j \in V} Y_{ij} s_i t_j$ with unit vectors supported on V , or with $I' \supseteq I$ deleted, is one of the forms admitted in $R_{Y,I}(p)$, so the suprema are nested. (iii) By independence and centering, $\mathbb{E}(\sum_{i,j \notin I} Y_{ij} s_i t_j)^2 = \sum_{i,j \notin I} v_{ij} s_i^2 t_j^2$, a bilinear form in the probability vectors $(s_i^2), (t_j^2)$, whose supremum is attained at vertices and equals $\max_{i,j \notin I} v_{ij}$; every v_{ij} is at most σ_r^2 and at most σ_c^2 , and $\min\{\sigma_r, \sigma_c\} \leq (\sigma_r + \sigma_c)/2 = M/2$. $\square$

Part (ii) is what carries the deletion parameter to the blocks $a \geq 3$: $V_a \cap S_{a-1} = \emptyset$ because $\ell_i \geq a$ on V_a , so the weak moment of the block at every order is at most $R_{Y,S_{a-1}}$ at that order, which Lemma G.1(iii) bounds by D at the order π_{a-2} . Part (iii) carries the variance to the blocks $a \in \{1, 2\}$. In both cases the order is then raised to P_a by (79), exactly as displayed in (128) and (129).

G.4 The maximum over the blocks of a mask

Lemma G.4 (Maximum with moment control). Let $Z_1, Z_2, \dots$ be finitely many nonnegative random variables with $\|Z_i\|_{L_{p_i}} \leq L$ and $p_i > 1$. Then

$$\mathbb{E} \max_i Z_i \leq L \left(1 + \sum_i \frac{1}{p_i - 1} \right),$$

with no independence assumption.

Proof. If $L = 0$, each variable is zero almost surely. Otherwise integrate the upper tail, split at L , and apply a union bound:

$$\begin{aligned} \mathbb{E} \max_i Z_i &\leq L + \int_L^\infty \mathbb{P}\{\max_i Z_i > t\} dt \\ &\leq L + \sum_i \int_L^\infty \mathbb{P}\{Z_i > t\} dt \\ &\leq L + \sum_i \int_L^\infty (L/t)^{p_i} dt = L \left(1 + \sum_i \frac{1}{p_i - 1} \right). \end{aligned}$$

The third line is Markov's inequality. Convergence uses $p_i > 1$; independence is not used at any step. $\square$

In Section 5.4 the lemma is applied with $Z_a = \|Y[V_a, V_a]\|$, or its nearCross restriction, $p_a = P_a$ and the common bound L of (131); since $P_a - 1 \geq (50c \log 2 - 1)5^{a-1}$ and each $a \geq 1$ occurs in exactly one mask, the sum over both masks is at most $2 + \sum_{a \geq 1} (50c \log 2 - 1)^{-1} 5^{1-a} = N_c$, the constant of (132). The nearLow block is a single matrix and needs no maximum.

G.5 The far remainder

Lemma G.5 (Centered Gram estimate). *Let A be the lower far mask, $A_{ij} = Y_{ij}\mathbf{1}\{\ell_j+1 < \ell_i\}$, and $w_{ij} = \mathbb{E}A_{ij}^2$. Then*

$$\sum_{j,k} \sum_i w_{ij}w_{ik} \leq 2M^4, \quad \mathbb{E}\|A\| \leq M\sqrt{1 + \sqrt{2}\alpha^2}. \quad (270)$$

The transpose of the same argument bounds the upper far mask.

Proof. 1. Use the level separation to control pairs of columns.

Fix an ordered pair (j, k) and, interchanging their names if needed, assume $\ell_j \leq \ell_k =: r$. If $A_{ik} \neq 0$, then $\ell_i > r + 1$. Thus $i \notin S_{r+1}$ and $j \in S_r$, so Lemma G.1(ii) gives

$$w_{ij} \leq v_{ij} \leq \frac{M^2}{b_r^3 + 1}.$$

It follows that

$$\sum_i w_{ij}w_{ik} \leq \frac{M^2}{b_r^3 + 1} \sum_i w_{ik} \leq \frac{M^4}{b_r^3 + 1}.$$

There are at most b_r^2 ordered column pairs with maximum level r . Consequently

$$\sum_{j,k} \sum_i w_{ij}w_{ik} \leq M^4 \sum_{r \geq 0} \frac{b_r^2}{b_r^3 + 1} < 2M^4.$$

For the last inequality, the $r = 0$ term is $1/2$, the $r = 1$ term is $31^2/(31^3 + 1)$, and the remaining terms are bounded by $\sum_{r \geq 2} b_r^{-1}$.

2. Center the Gram and bound its Frobenius second moment.

Set

$$G = A^\top A - \text{diag} \left(\sum_i w_{ij} \right).$$

For fixed j, k , the summands in $G_{jk} = \sum_i (A_{ij}A_{ik} - \mathbb{E}A_{ij}A_{ik})$ are independent and centered across i . Therefore

$$\mathbb{E}G_{jk}^2 \leq \sum_i \mathbb{E}(A_{ij}^2 A_{ik}^2).$$

For $j \neq k$, independence within a row makes this $\sum_i w_{ij}w_{ik}$. For $j = k$, regularity gives $\mathbb{E}A_{ij}^4 \leq \alpha^4 w_{ij}^2$. Since $\alpha \geq 1$,

$$\mathbb{E}\|G\|_F^2 = \sum_{j,k} \mathbb{E}G_{jk}^2 \leq \alpha^4 \sum_{j,k} \sum_i w_{ij}w_{ik} \leq 2\alpha^4 M^4.$$

No independence among the different entries of G is asserted or needed.

3. Convert back to the expected operator norm.

Each deterministic diagonal entry $\sum_i w_{ij}$ is at most M^2 . For every realization,

$$\|A\|^2 = \|A^\top A\| \leq M^2 + \|G\| \leq M^2 + \|G\|_F.$$

Cauchy–Schwarz first over A , then over G , yields

$$\mathbb{E}\|A\| \leq \left(M^2 + \mathbb{E}\|G\|_F \right)^{1/2} \leq \left(M^2 + (\mathbb{E}\|G\|_F^2)^{1/2} \right)^{1/2} \leq M\sqrt{1 + \sqrt{2}\alpha^2}.$$

Transposition gives the upper-far estimate with the same constant. $\square$

No independence among the entries of G is used, and no general matrix inequality; this is the elementary replacement, in the present version, for the variance-envelope theorem used in the earlier version of the assembly.

G.6 Symmetrization

Lemma G.6 (Symmetrization). *Let X have independent centered entries with finite absolute moments satisfying (7) with the constant α , and let $Y = X - X'$ with X' an independent copy. Then Y has independent entries symmetric in law satisfying (7) with the constant 2α , $M(Y) = \sqrt{2}M(X)$, $\mathbb{E}\|X\| \leq \mathbb{E}\|Y\|$, and*

$$M(Y) + D(Y) \leq 2\{M(X) + D(X)\}. \quad (271)$$

Proof. 1. Transfer the law, variance, and expected norm.

The entries of $Y = X - X'$ are independent and symmetric in law. Since X is centered,

$$\mathbb{E}Y_{ij}^2 = 2\mathbb{E}X_{ij}^2, \quad M(Y) = \sqrt{2}M(X).$$

Conditional Jensen gives $\mathbb{E}\|X\| = \mathbb{E}\|\mathbb{E}[Y \mid X]\| \leq \mathbb{E}\|Y\|$. For $p \geq 1$, Minkowski, regularity, and conditional Jensen also give

$$\|Y_{ij}\|_{L_{2p}} \leq 2\|X_{ij}\|_{L_{2p}} \leq 2\alpha\|X_{ij}\|_{L_p} \leq 2\alpha\|Y_{ij}\|_{L_p}.$$

Thus Y has regularity constant 2α .

2. Treat the two weak-moment ranges separately.

For each deterministic deleted set I , apply the same argument to each bilinear form and then take the supremum. Subadditivity of x^p below one, and Minkowski at and above one, give respectively

$$R_{Y,I}(p) \leq \begin{cases} 2^{1/p}R_{X,I}(p), & 0 < p < 1, \\ 2R_{X,I}(p), & p \geq 1. \end{cases}$$

The budgets $k \geq 2$ in D use orders $\log(k+1) > 1$, so their contribution is at most $2D(X)$. Only budget $k = 1$, at order $\log 2$, has the larger factor $2^{1/\log 2} = e$.

3. Use variance as an alternative at budget one.

Write

$$u_1 = \min_{|I| \leq 1} R_{Y,I}(\log 2).$$

The preceding estimate gives $u_1 \leq eD(X)$. Monotonicity in the moment order and Lemma G.3(iii) give another estimate:

$$u_1 \leq R_{Y,\emptyset}(2) \leq \frac{M(Y)}{2} = \frac{M(X)}{\sqrt{2}}.$$

A convex combination of these two upper bounds, with weights $2/e$ and $1 - 2/e$, yields

$$u_1 \leq 2D(X) + \frac{1 - 2/e}{\sqrt{2}}M(X).$$

The same right side bounds every other budget, because those are already at most $2D(X)$.

4. Add the variance term.

Therefore

$$M(Y) + D(Y) \leq \frac{3 - 2/e}{\sqrt{2}}M(X) + 2D(X) \leq 2\{M(X) + D(X)\},$$

since $(3 - 2/e)/\sqrt{2} < 2$. This proves the claimed bound without using a triangle inequality for a subunit moment. $\square$

G.7 Assembly

The preceding lemmas close the upper proof in Section 5.4. The estimates are used in the following order.

1. Decompose the original matrix, without changing its law.

Lemma G.2 partitions its entries into five masks. The nearLow block has at most 31 indices and contributes at most $\sqrt{31}M$ by the Frobenius second-moment bound.

2. Estimate each larger near block at its scheduled order.

Use Theorem 4.11 with moment order P_a from (126). The size bound $|V_a| \leq b_{a+1}$ gives the factor $e^{1/(2c)}$, and the row and column variance scales are at most M . For $a = 1, 2$, start the weak-moment comparison from the exact second-moment bound in Lemma G.3(iii). For $a \geq 3$, start from the deletion moment at order π_{a-2} , using Lemma G.1(iii) and principal-restriction monotonicity. The anchor comparison raises these input orders to P_a , producing (128), (129), and then the common bound (131).

3. Control maxima and the two far masks.

Lemma G.4 sums the upper tails at their actual orders; the two near-mask maxima together cost $N_c L$, not two unqualified applications of a constant-3 maximum bound. Lemma G.5 bounds each far mask by $M\sqrt{1 + \sqrt{2}\alpha^2}$. No independence between the mask norms is needed in the final triangle inequality.

4. Combine the coefficients and symmetrize.

These contributions give (133), retaining separate coefficients of M and D . Appendix D verifies the tail comparisons and $A_1 < C_c$ for each stated policy. Finally, Lemma G.6 transfers the symmetric-law upper estimate to centered entries.

The lower half is Theorem 4.1 of Latała and Świątkowski [48]. It is an external result used with attribution; none of the preceding upper-bound computations is a replacement proof of that lower bound.

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